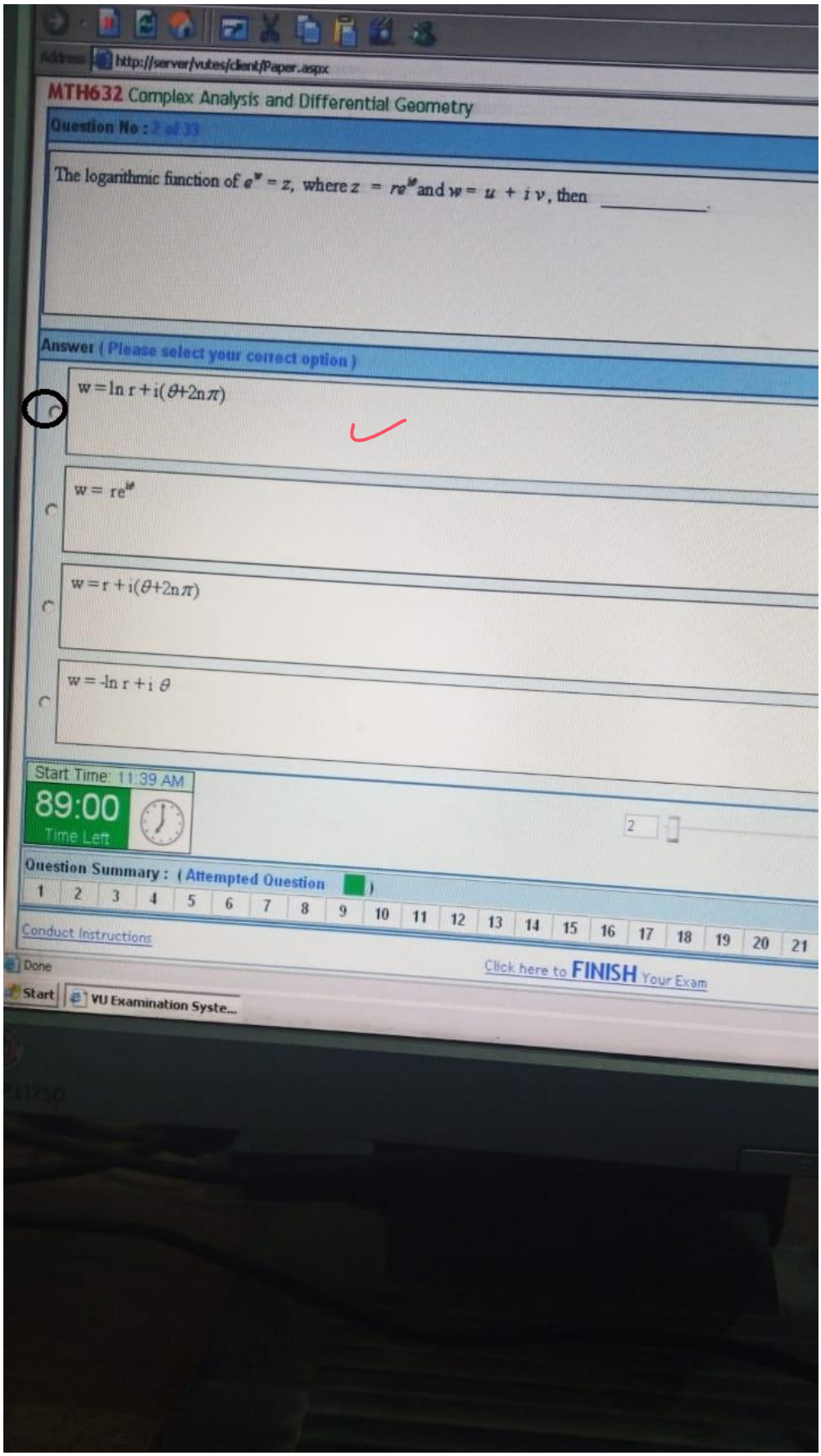
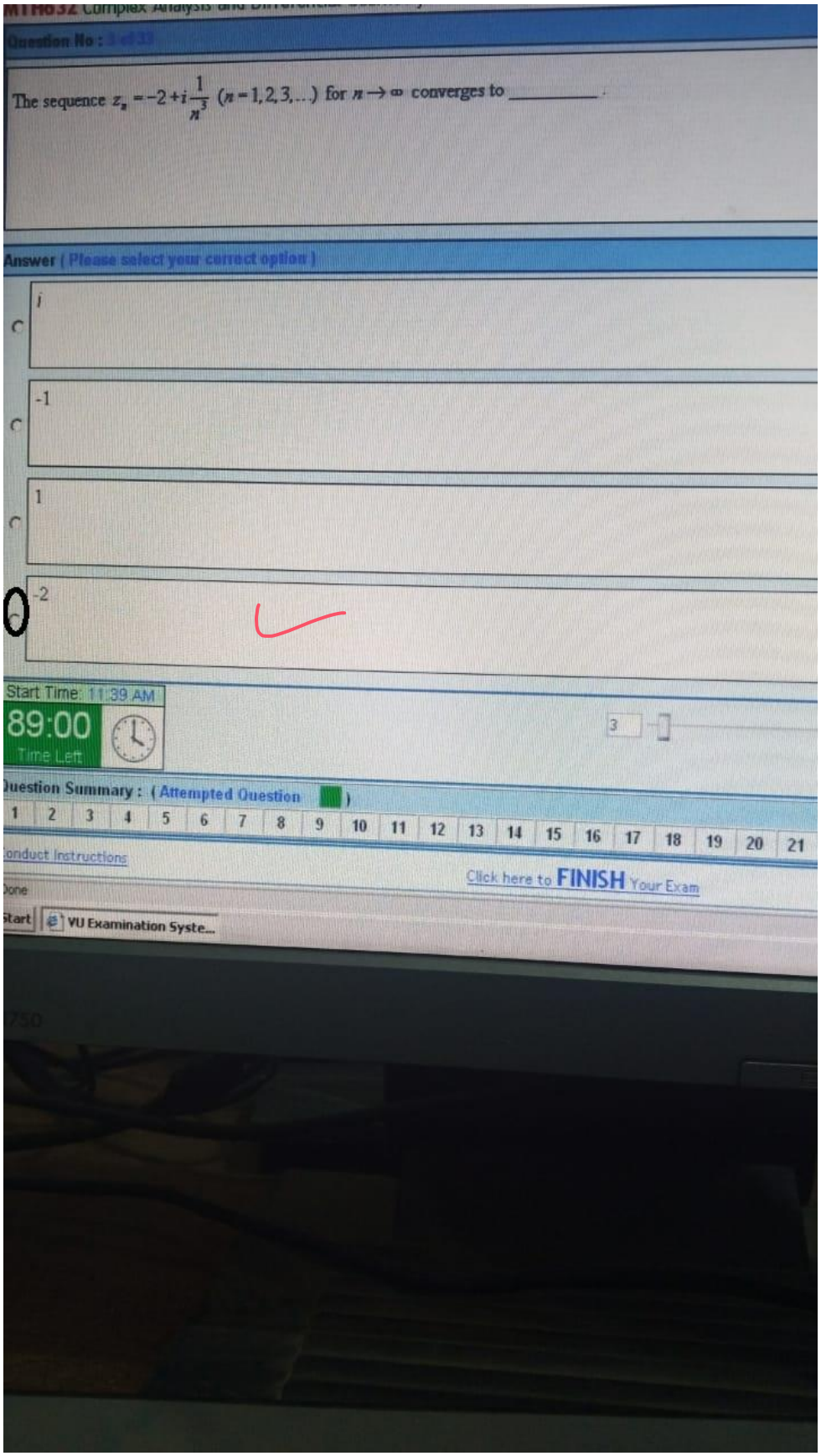
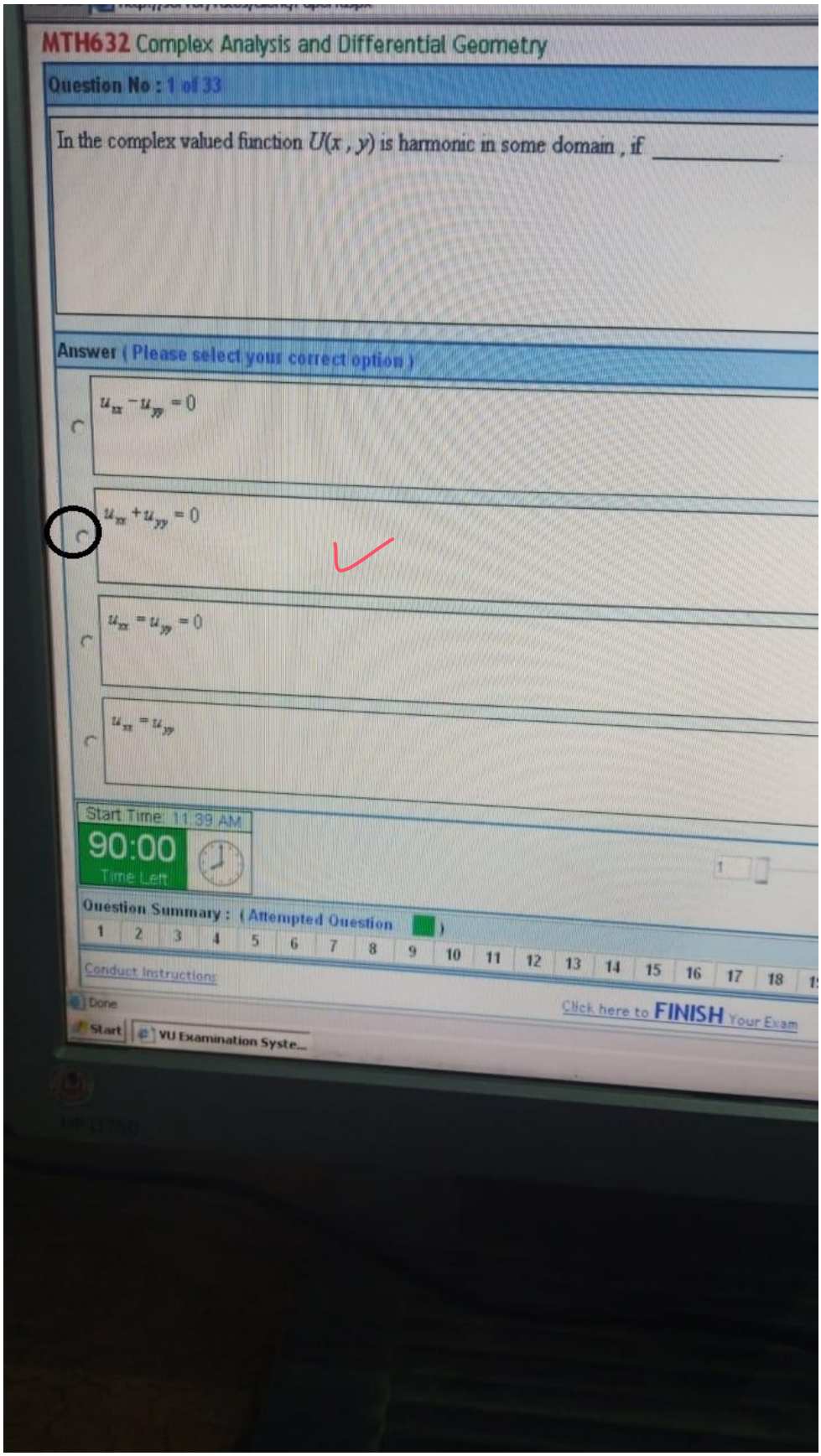
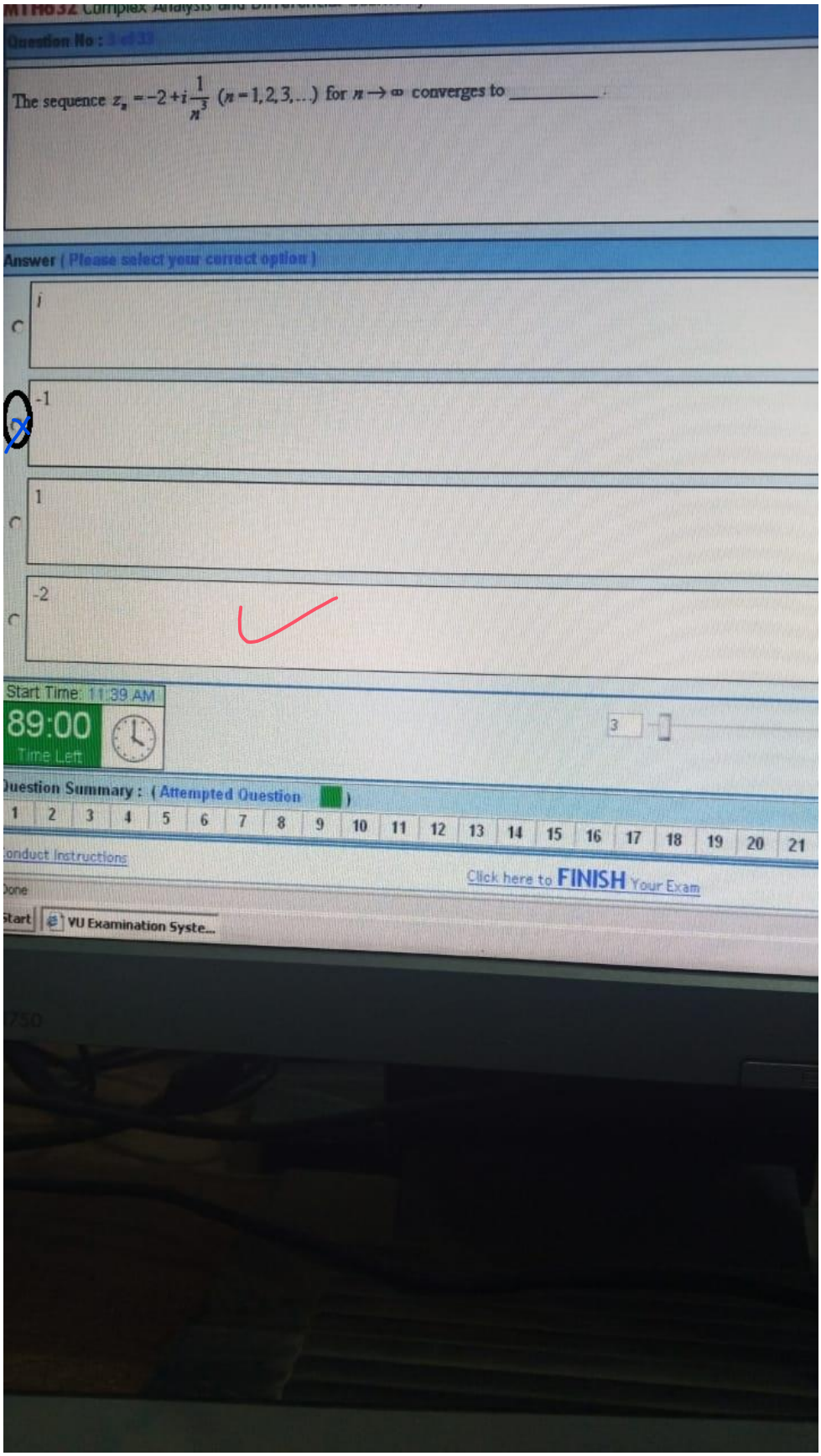
MTH632

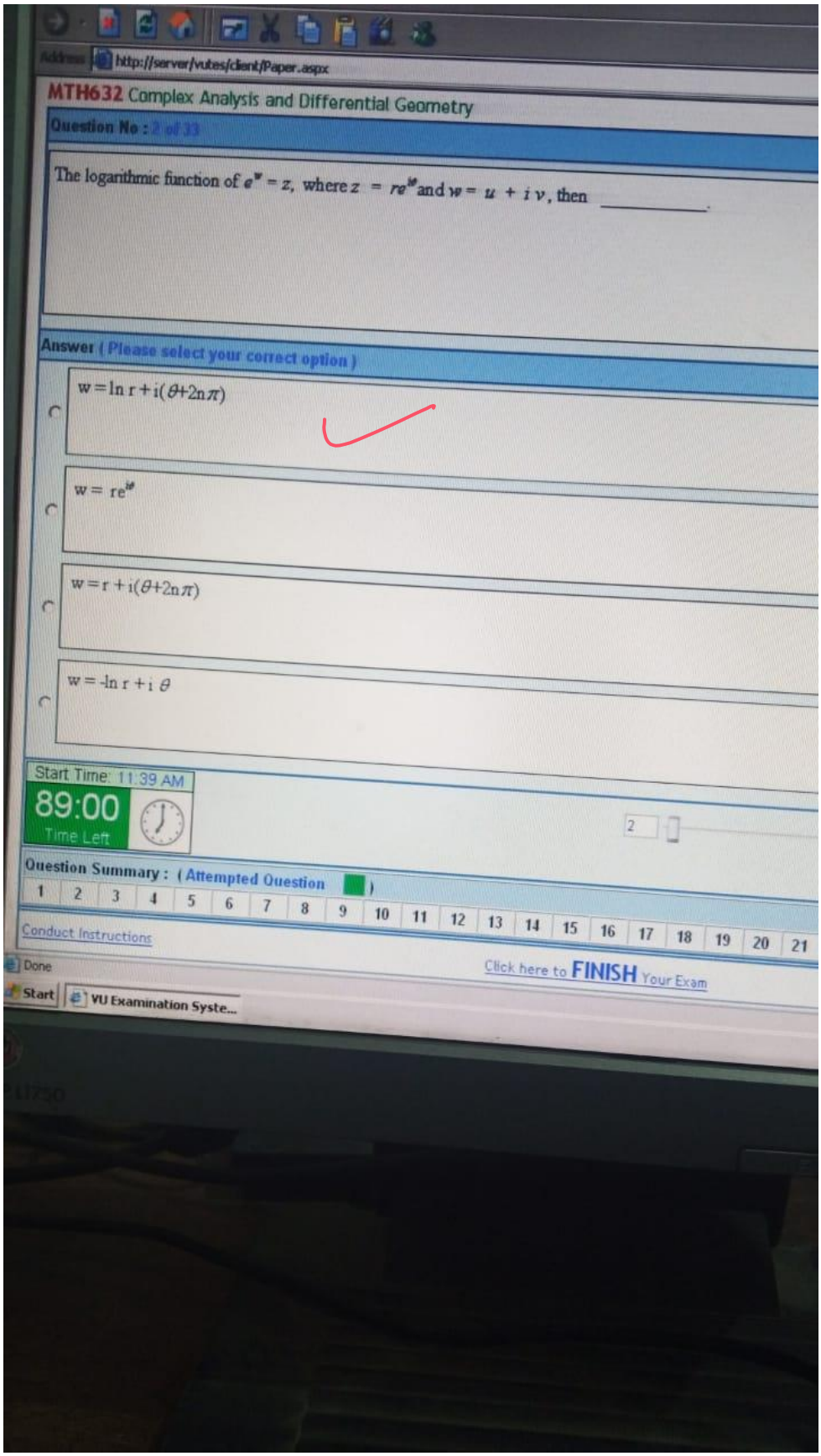
6-3-21

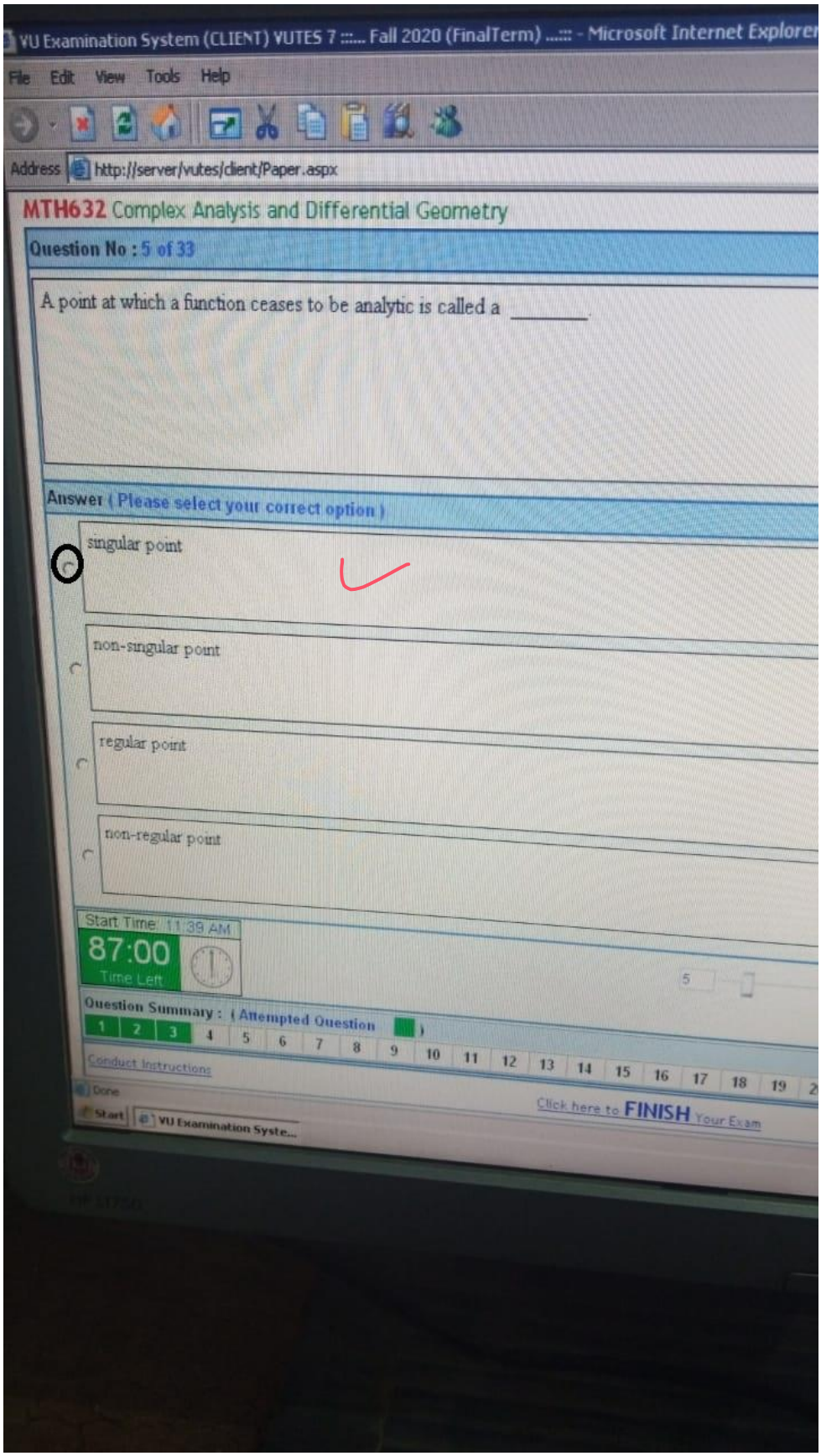

http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\,\theta$
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

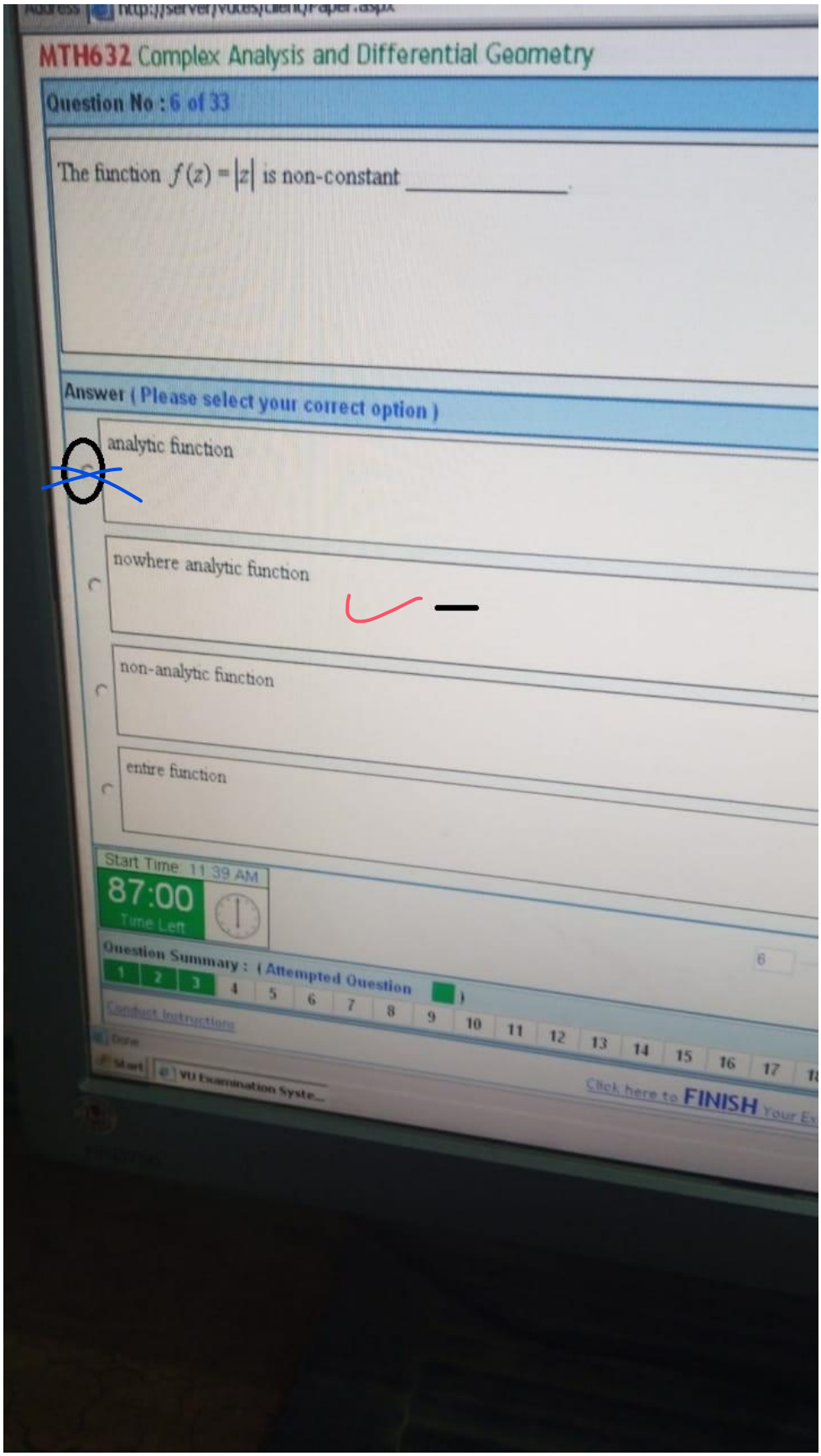

Question No :
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

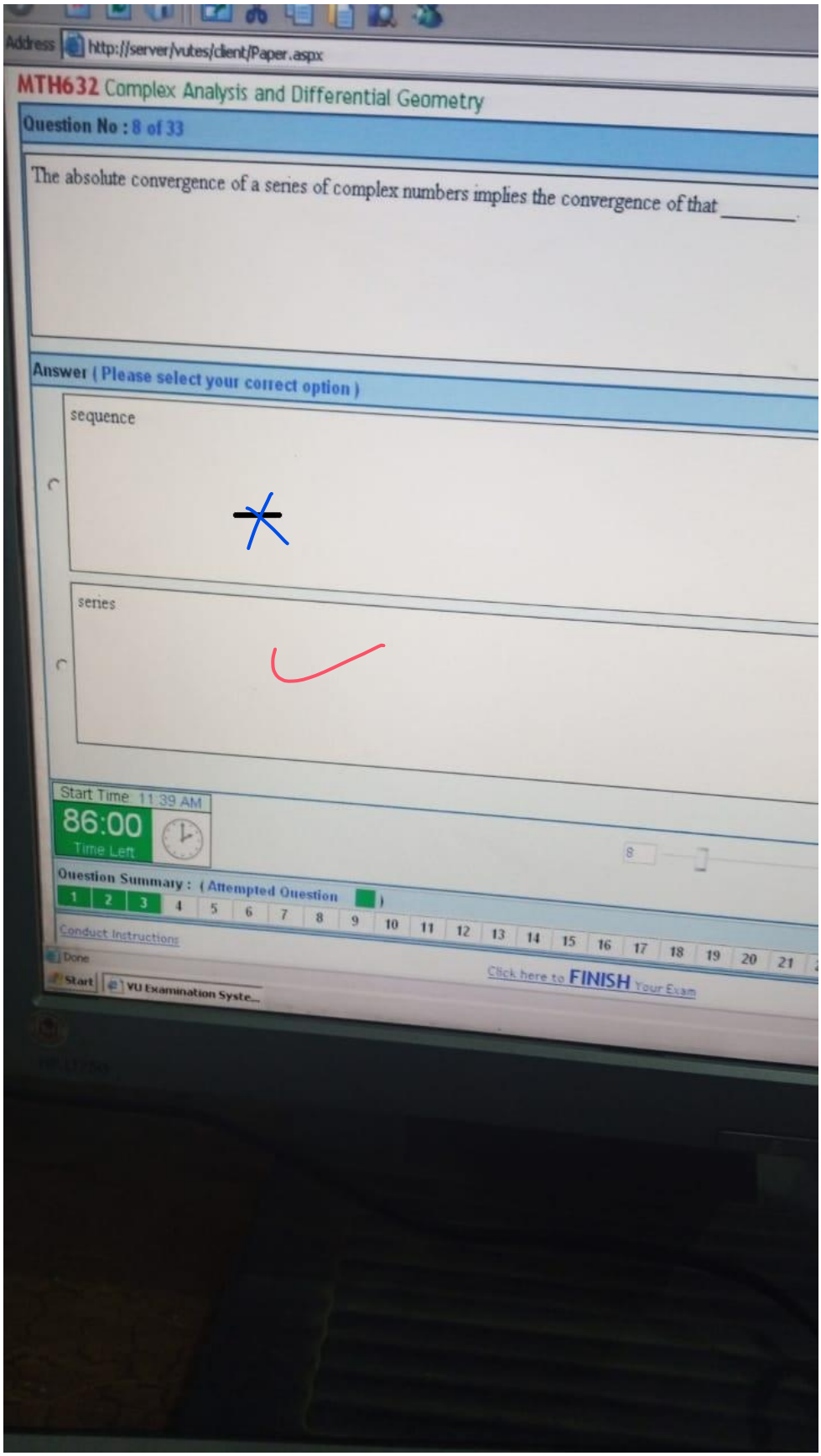

MTH632 Complex Analysis and Differential Geometry
Question No : 1 of 33
In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.
Answer (Please select your correct option)
$u_{xx} - u_{yy} = 0$
$u_{xx} + u_{yy} = 0$
$u_{xx} = u_{yy} = 0$
$u_{xx} = u_{yy}$
Start Time: 11:39 AM
90:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instruction
Done
Click here to FINISH Your Exam
Start
VU Examination Syste...

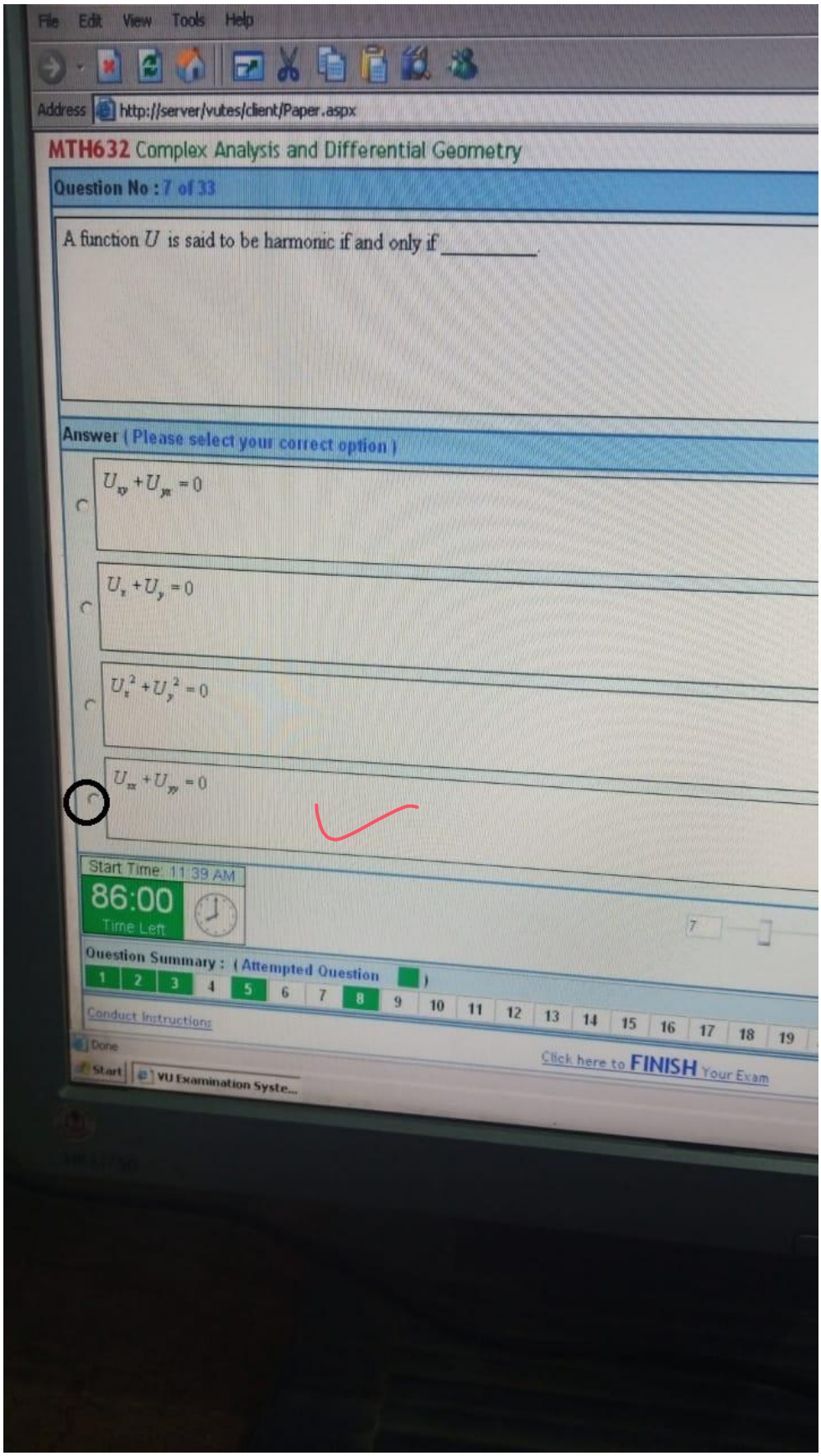

Question No :
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

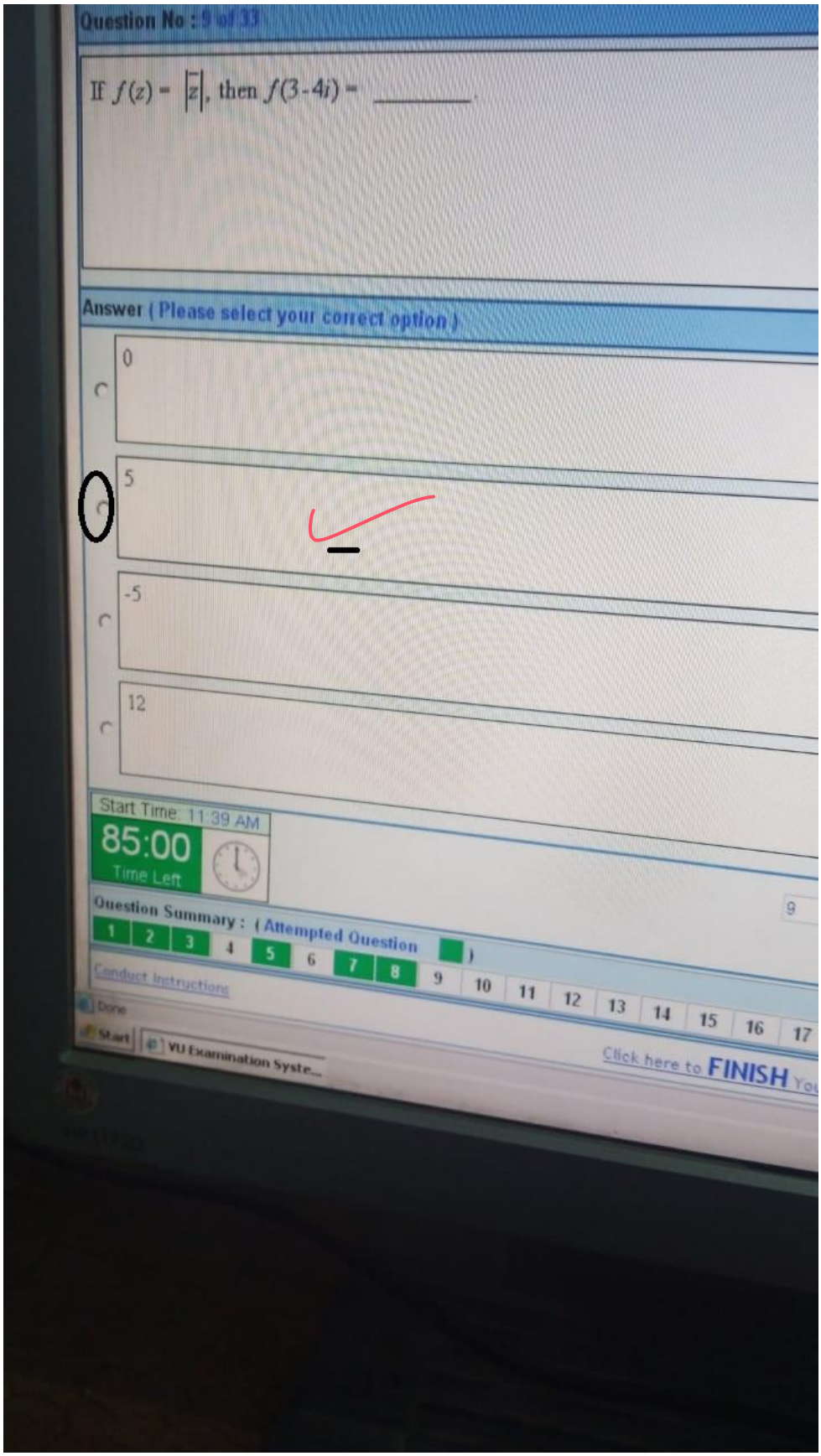

http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\,\theta$
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

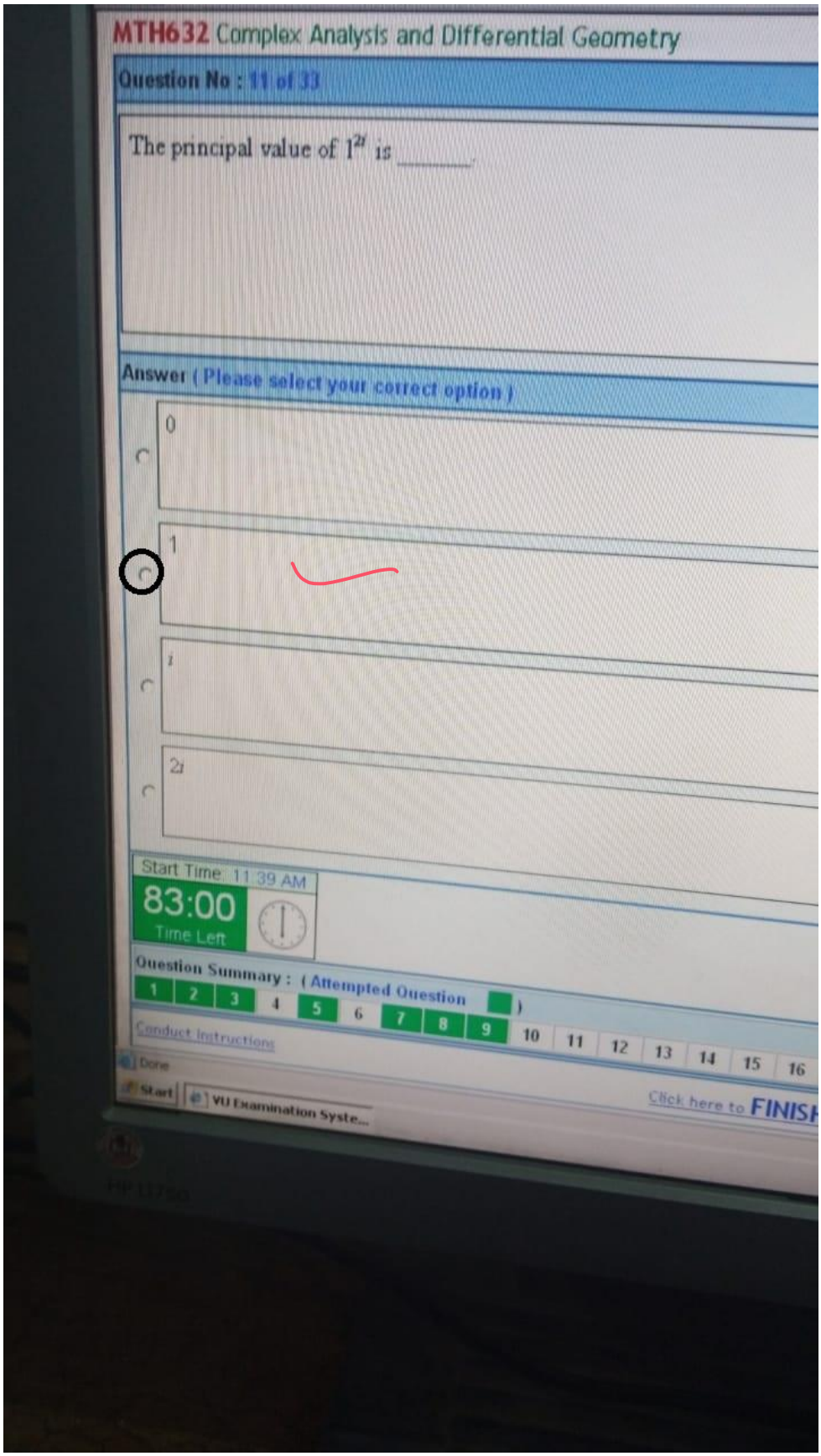

VU Examination System (CLIENT) VUTES 7 ::::: Fall 2020 (FinalTerm) ....:: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
A point at which a function ceases to be analytic is called a _____.
Answer (Please select your correct option)
singular point
non-singular point
regular point
non-regular point
Start Time 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
Done
Click here to FINISH Your Exam
Start
VU Examination Syste...

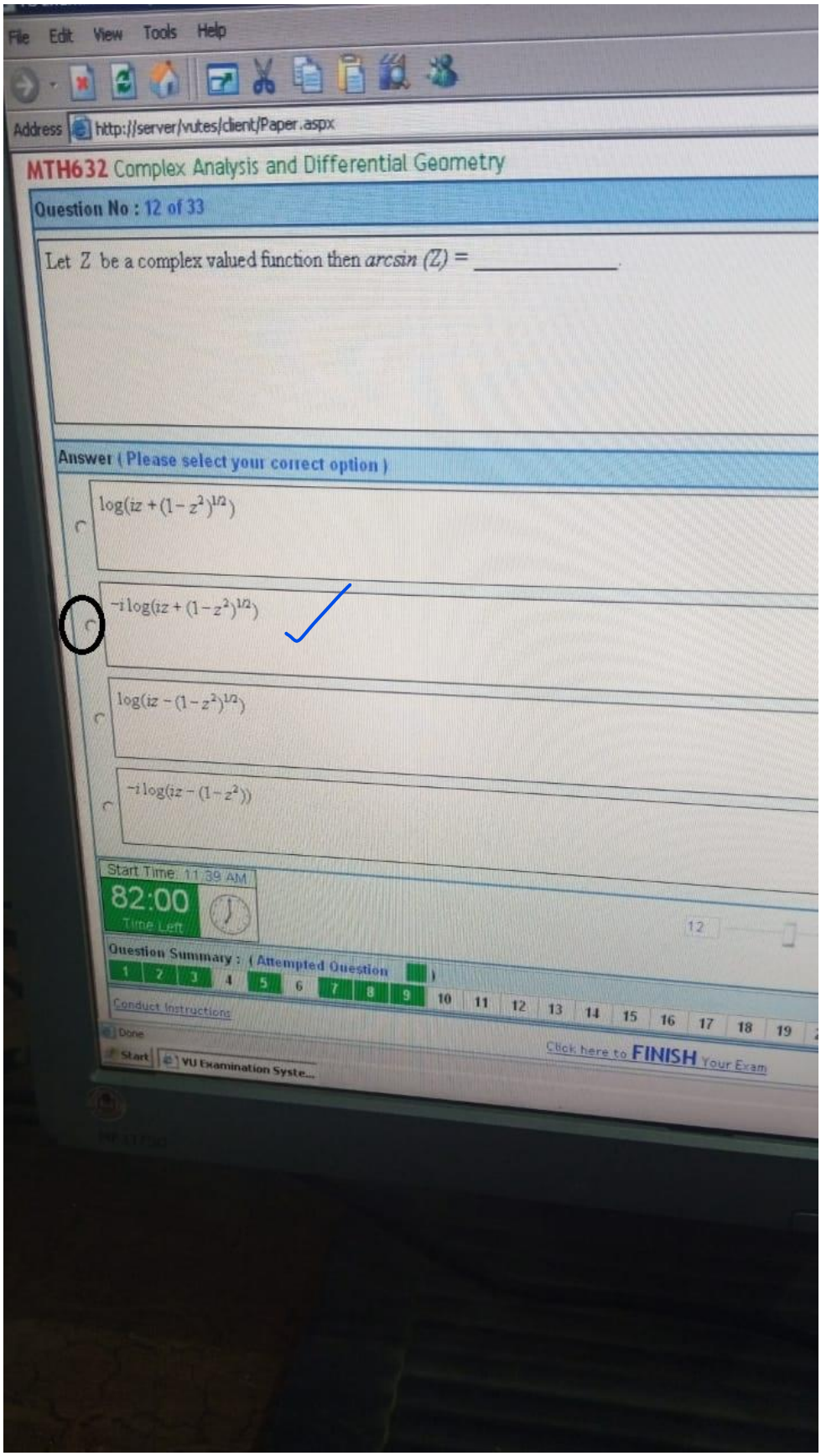

MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH

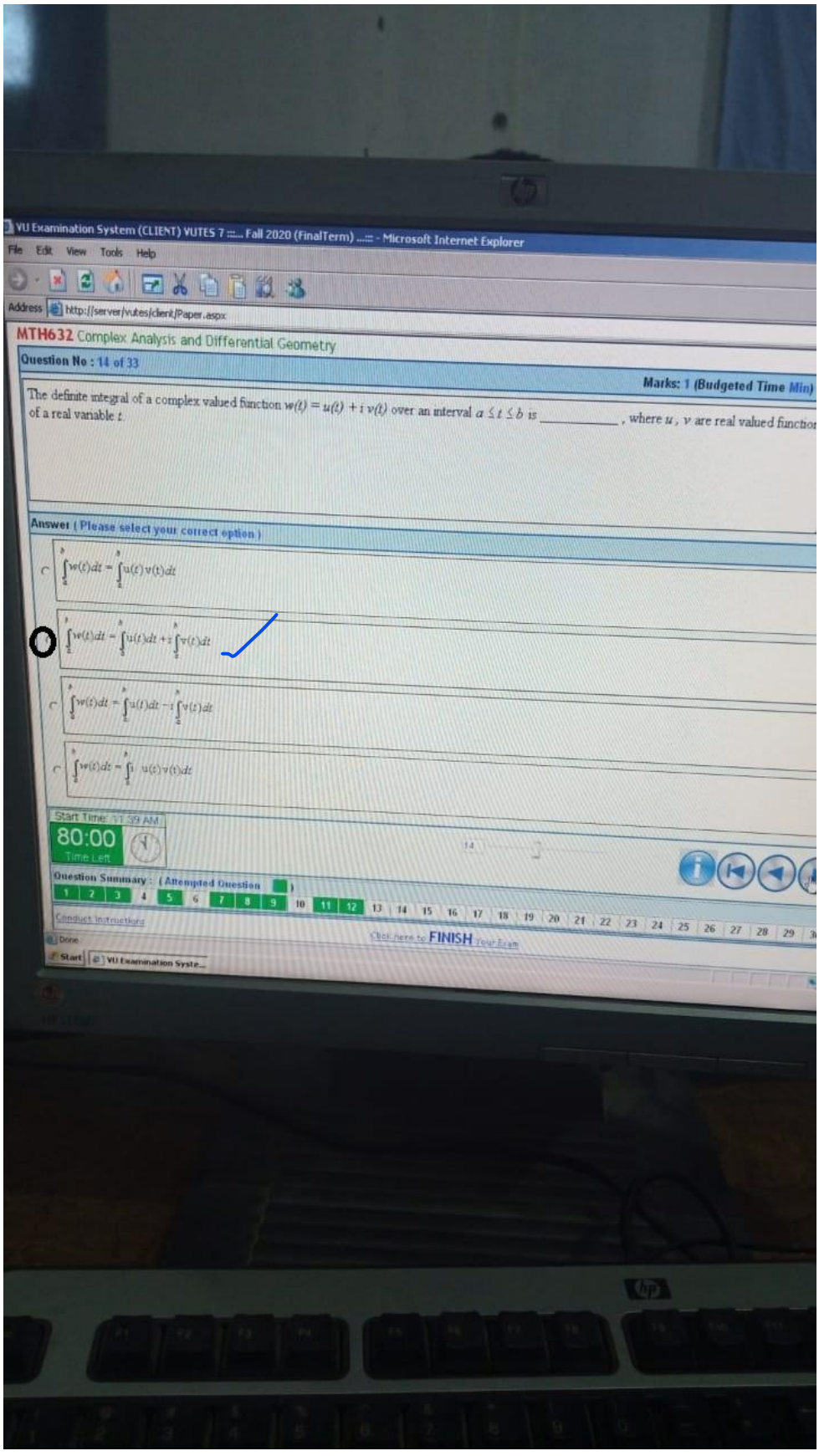

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
The absolute convergence of a series of complex numbers implies the convergence of that _____.
Answer (Please select your correct option)
sequence
series
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

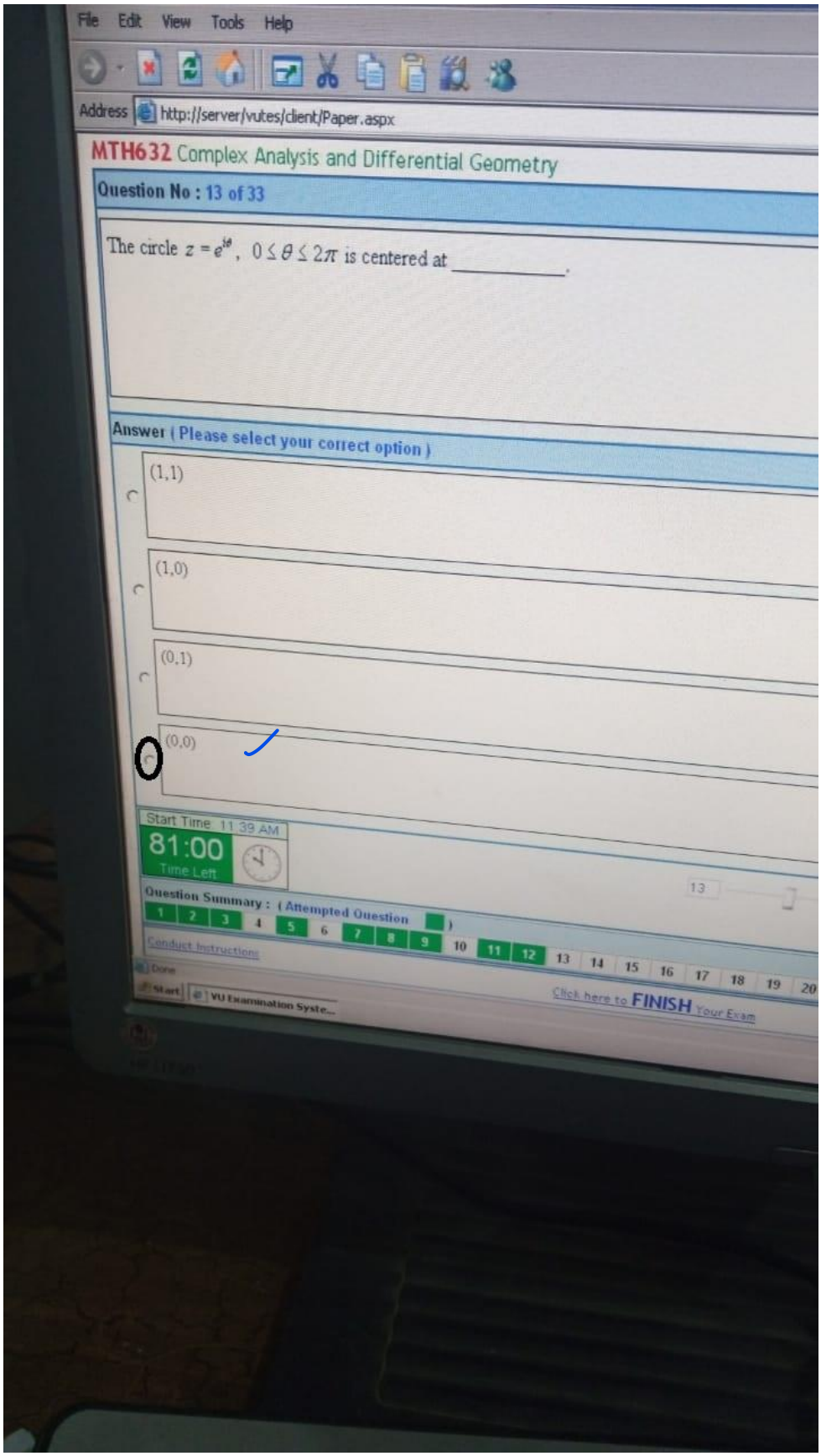

File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 7 of 33
A function U is said to be harmonic if and only if _____.
Answer (Please select your correct option)
$U_{xy} + U_{yx} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

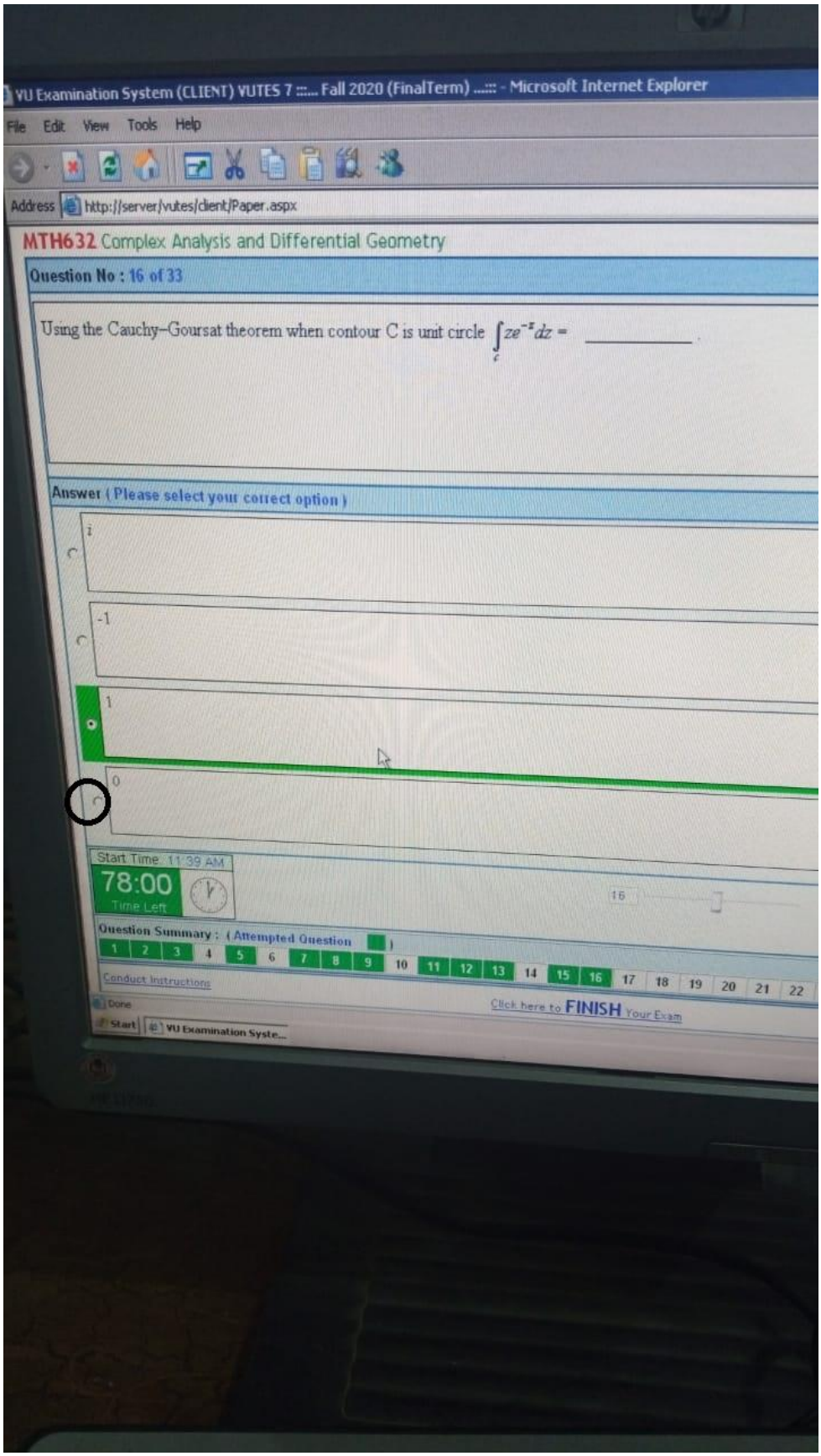

Question No : 9 of 33
If $f(z) = |\bar{z}|$, then $f(3-4i) =$ _____.
Answer (Please select your correct option)
0
5
-5
12
Start Time 11:39 AM
85:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH

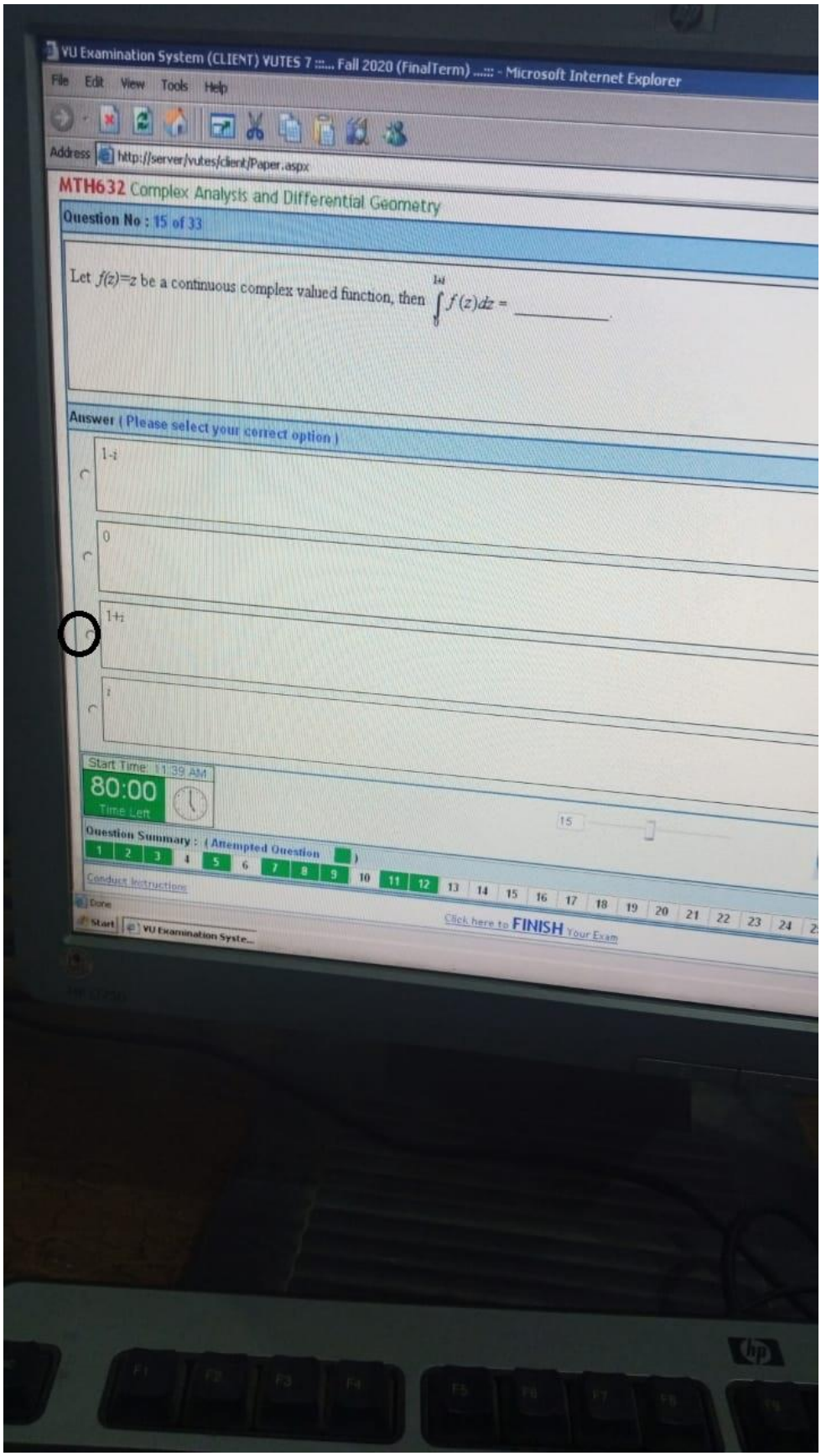

MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____.
Answer (Please select your correct option)
0
1
i
2i
Start Time: 11:39 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH

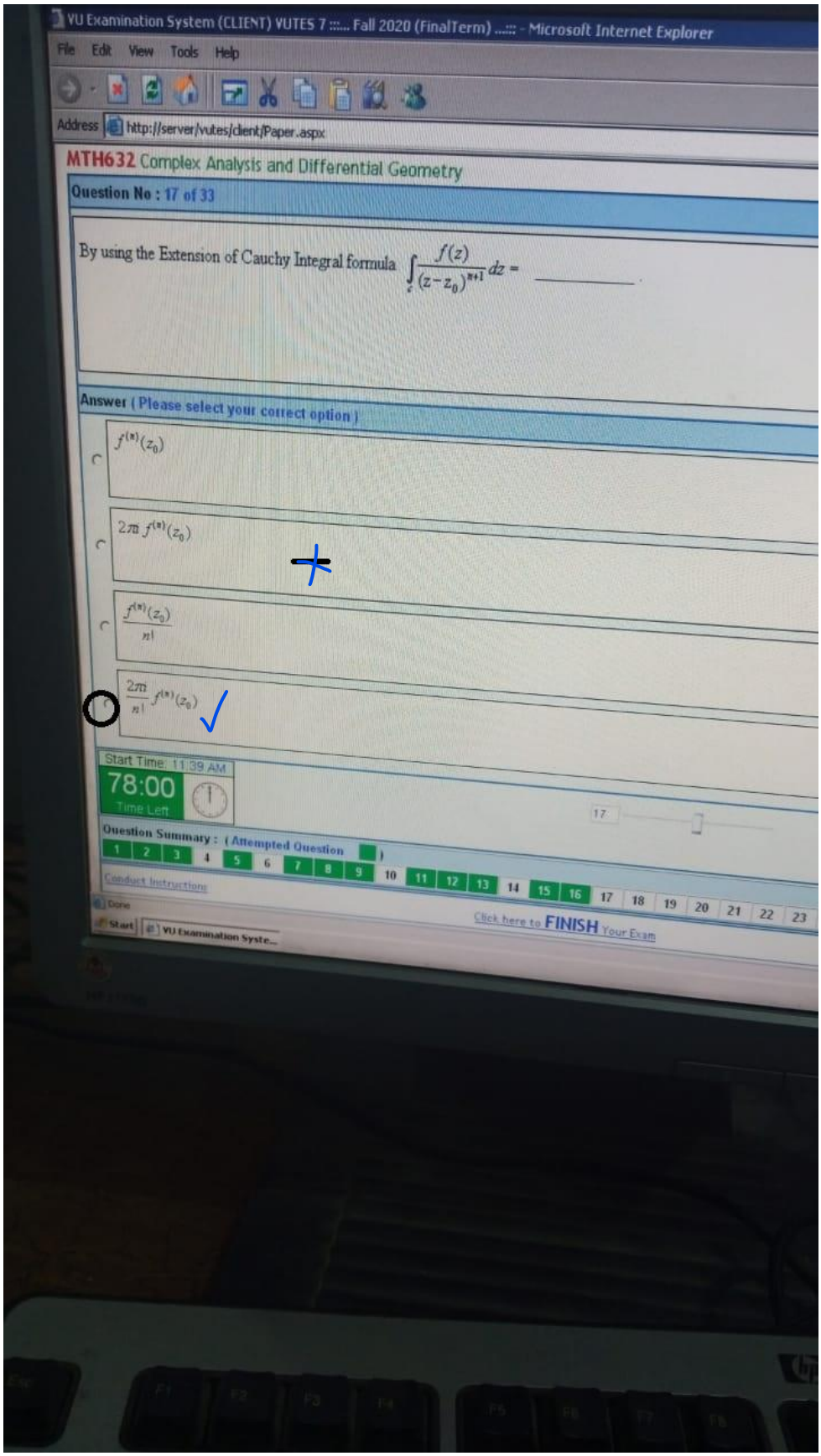

File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____.
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time: 11:39 AM
82:00
Time Left
12
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

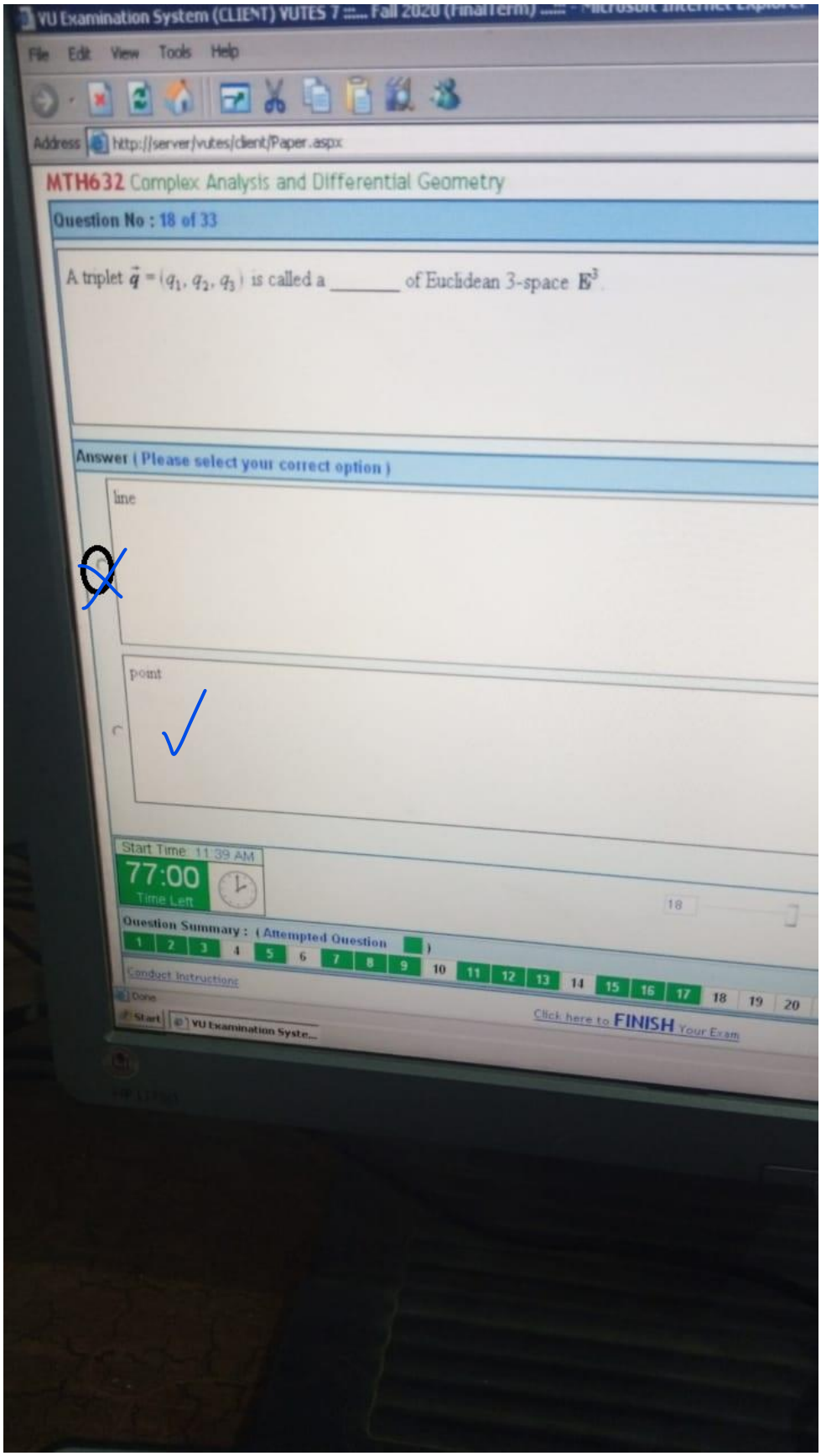

VU Examination System (CLIENT) VUTES 7 ::::.. Fall 2020 (FinalTerm) ...::: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function w(t) = u(t) + i v(t) over an interval a ≤ t ≤ b is _____ , where u , v are real valued functio
of a real variable t.
Answer (Please select your correct option)
Start Time: 11:39 AM
80:00
Time Left
Question Summary : (Attempted Question)
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

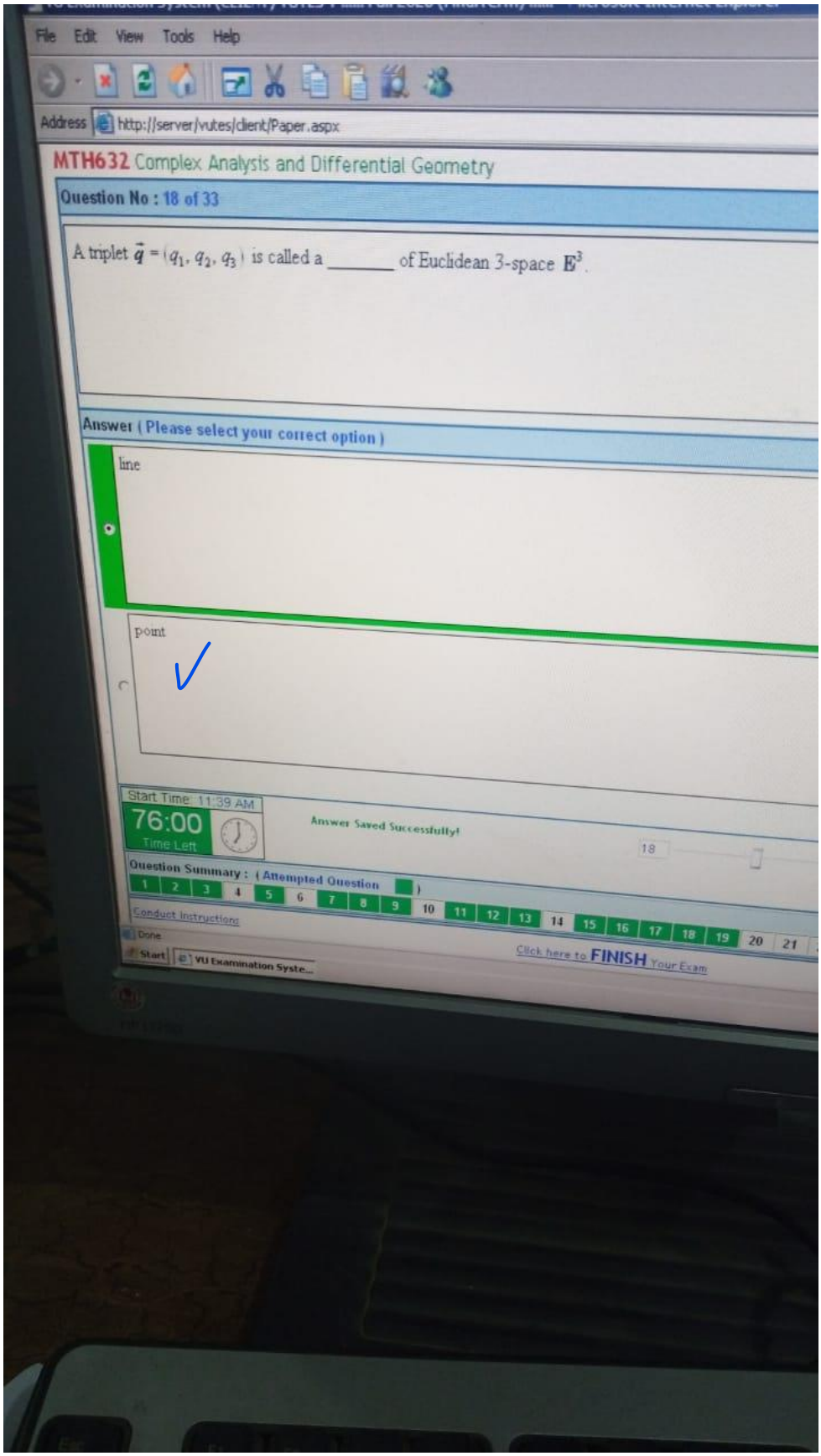

File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time 11:39 AM
81:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Conduct Instructions
Click here to FINISH Your Exam

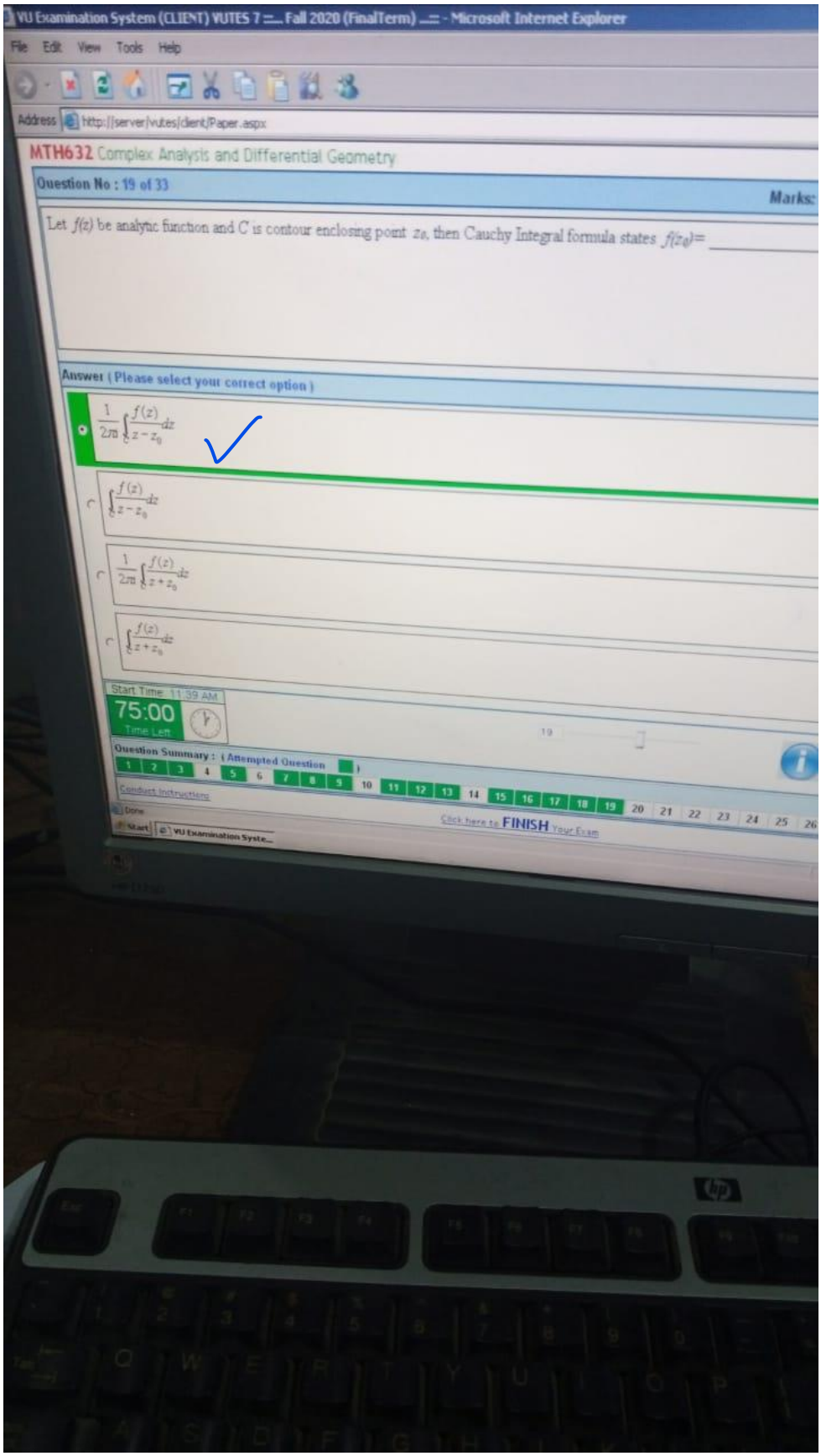

VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____ .
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:39 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

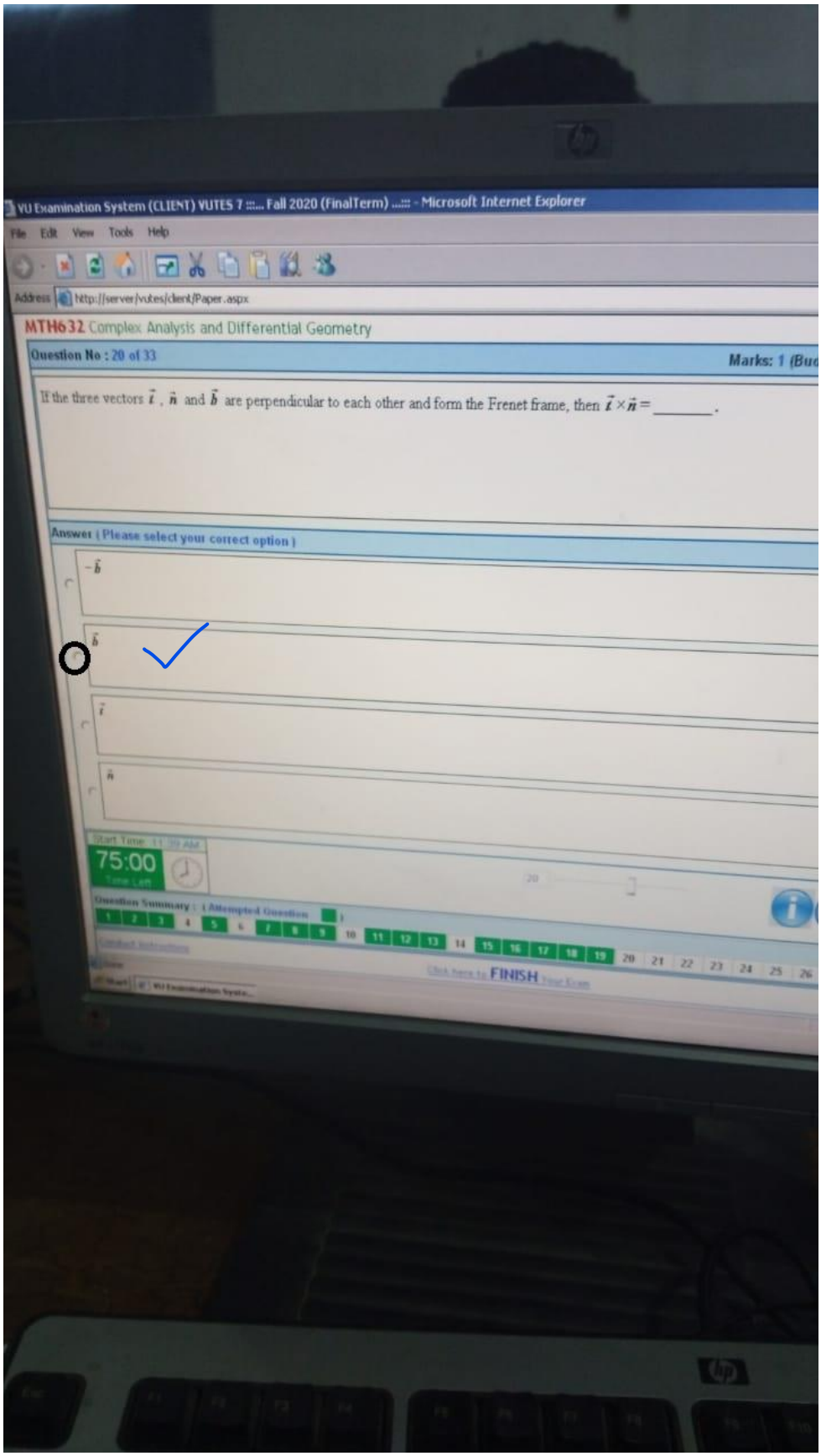

VU Examination System (CLIENT) VUTES 7 ::::.. Fall 2020 (FinalTerm) .....:: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:39 AM
80:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

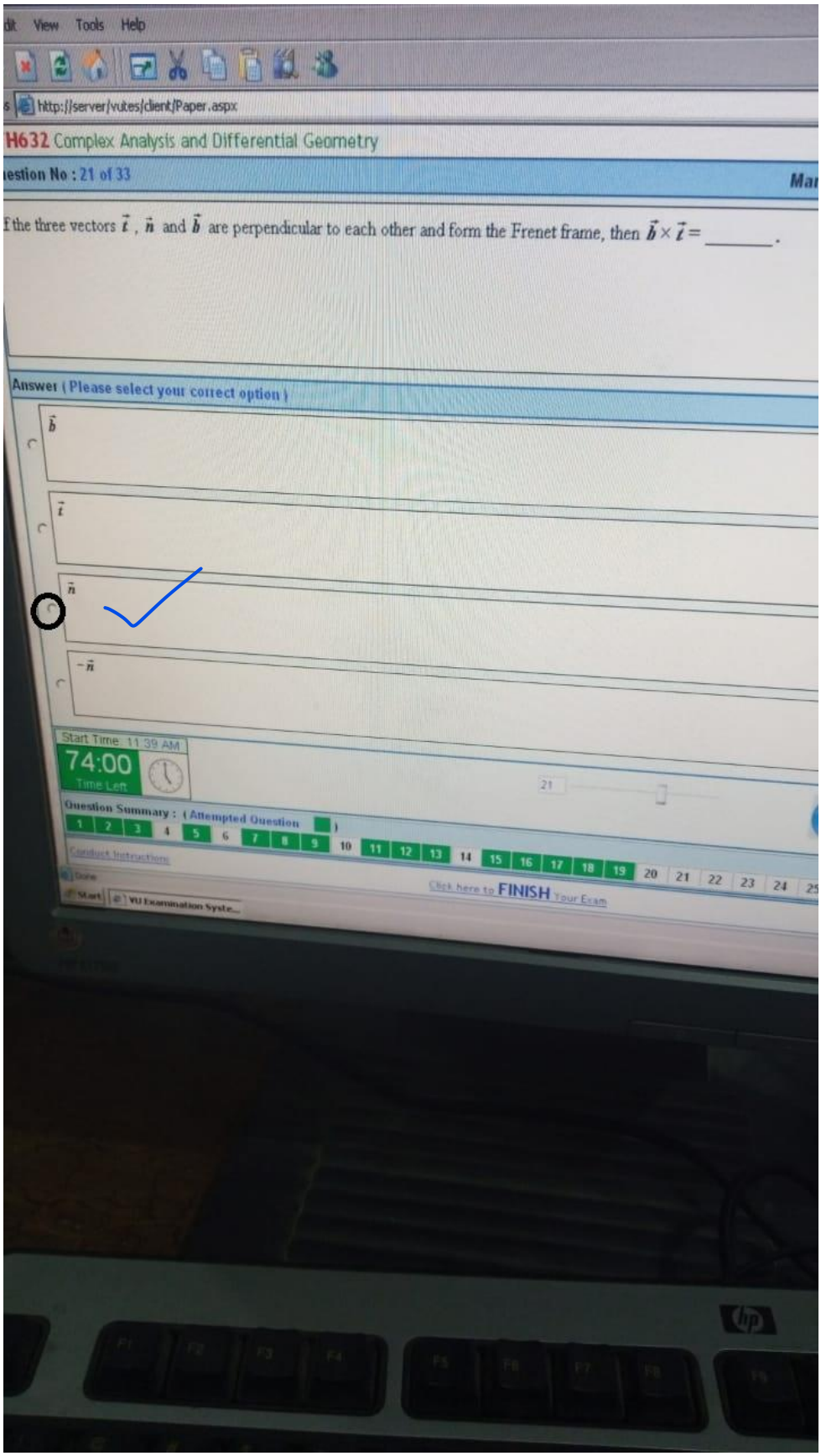

VU Examination System (CLIENT) VUTES 7 ::::.. Fall 2020 (FinalTerm) .....:: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:39 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

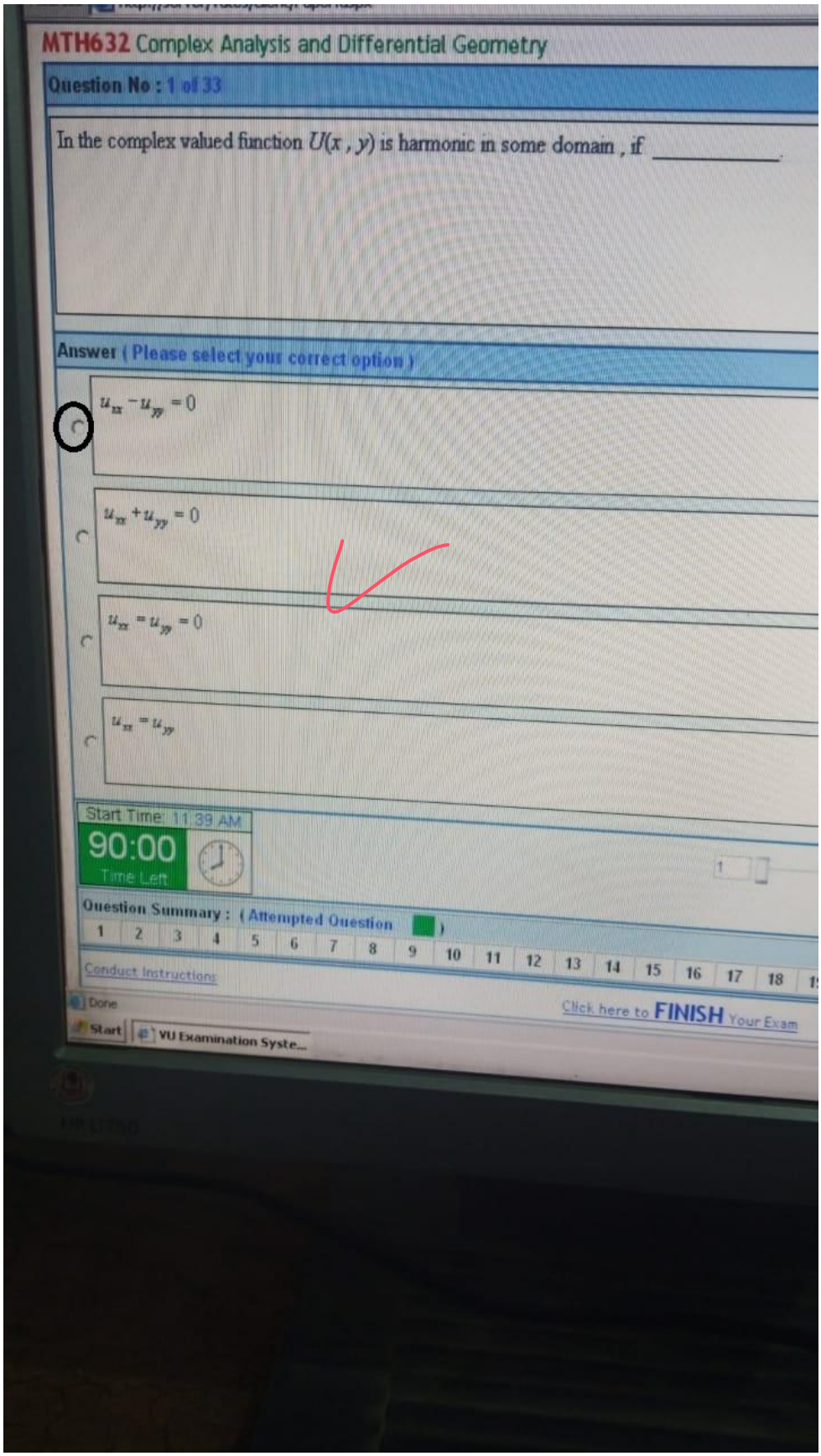

VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) .....
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
77:00
Time Left
18
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

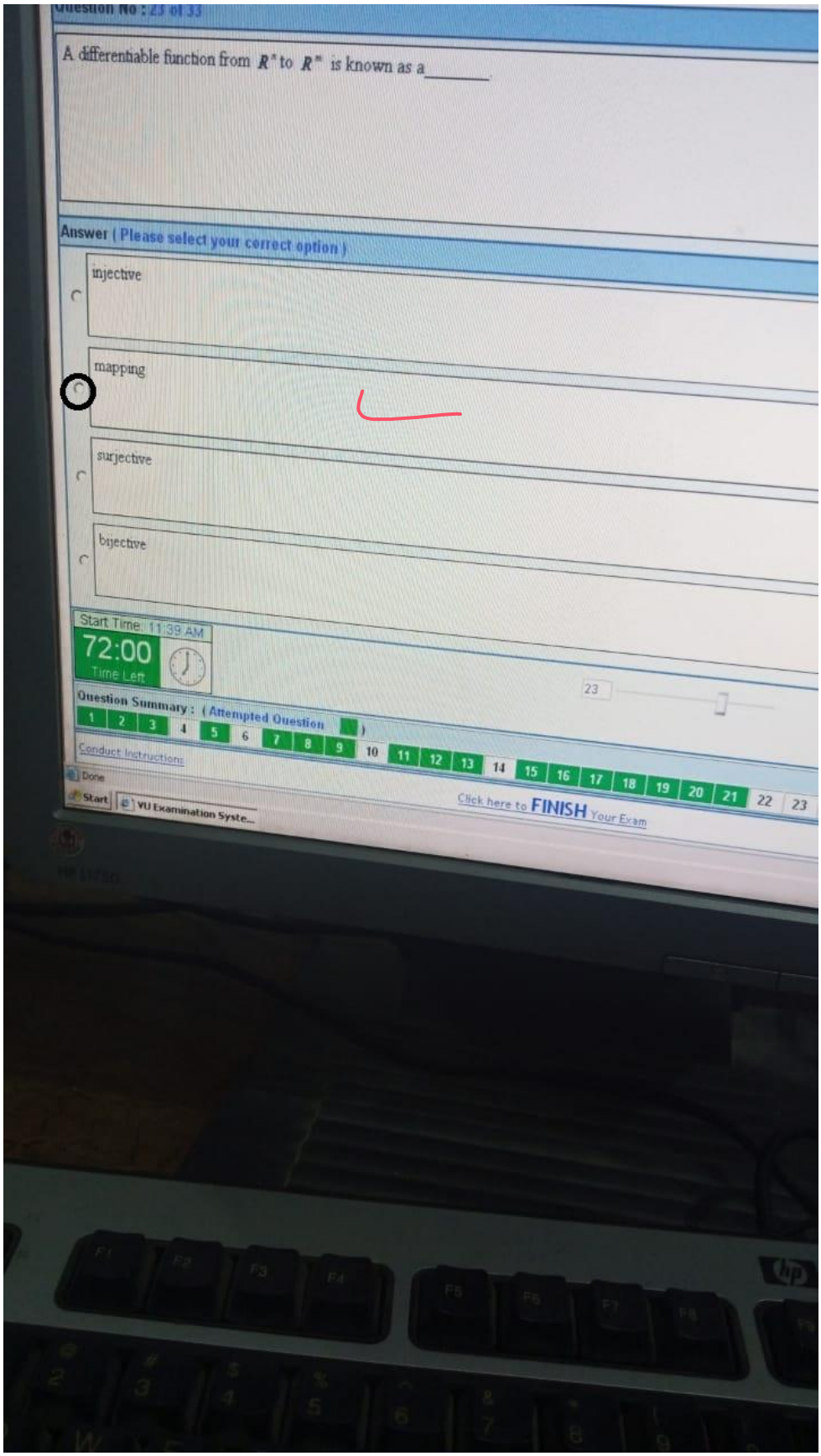

File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
76:00
Time Left
Answer Saved Successfully!
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

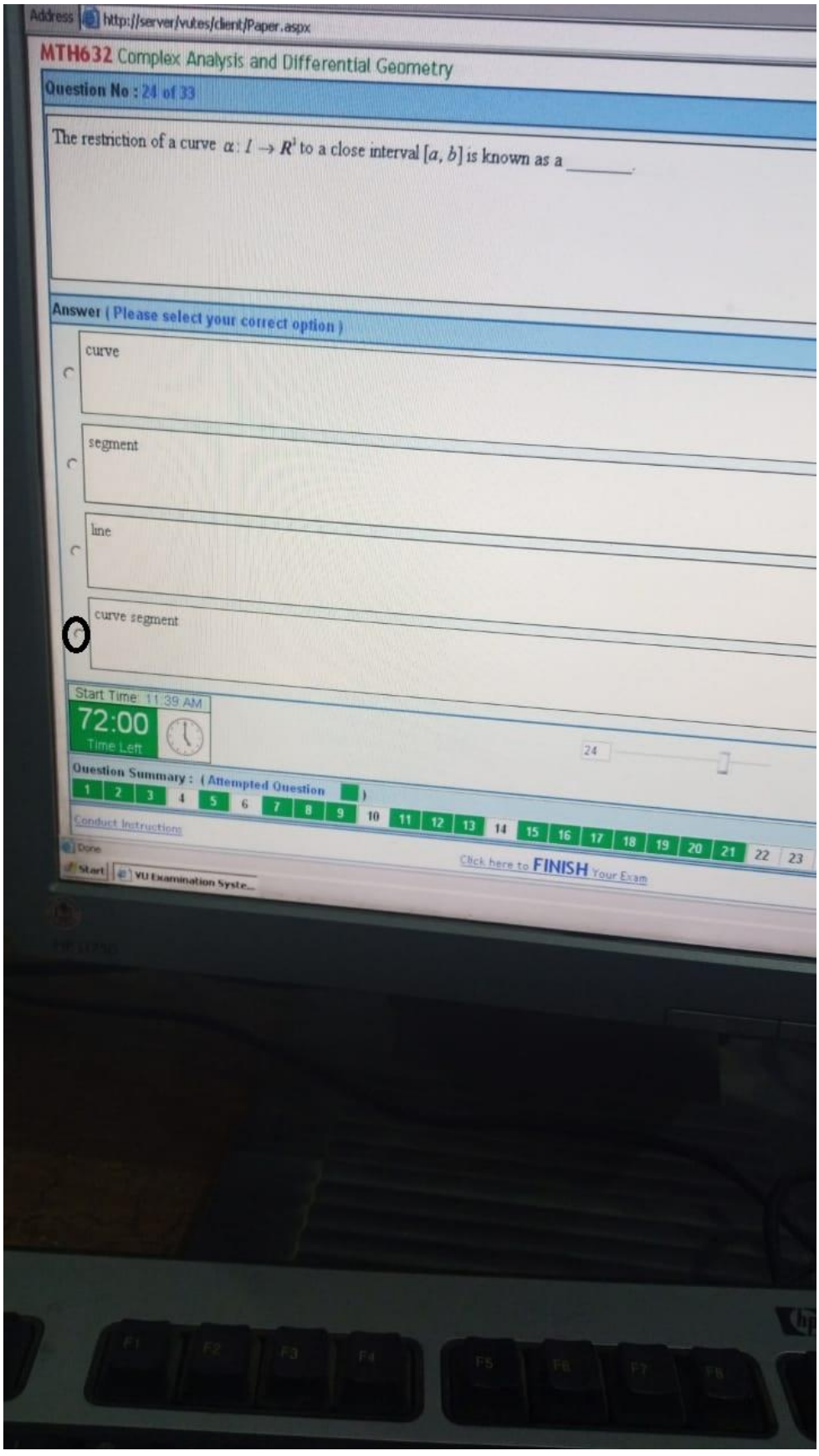

VU Examination System (CLIENT) VUTES 7 :::.. Fall 2020 (FinalTerm) ....:: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 19 of 33
Marks:
Let $f(z)$ be analytic function and C is contour enclosing point z_0, then Cauchy Integral formula states $f(z_0)=$ _____
Answer (Please select your correct option)
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}dz$
$\int_C \frac{f(z)}{z-z_0}dz$
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z+z_0}dz$
$\int_C \frac{f(z)}{z+z_0}dz$
Start Time 11:39 AM
75:00
Time Left
Question Summary : (Attempted Question)
Conduct Instructions
Click here to FINISH Your Exam

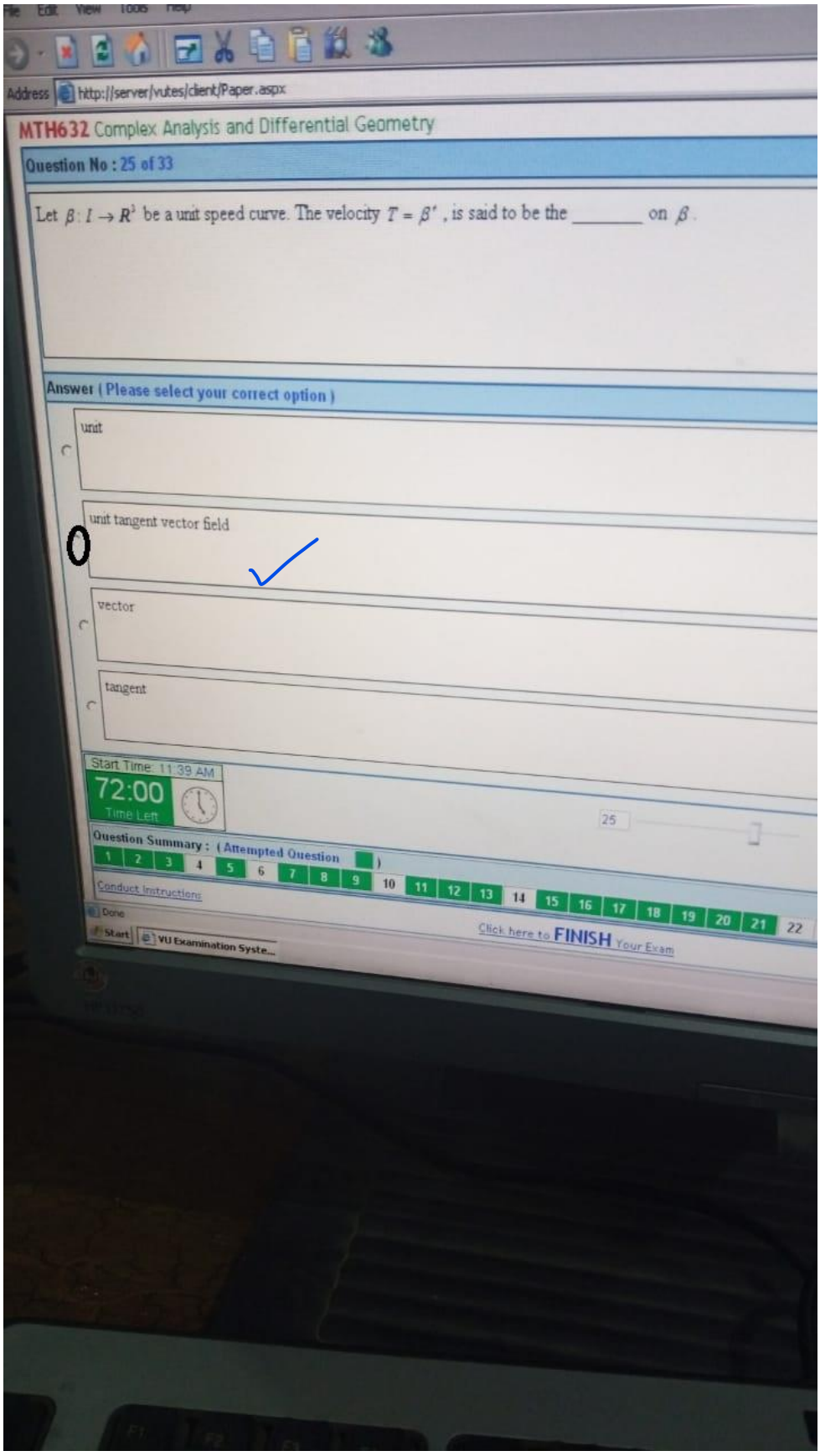

VU Examination System (CLIENT) VUTES 7 ::::.. Fall 2020 (FinalTerm) .....:: - Microsoft Internet Explorer
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 20 of 33
Marks: 1 (Bu
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$ _____.
Answer (Please select your correct option)
$-\vec{b}$
$\vec{b}$
$\vec{t}$
$\vec{n}$
75:00
Question Summary : 1 Attempted Question
FINISH

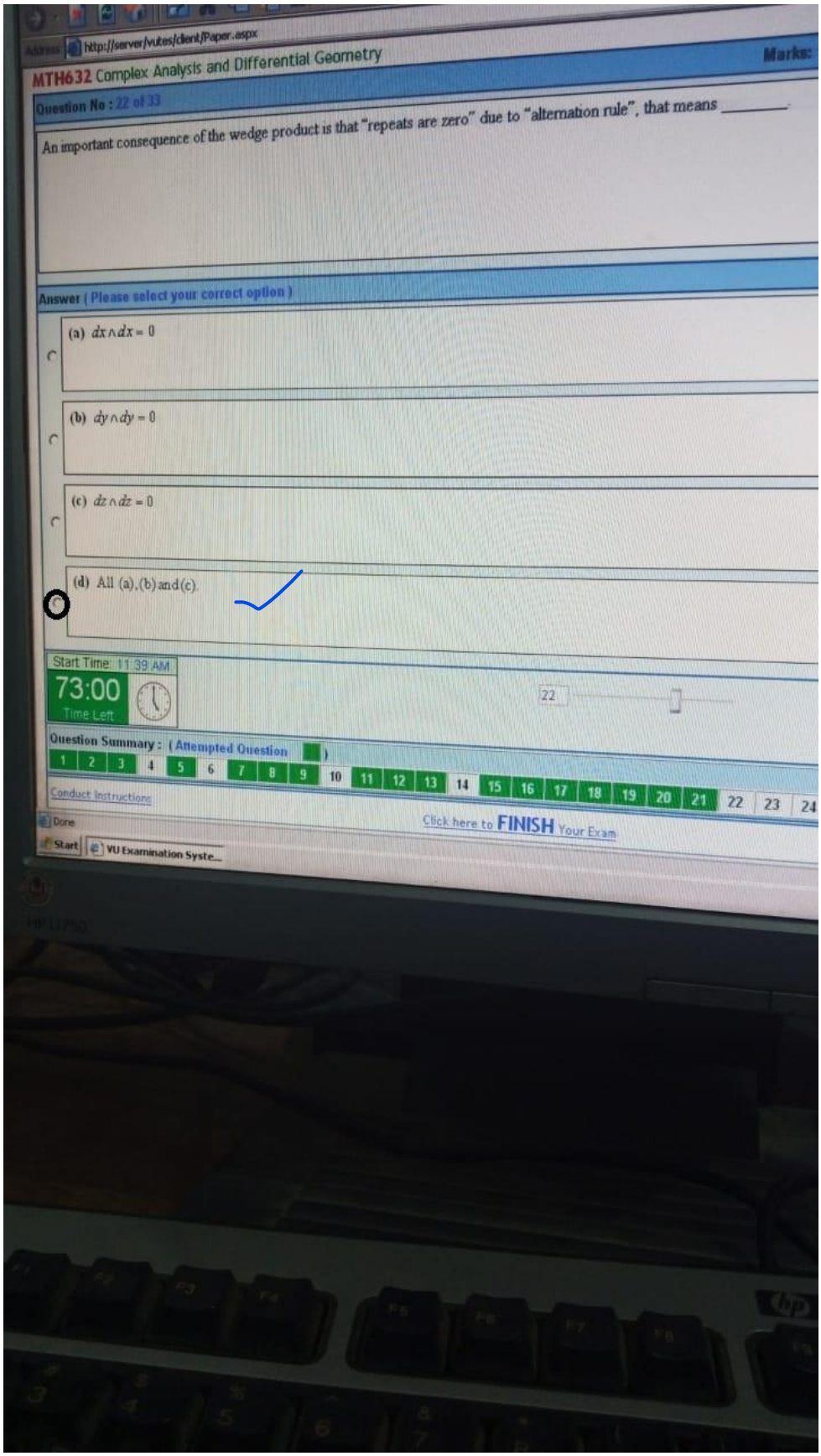

View Tools Help
http://server/vutes/client/Paper.aspx
H632 Complex Analysis and Differential Geometry
Question No : 21 of 33
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time 11:39 AM
74:00
Time Left
Question Summary : (Attempted Question)
Click here to FINISH Your Exam

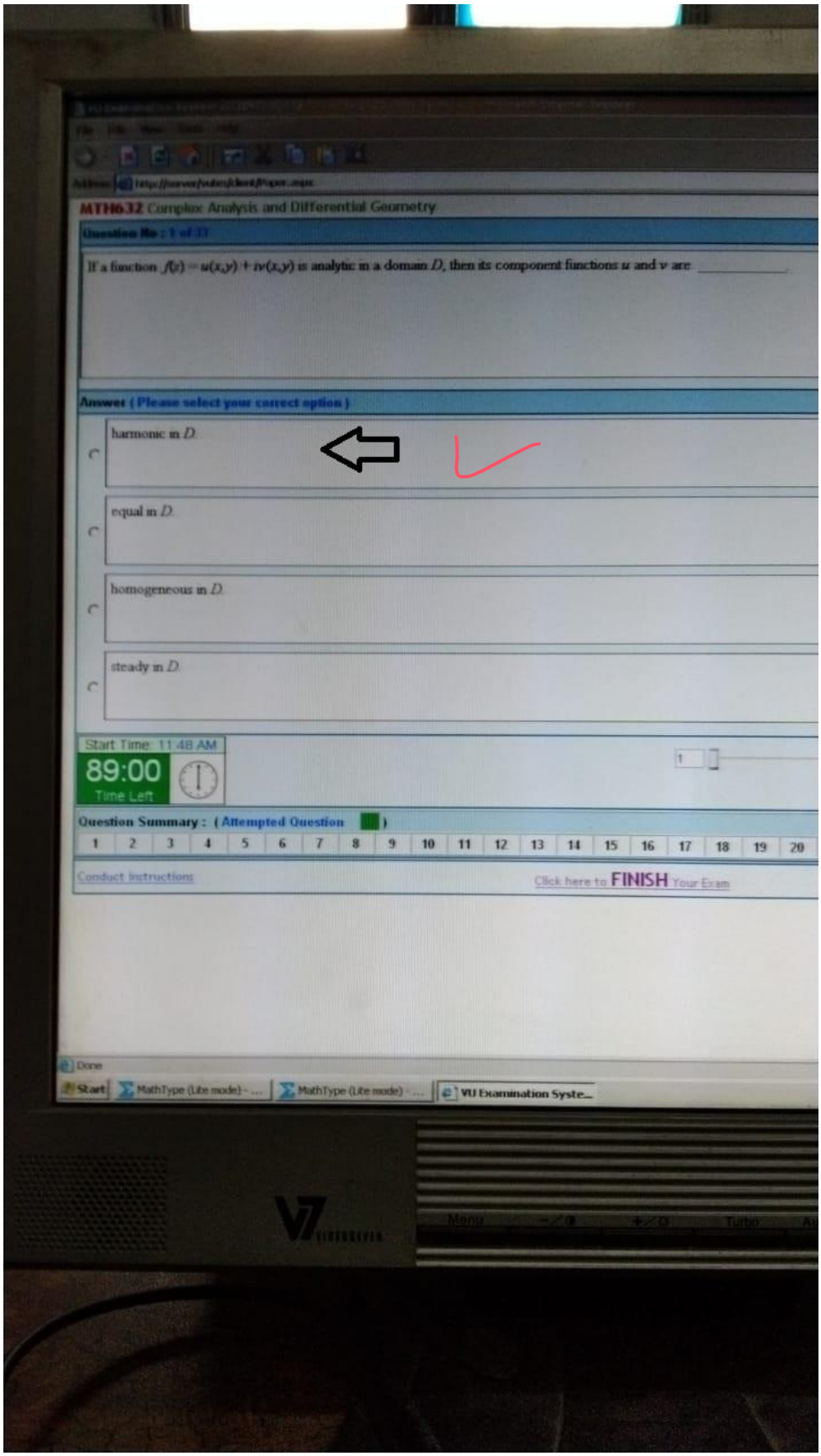

MTH632 Complex Analysis and Differential Geometry
Question No : 1 of 33
In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.
Answer (Please select your correct option)
$u_{xx} - u_{yy} = 0$
$u_{xx} + u_{yy} = 0$
$u_{xx} = u_{yy} = 0$
$u_{xx} = u_{yy}$
Start Time: 11:39 AM
90:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instruction
Done
Click here to FINISH Your Exam
Start
VU Examination Syste...

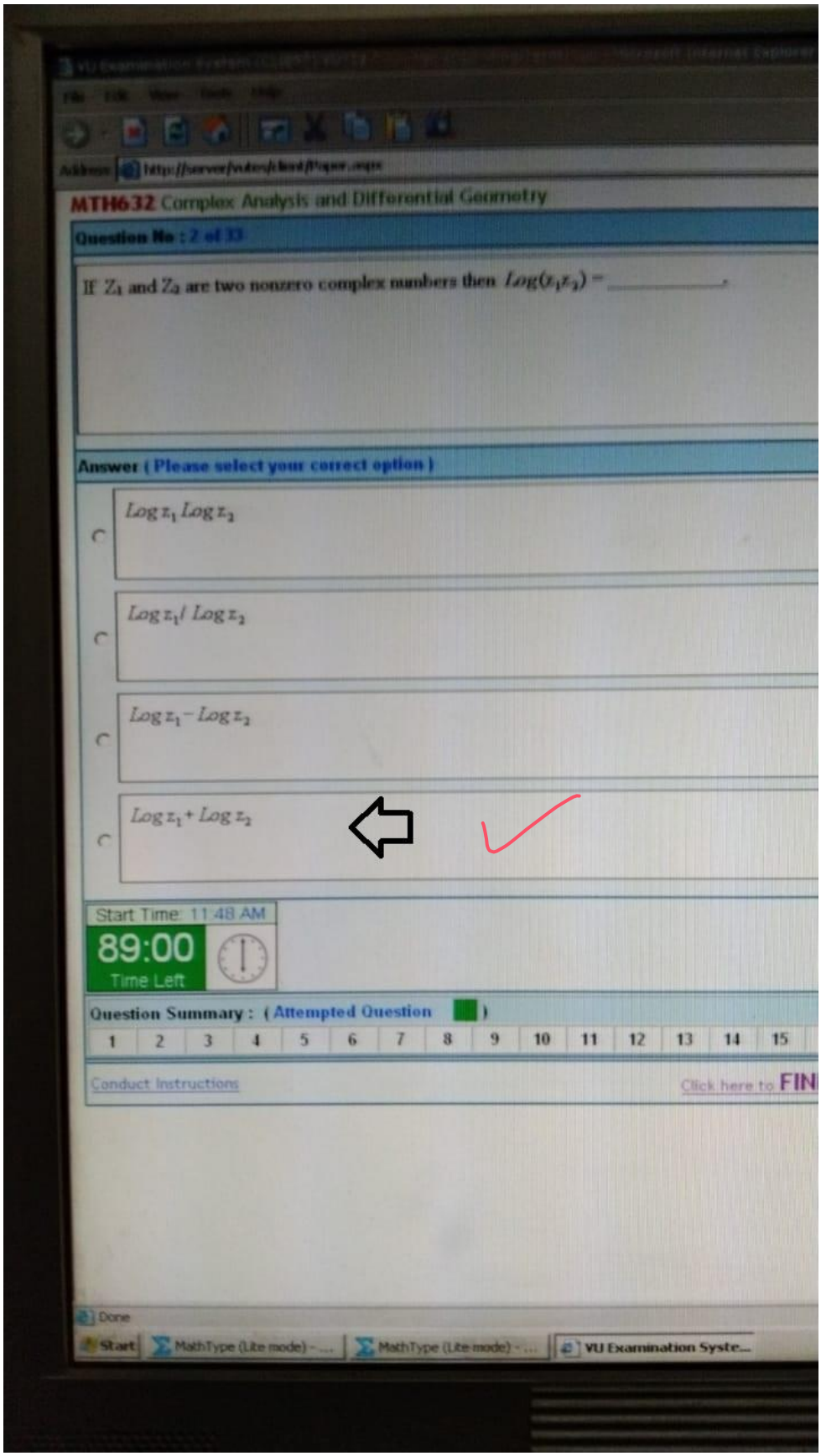

Question No : 23 of 33
A differentiable function from R^n to R^m is known as a_____.
Answer (Please select your correct option)
injective
mapping
surjective
bijective
Start Time 11:39 AM
72:00
Time Left
23
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

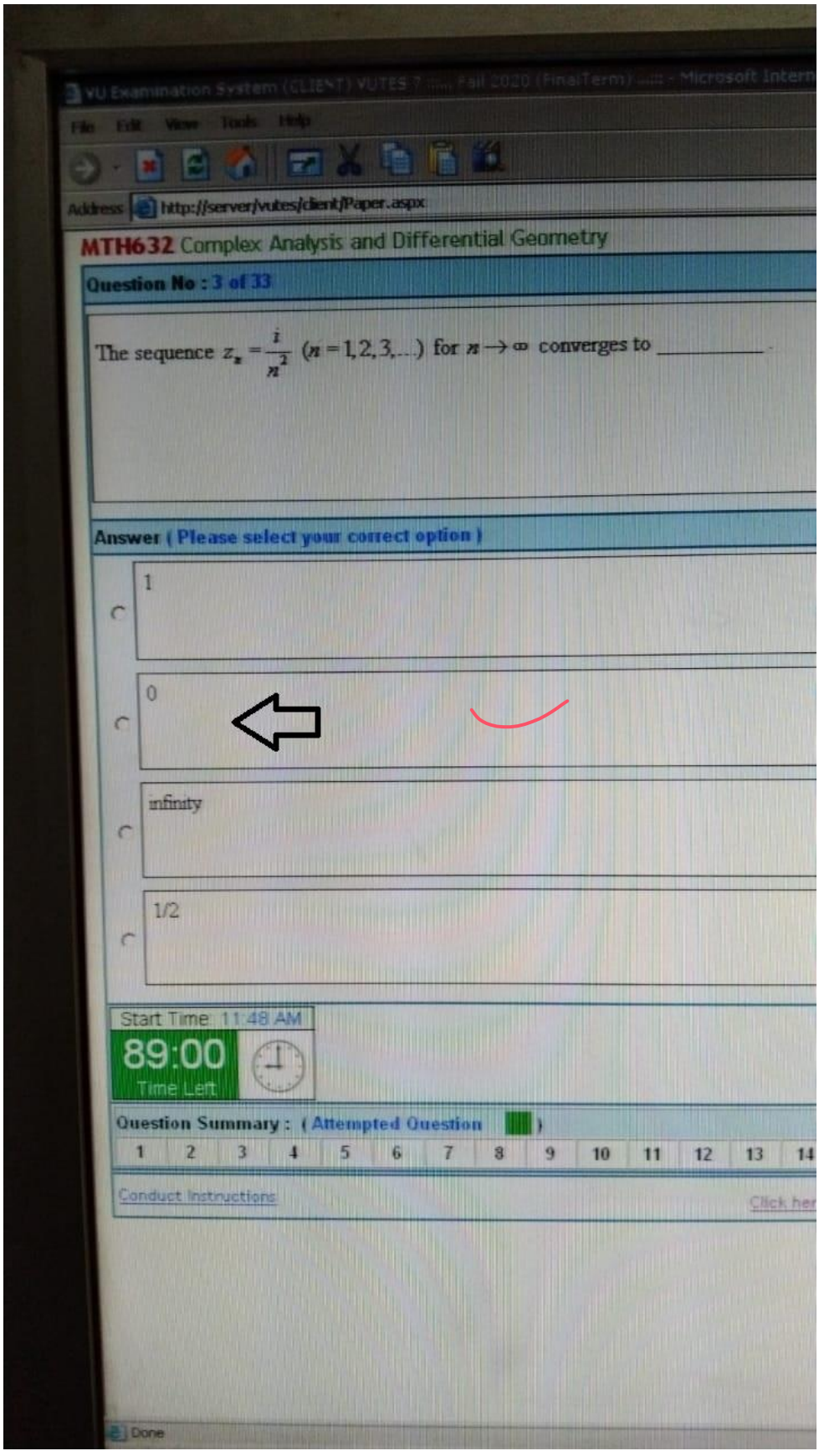

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The restriction of a curve $\alpha : I \to R^1$ to a close interval $[a, b]$ is known as a _____.
Answer (Please select your correct option)
curve
segment
line
curve segment
Start Time: 11:39 AM
72:00
Time Left
24
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Conduct Instructions
Done
Click here to FINISH Your Exam
Start
VU Examination Syste...

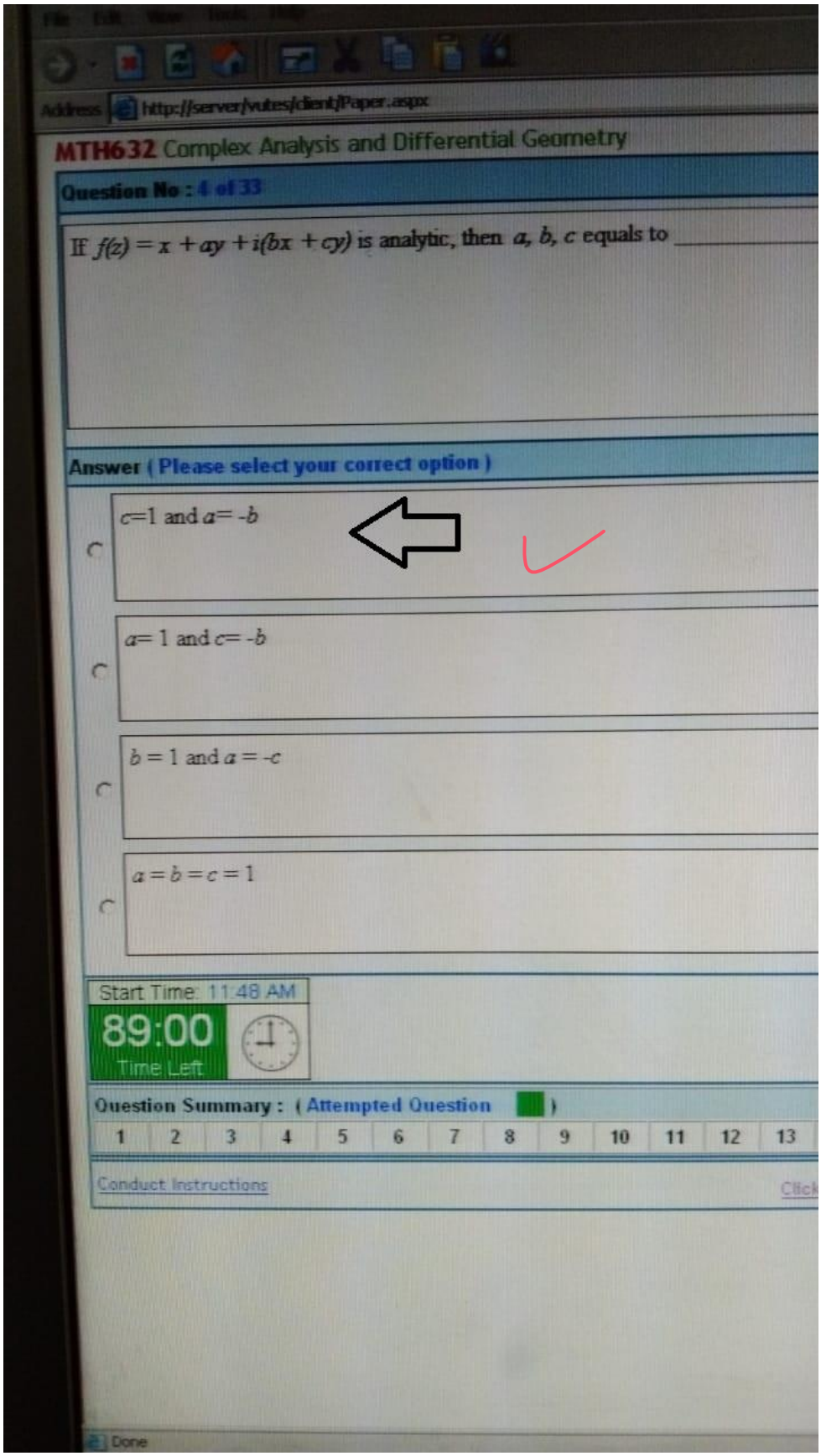

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
0
Start Time: 11:39 AM
72:00
Time Left
25
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
Conduct Instructions
Done
Start
VU Examination Syste...
Click here to FINISH Your Exam

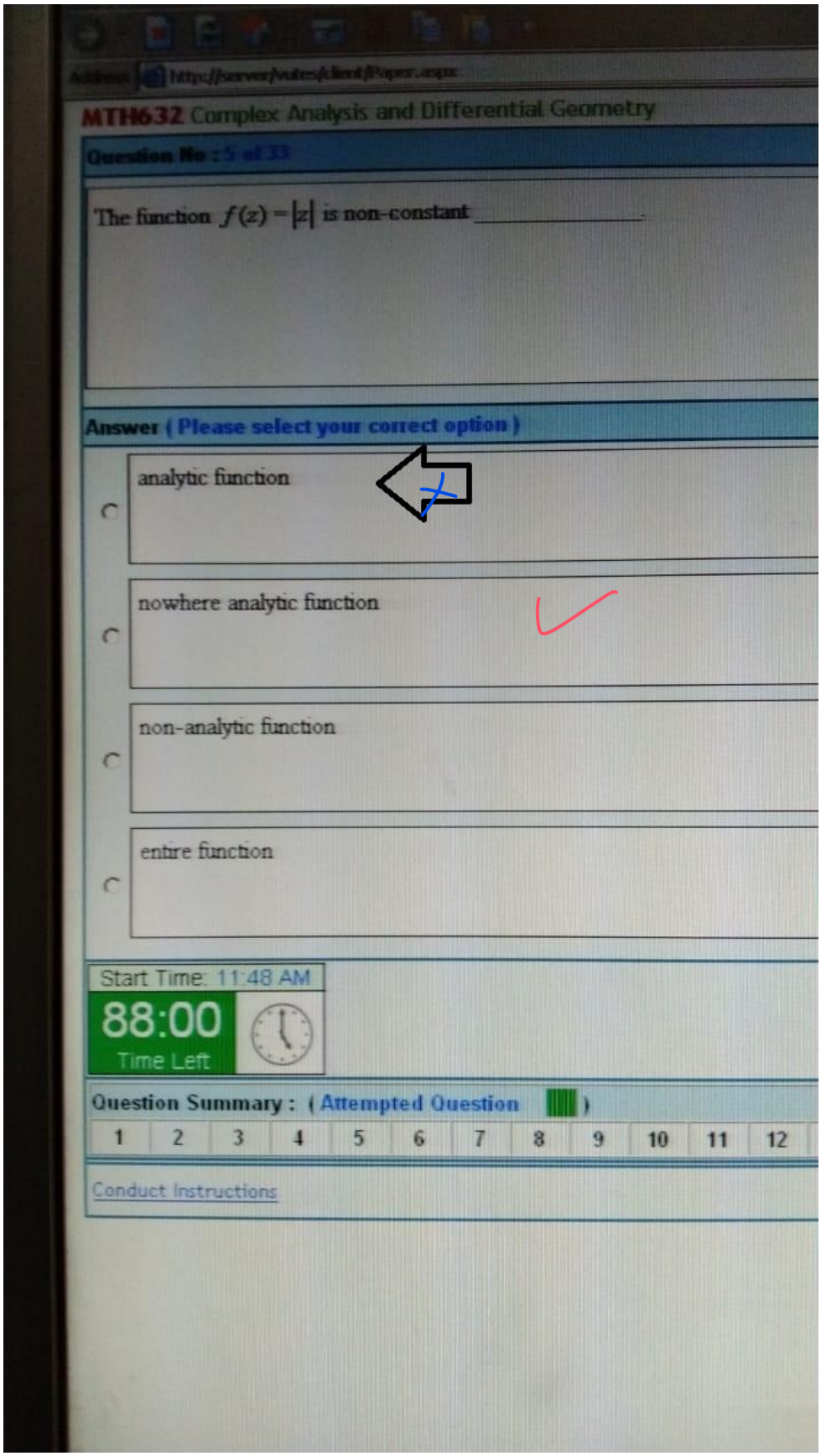

http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Marks:
Question No : 22 of 33
An important consequence of the wedge product is that "repeats are zero" due to "alternation rule", that means _____.
Answer (Please select your correct option)
(a) $dx \wedge dx = 0$
(b) $dy \wedge dy = 0$
(c) $dz \wedge dz = 0$
(d) All (a), (b) and (c).
Start Time: 11:39 AM
73:00
Time Left
22
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...

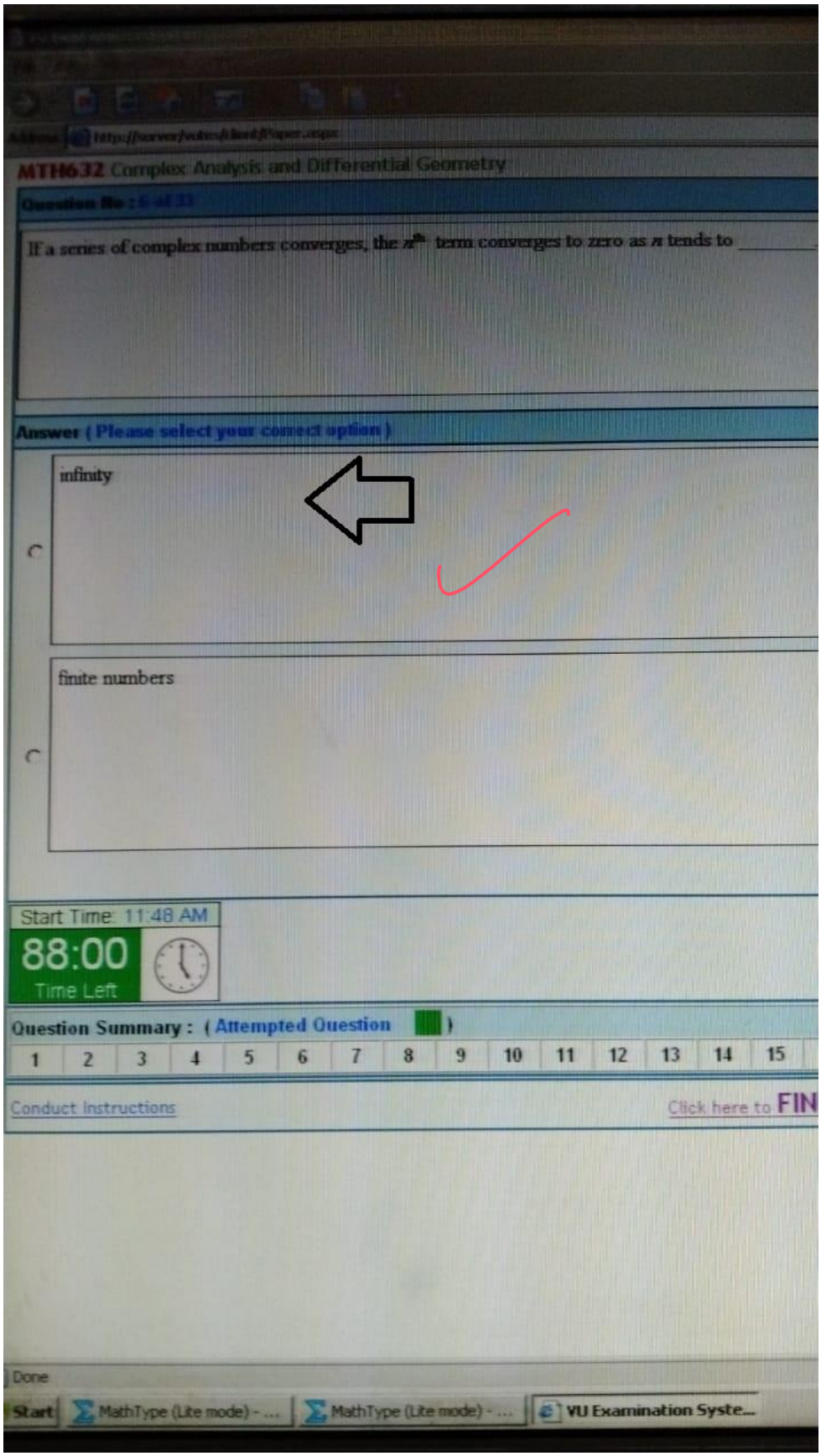

MTH632 Complex Analysis and Differential Geometry
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Intern
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click her
Done


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
$c=1$ and $a=-b$
$a=1$ and $c=-b$
$b=1$ and $a=-c$
$a=b=c=1$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done


MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1 ✓
- 2
- i ⇦

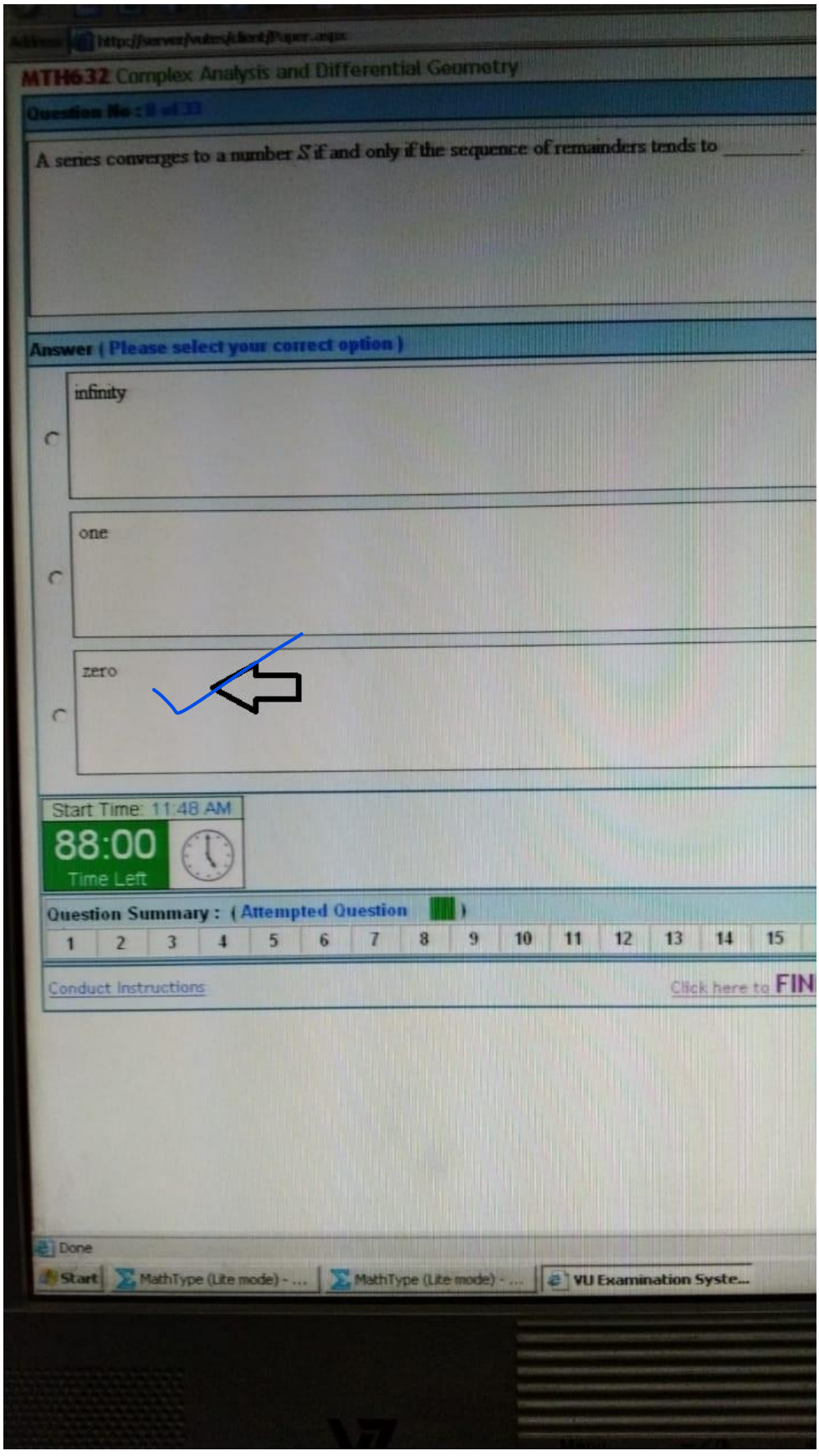
Start Time: 11:48 AM

88:00 Time Left

Question Summary : (Attempted Question ■)

1	2	3	4	5	6	7	8	9	10

Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A series converges to a number S if and only if the sequence of remainders tends to _____.
Answer (Please select your correct option)
infinity
one
zero
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 20

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

Answer (Please select your correct option)

- 1 and -1
- i and $-i$
- 1 and i
- -1 and $-i$

Start Time: 11:48 AM

88:00
Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	1

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ x = _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

88:00

Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Done

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

Start Time: 11:48 AM

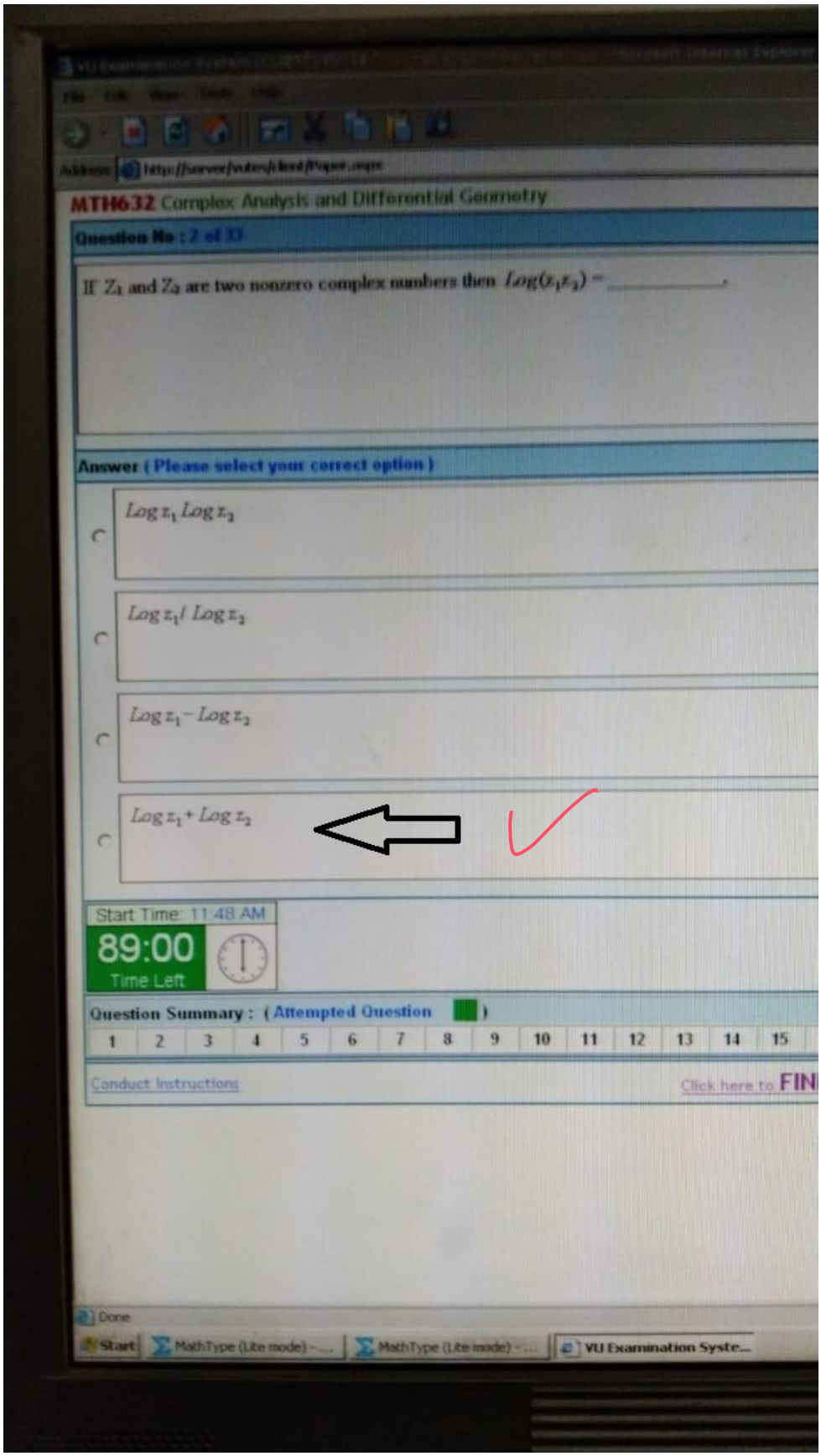
88:00

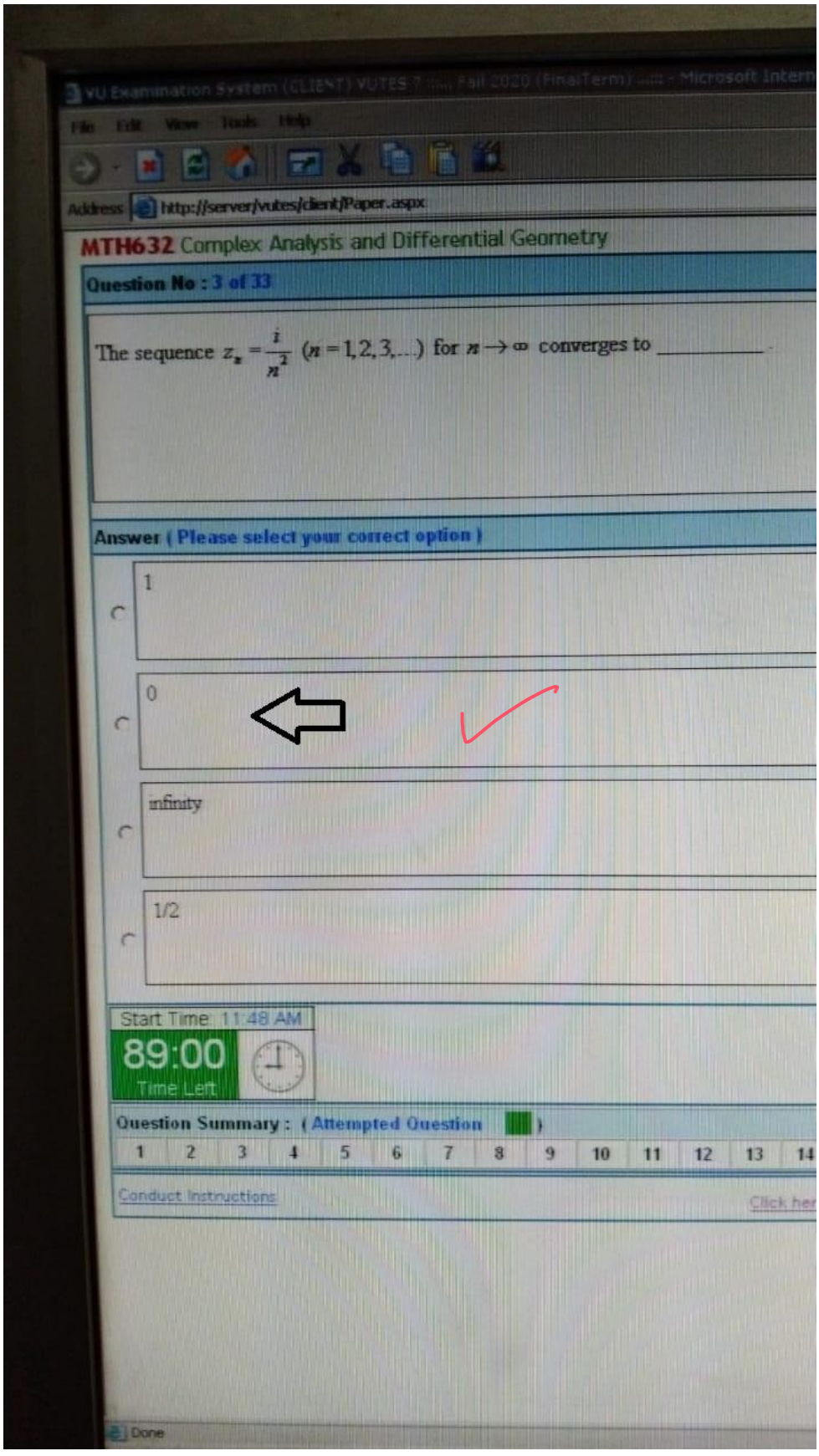
Time Left

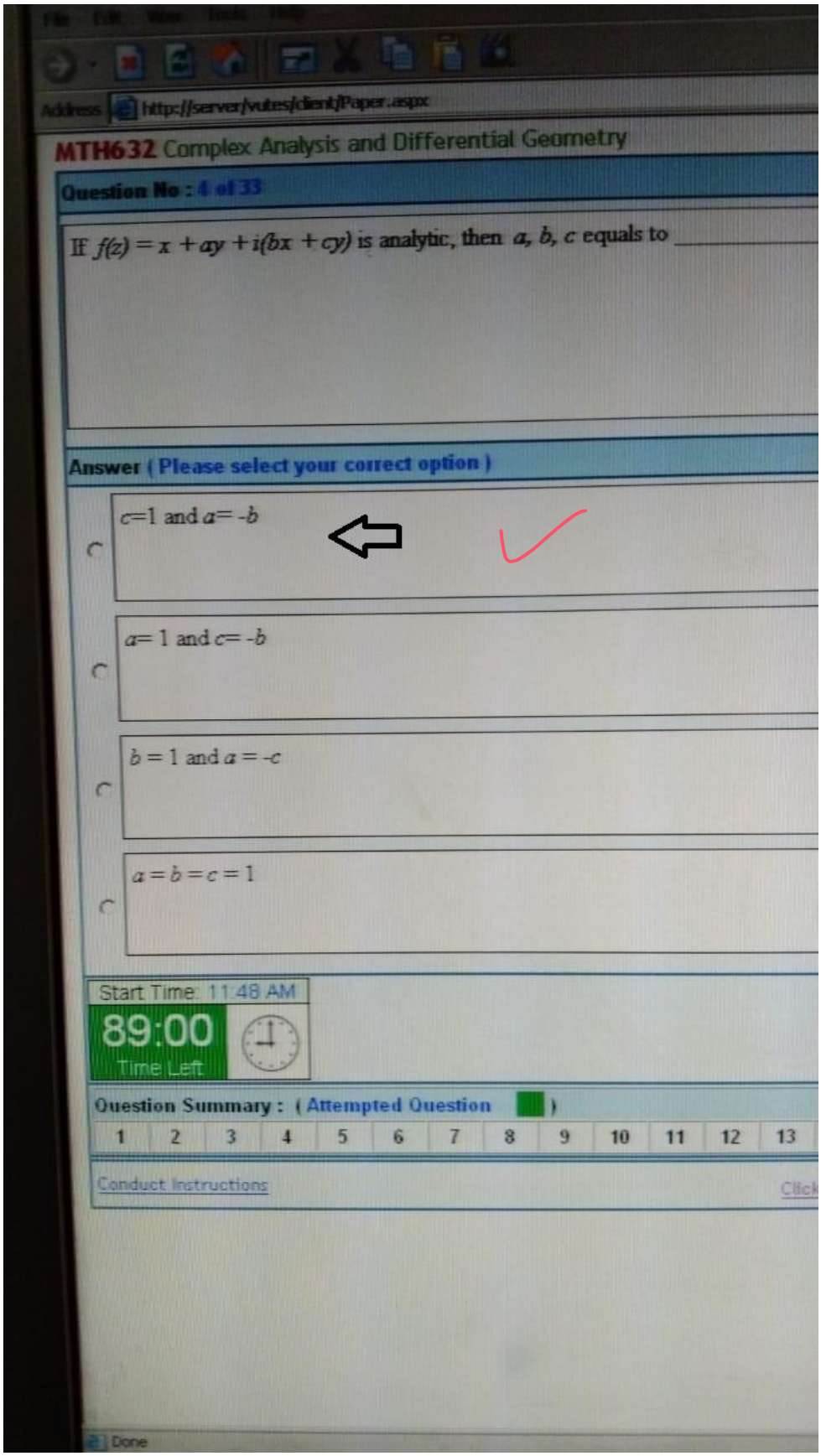
Question Summary : (Attempted Question)

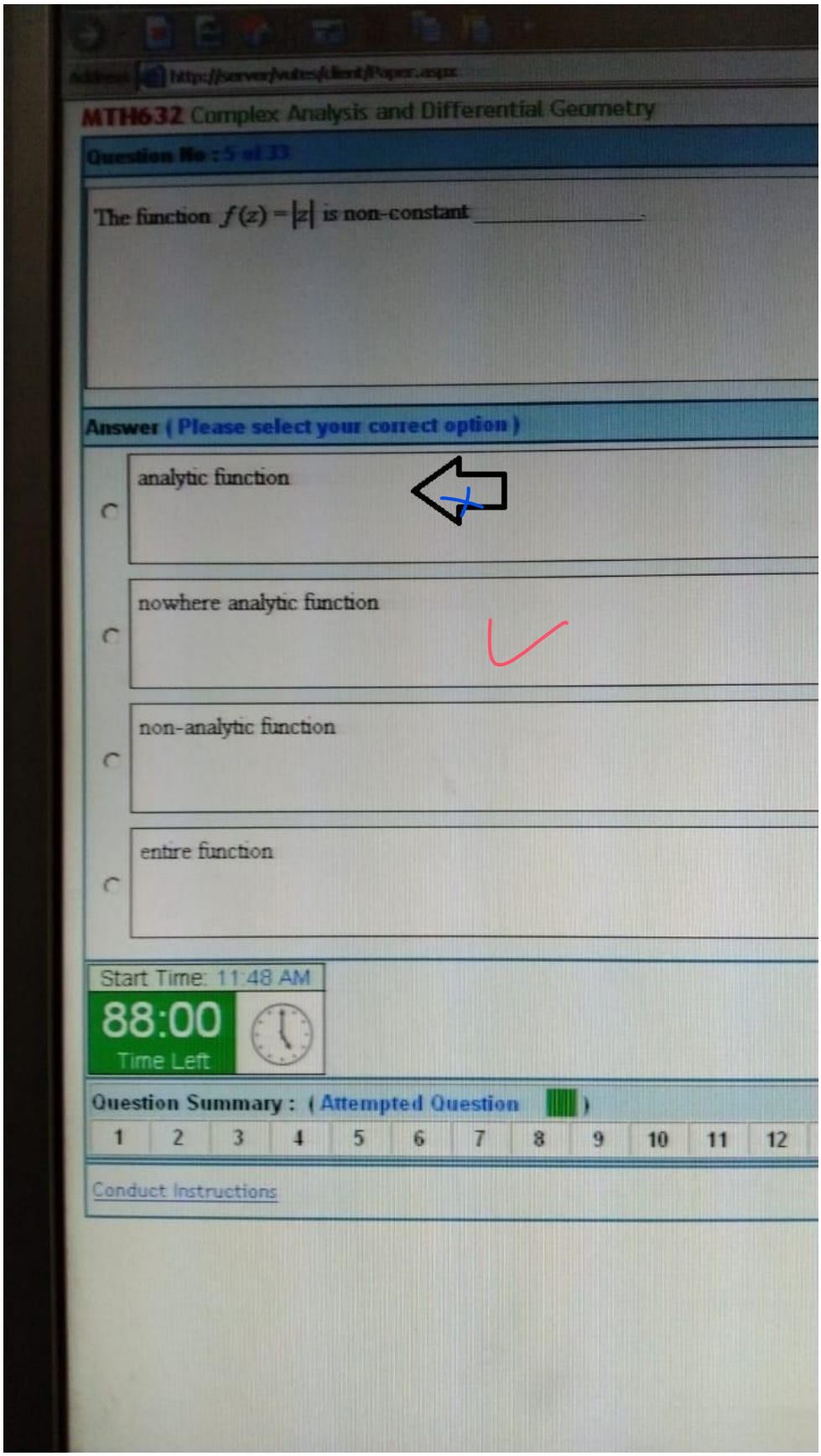
1	2	3	4	5	6	7	8	9	10

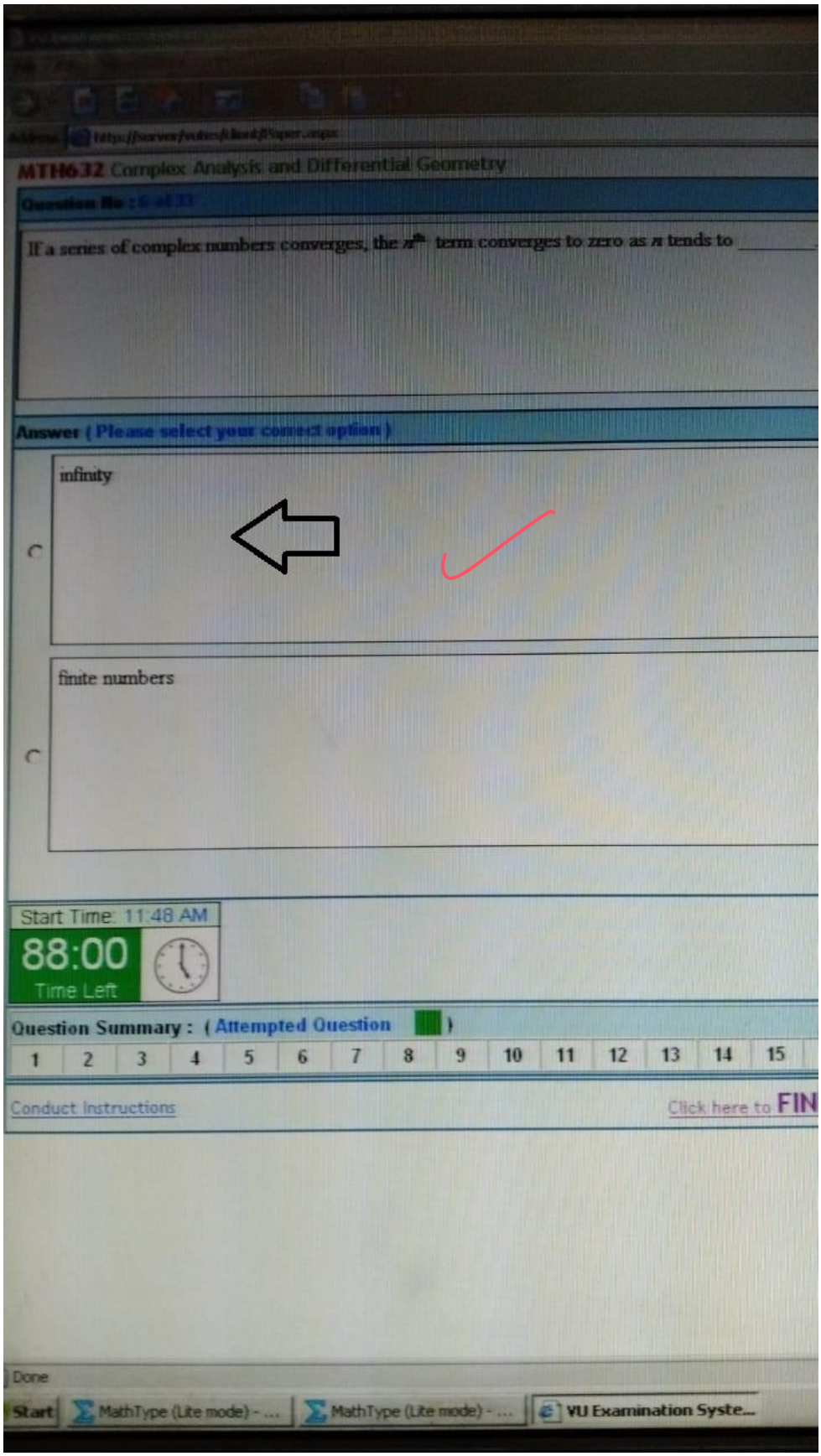
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Intern
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click her
Done


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
c=1 and a= -b
a= 1 and c= -b
b = 1 and a = -c
a = b = c = 1
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done


MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 10

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1 ✓
- 2
- i

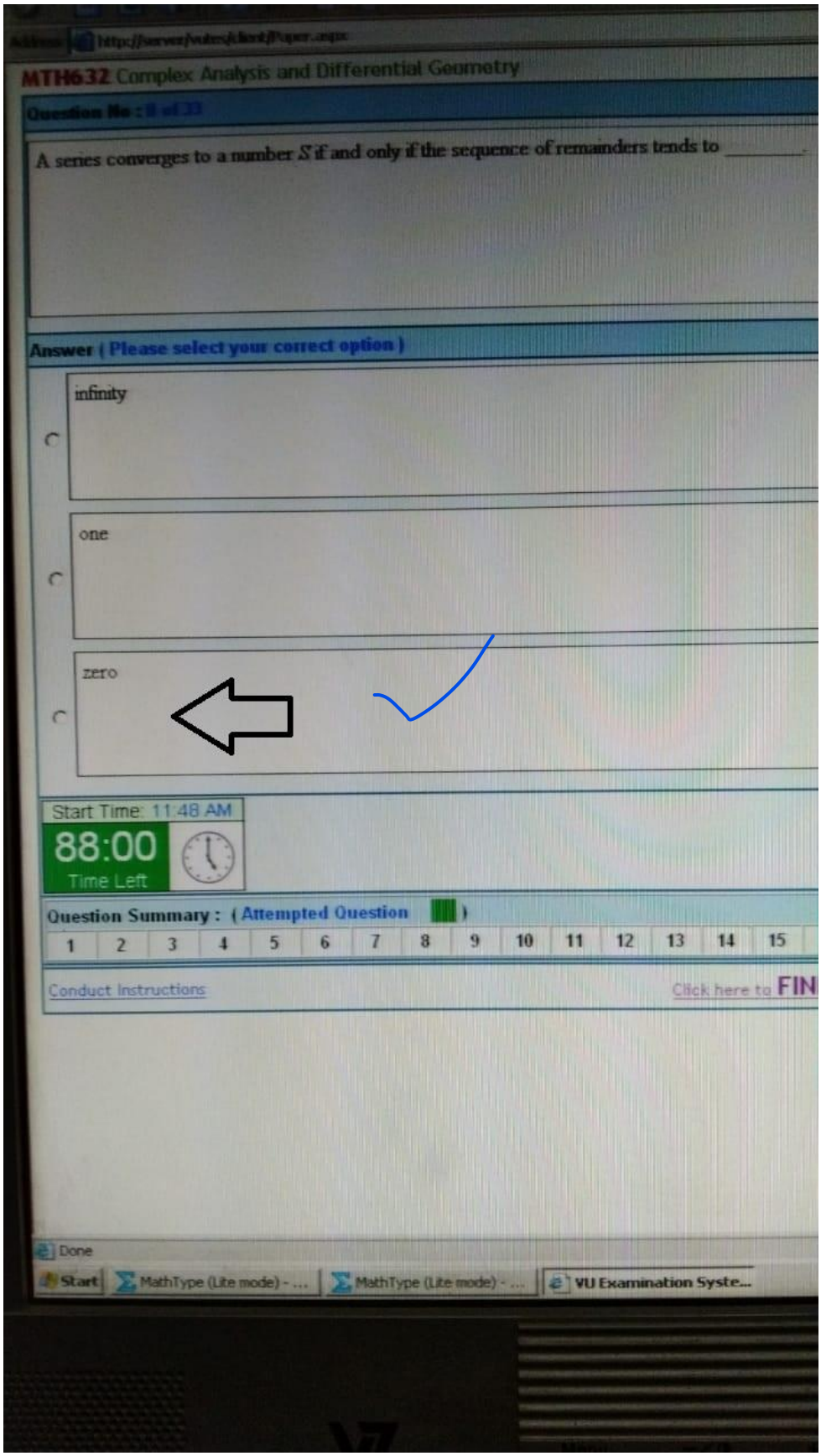
Start Time: 11:48 AM

88:00 Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A series converges to a number S if and only if the sequence of remainders tends to _____.
Answer (Please select your correct option)
infinity
one
zero
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 20

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

Answer (Please select your correct option)

- 1 and -1
- i and $-i$ 
- 1 and i
- -1 and $-i$

Start Time: 11:48 AM

88:00
Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	1

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ $x =$ _____

Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

Start Time: 11:48 AM

88:00

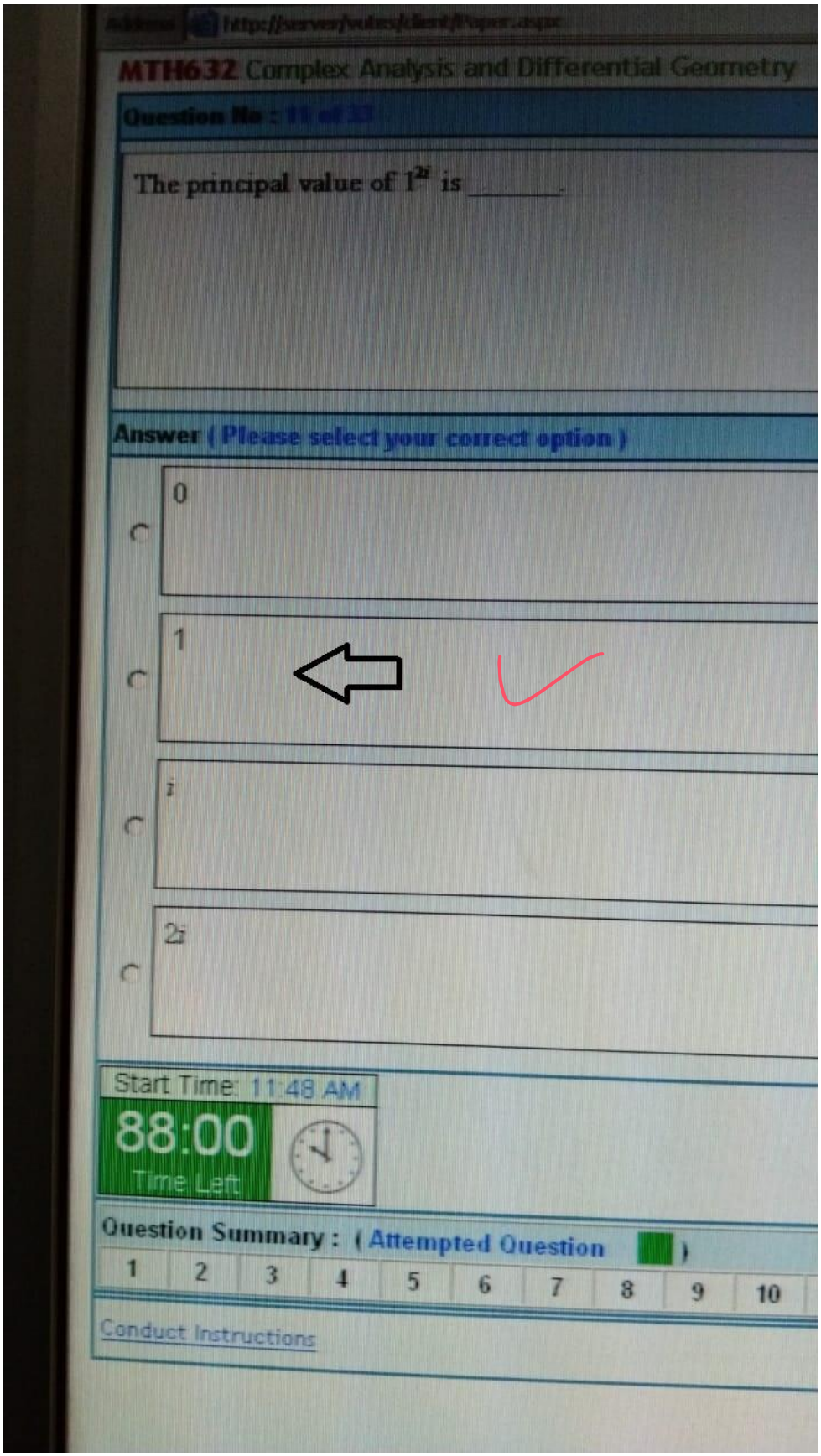
Time Left

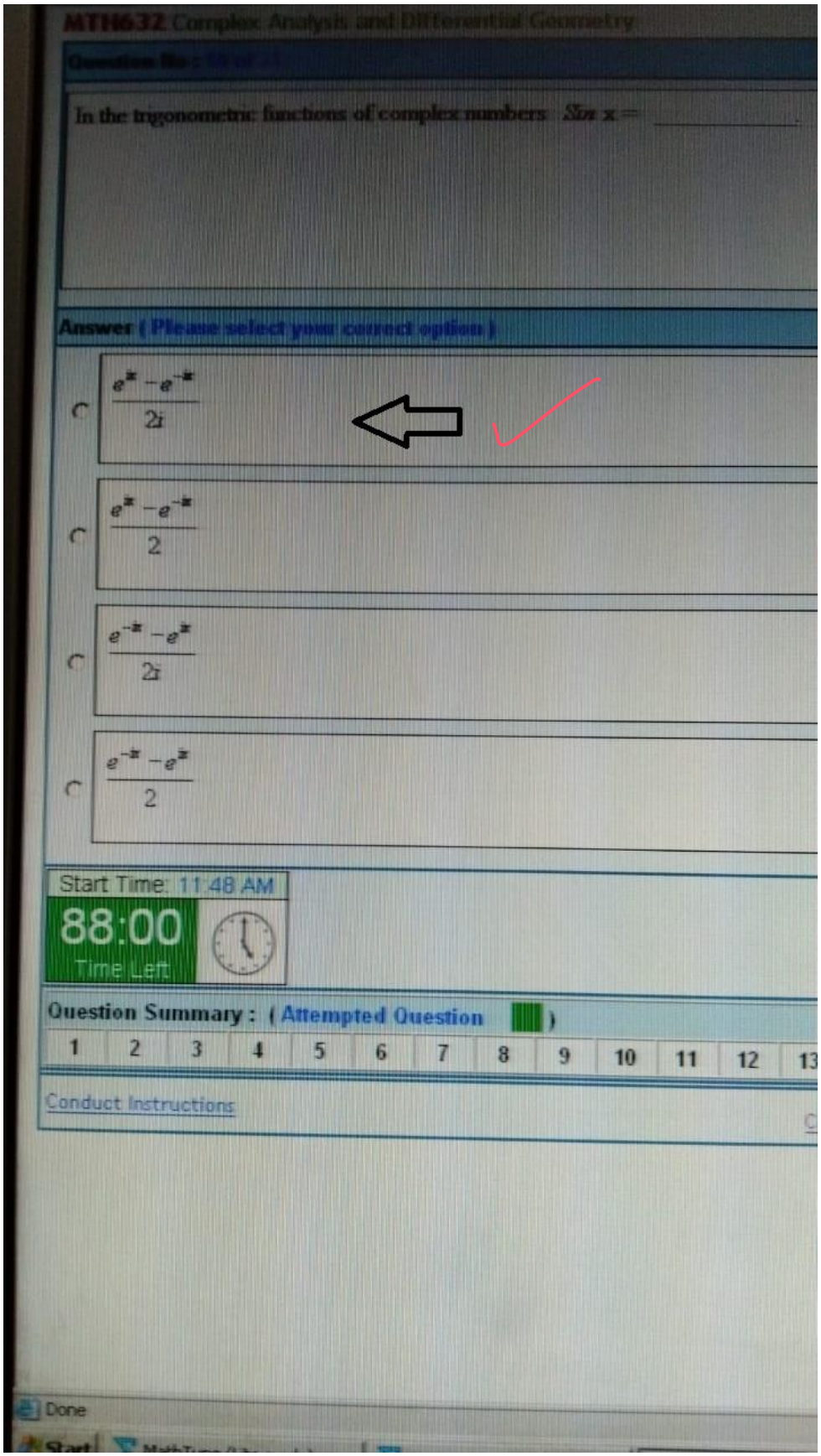
Question Summary : (Attempted Question)

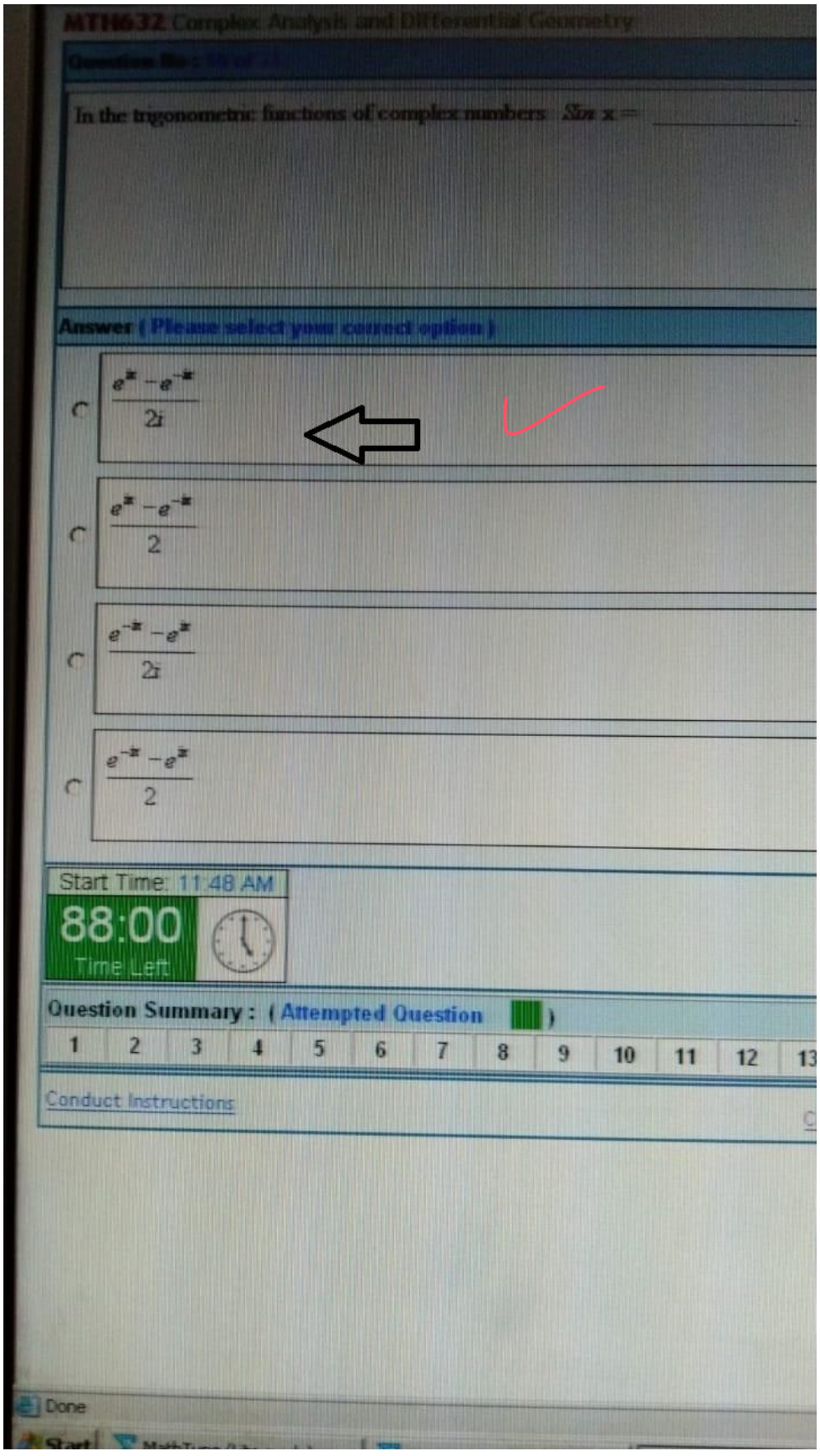
1	2	3	4	5	6	7	8	9	10	11	12	13

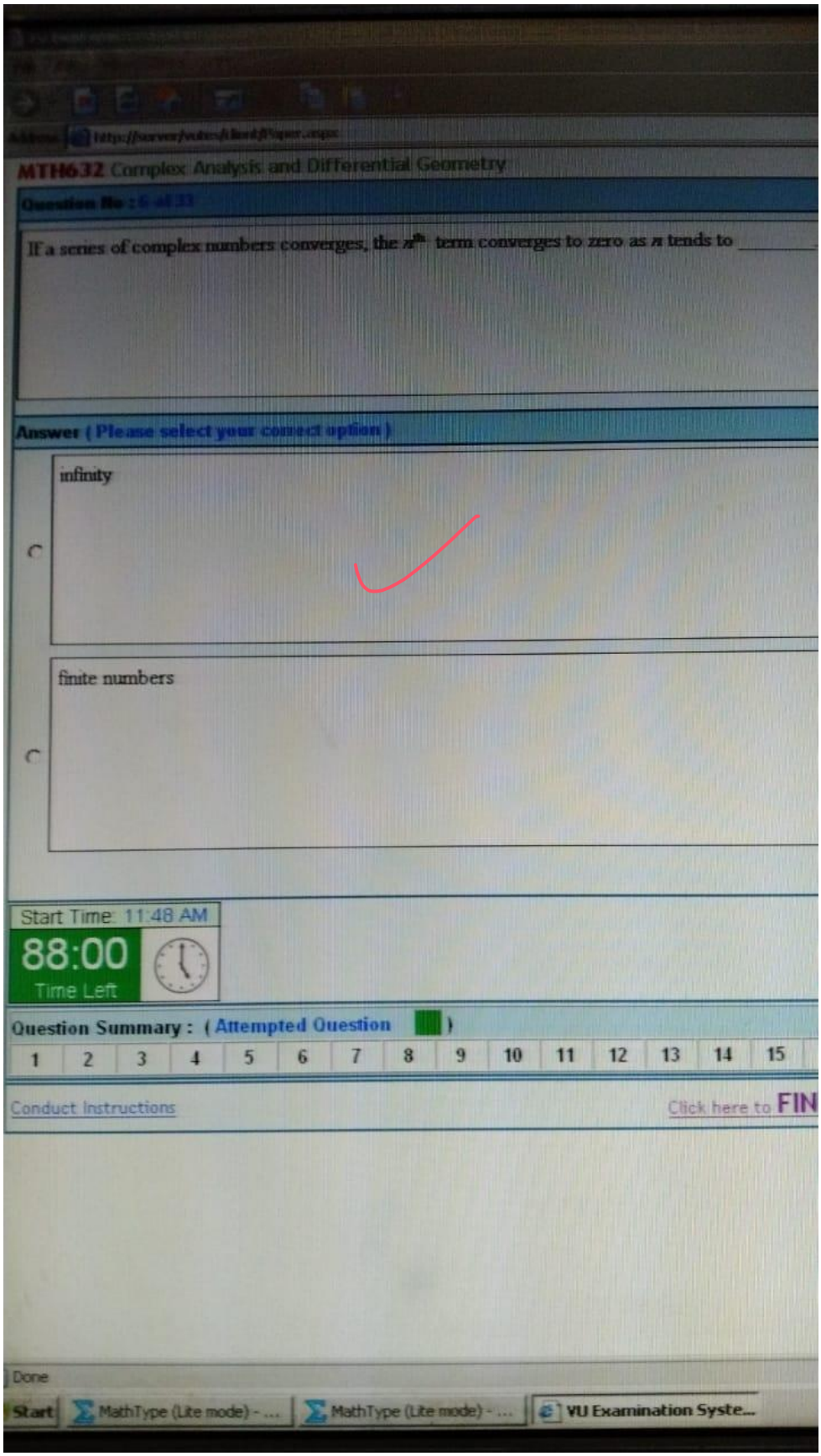
Conduct Instructions

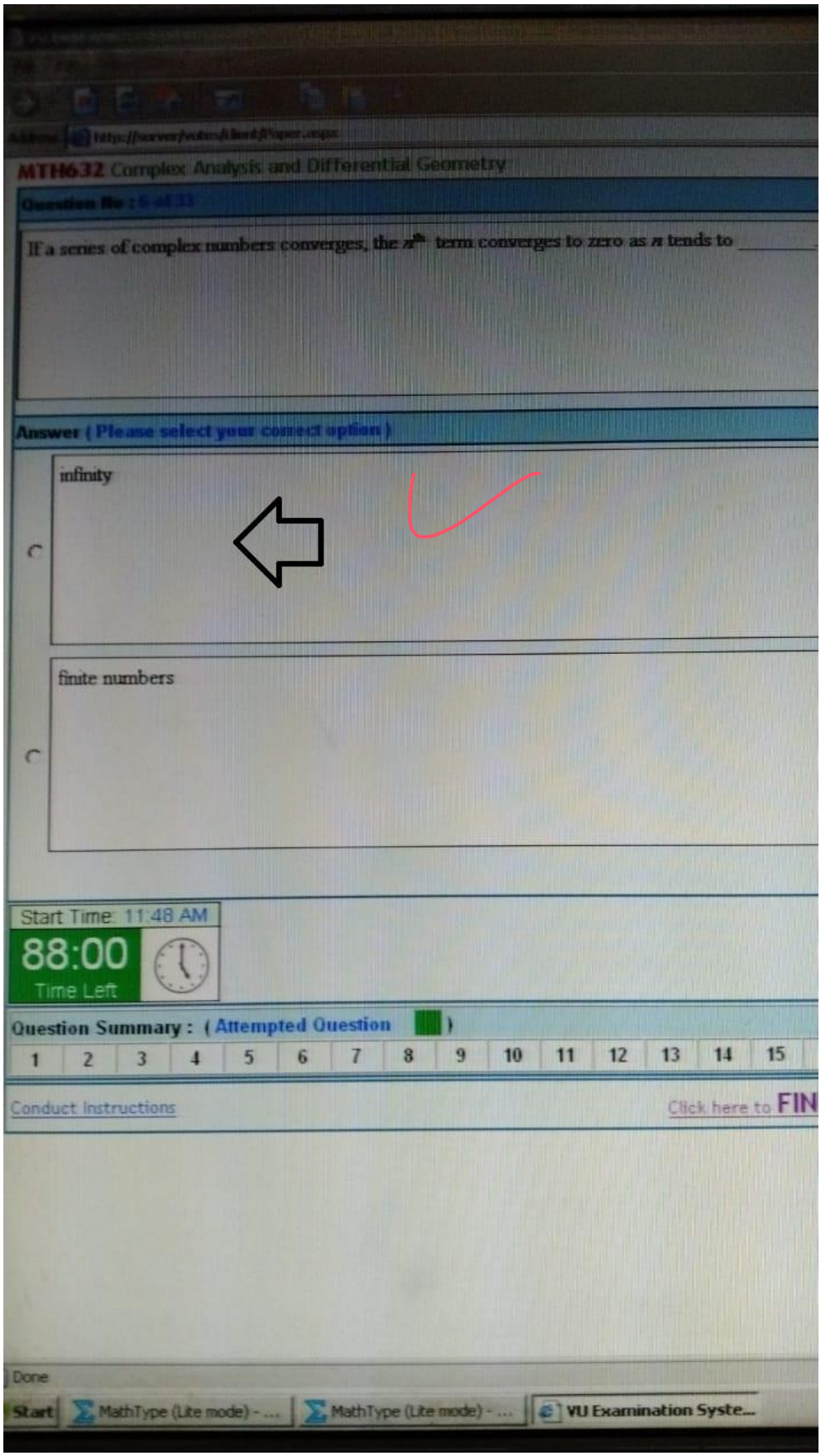
Done

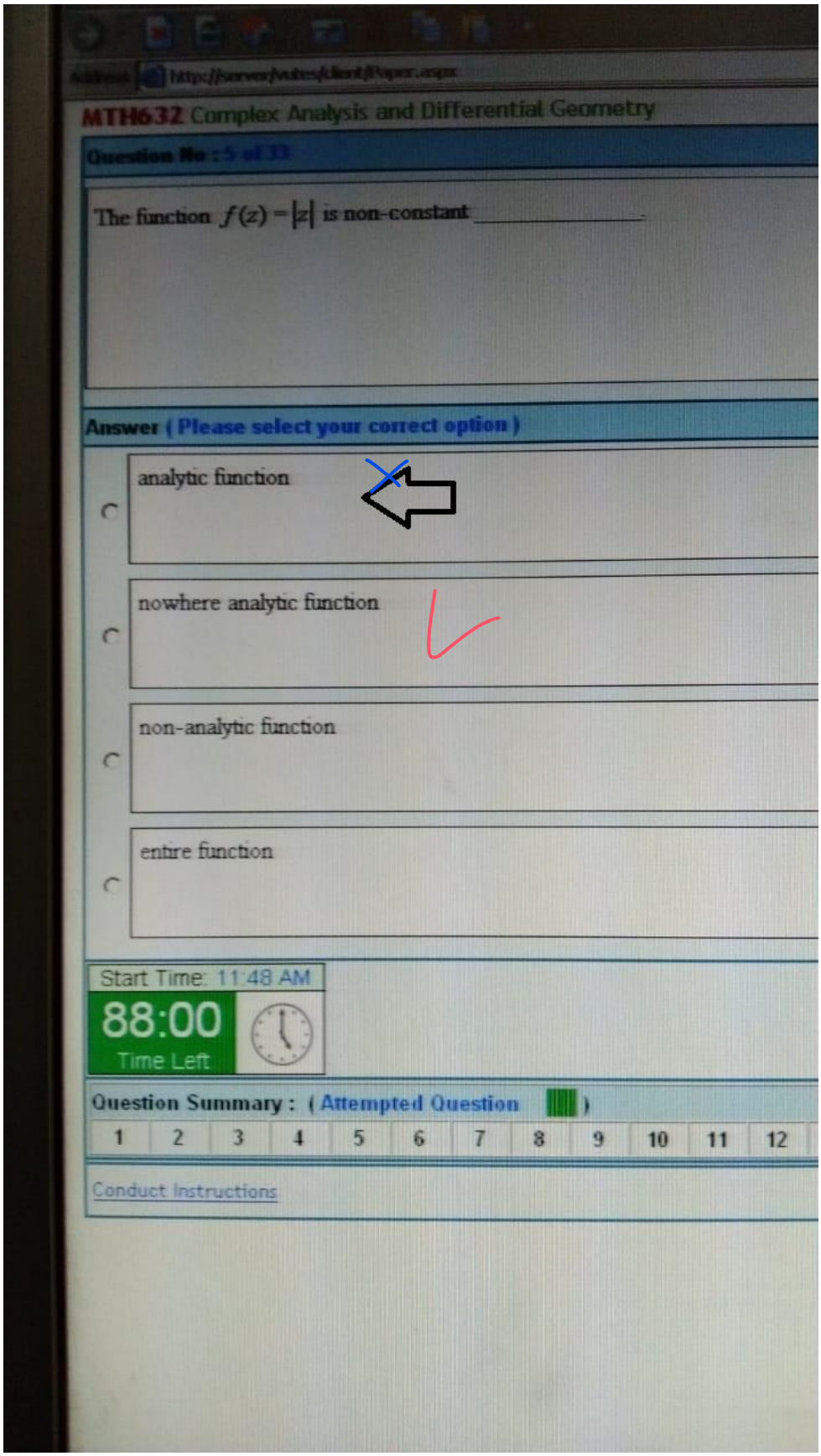

MTH632 Complex Analysis and Differential Geometry
Question No :
The principal value of 1^{2i} is _____.
Answer (Please select your correct option)
0
1
i
2i
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10
Conduct Instructions

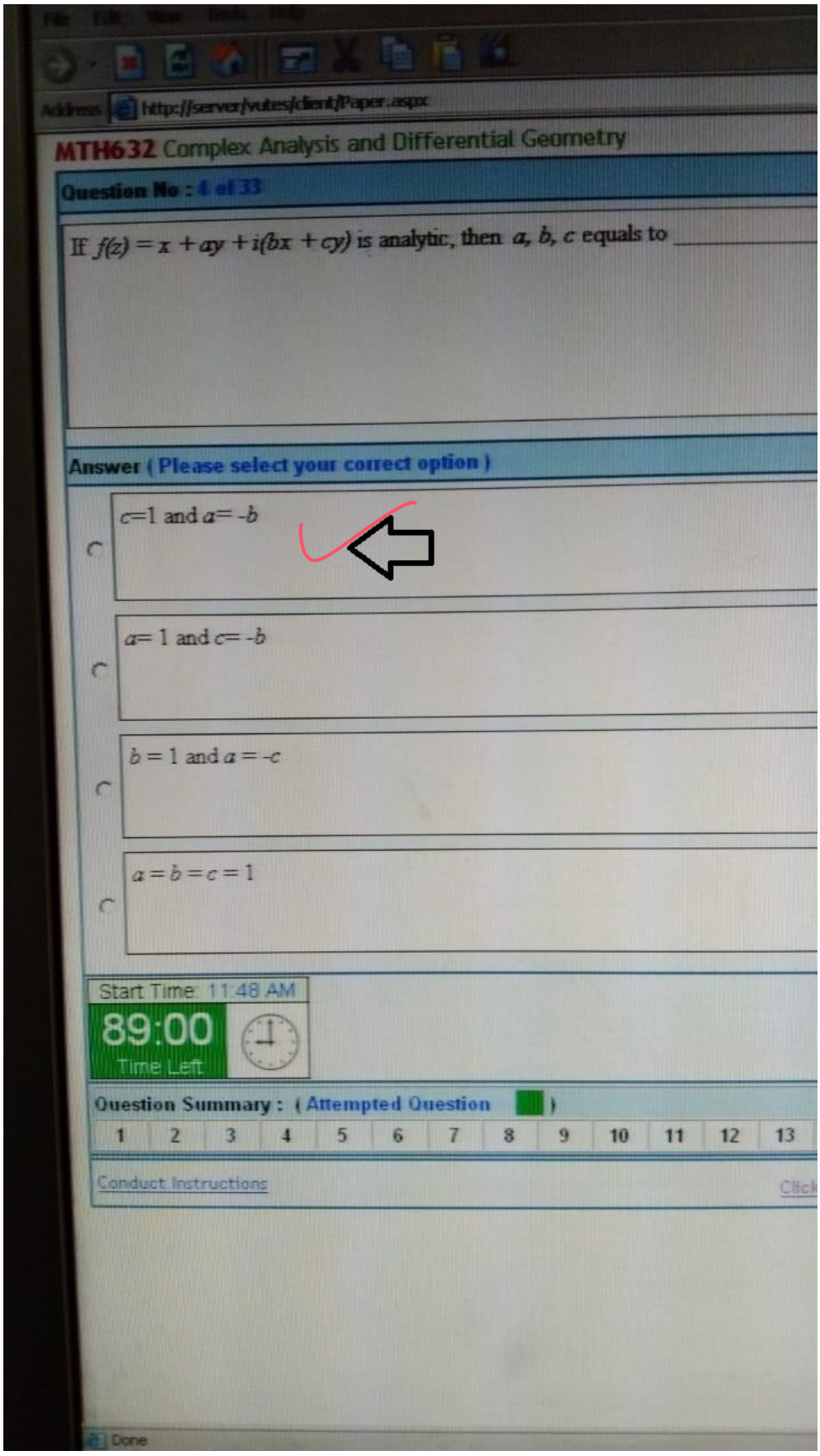

MTH632 Complex Analysis and Differential Geometry
In the trigonometric functions of complex numbers Sin x =
Answer (Please select your correct option)
$\frac{e^{ix}-e^{-ix}}{2i}$
$\frac{e^{ix}-e^{-ix}}{2}$
$\frac{e^{-ix}-e^{ix}}{2i}$
$\frac{e^{-ix}-e^{ix}}{2}$
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done

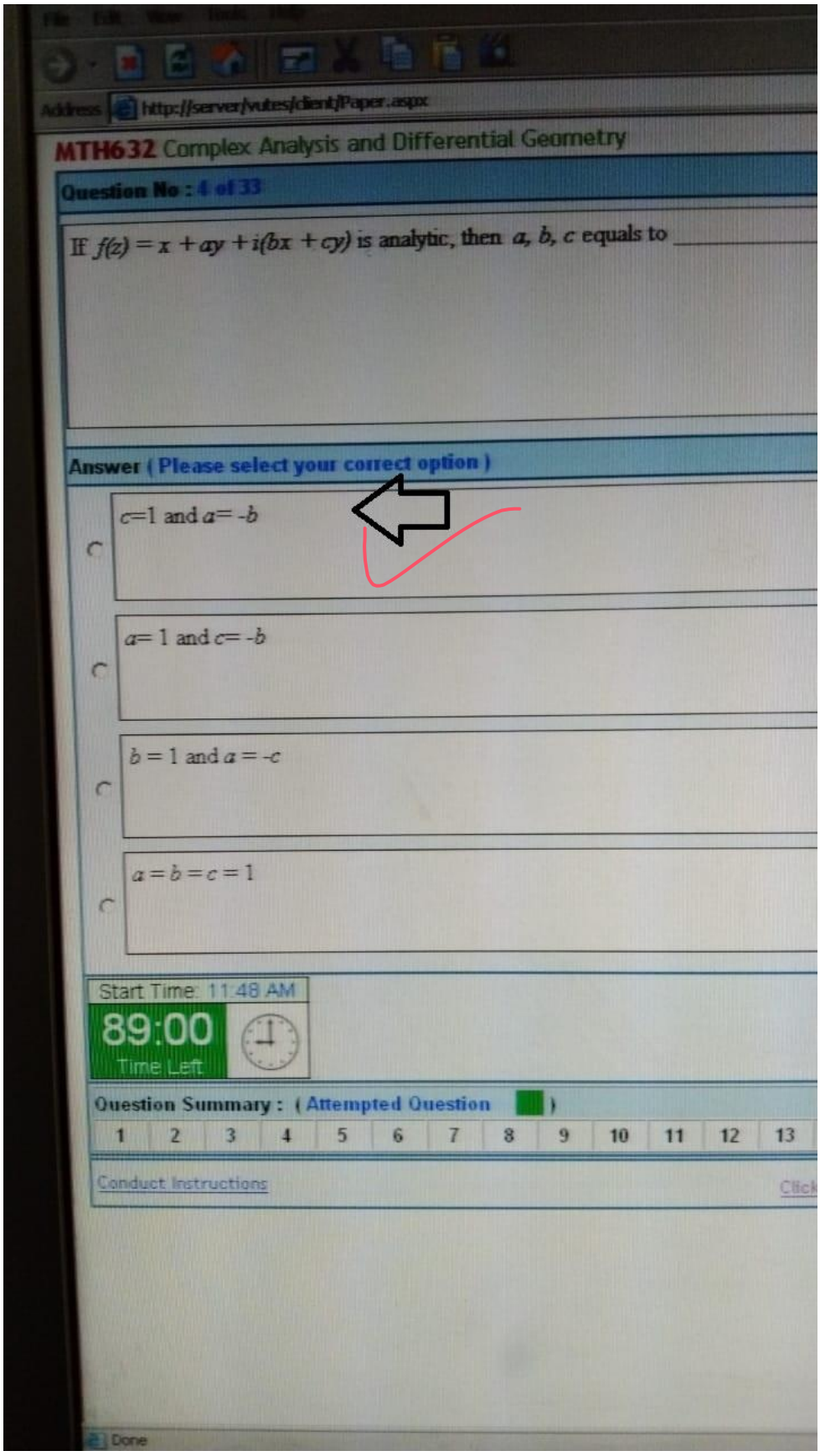

MTH632 Complex Analysis and Differential Geometry
In the trigonometric functions of complex numbers Sin x = _____
Answer (Please select your correct option)
$\frac{e^{ix}-e^{-ix}}{2i}$
$\frac{e^{ix}-e^{-ix}}{2}$
$\frac{e^{-ix}-e^{ix}}{2i}$
$\frac{e^{-ix}-e^{ix}}{2}$
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done

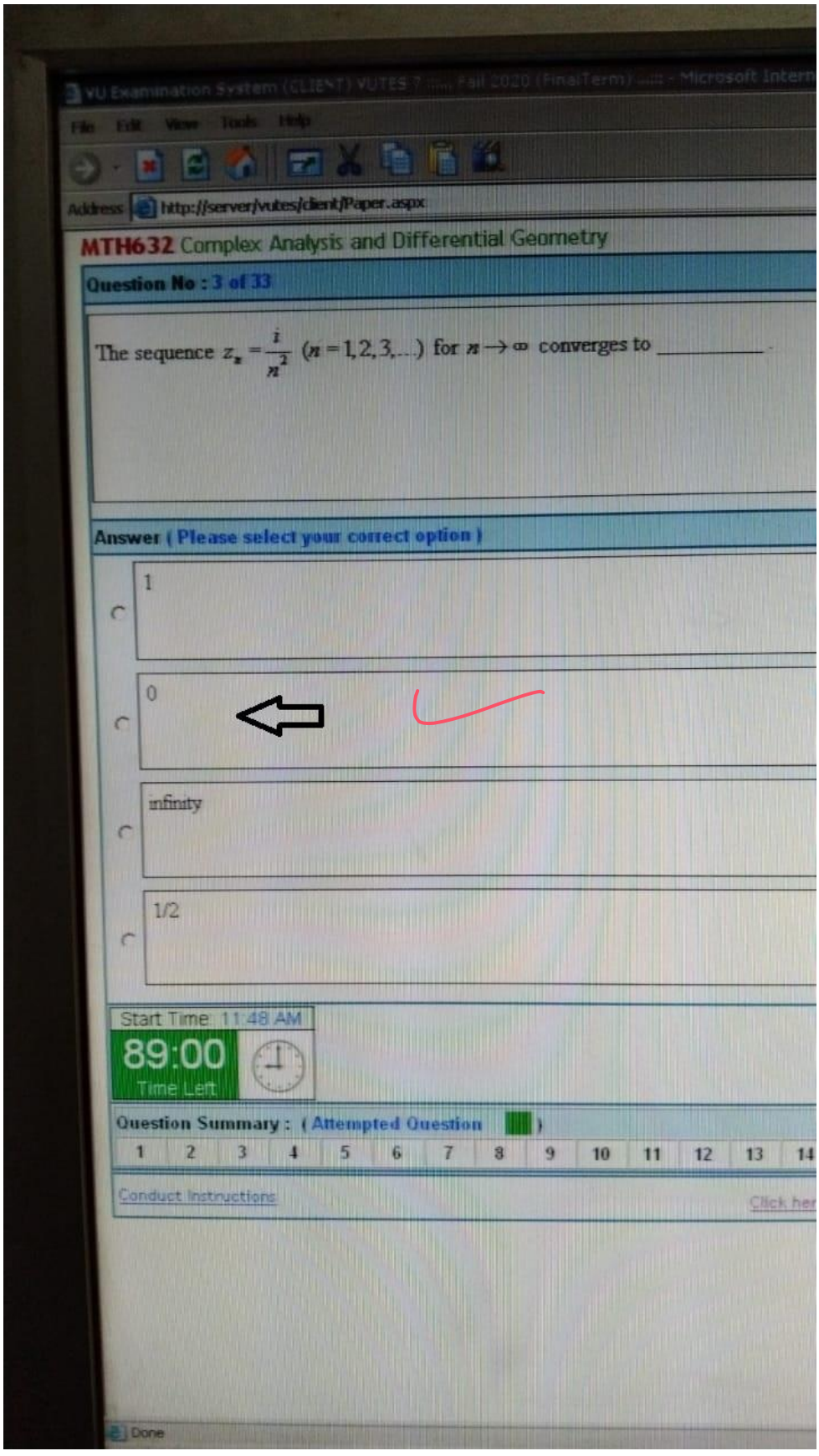

MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

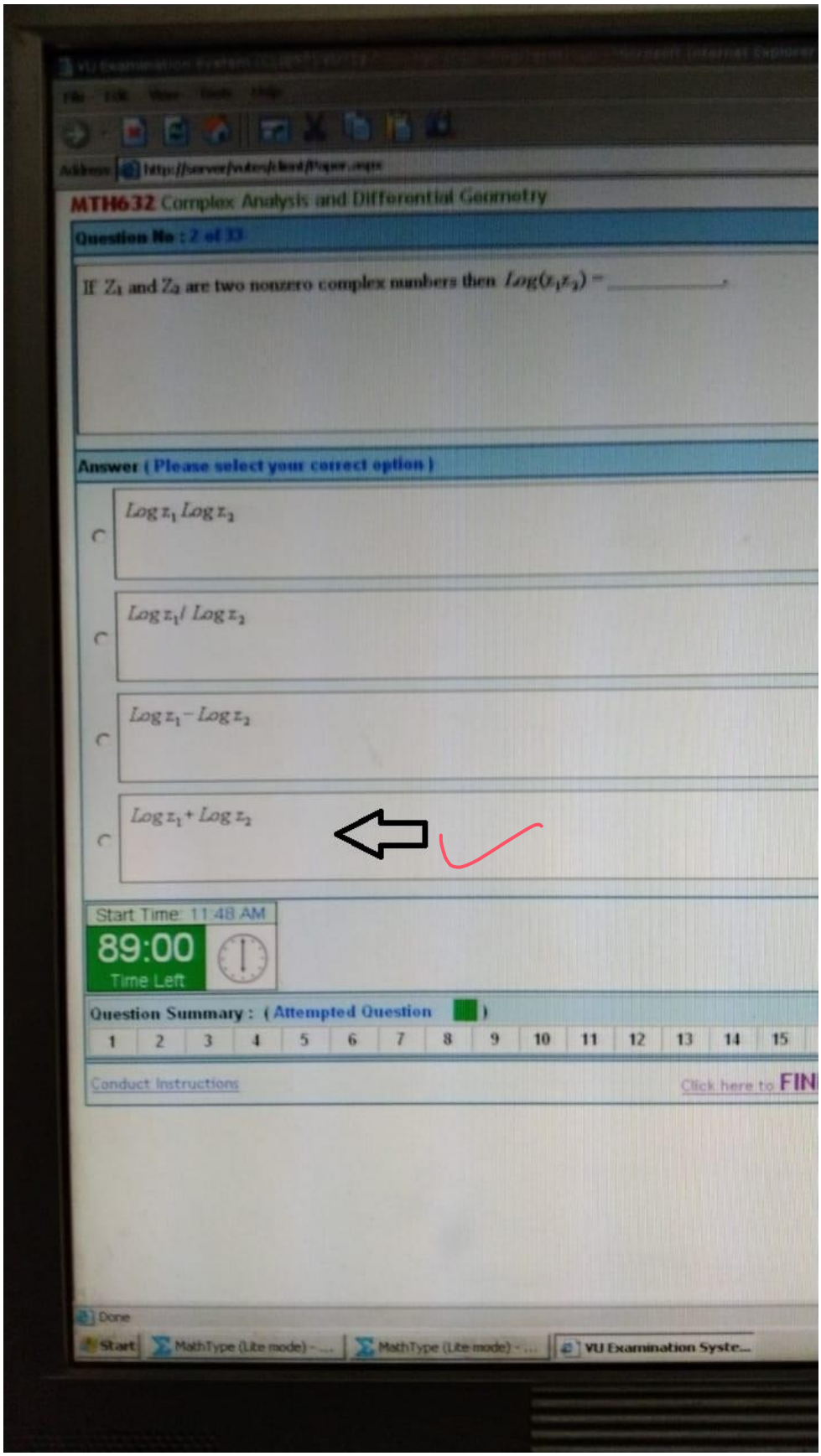

MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

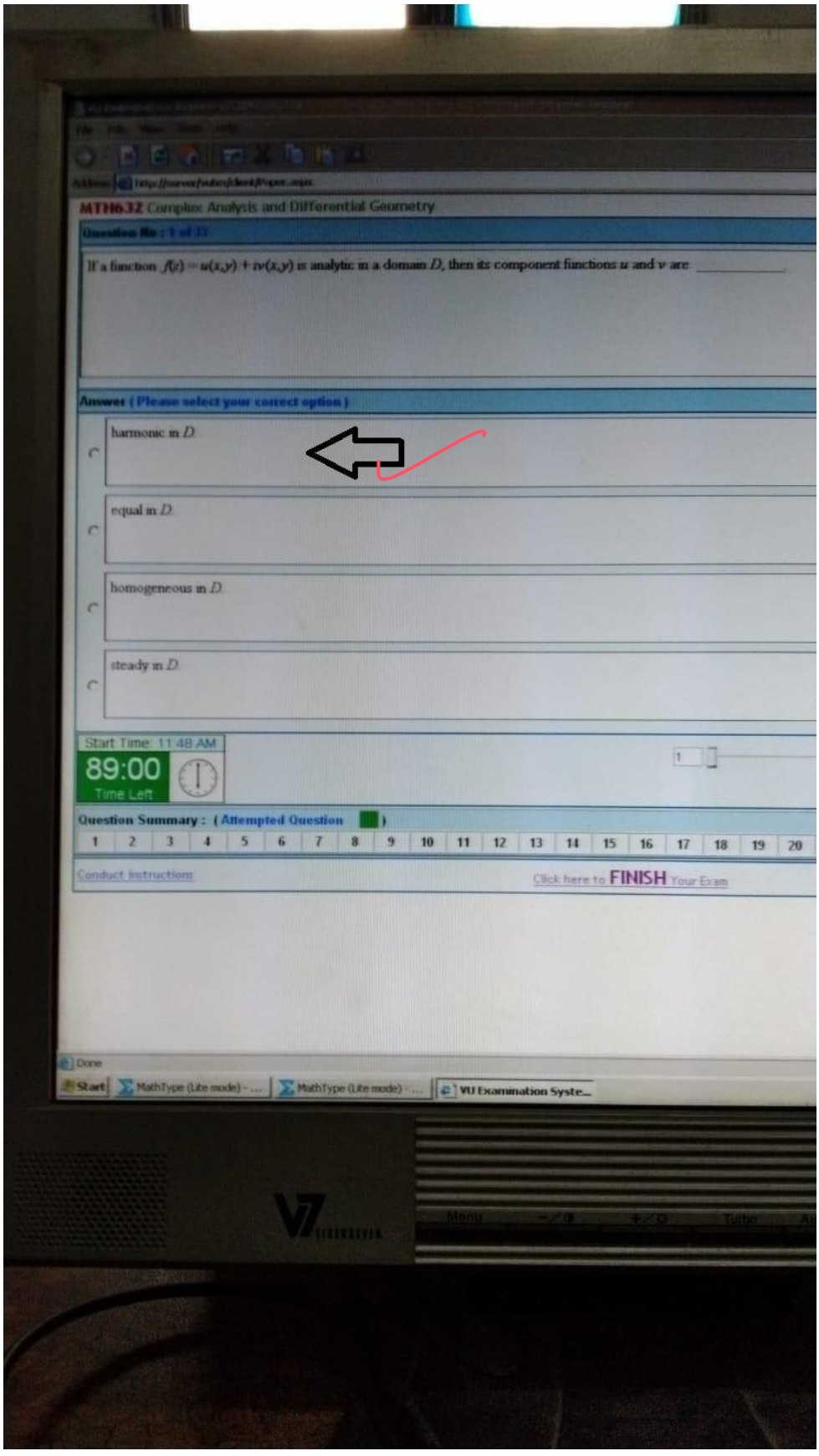

MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
The function $f(z) = |z|$ is non-constant _____.
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
$c=1$ and $a=-b$
$a=1$ and $c=-b$
$b=1$ and $a=-c$
$a=b=c=1$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
If f(z) = x + ay + i(bx + cy) is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
c=1 and a= -b
a= 1 and c= -b
b = 1 and a = -c
a = b = c = 1
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions
Done


VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Intern
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click her
Done


MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\sinh Z =$ _____

Answer (Please select your correct option)

- $\frac{e^{z}-e^{-z}}{2}$ 
- $\frac{e^{z}+e^{-z}}{2}$
- $\frac{e^{z}-e^{-z}}{2i}$
- $\frac{e^{z}+e^{-z}}{2i}$

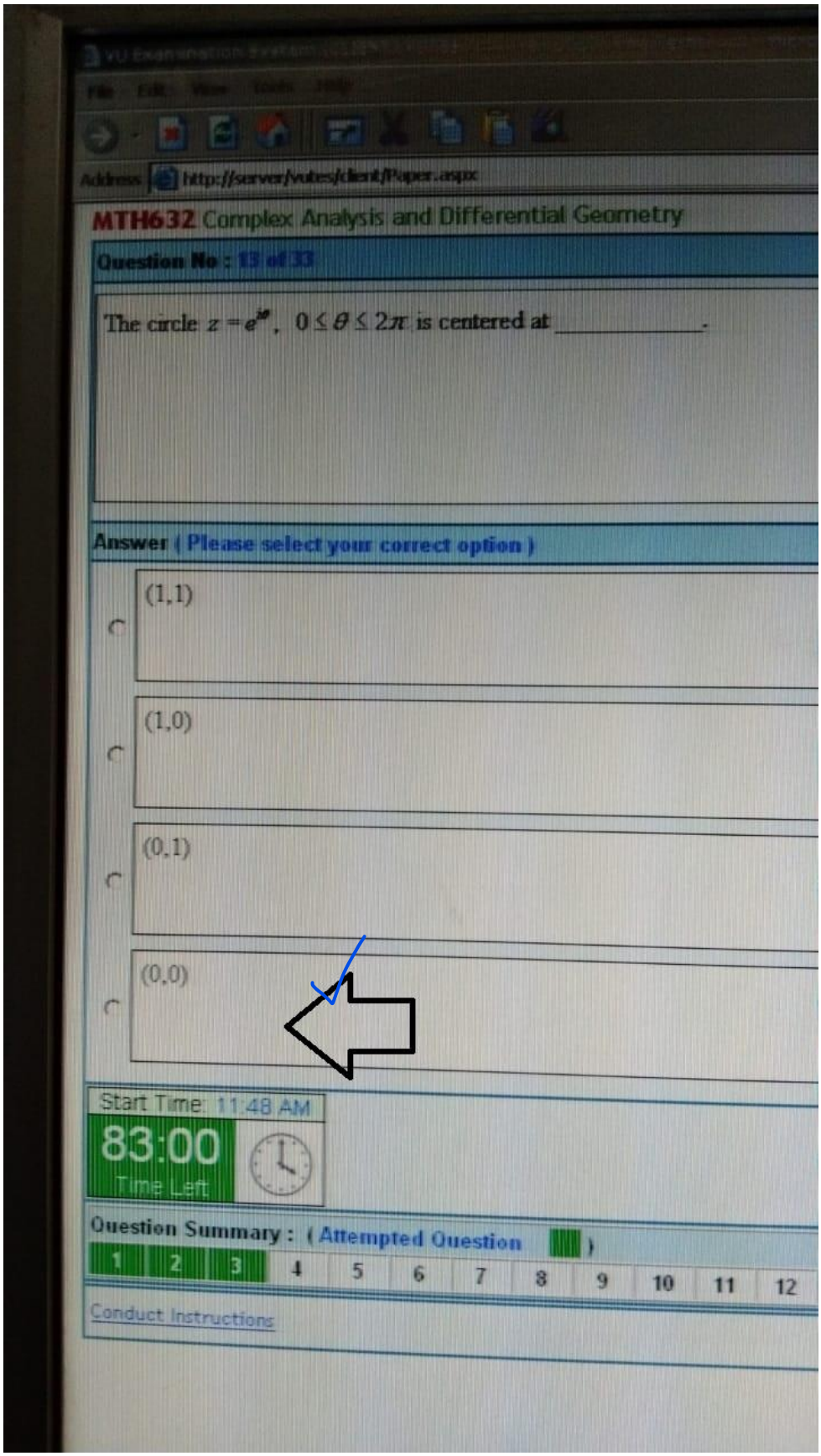
Start Time: 11:48 AM

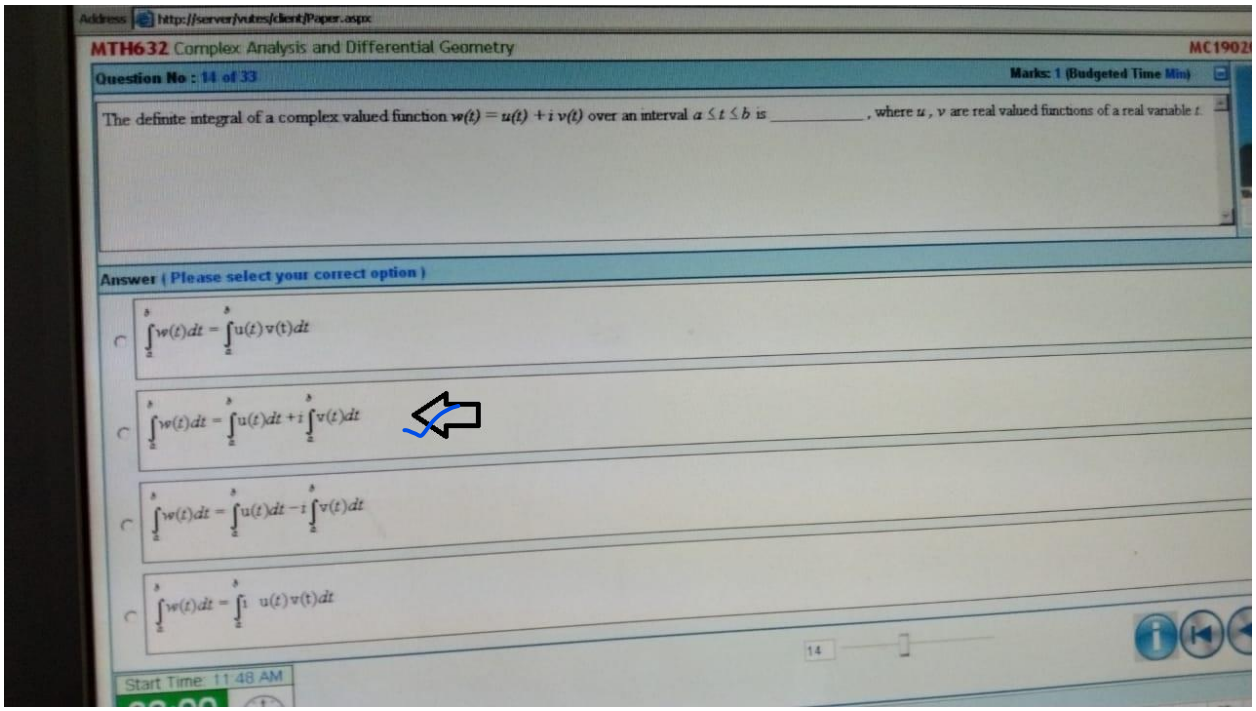
83:00
Time Left

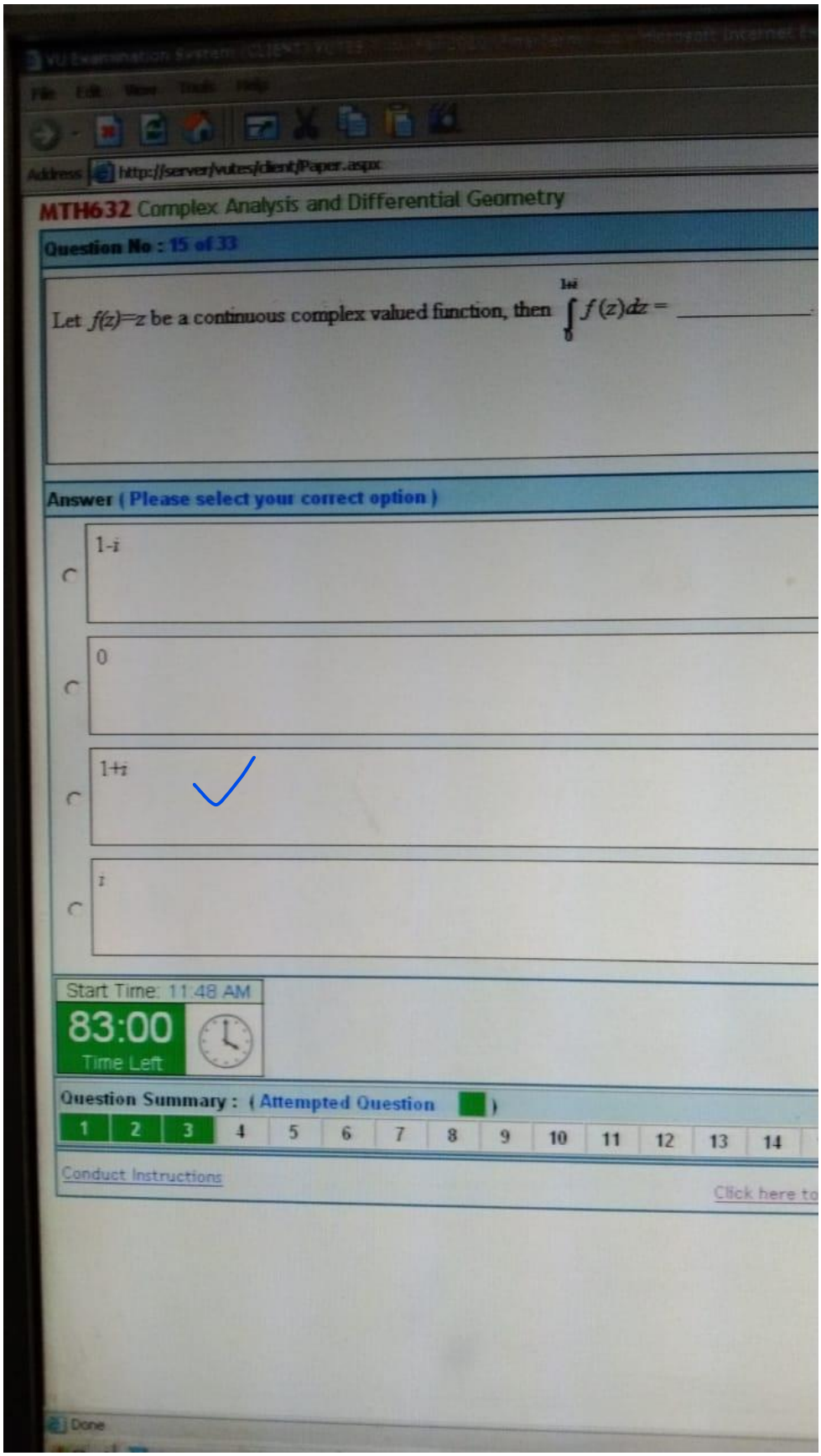
Question Summary : (Attempted Question)

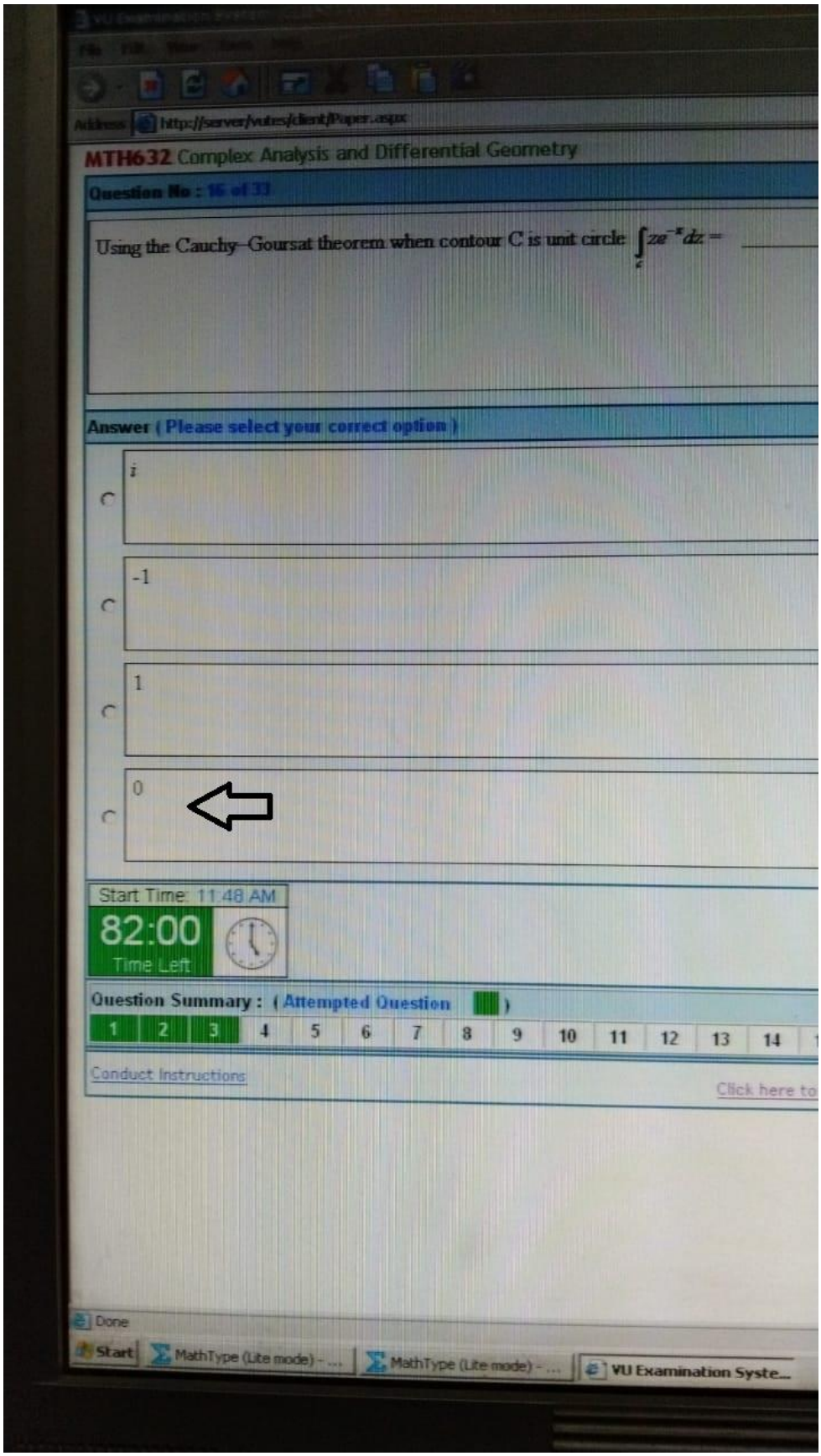
1	2	3	4	5	6	7	8	9	10	11

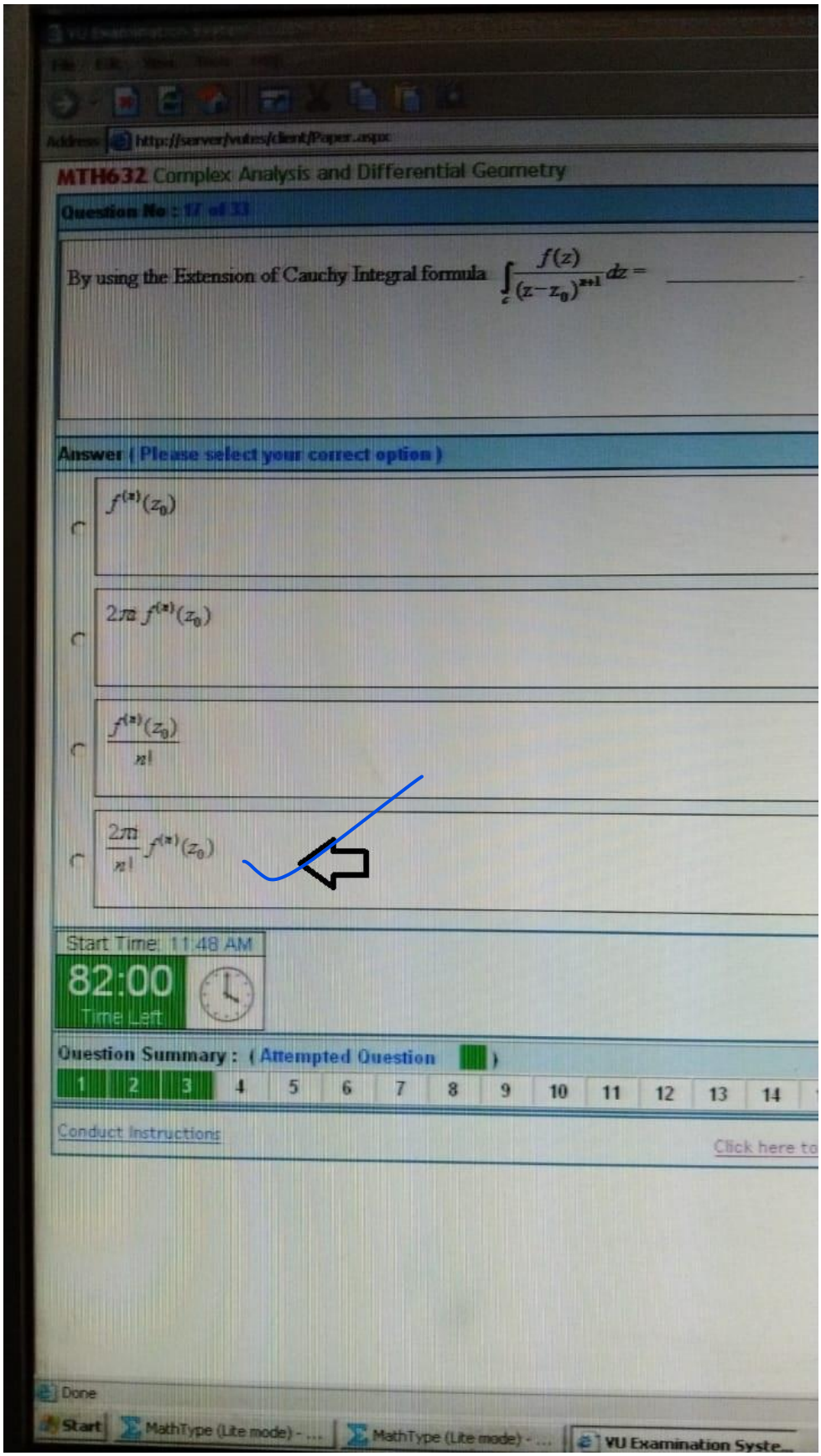
Conduct Instructions

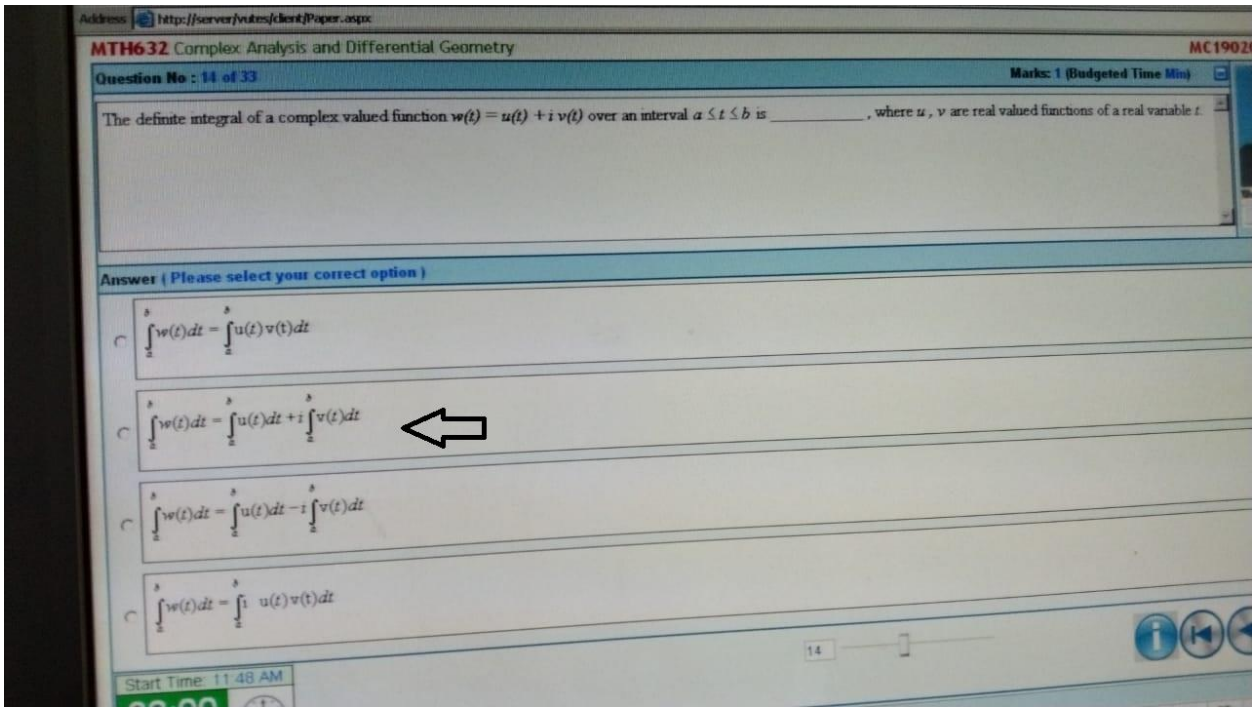

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

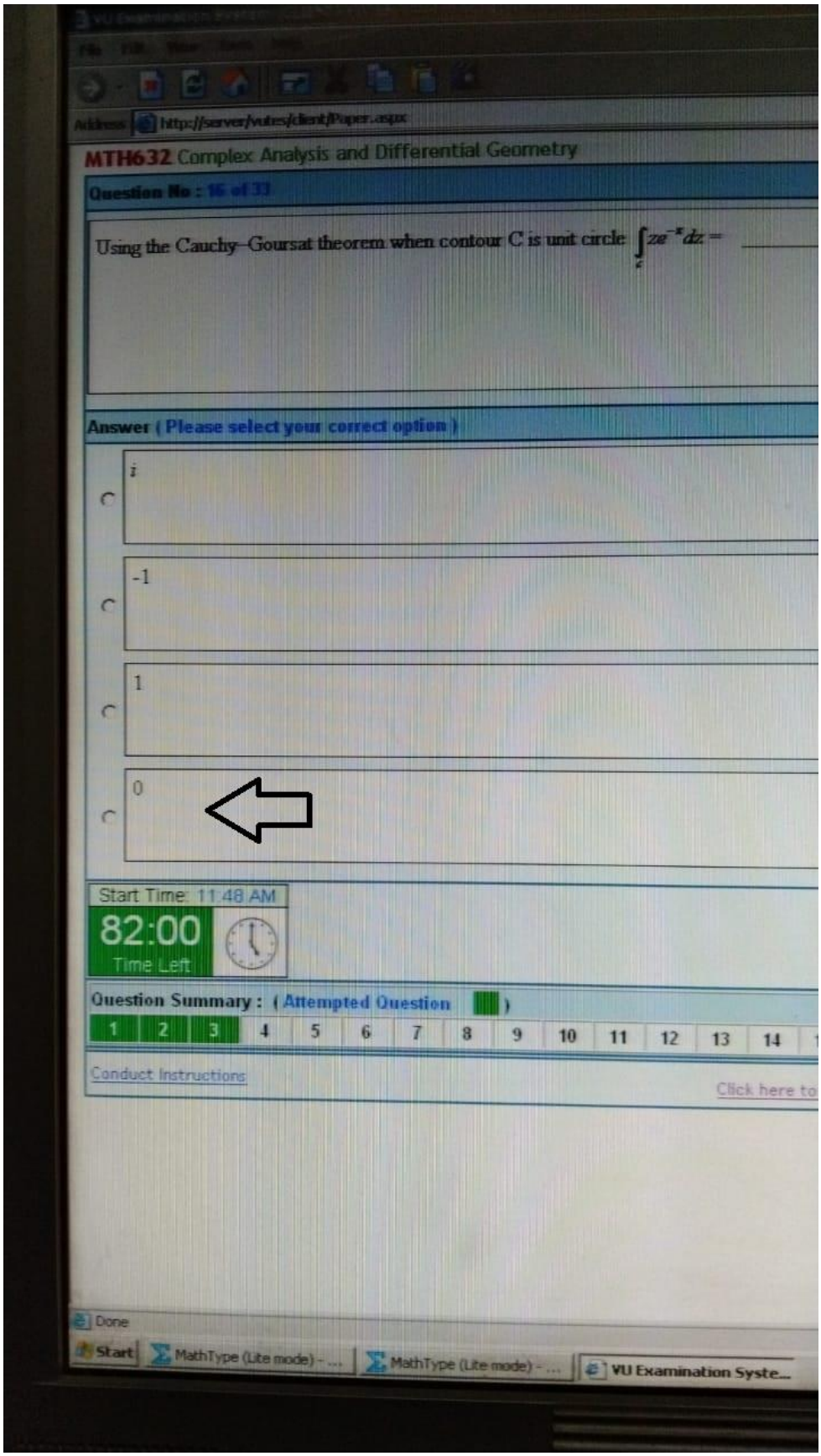

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b 1\ u(t)v(t)dt$
14
Start Time: 11:48 AM

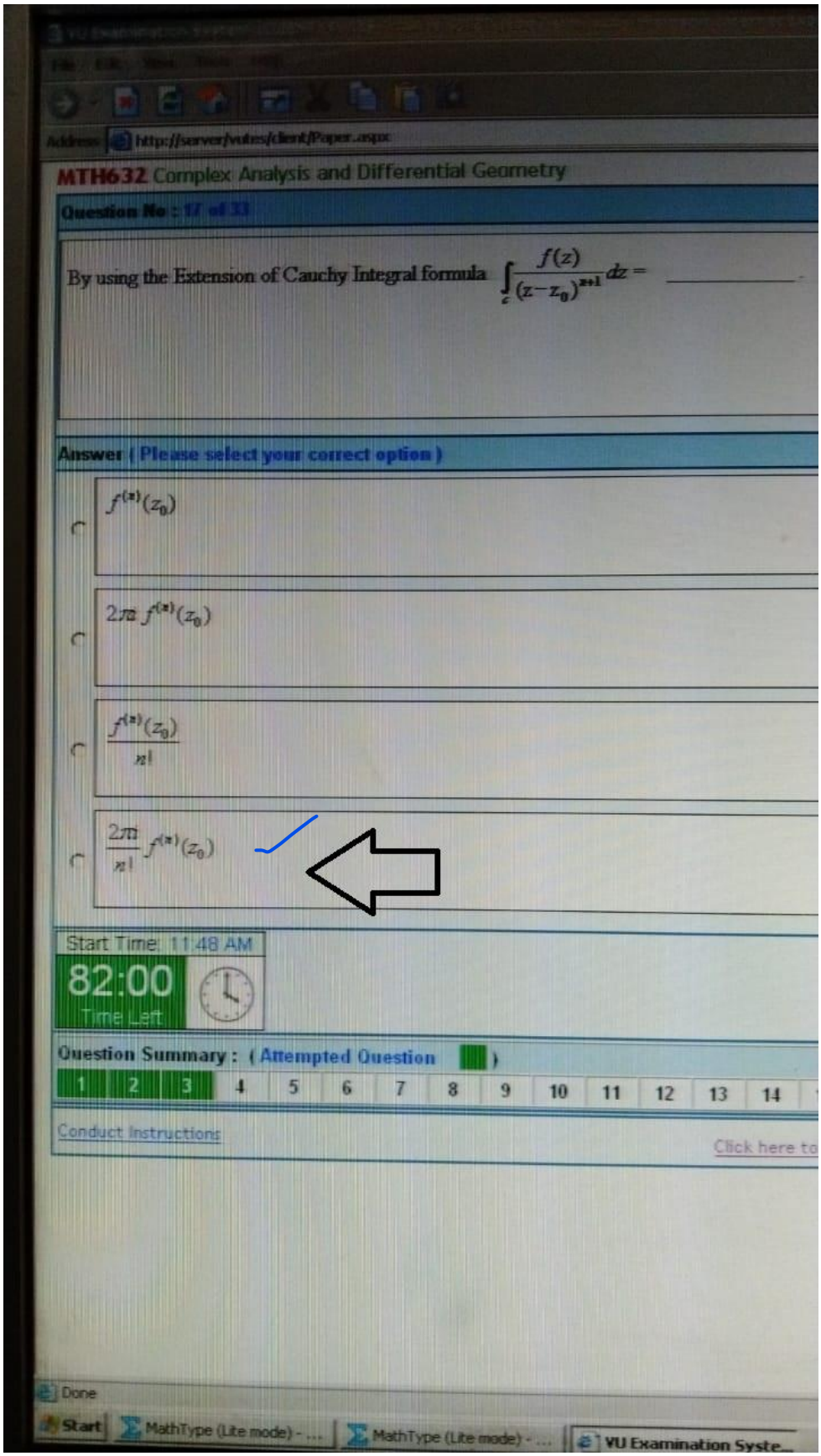

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done

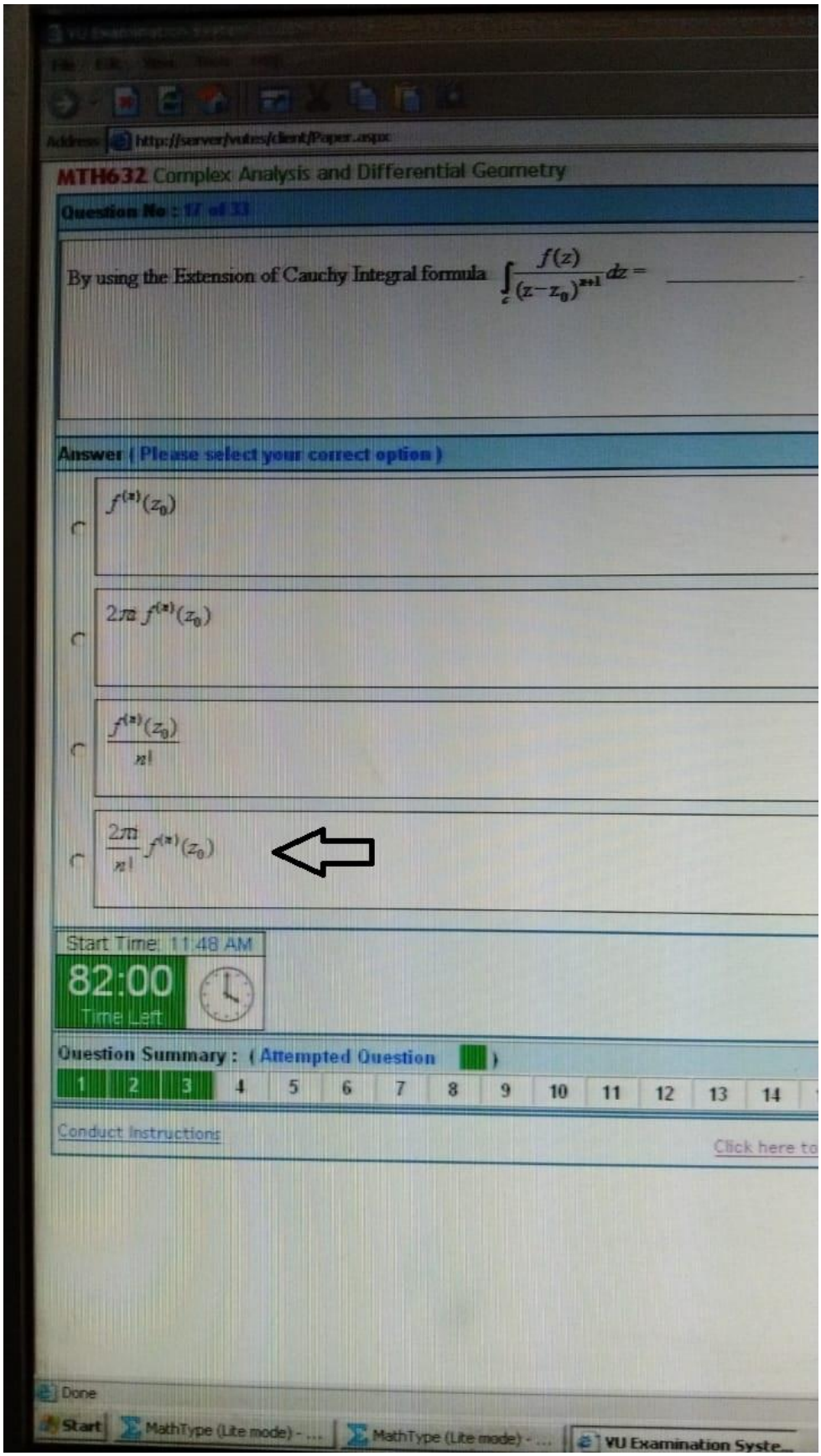

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

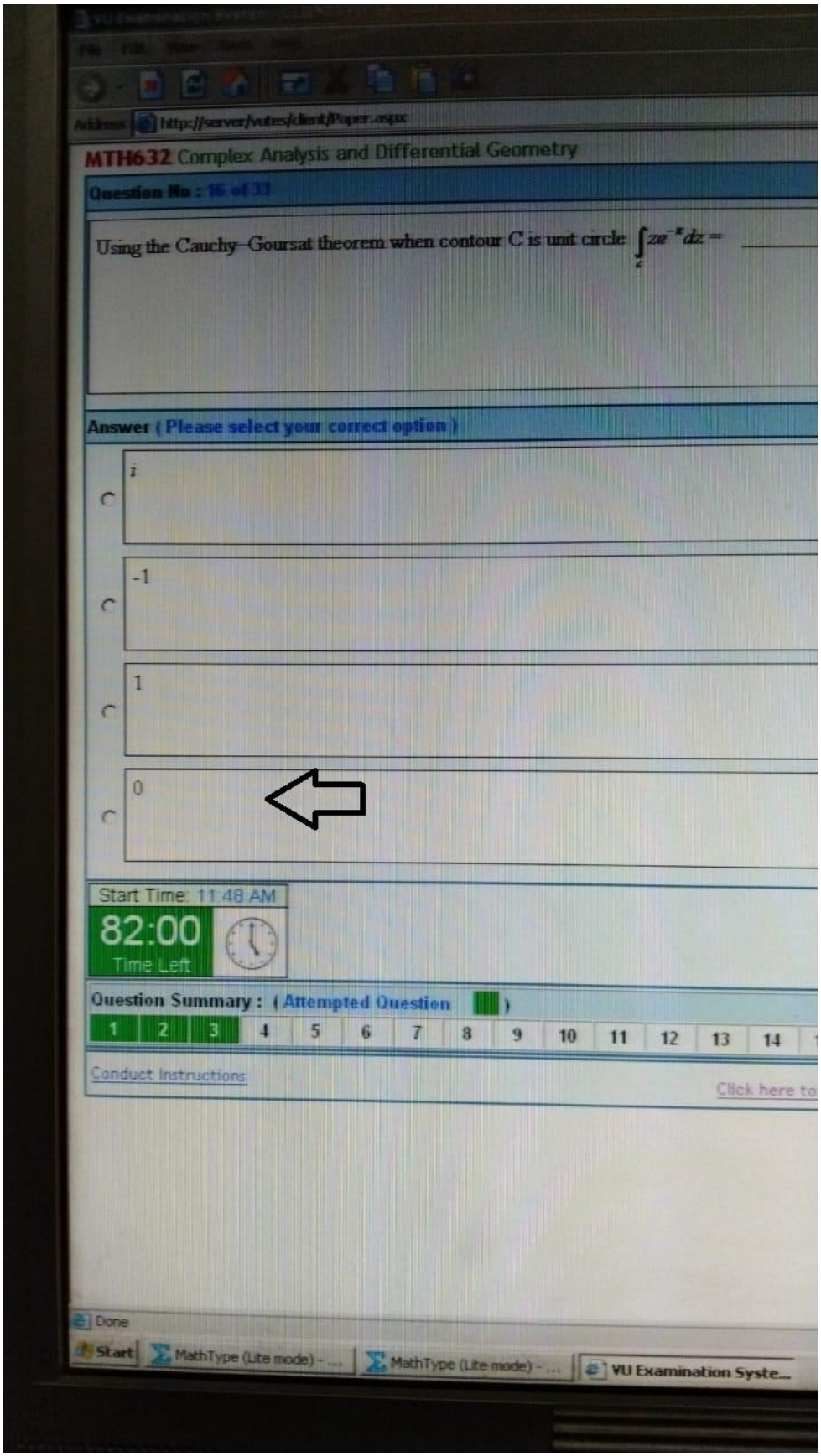

MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

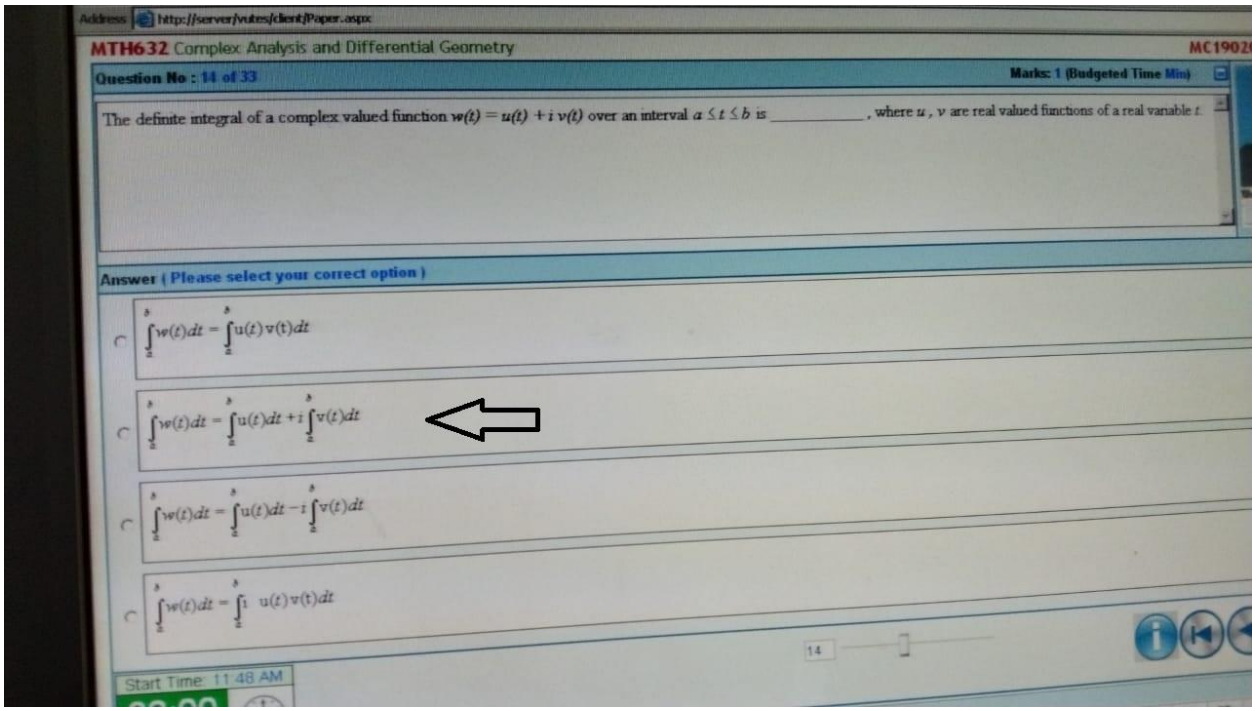

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
14
Start Time: 11:48 AM

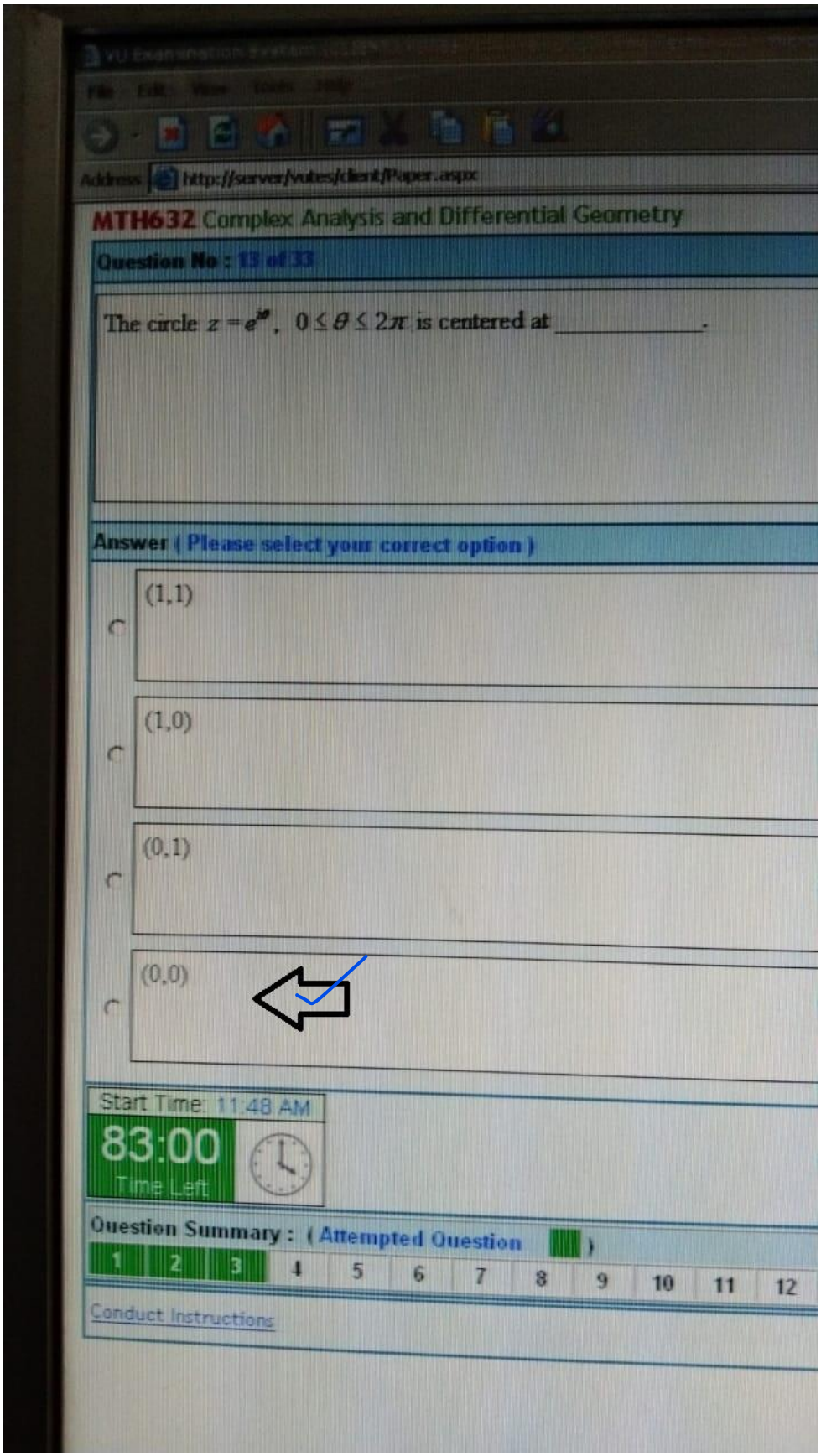

http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____
Answer (Please select your correct option)
i
-1
1
0
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b 1\ u(t)v(t)dt$
14
Start Time: 11:48 AM


Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\sinh Z =$ _____

Answer (Please select your correct option)

- $\frac{e^{z}-e^{-z}}{2}$
- $\frac{e^{z}+e^{-z}}{2}$
- $\frac{e^{z}-e^{-z}}{2i}$
- $\frac{e^{z}+e^{-z}}{2i}$

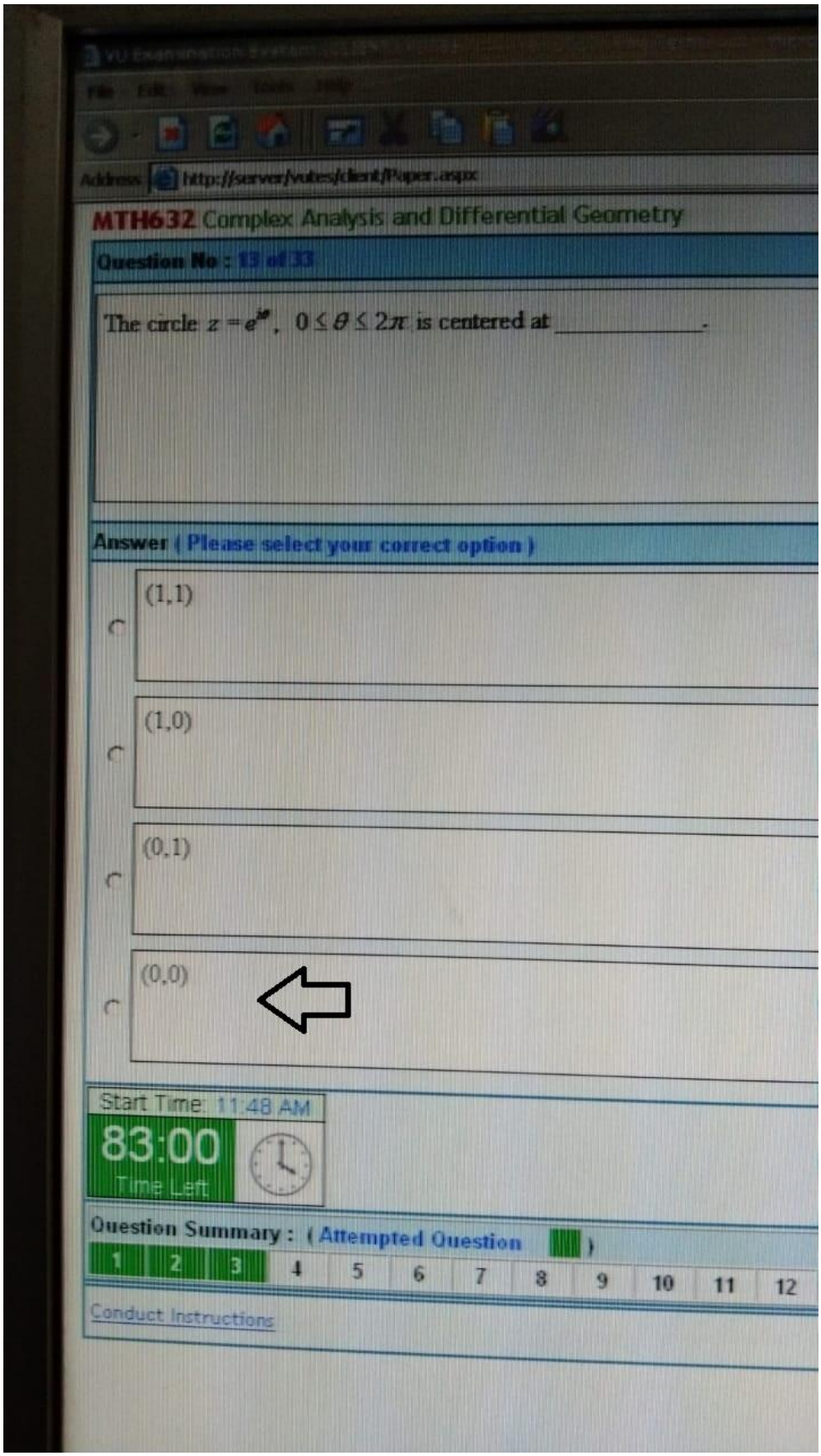
Start Time: 11:48 AM

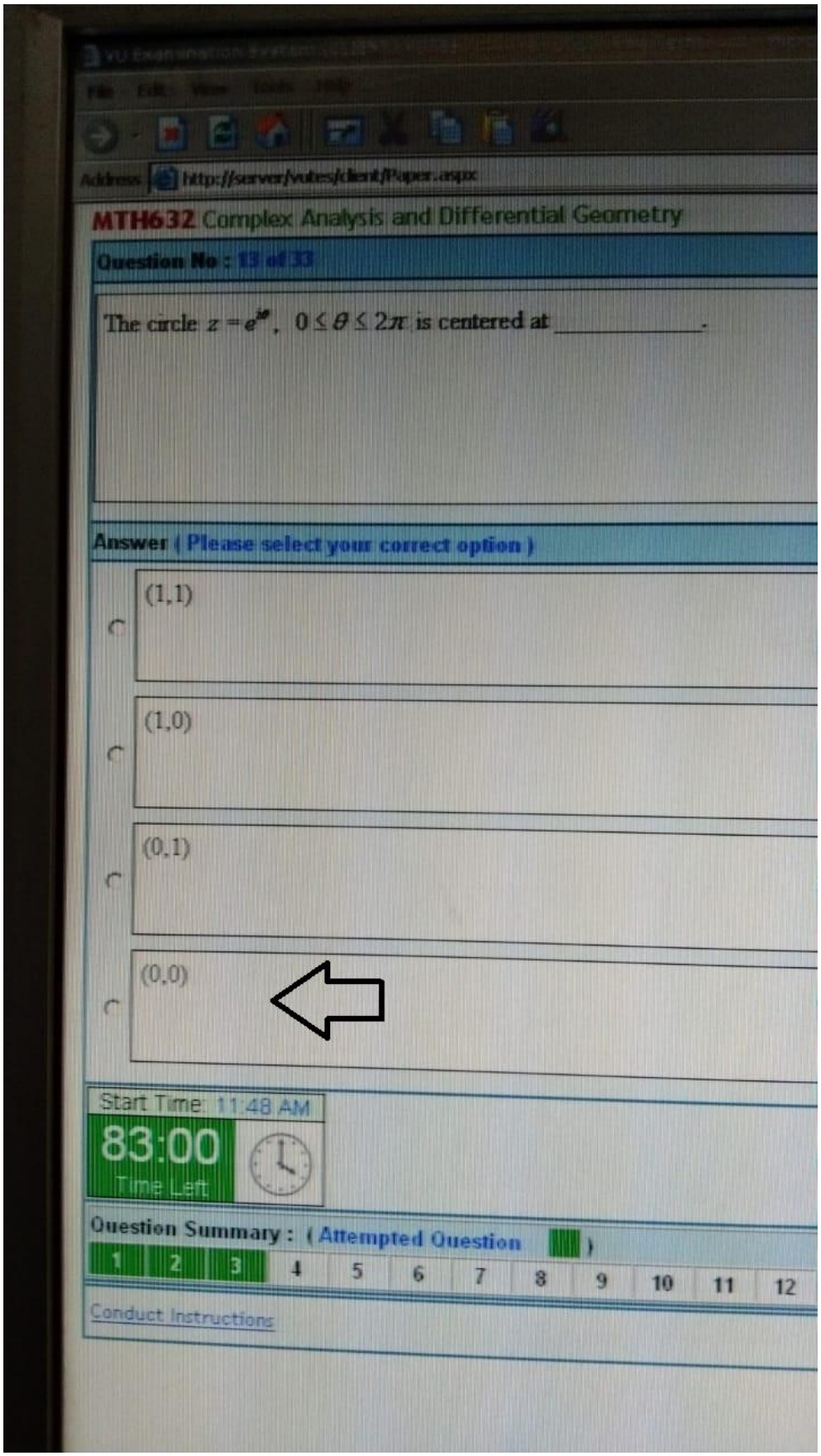
83:00 Time Left

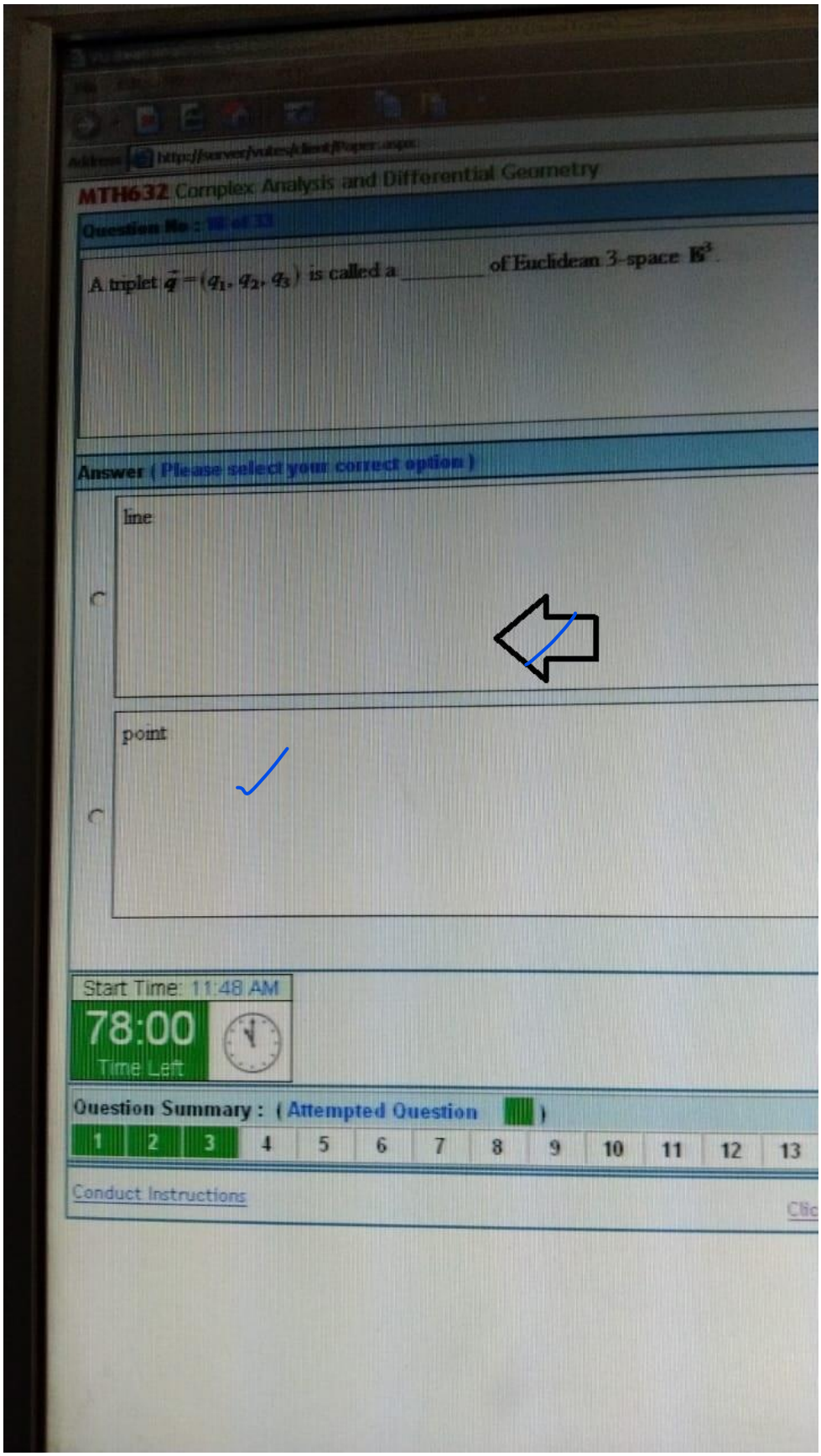
Question Summary : (Attempted Question)

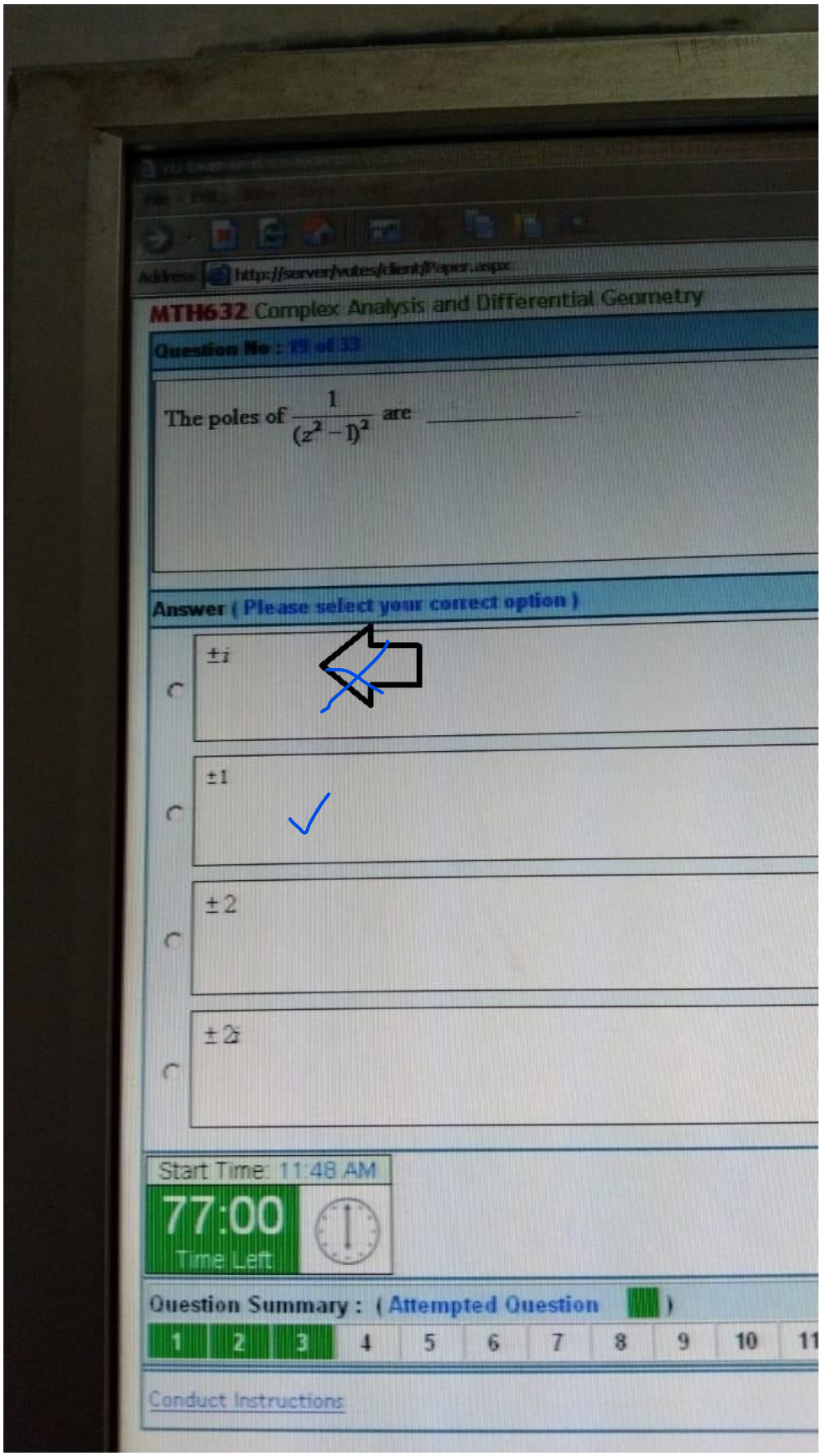
1	2	3	4	5	6	7	8	9	10	11

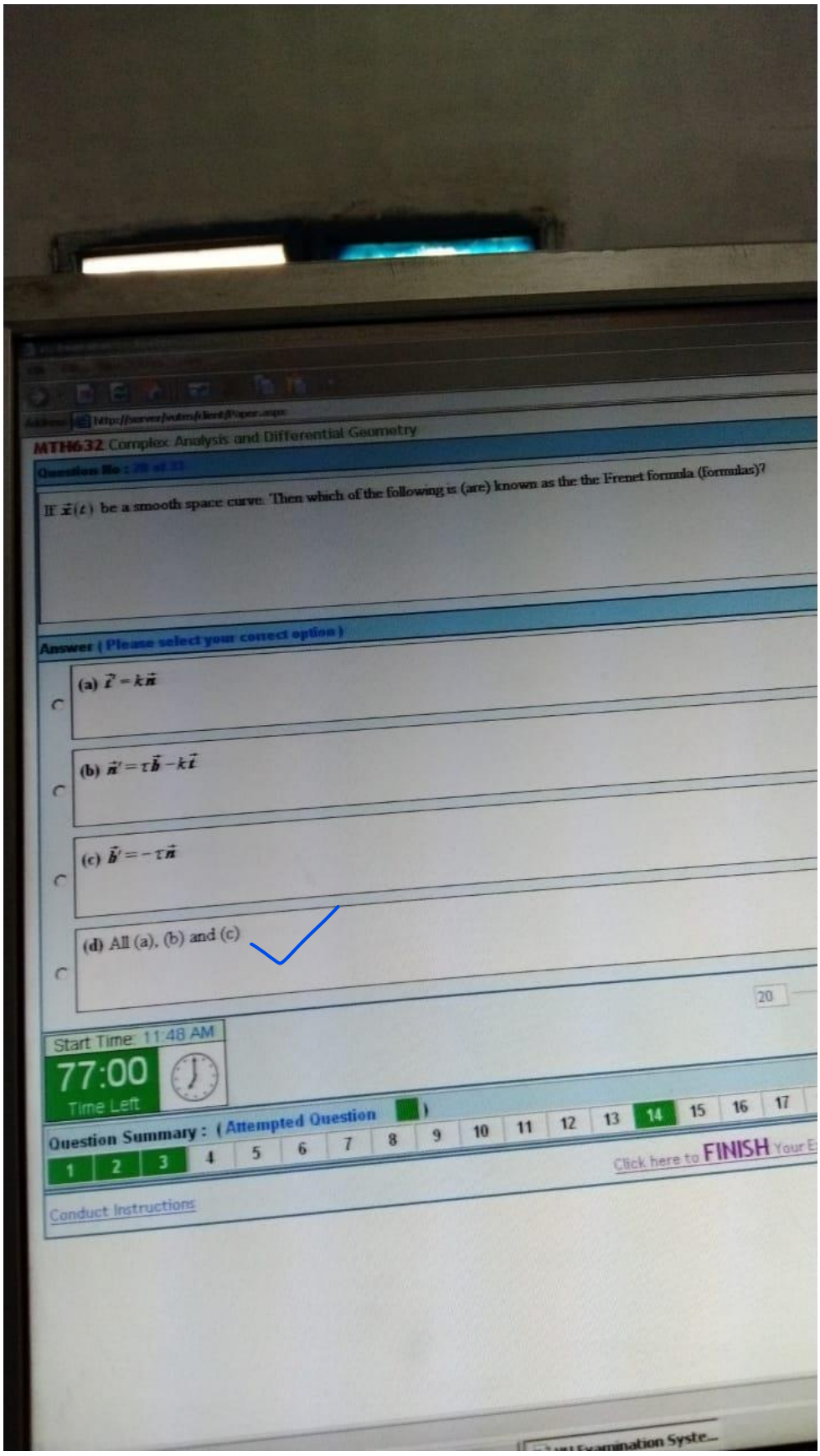
Conduct Instructions

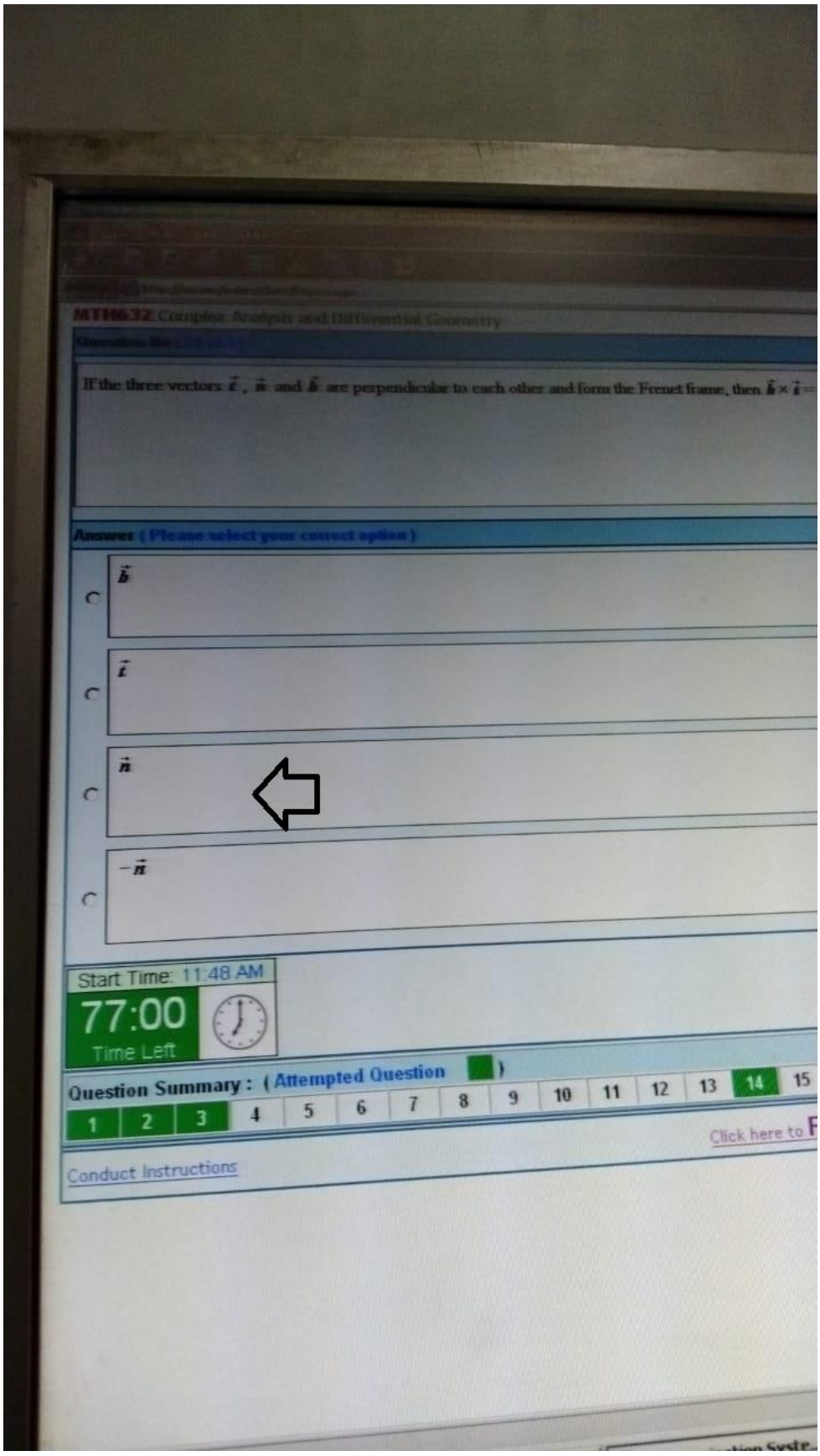

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

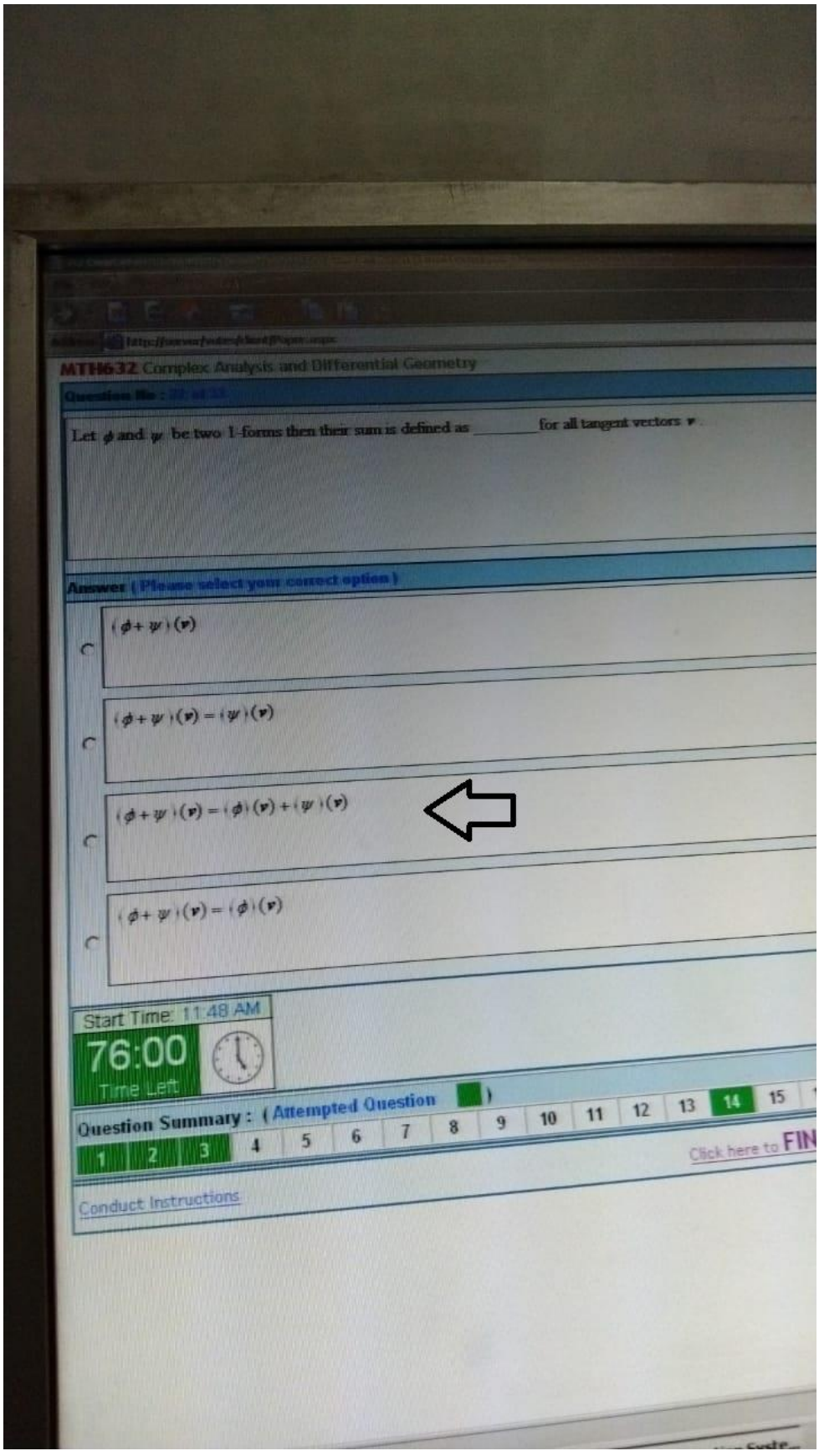

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

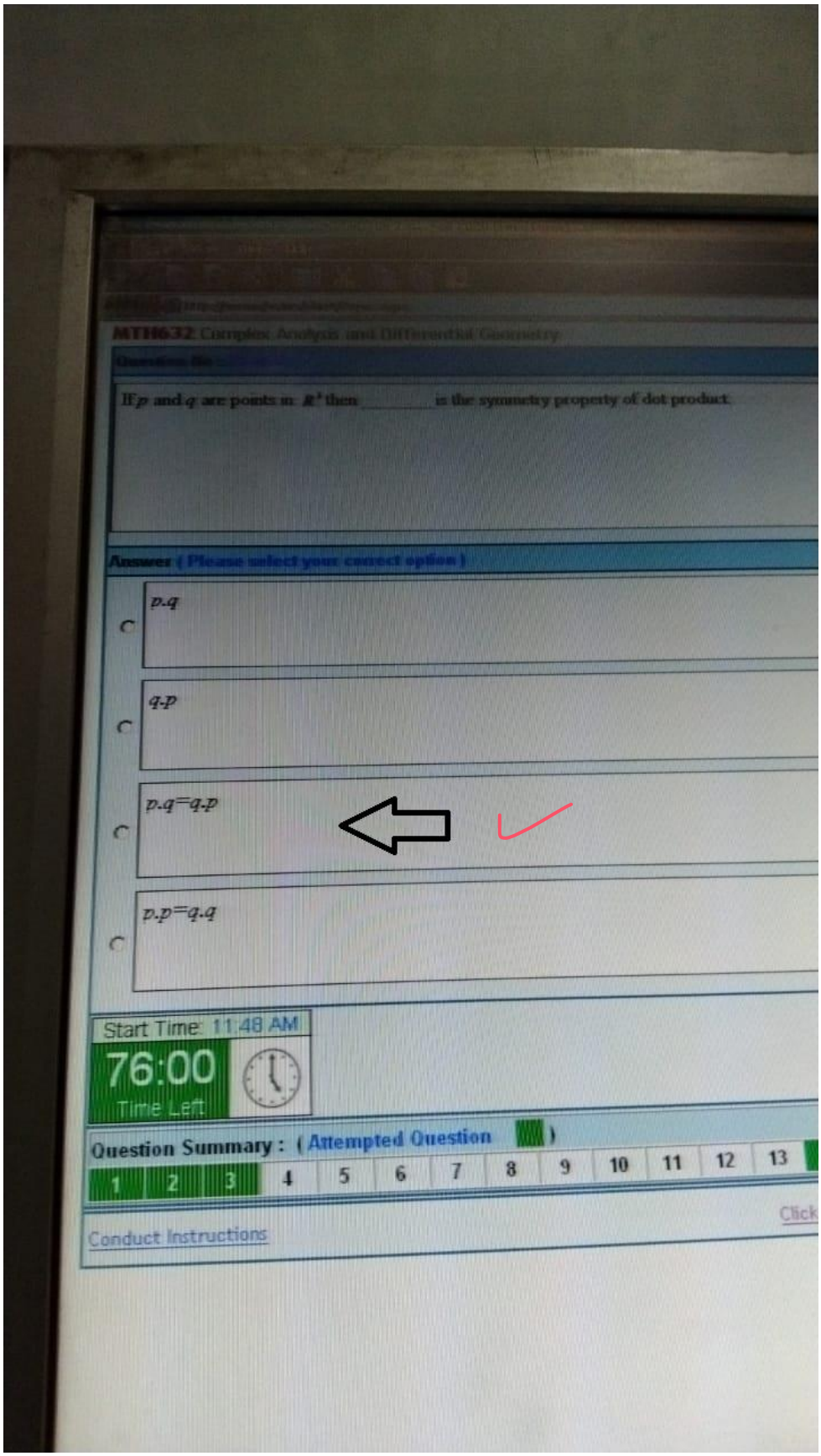

MTH632 Complex Analysis and Differential Geometry
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time: 11:48 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

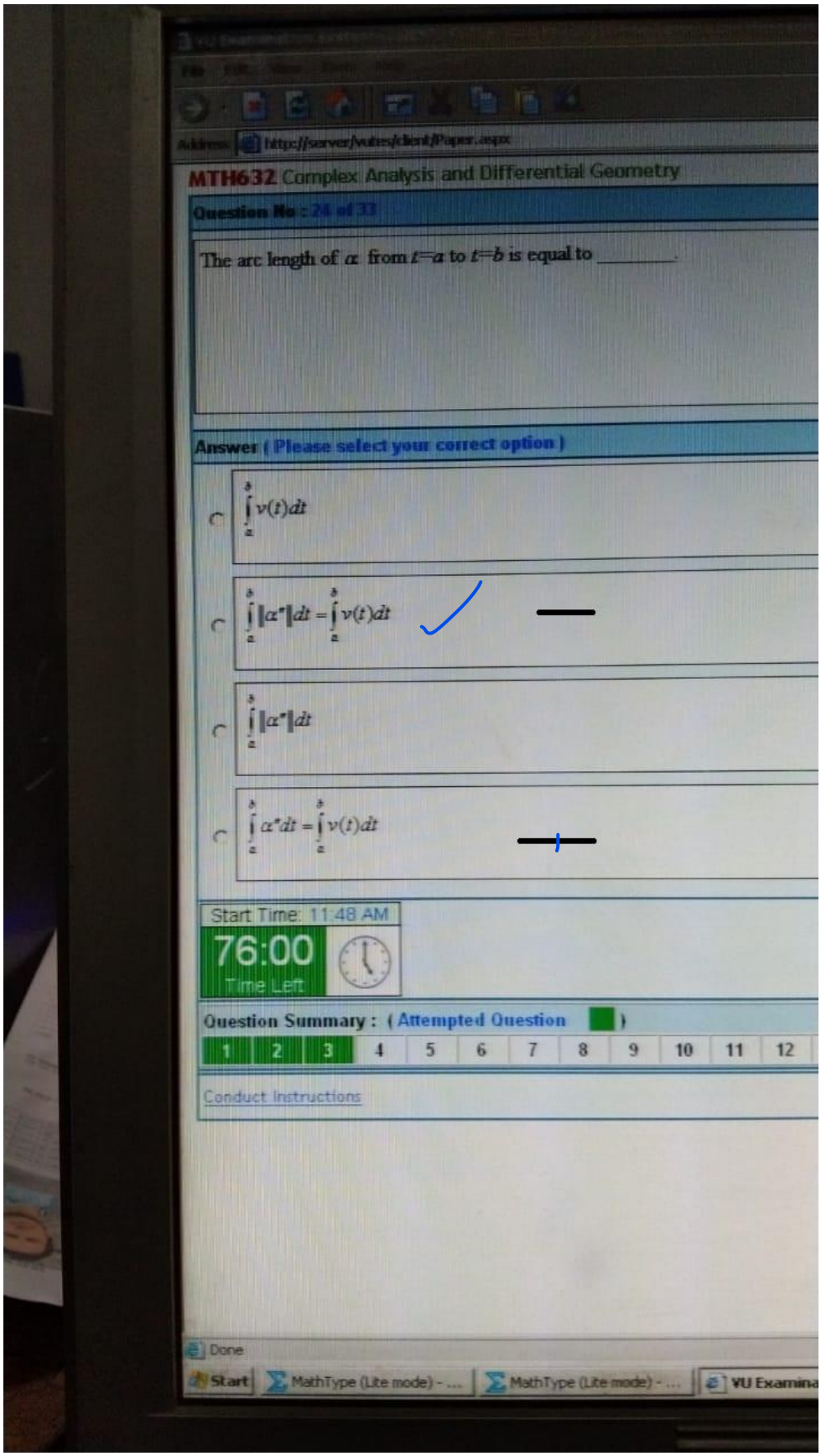

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632: Complex Analysis and Differential Geometry
If p and q are points in R³ then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions



MTH632 Complex Analysis and Differential Geometry

Question No : 24 of 33

The arc length of α from $t=a$ to $t=b$ is equal to _____.

Answer (Please select your correct option)

- $\int_a^b v(t)dt$
- $\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
- $\int_a^b \|\alpha'\|dt$
- $\int_a^b \alpha' dt = \int_a^b v(t)dt$

Start Time: 11:48 AM

76:00 Time Left

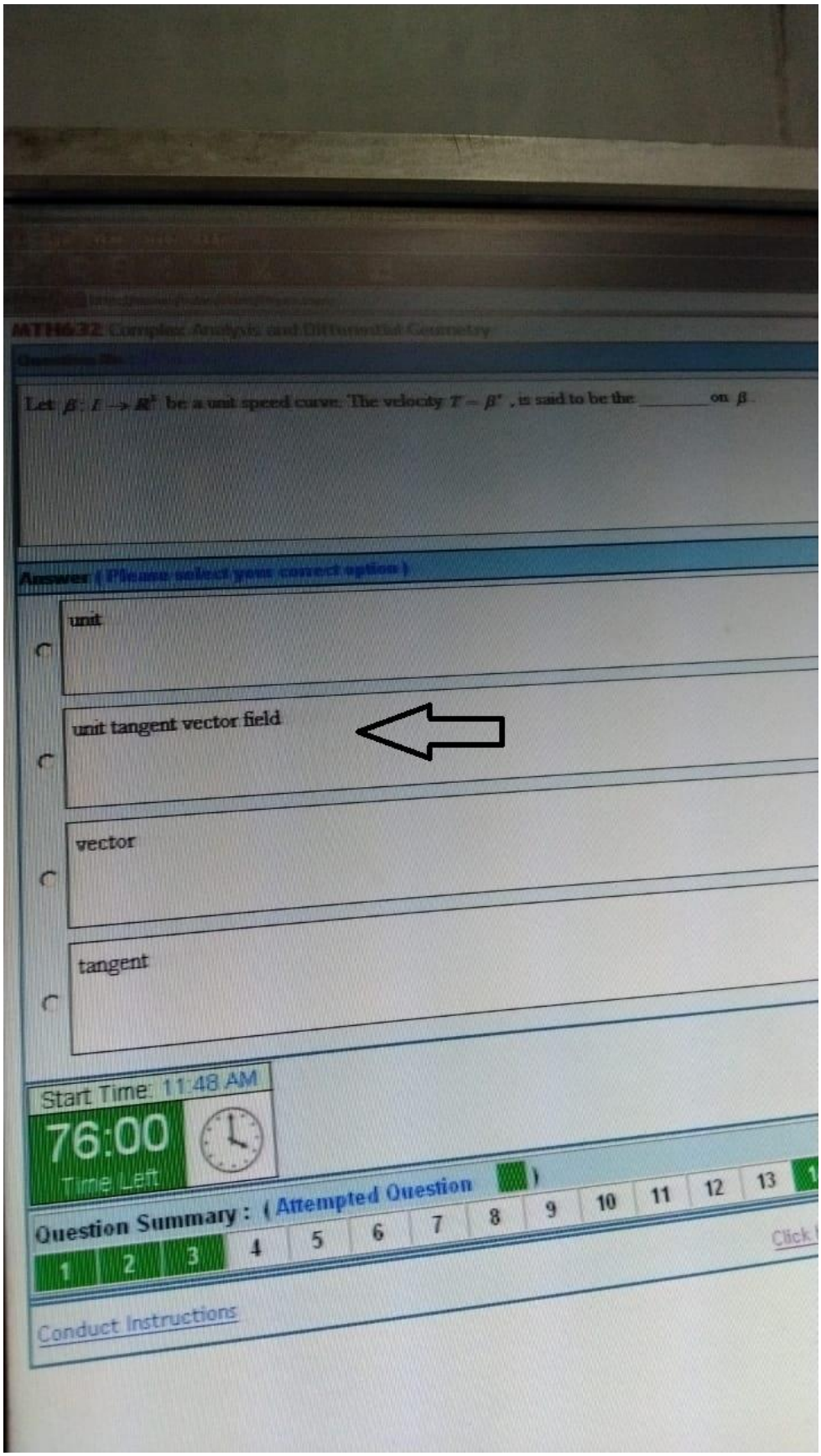
Question Summary : (Attempted Question)

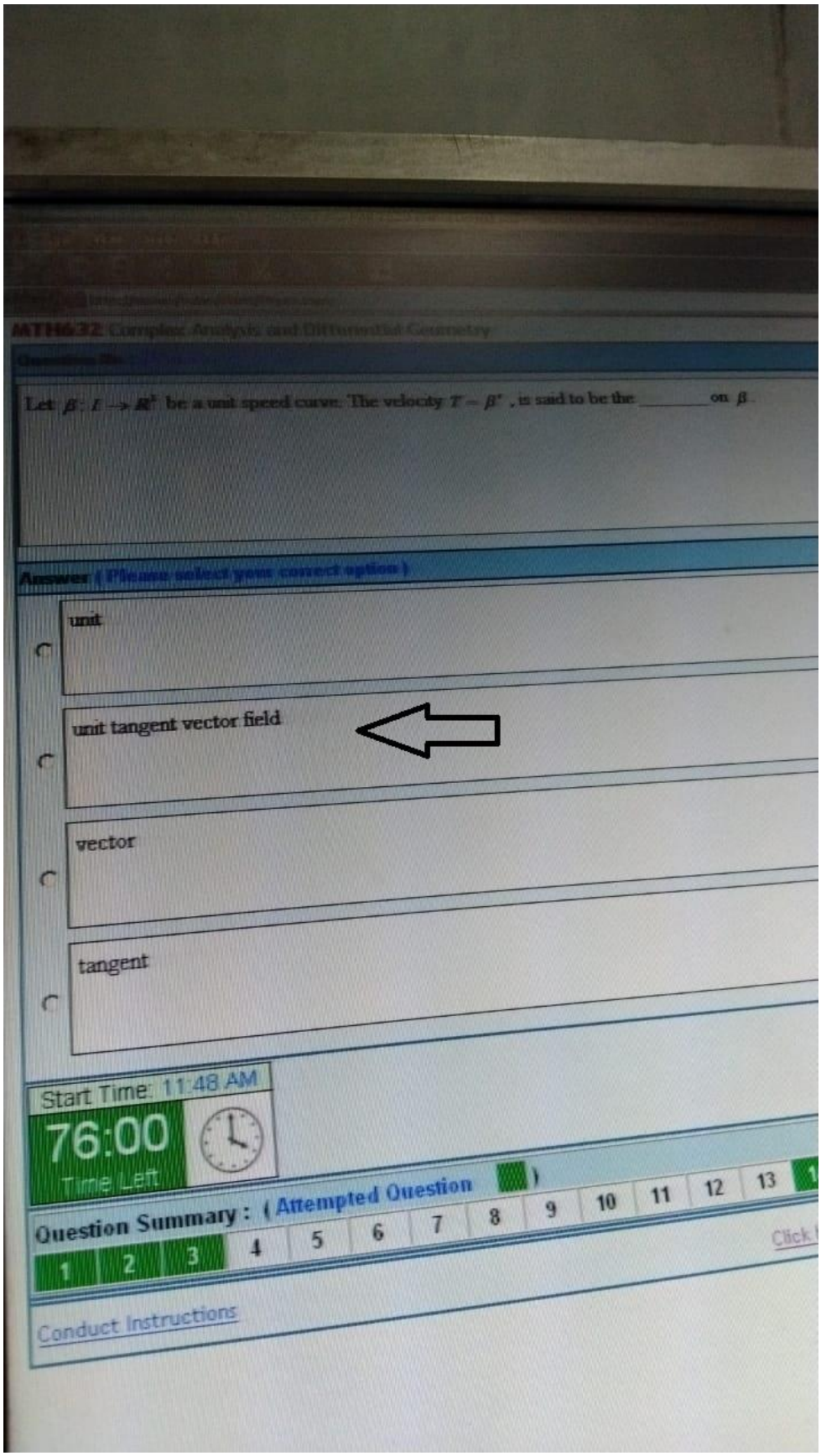
1 2 3 4 5 6 7 8 9 10 11 12

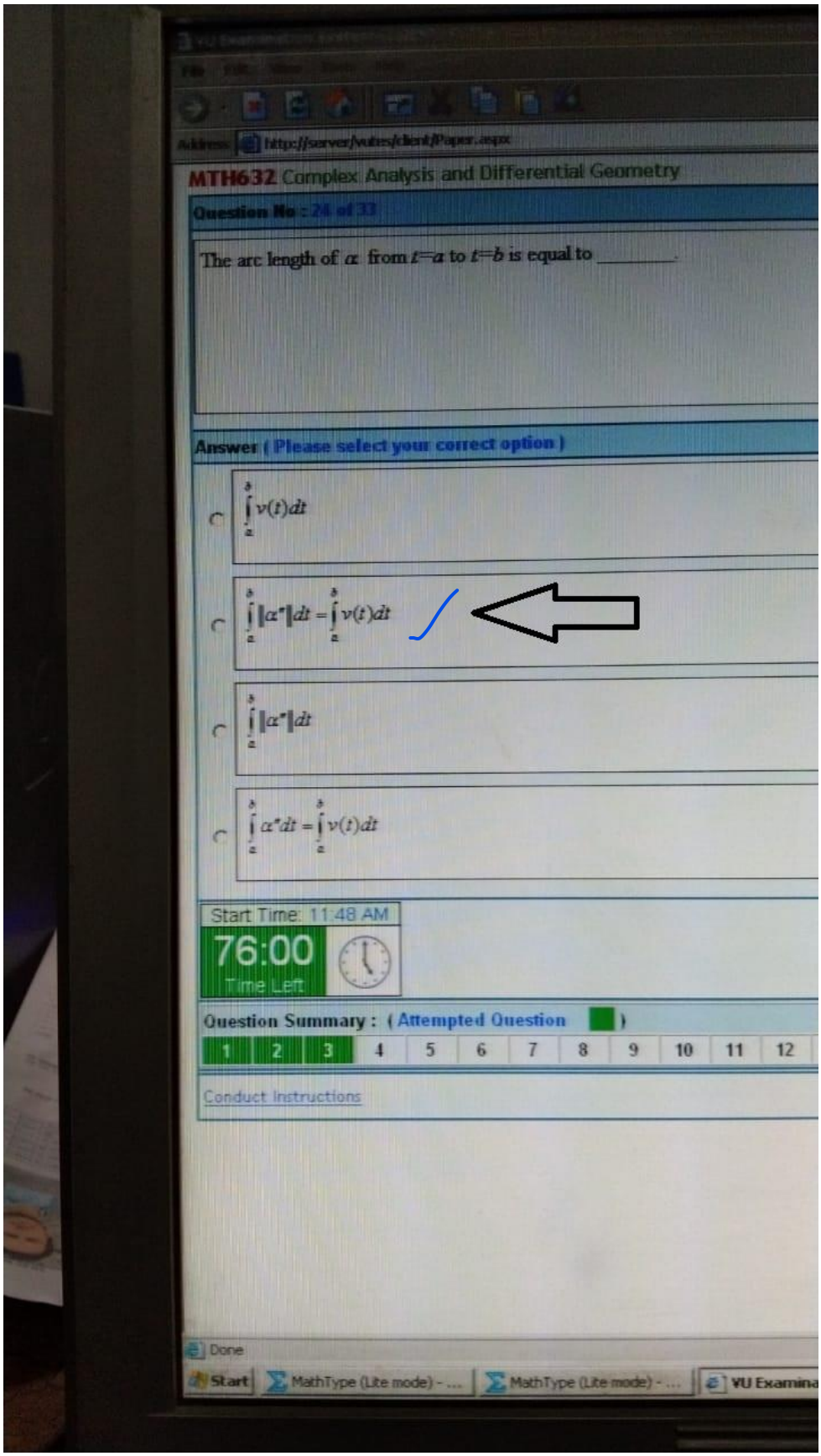
Conduct Instructions

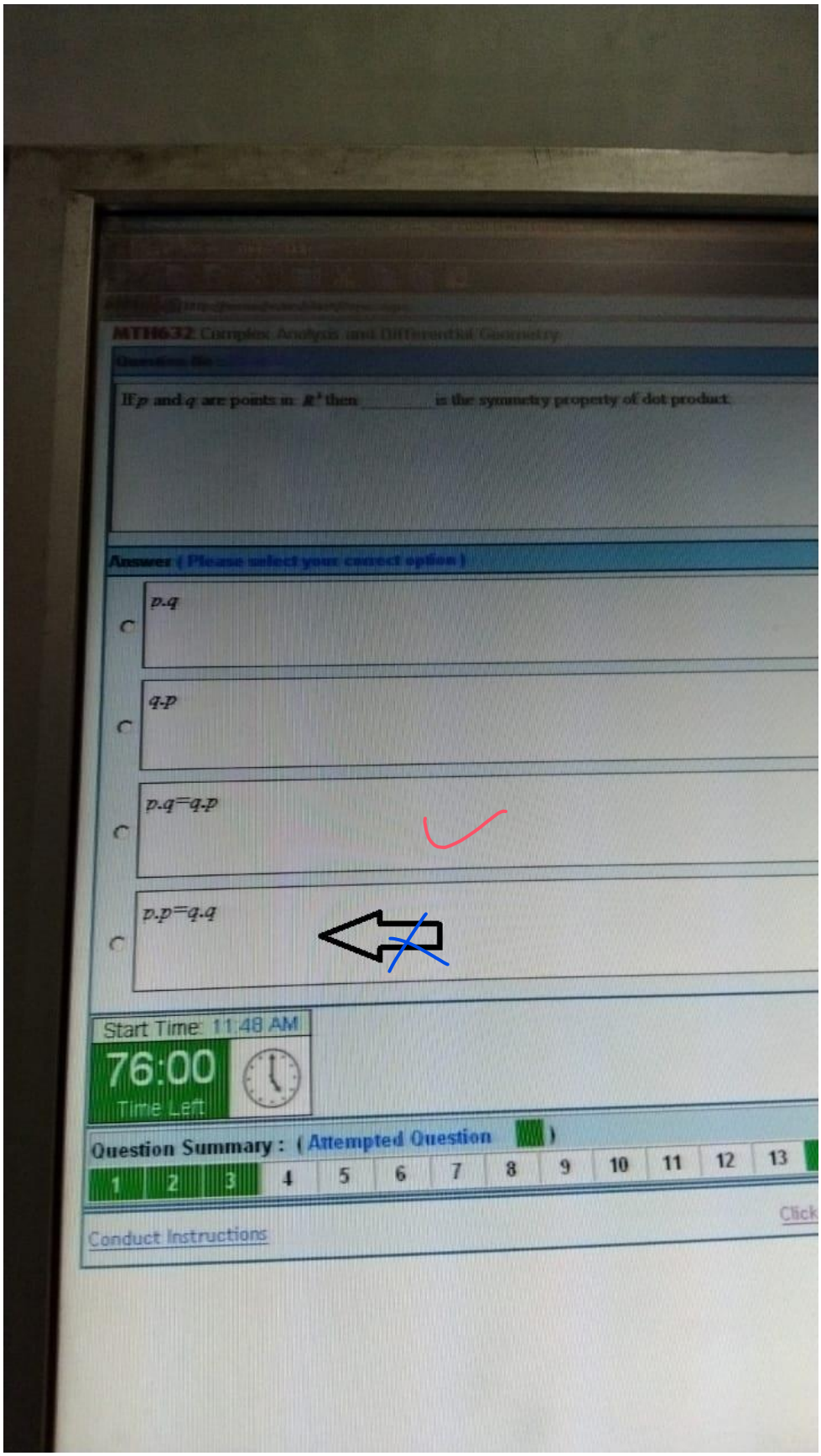
Done

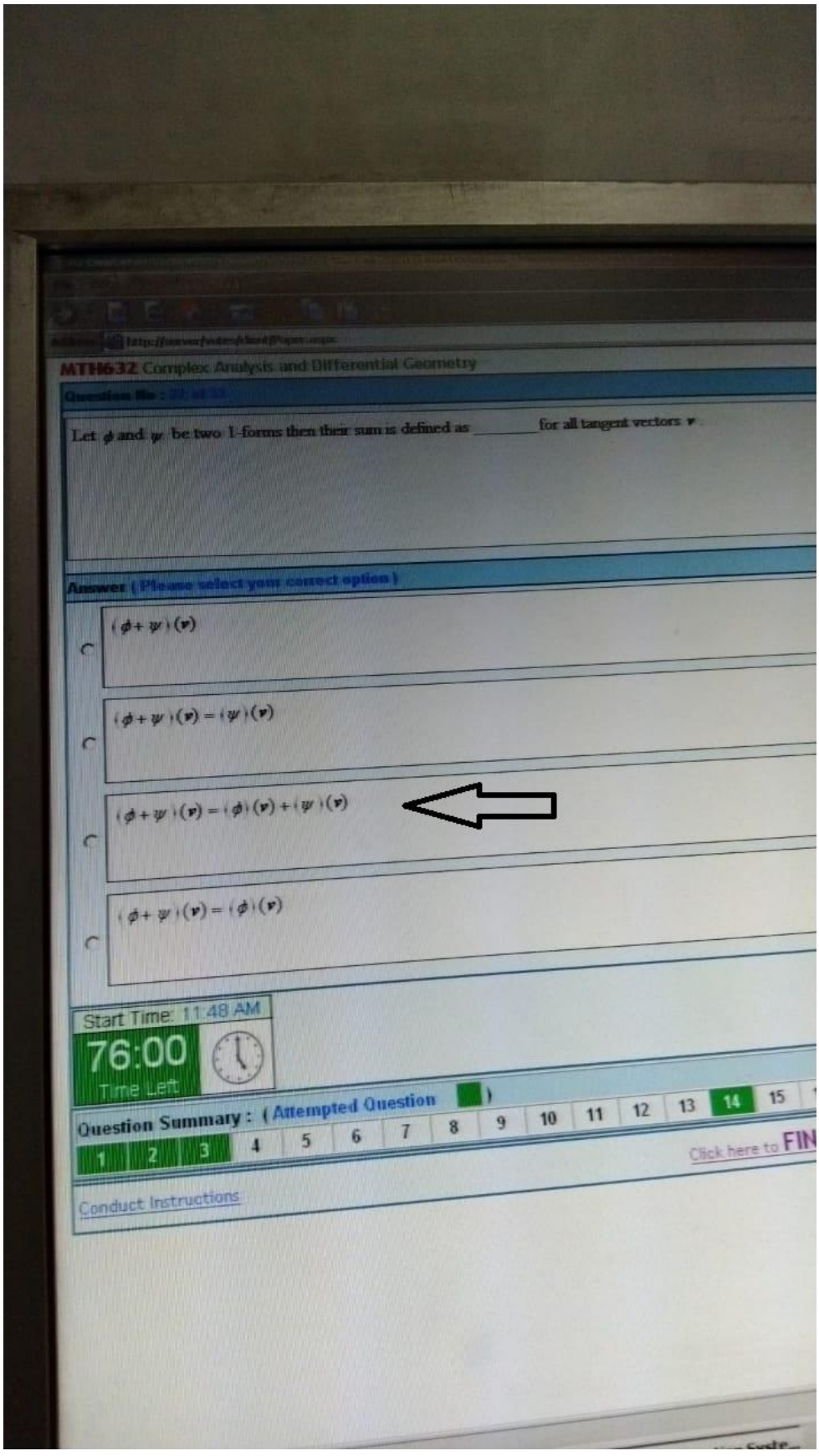
Start | MathType (Lite mode) - ... | MathType (Lite mode) - ... | VU Examina

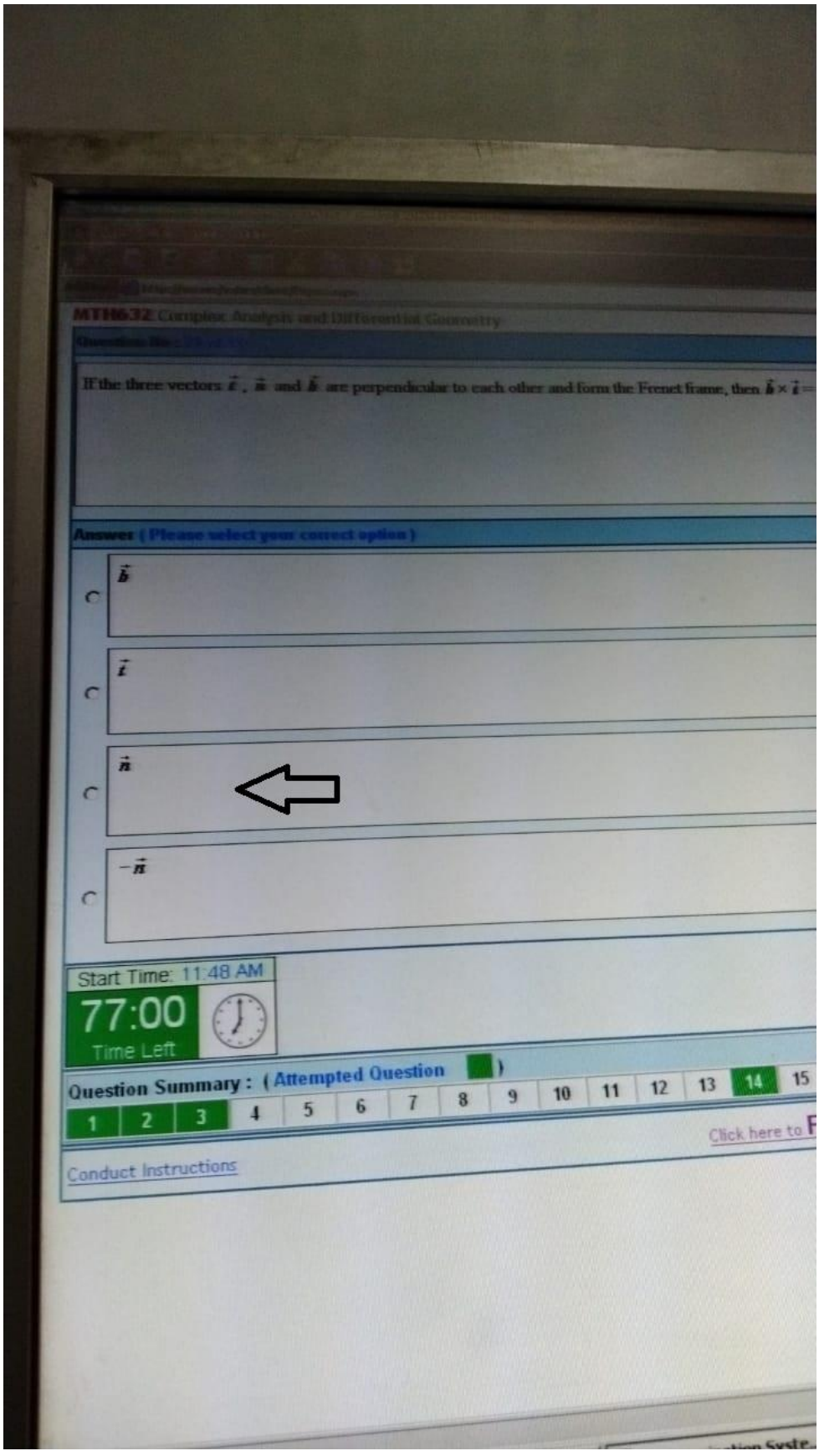

MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

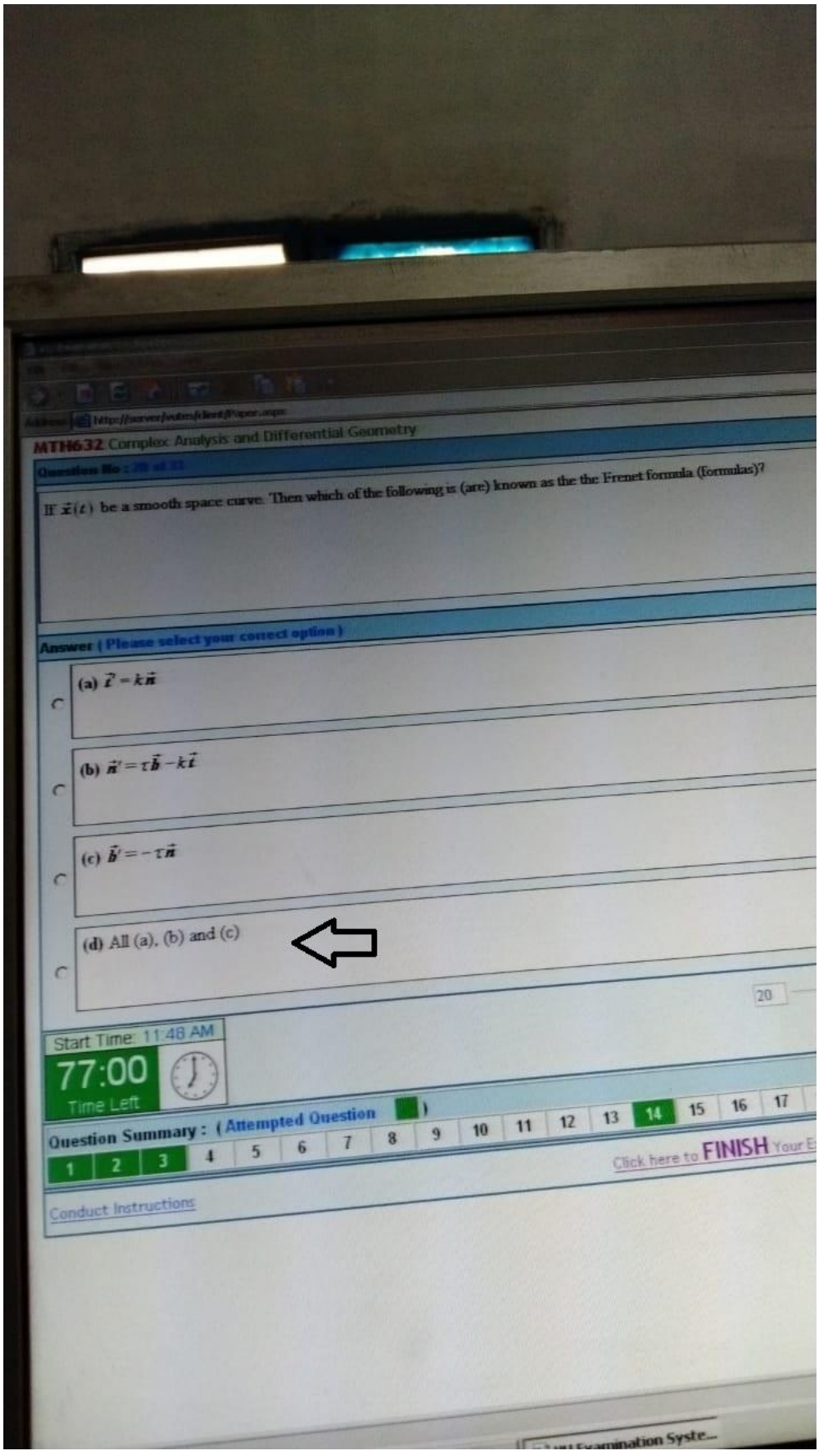

MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

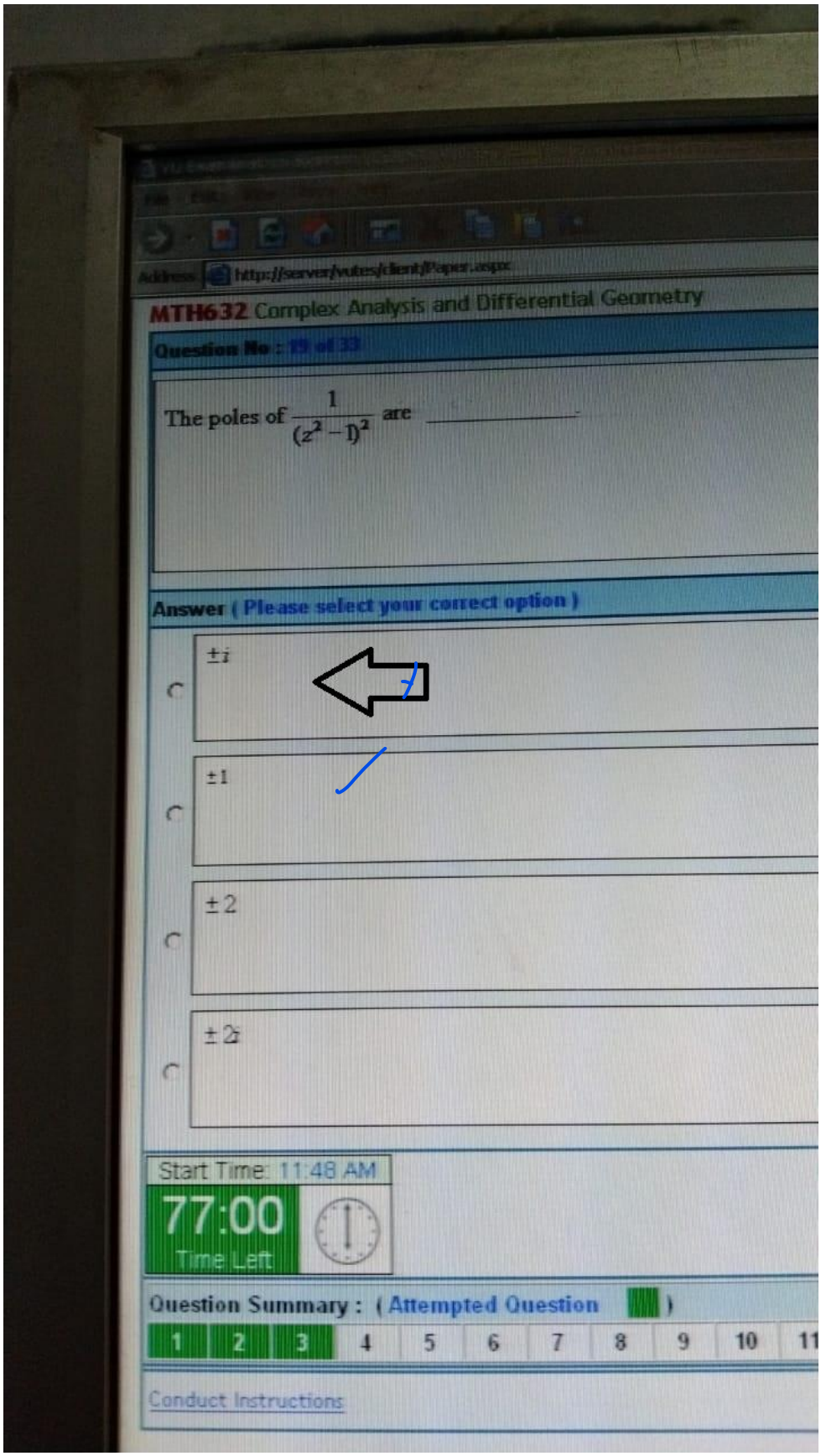

MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The arc length of α from $t=a$ to $t=b$ is equal to _____.
Answer (Please select your correct option)
$\int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt$
$\int_a^b \alpha' dt = \int_a^b v(t)dt$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examina

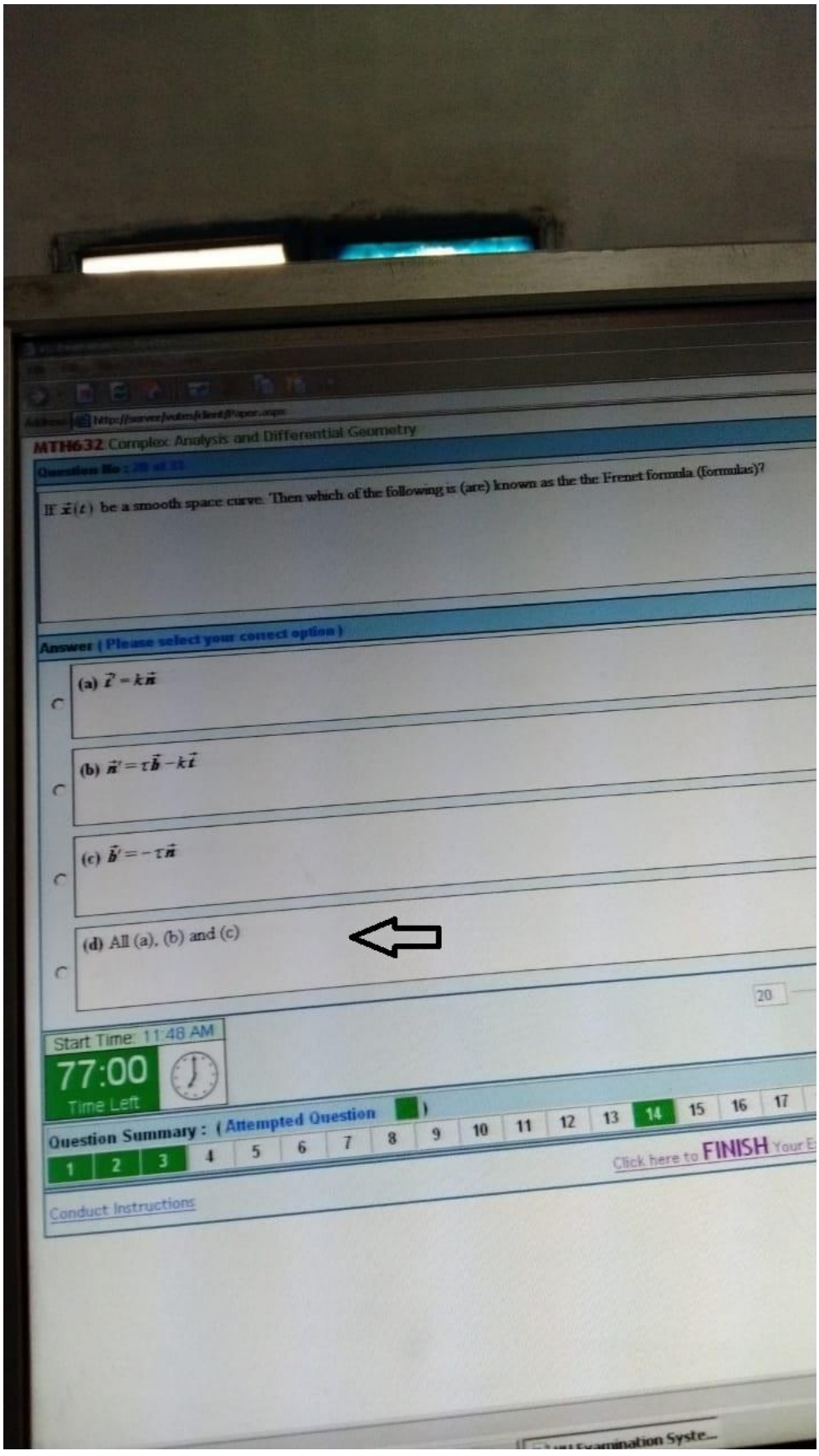

MTH632 Complex Analysis and Differential Geometry
If p and q are points in R^n then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

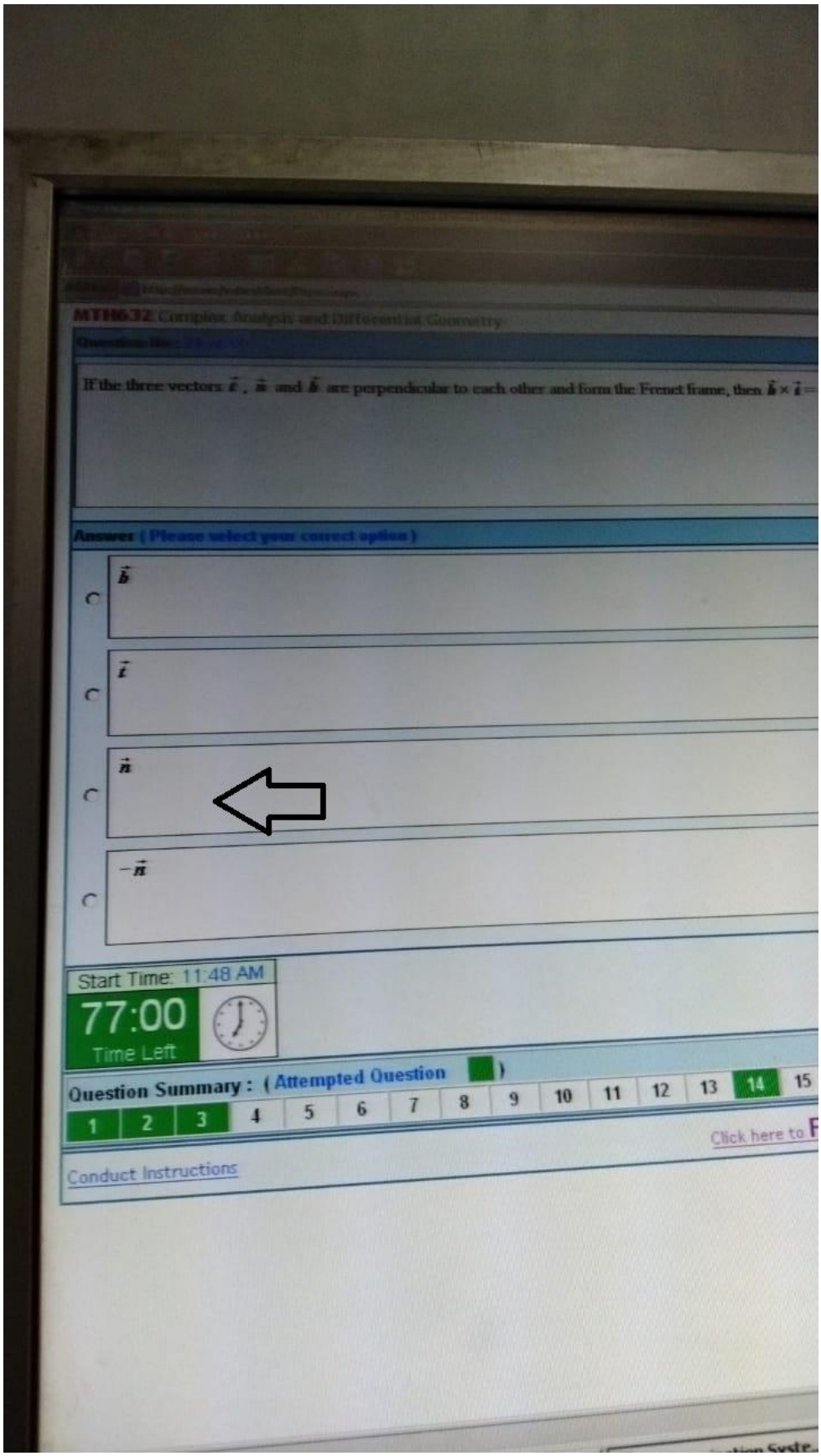

MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions

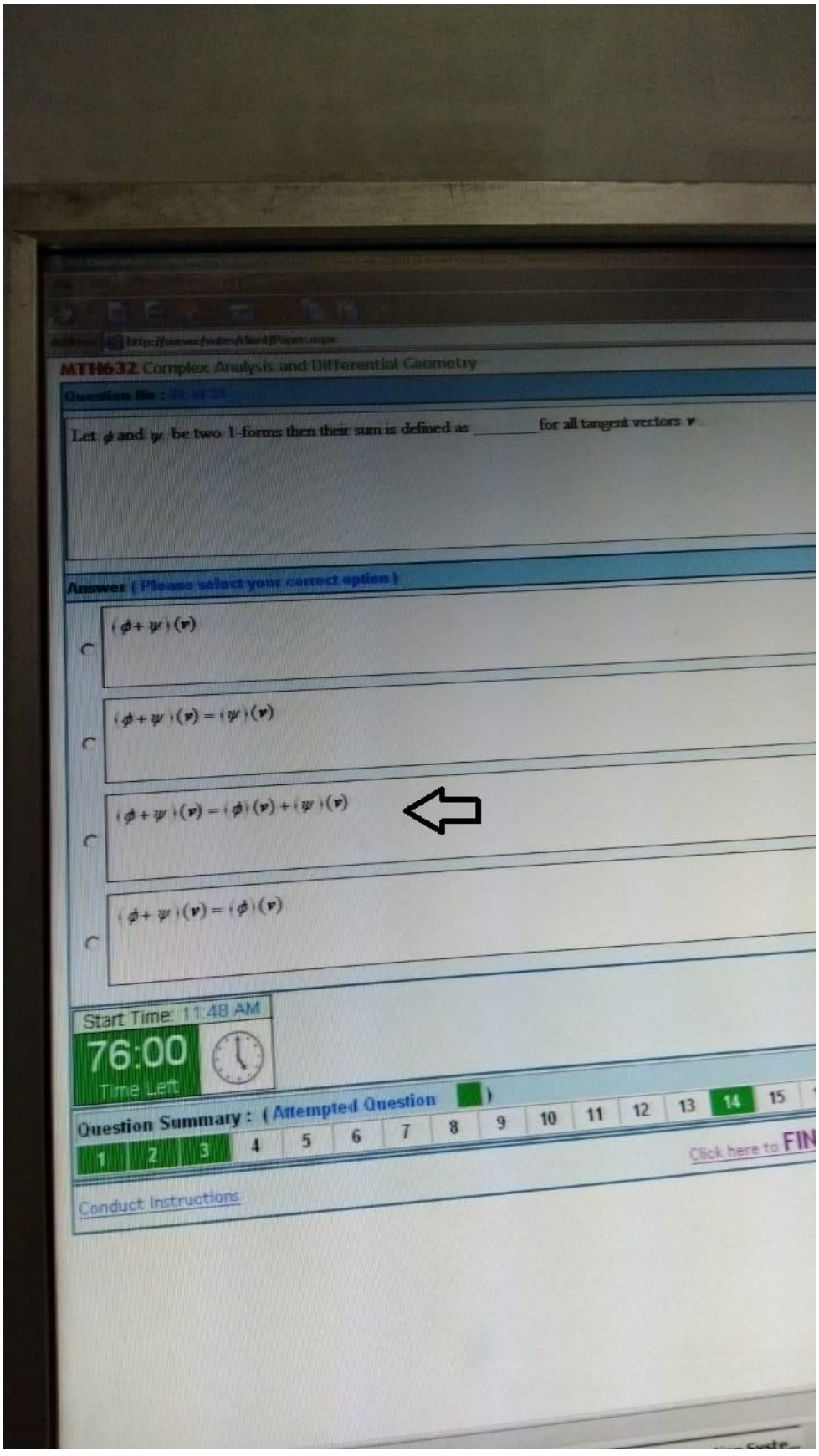

MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions

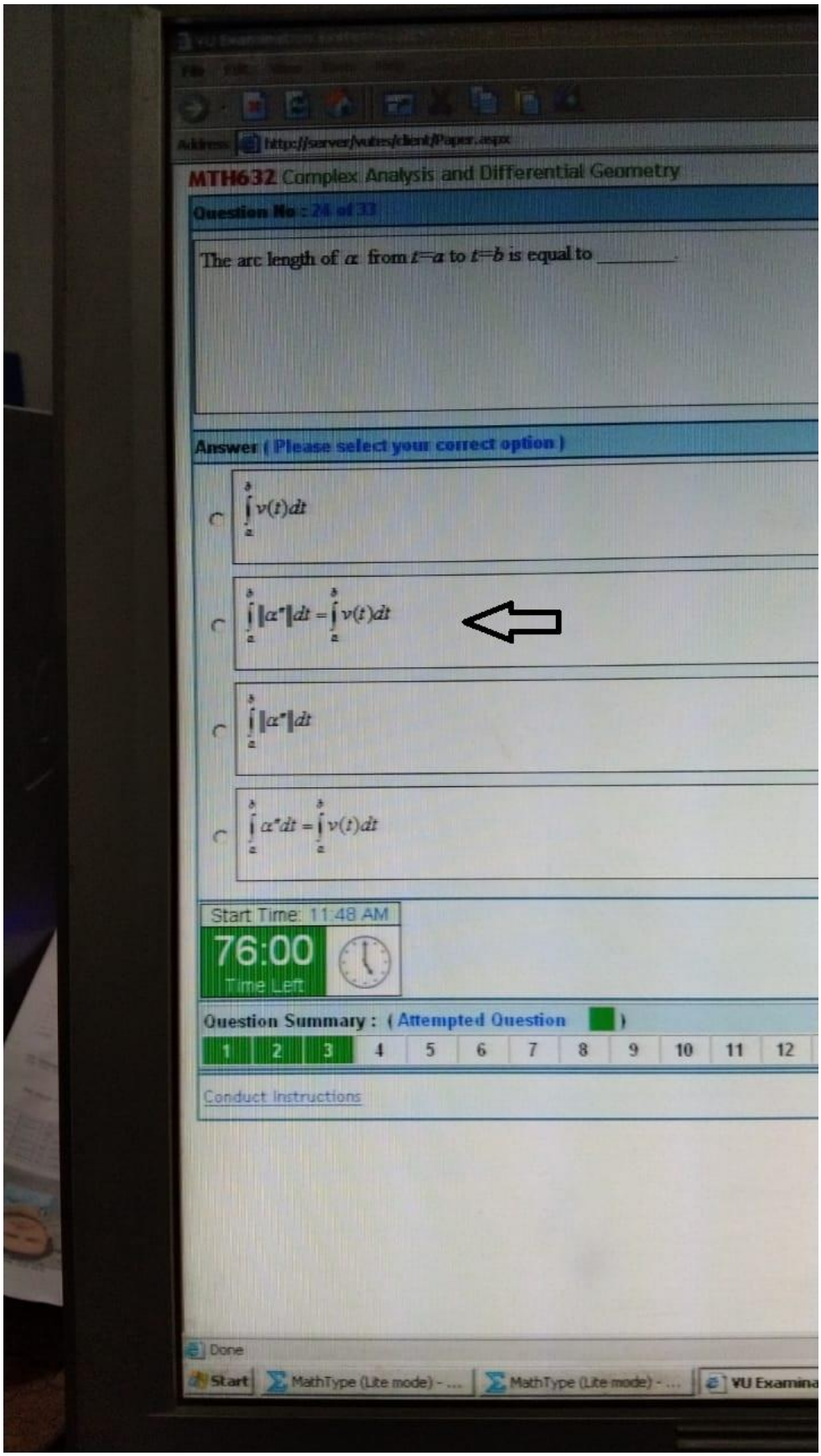

MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions

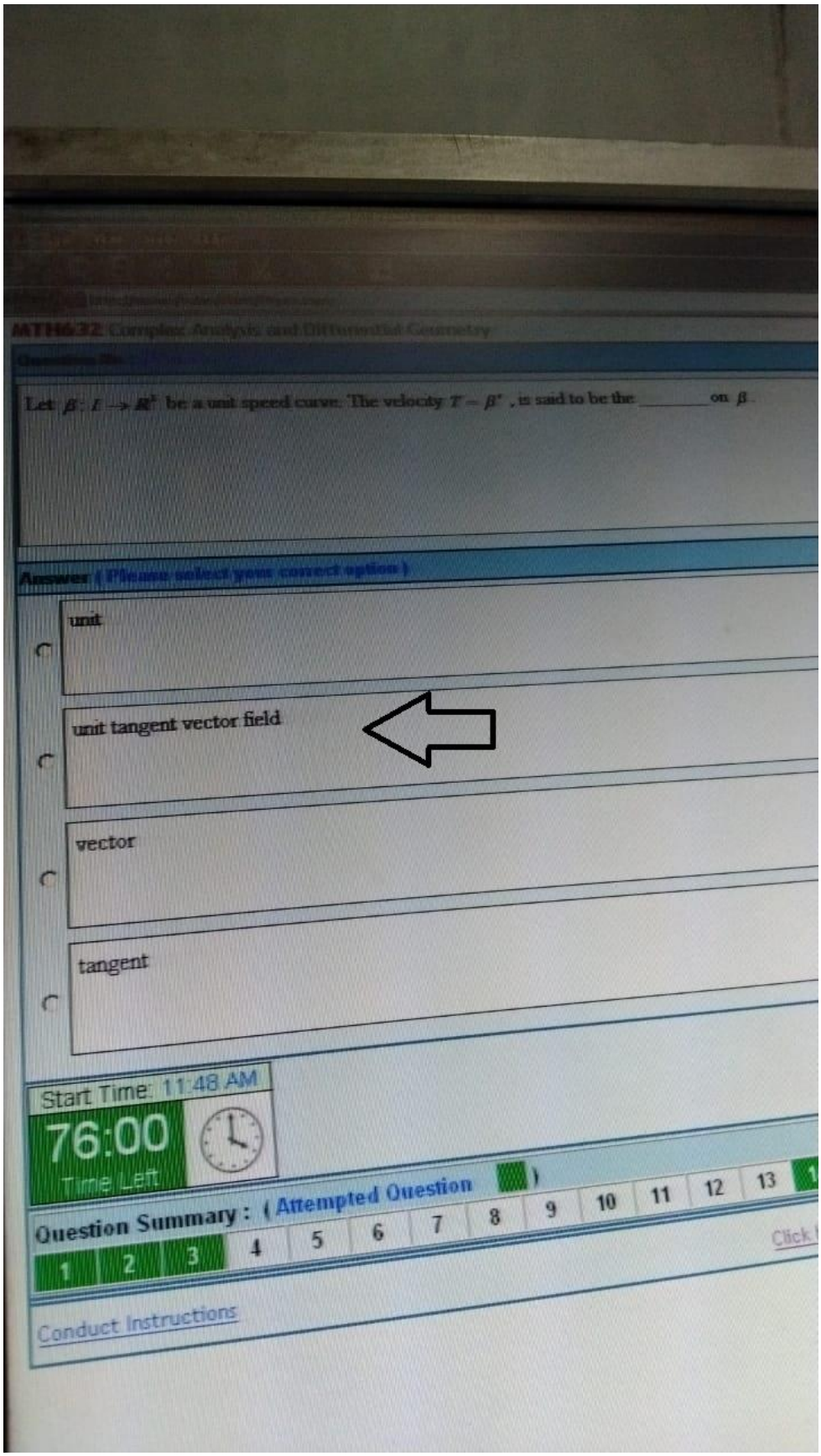

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH Your E
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The arc length of α from $t=a$ to $t=b$ is equal to _____.
Answer (Please select your correct option)
$\int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt$
$\int_a^b \alpha' dt = \int_a^b v(t)dt$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examina


MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : [illegible]

A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

Answer (Please select your correct option)

- line
- point

Start Time: 11:48 AM

78:00 Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Clic

MTH632 Complex Analysis and Differential Geometry

Question No : [illegible]

A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

Answer (Please select your correct option)

line



point



Start Time: 11:48 AM

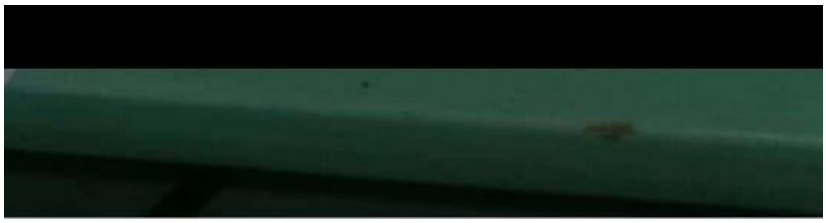
78:00
Time Left

Question Summary : (Attempted Question)

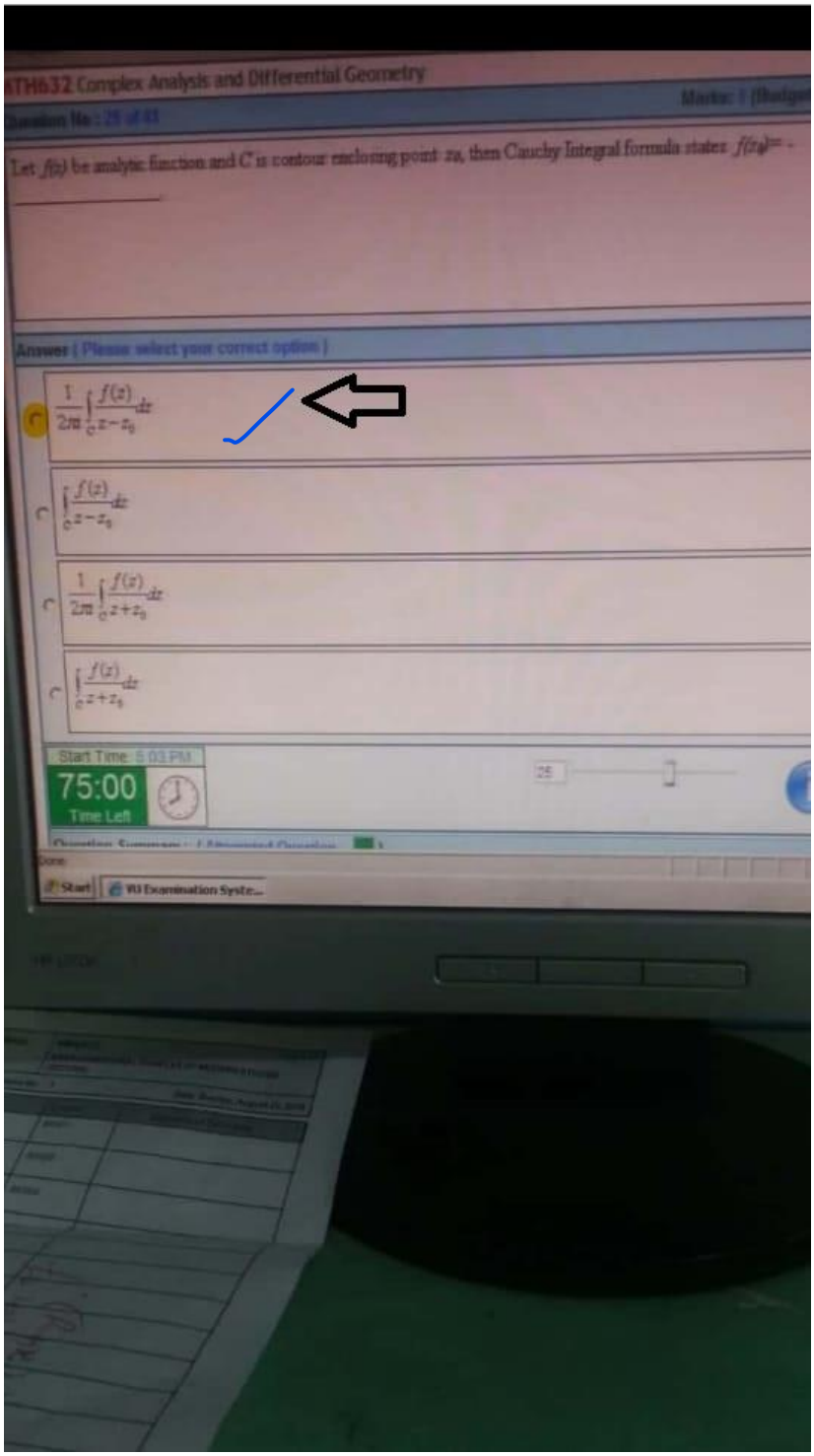
1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Clic

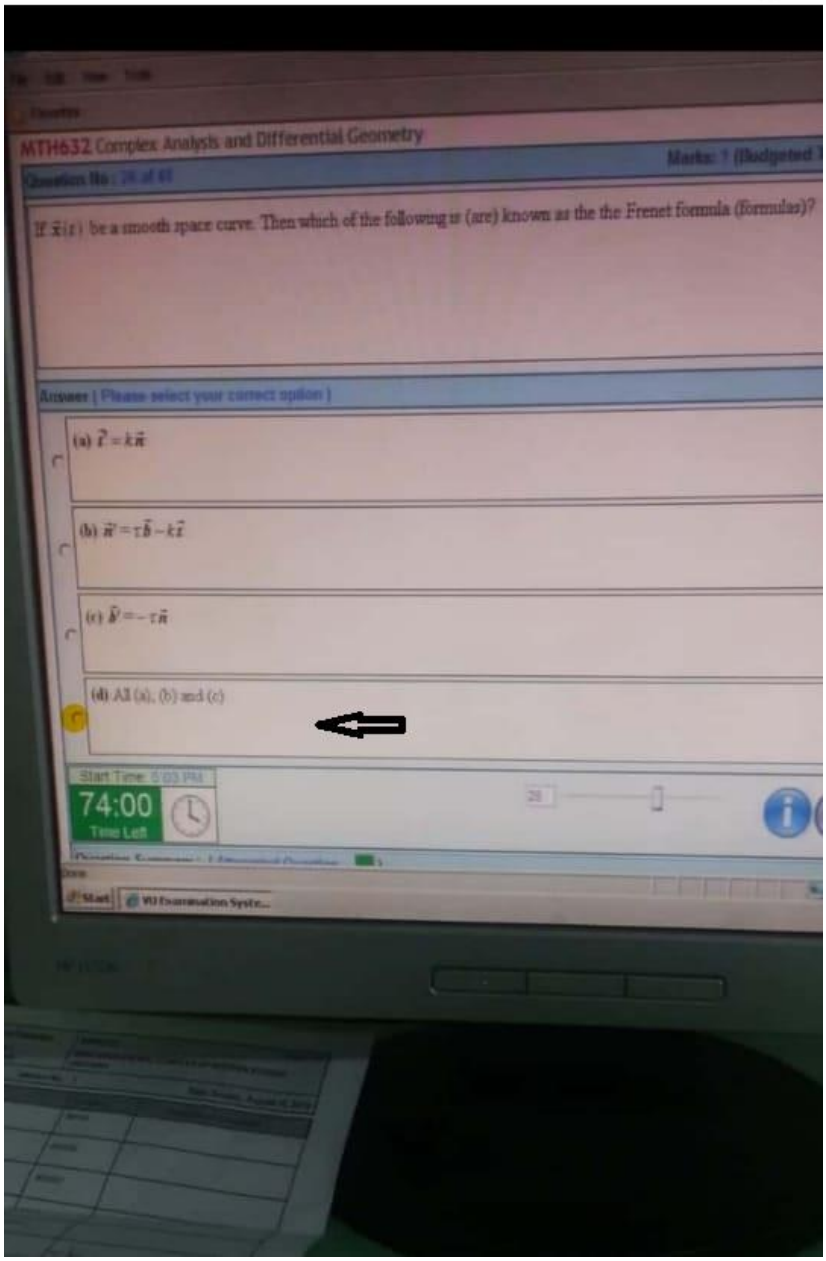


Scanned by Ca



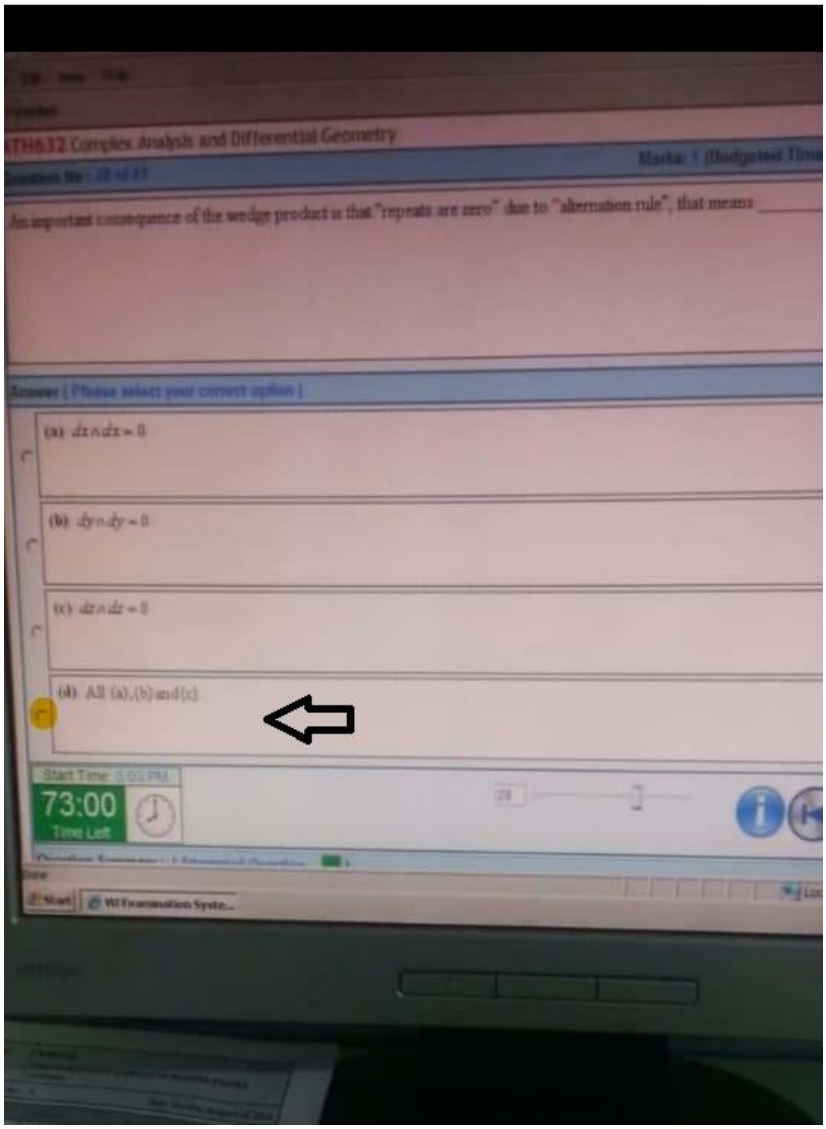


Scanned by Ca



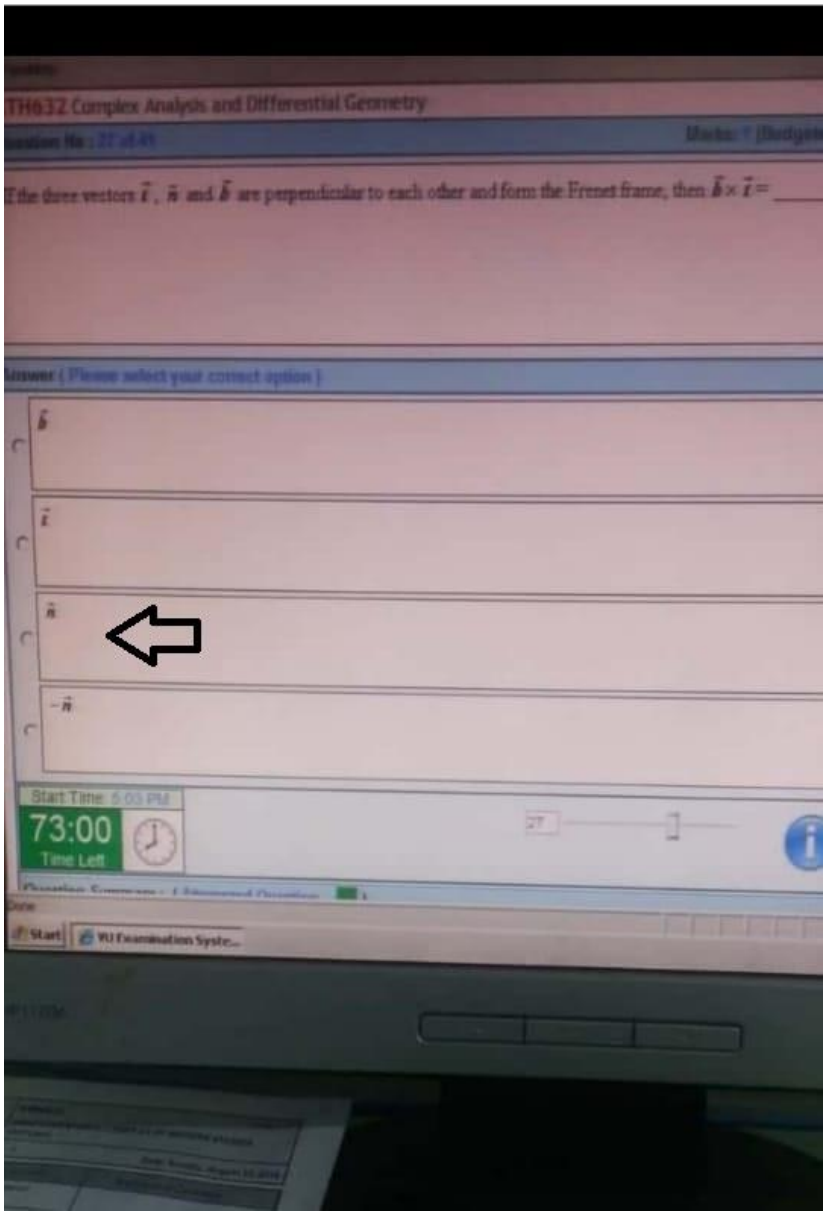


Scanned by CamS



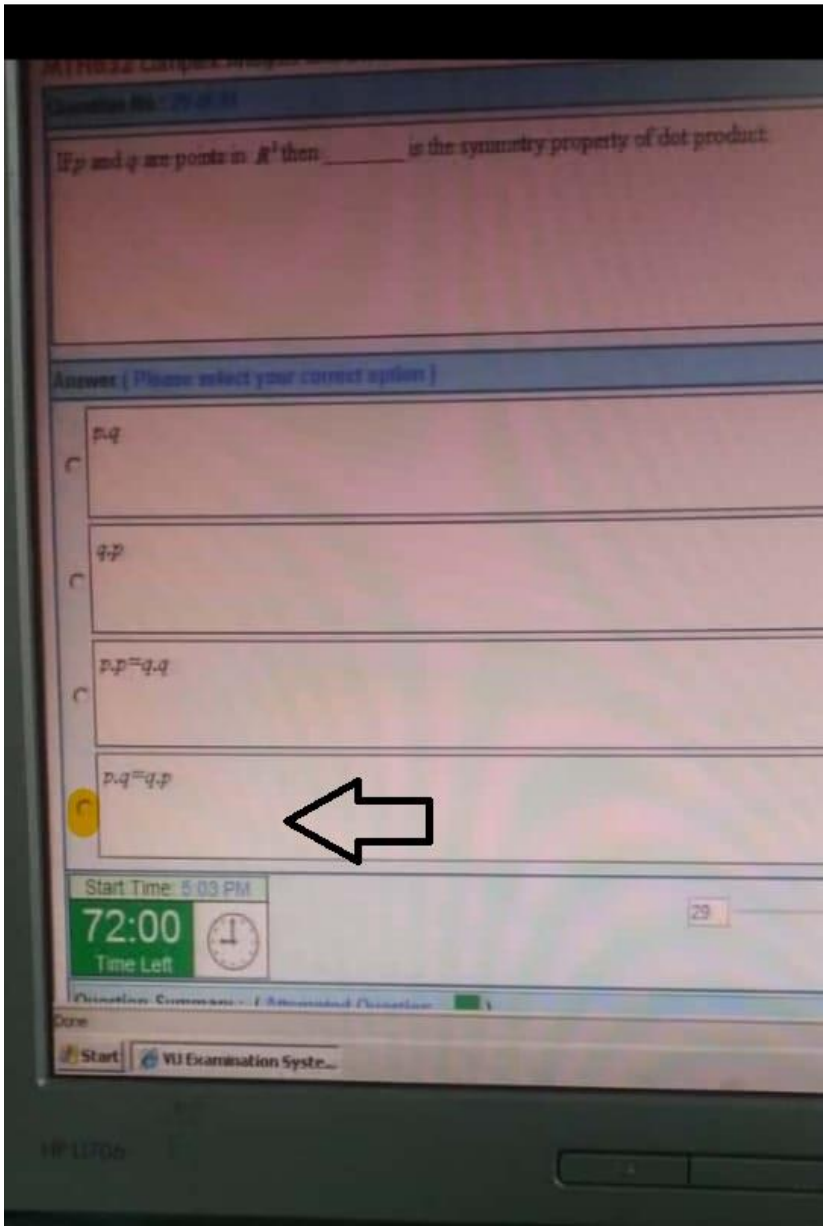


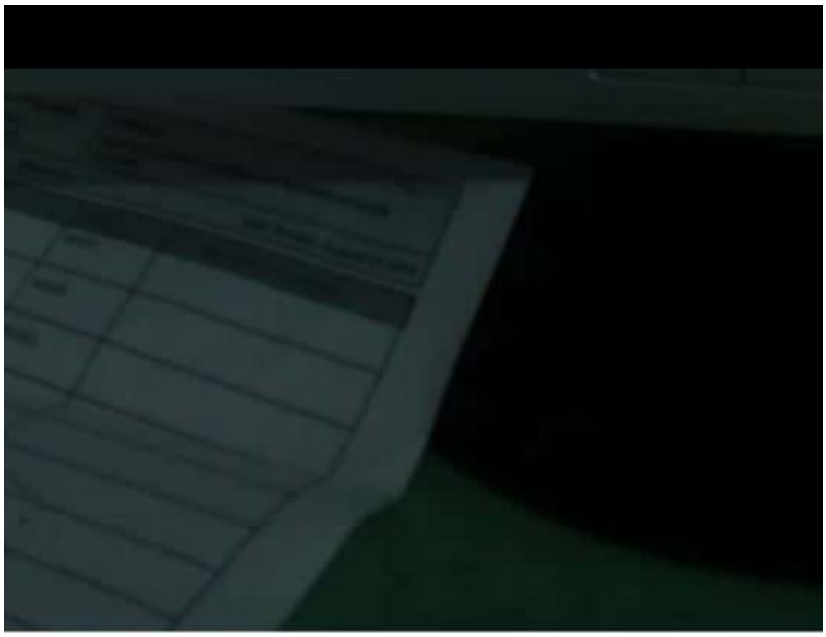
Scanned by


TH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ ____
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 6:03 PM
73:00
Time Left

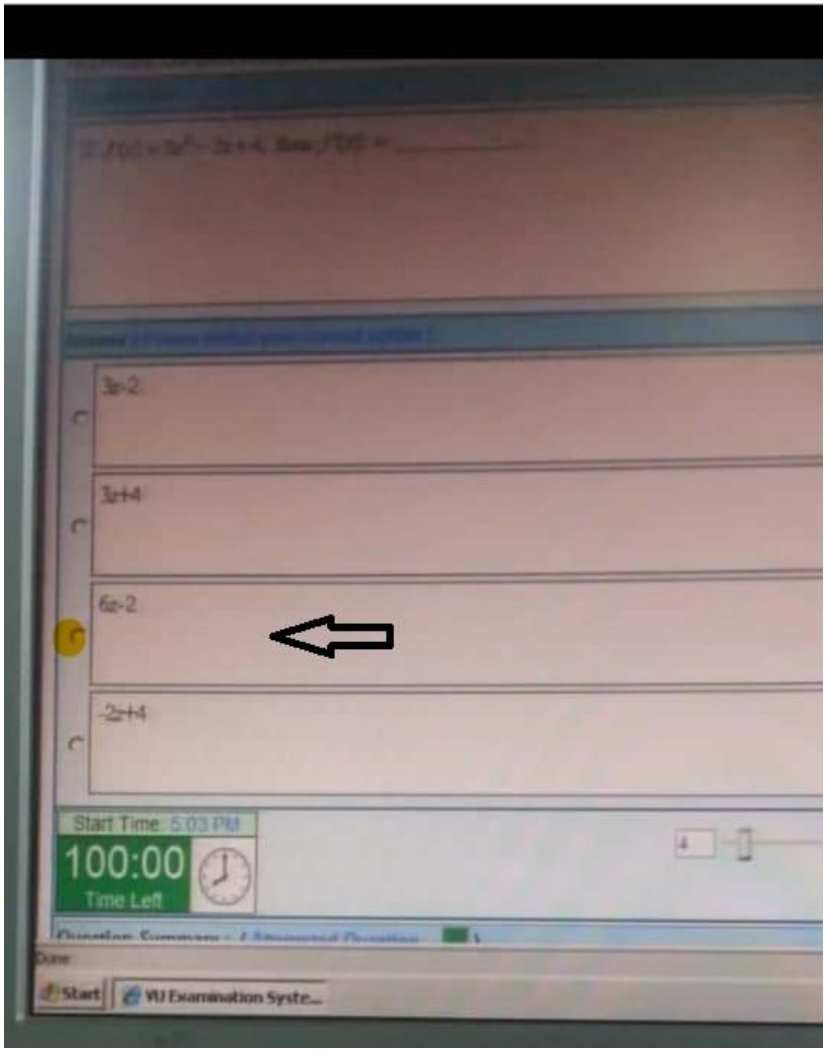


Scar


If p and q are points in R^3 then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...

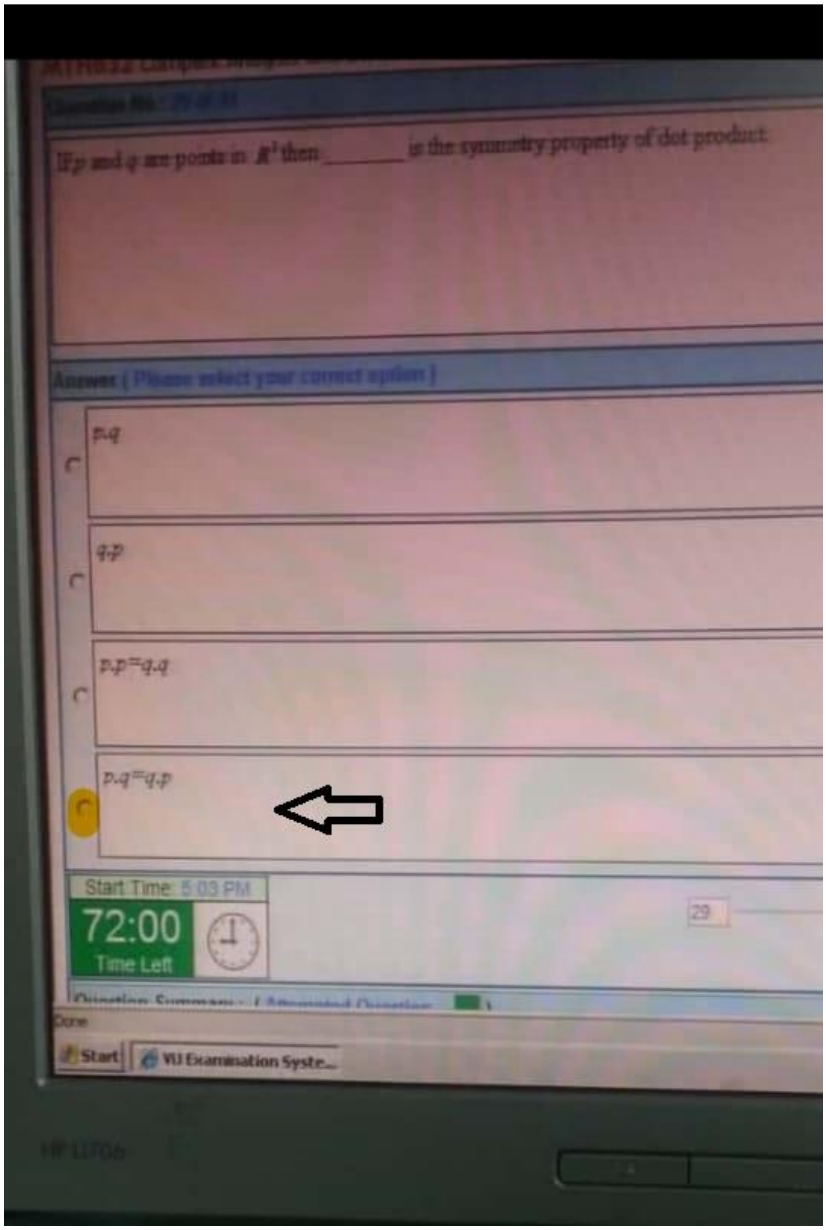


Scanne


3x-2
3x+4
6x-2
-2x+4
Start Time: 5:03 PM
100:00
Time Left
Start
VU Examination Syste...

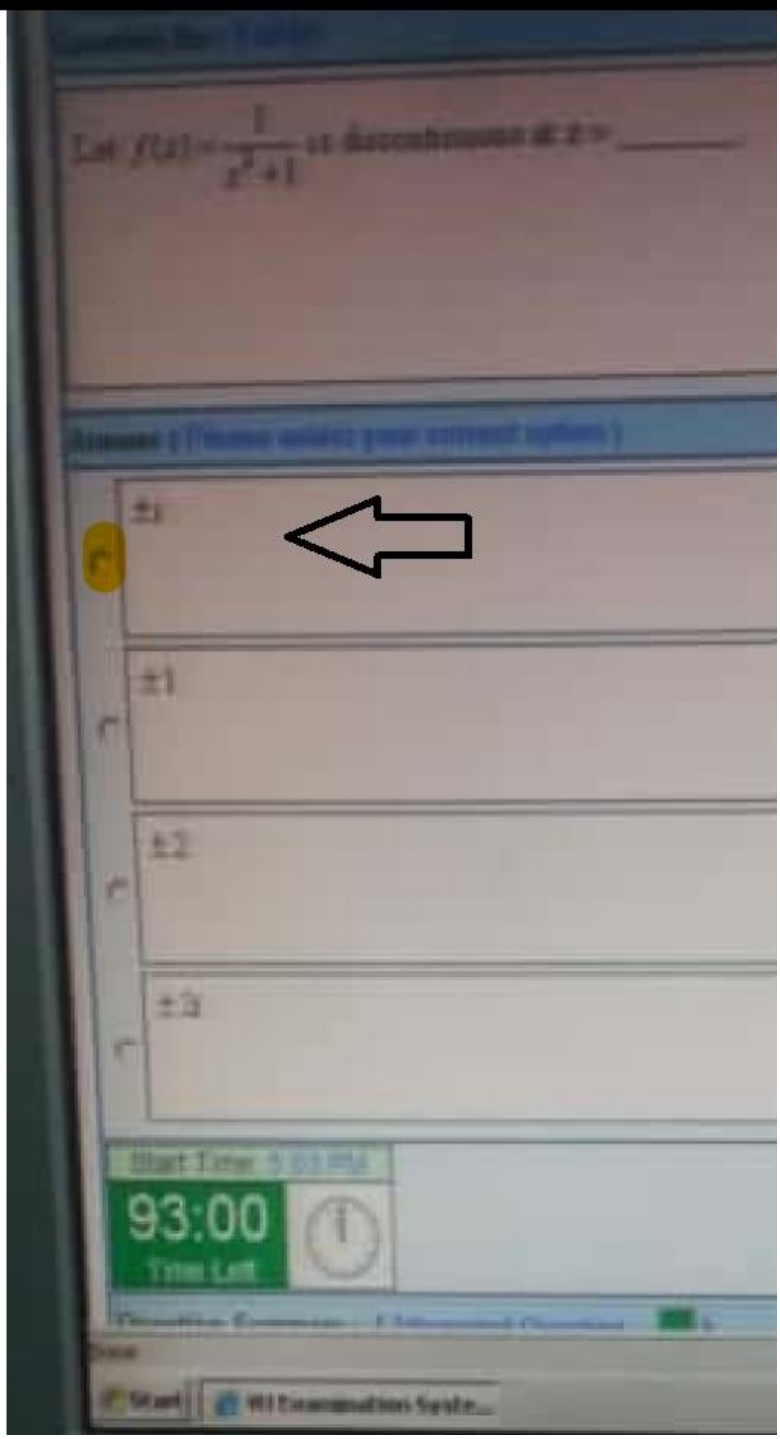


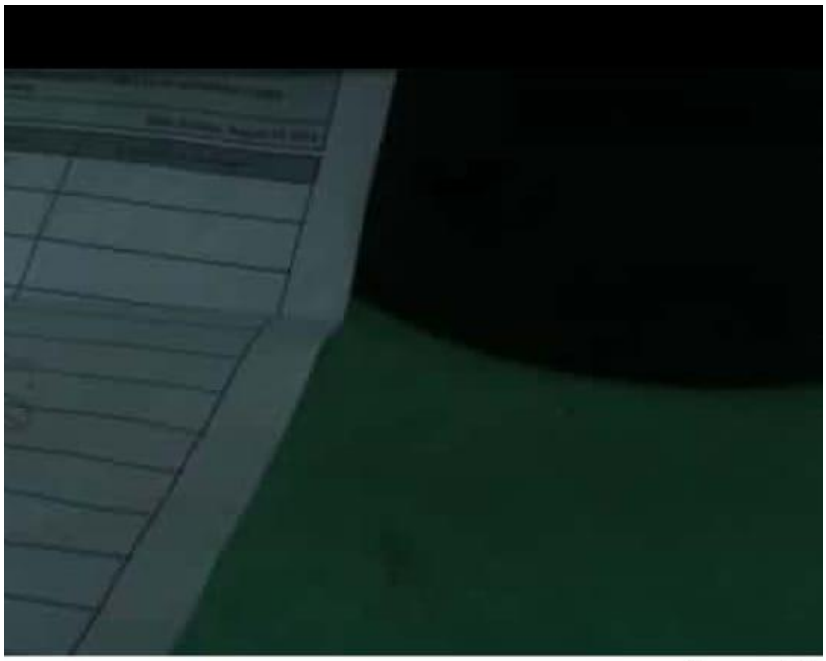
Scar


If p and q are points in R^2 then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...

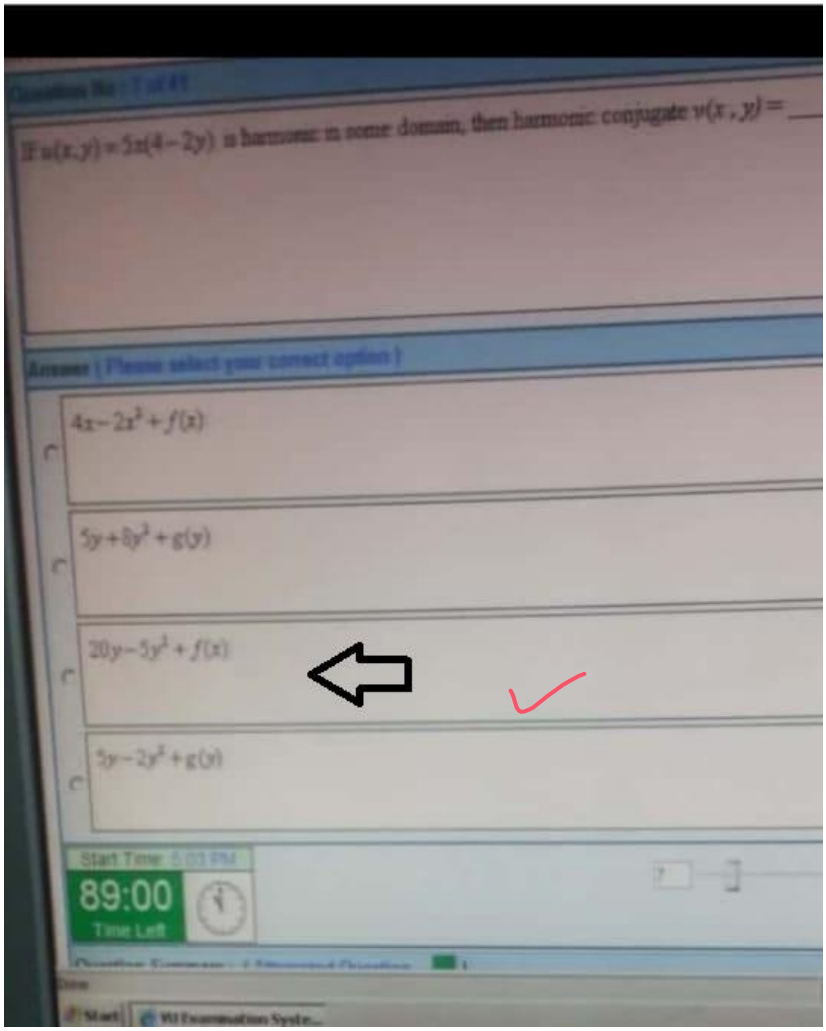
[redacted]

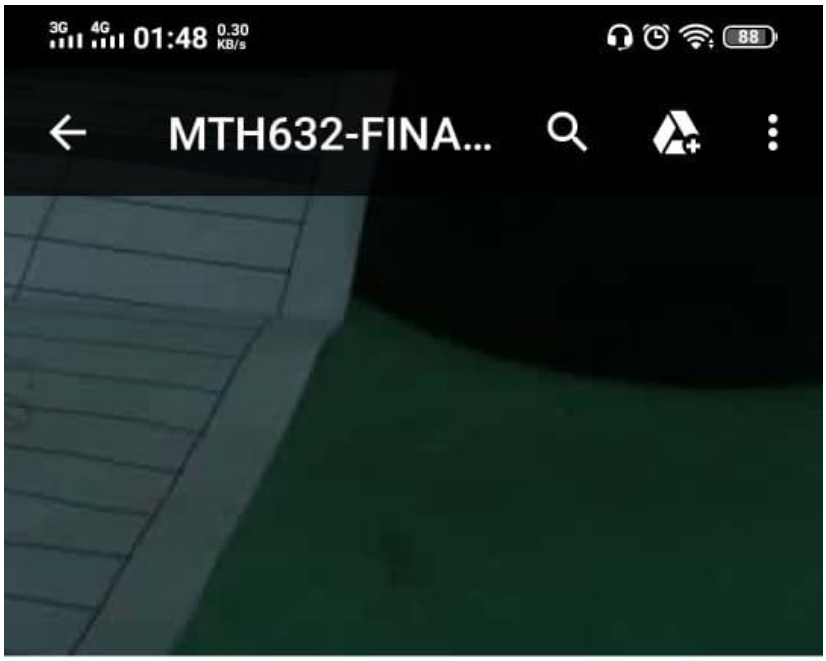
[redacted]



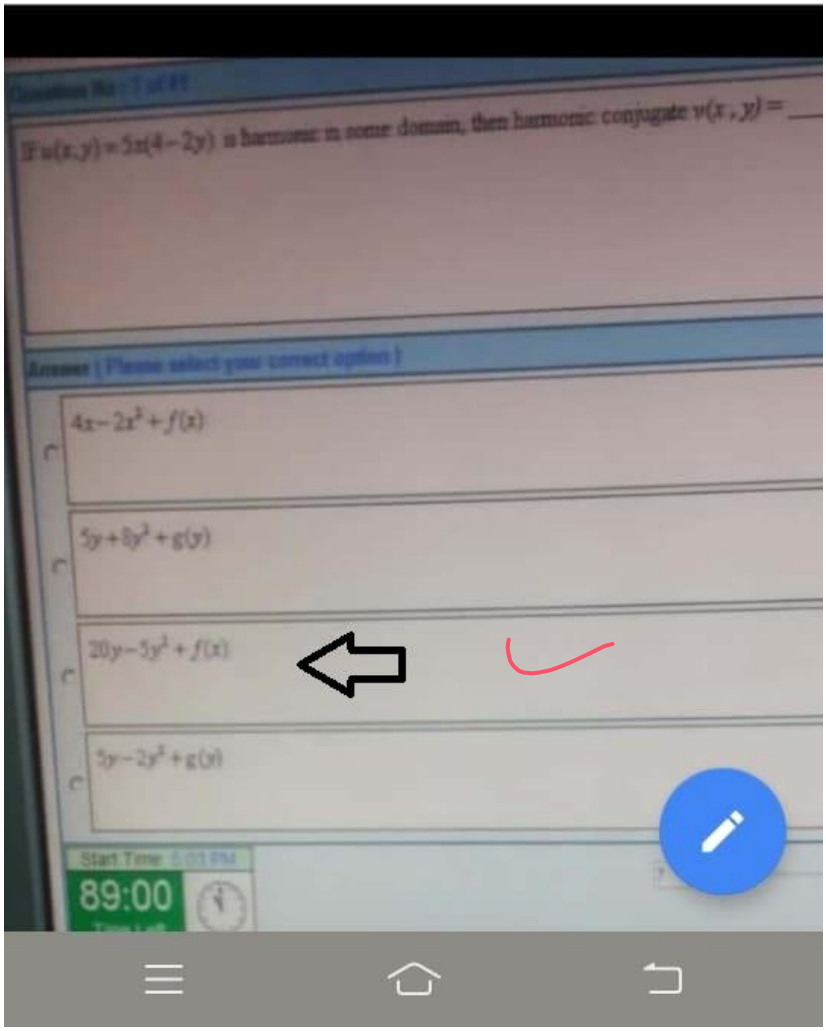


Scanned b


If u(x,y) = 5x(4−2y) is harmonic in some domain, then harmonic conjugate v(x, y) = ___
Answer (Please select your correct option)
4x−2x³ + f(x)
5y+8y³ + g(y)
20y−5y³ + f(x)
5y−2y³ + g(y)
Start Time: 6:03 PM
89:00
Time Left

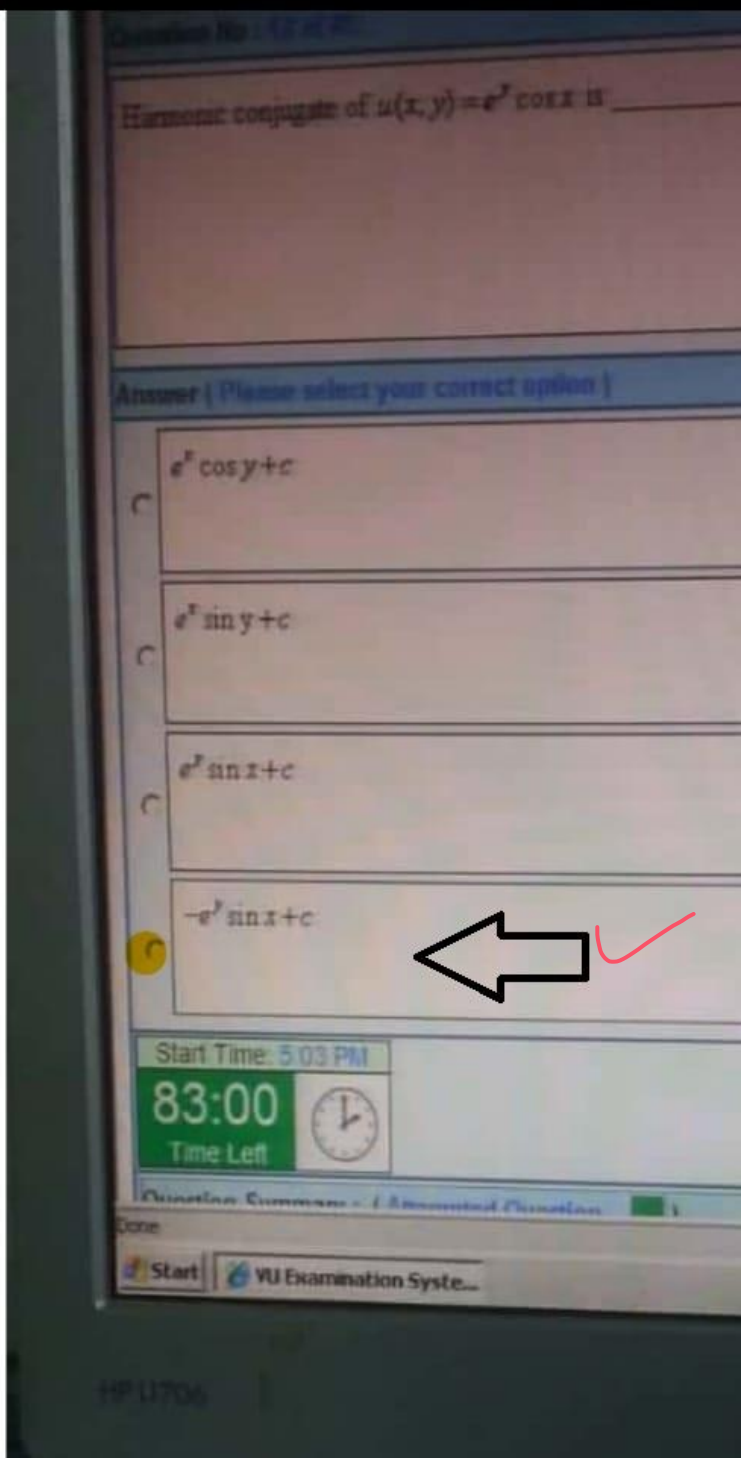

01:48
MTH632-FINA...

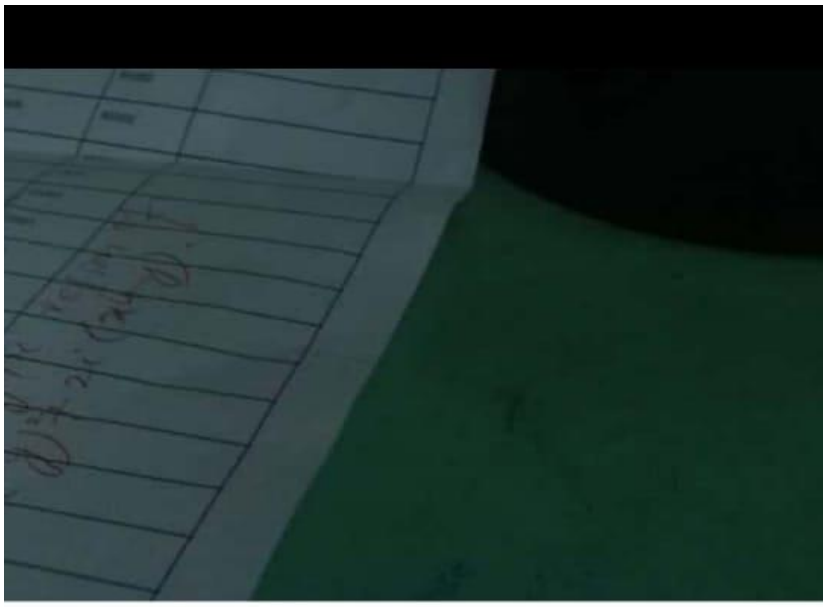
Scanned b


If u(x,y) = 5x(4−2y) is harmonic in some domain, then harmonic conjugate v(x, y) =
Answer (Please select your correct option)
4x−2x³ + f(x)
5y+8y³ + g(y)
20y−5y³ + f(x)
5y−2y³ + g(y)
89:00

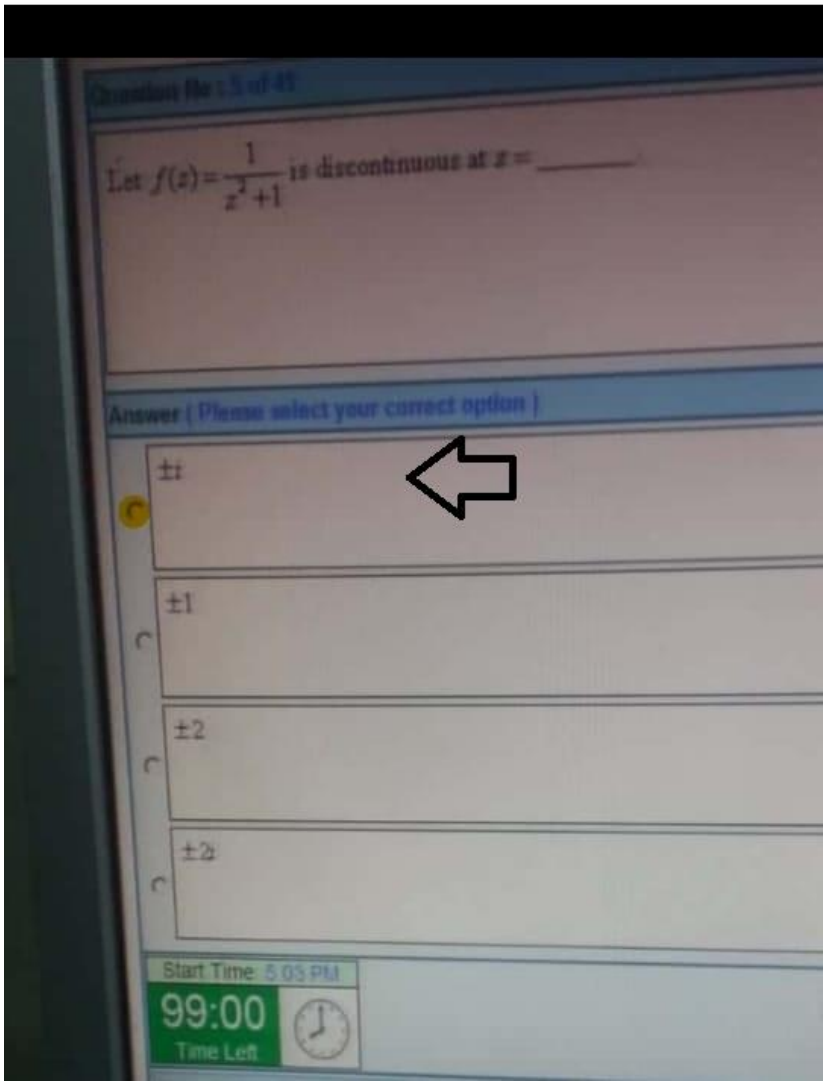
[redacted]

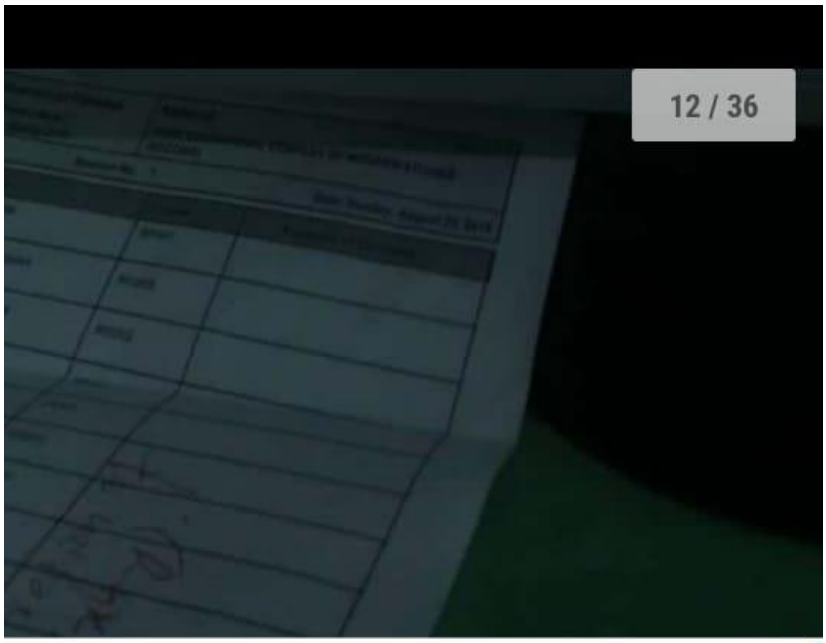
[redacted]



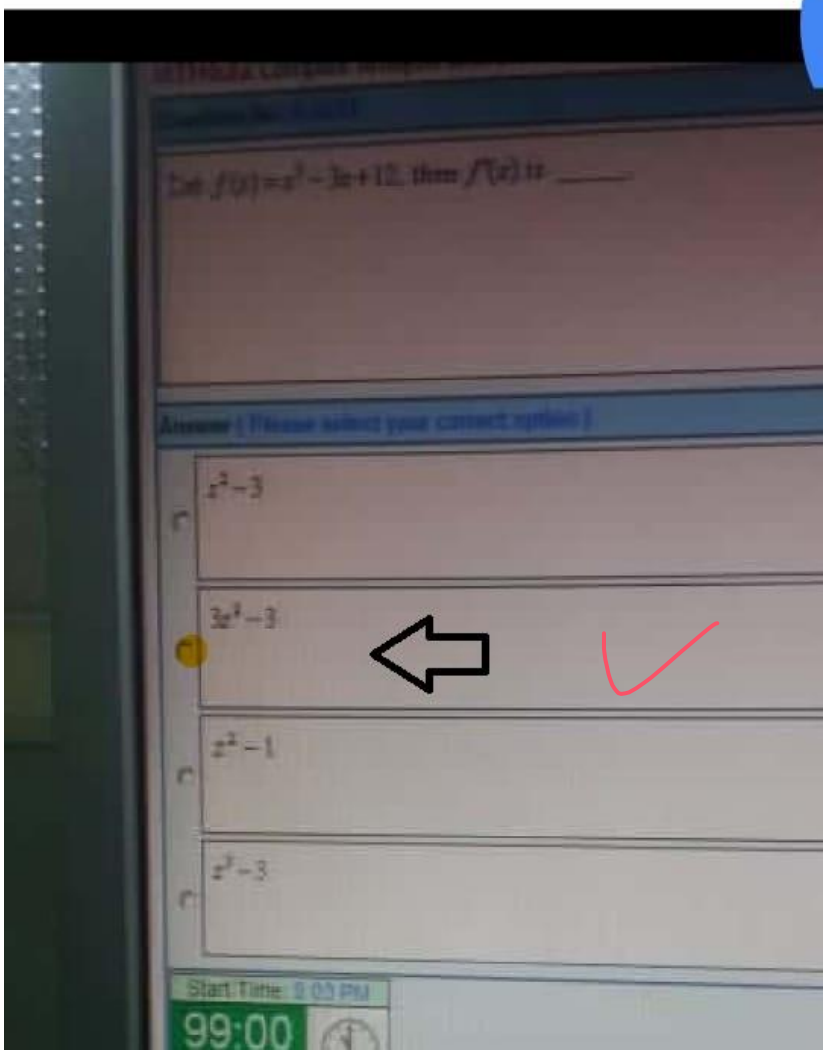


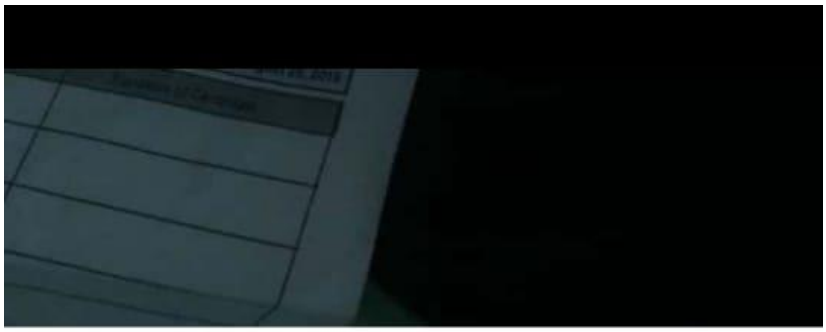
Scan


Let $f(x)=\frac{1}{x^2+1}$ is discontinuous at $x=$ _____
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time 5:03 PM
99:00
Time Left

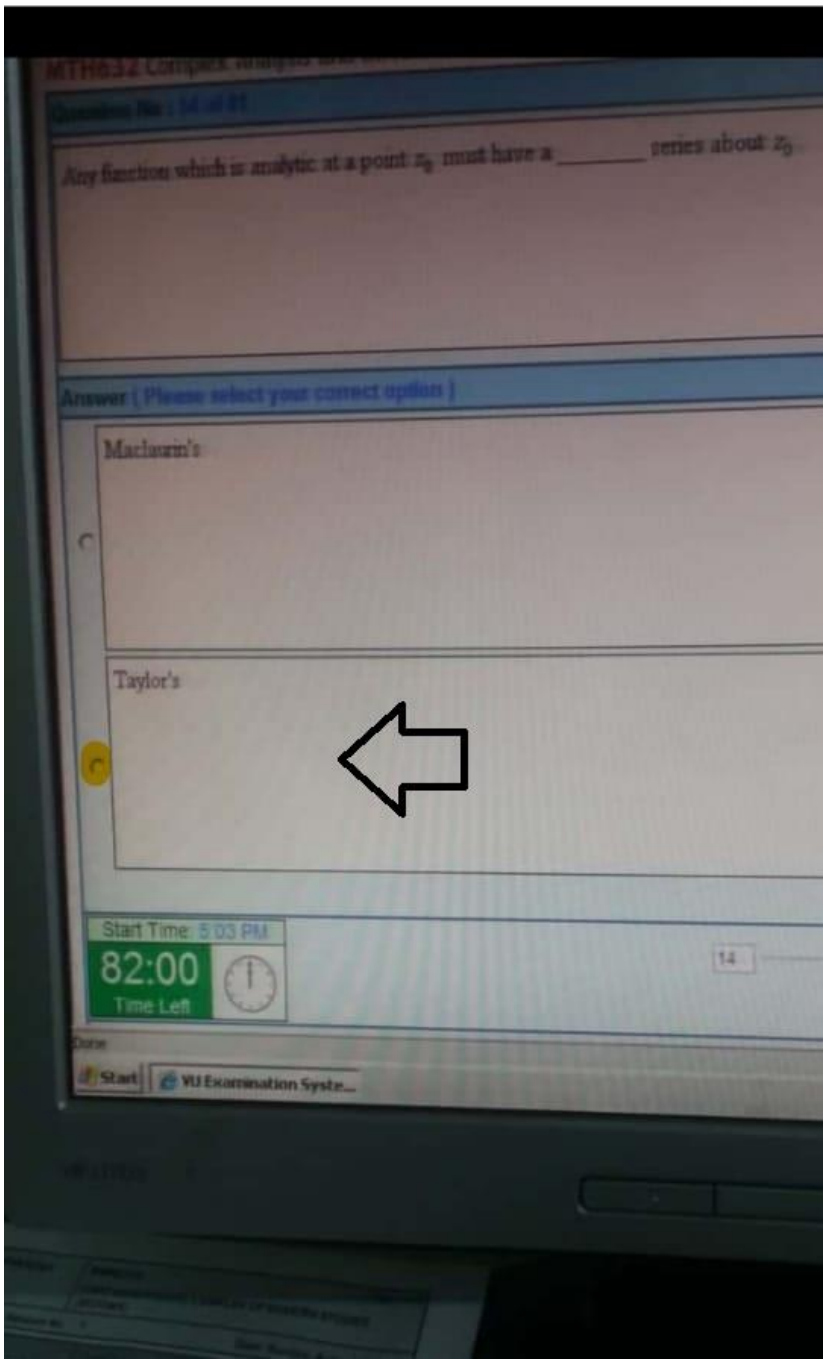


Sca

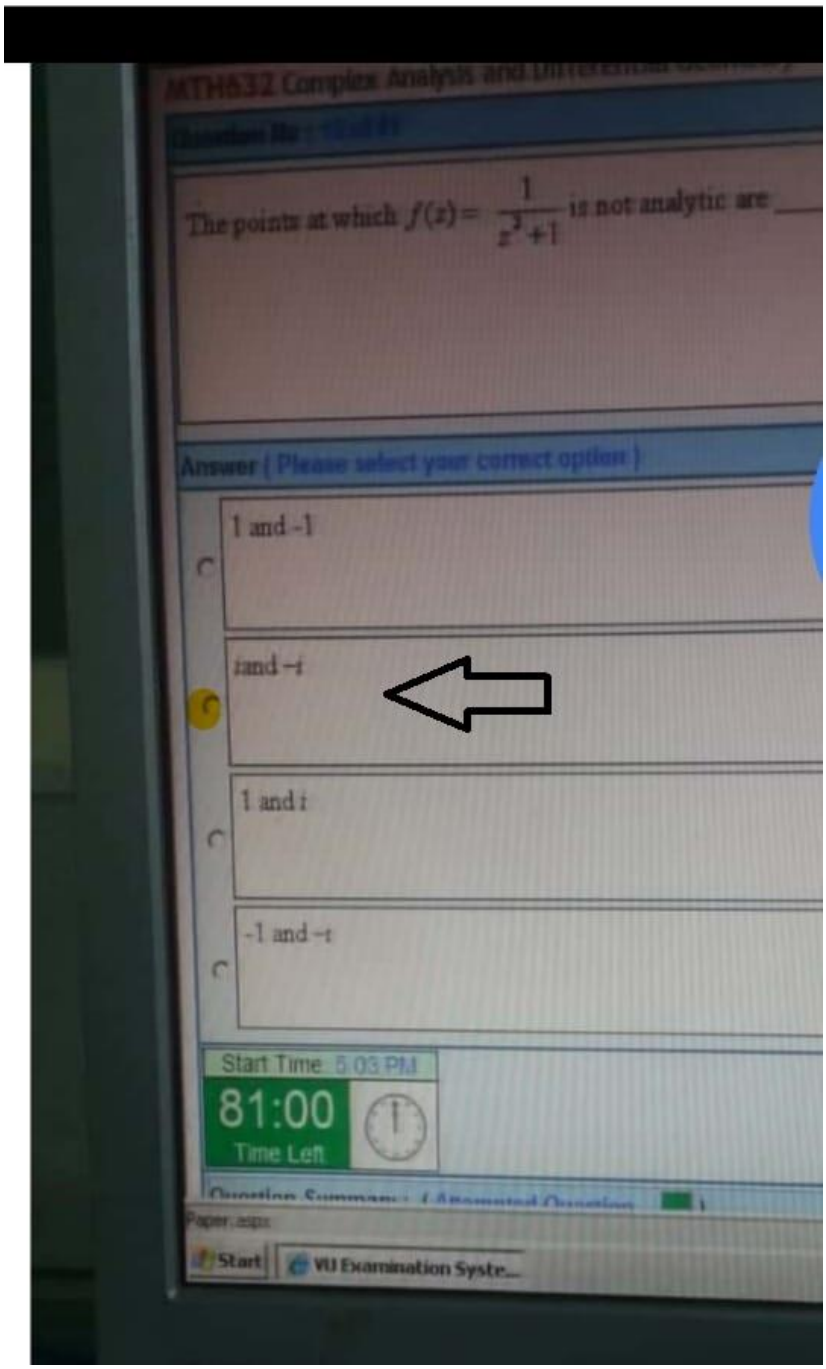




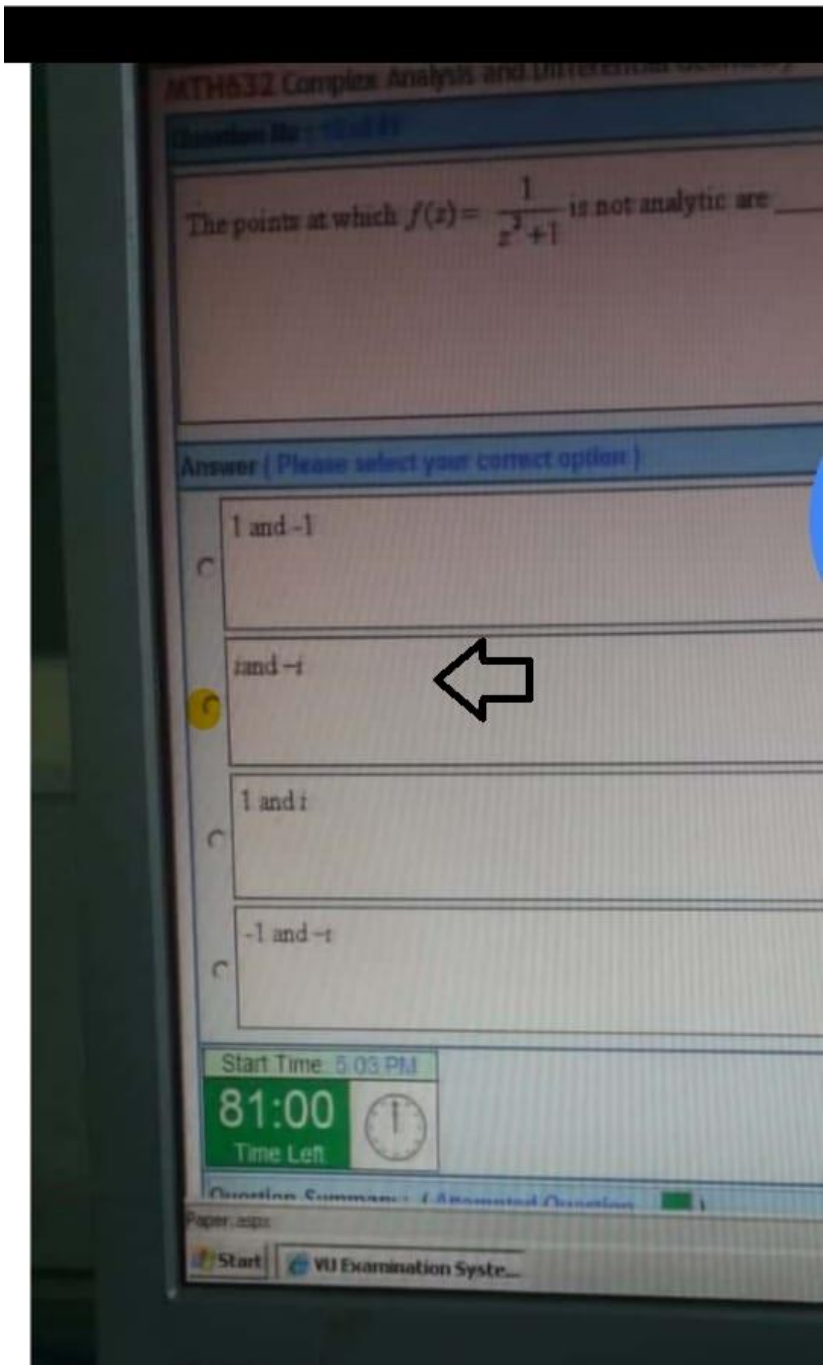
Scanned b

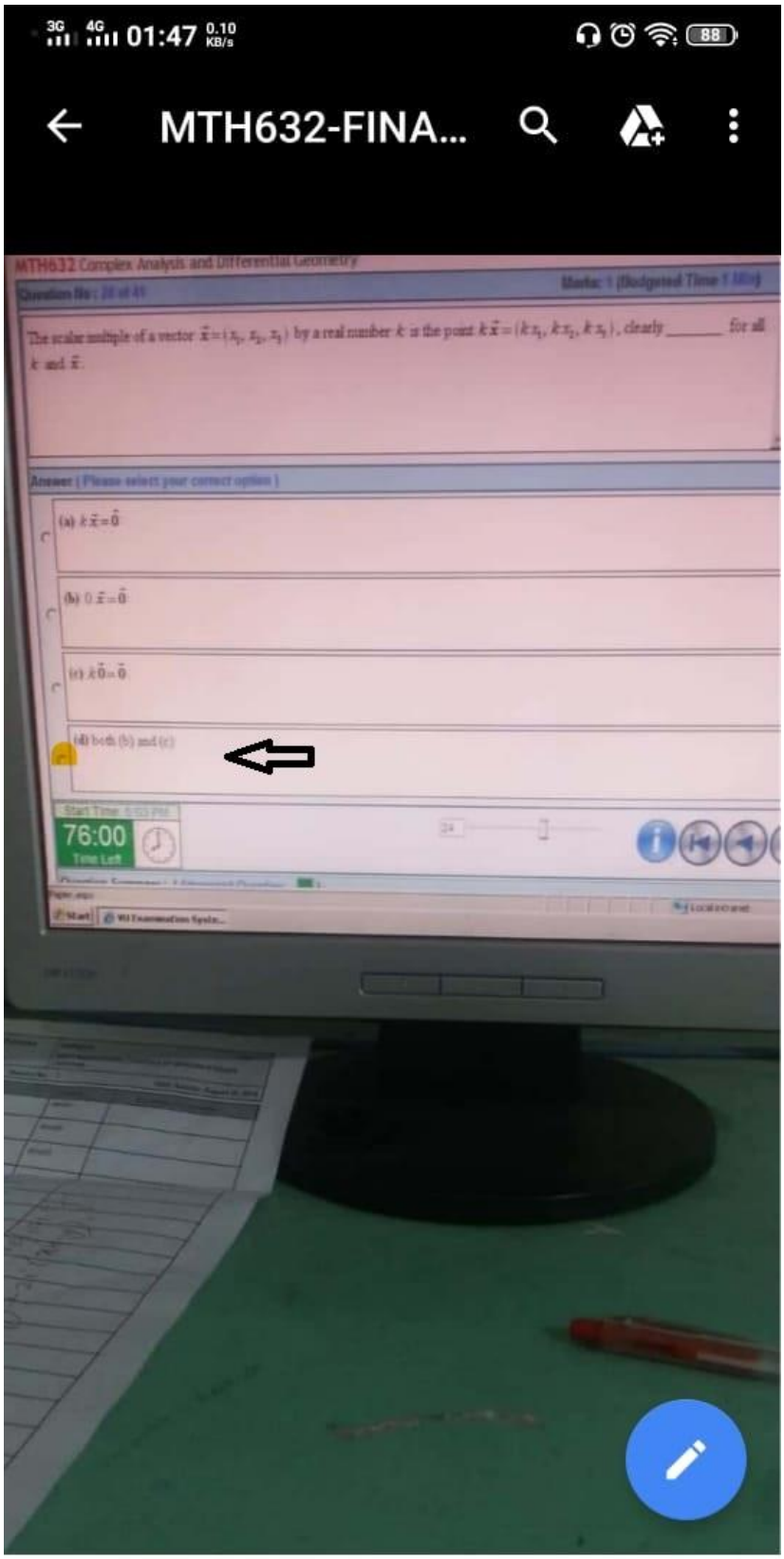

Any function which is analytic at a point z_0 must have a _____ series about z_0
Answer [Please select your correct option]
Maclaurin's
Taylor's
Start Time: 5:03 PM
82:00
Time Left
14
Start
VU Examination Syste...

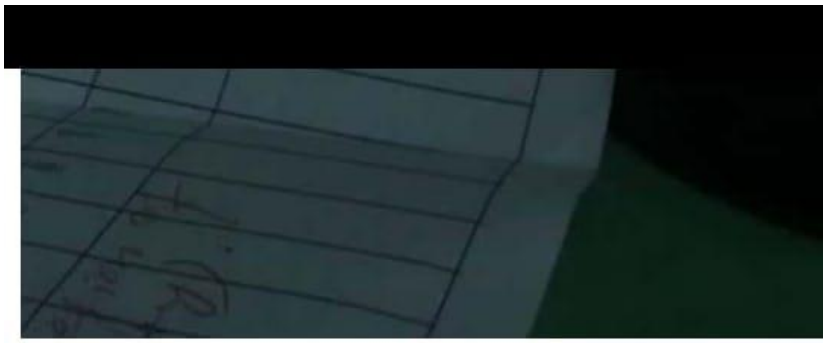


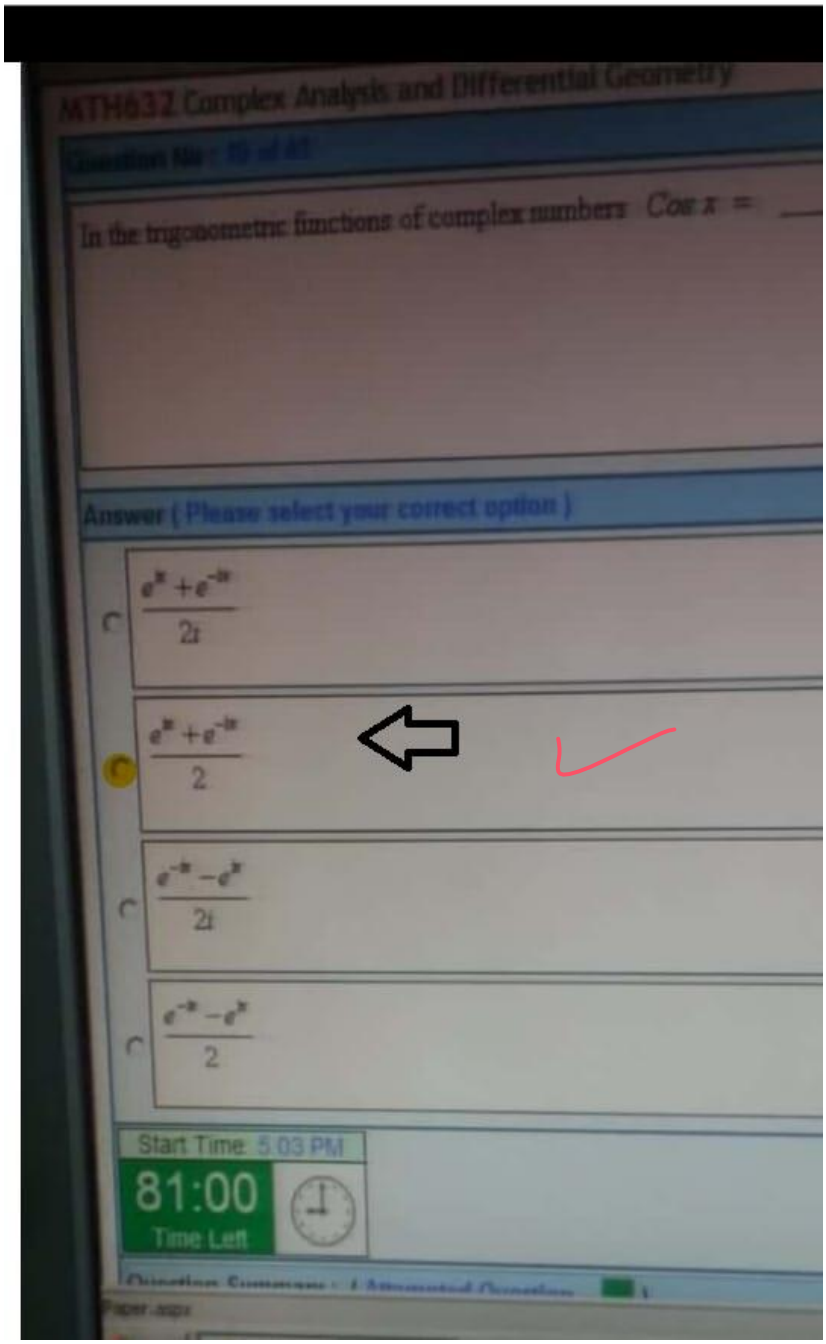

MTH632 Complex Analysis and Differential
The points at which $f(z) = \frac{1}{z^3+1}$ is not analytic are
Answer (Please select your correct option)
1 and -1
i and -i
1 and i
-1 and -i
Start Time 5:03 PM
81:00
Time Left
Start
VU Examination Syste...




MTH632 Complex Analysis and Differential
The points at which $f(z) = \frac{1}{z^3+1}$ is not analytic are
Answer (Please select your correct option)
1 and -1
i and -i
1 and i
-1 and -i
Start Time 5:03 PM
81:00
Time Left
Start
VU Examination Syste...

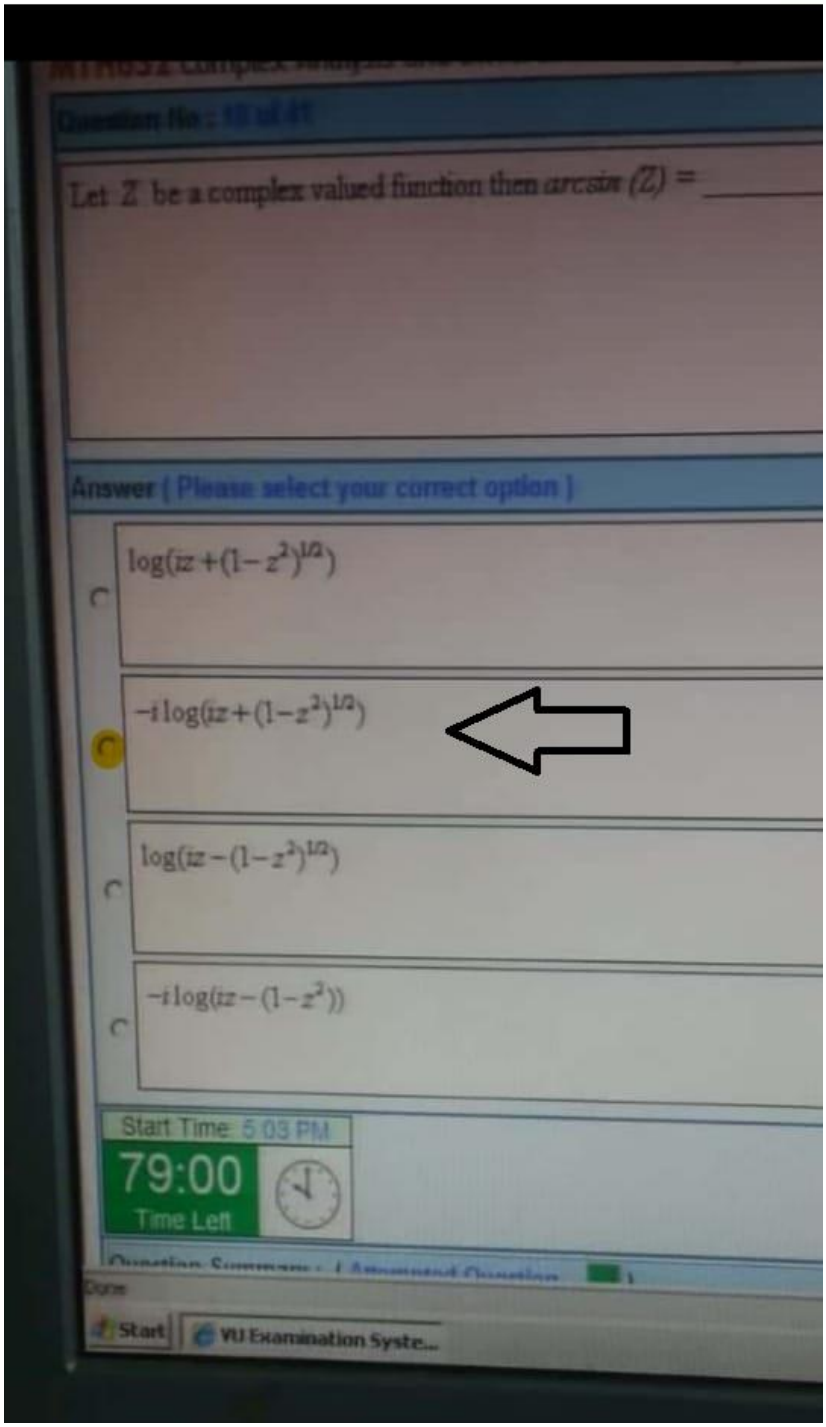

01:47
MTH632-FINA...
MTH632 Complex Analysis and Differential Geometry
The scalar multiple of a vector $\vec{x}=(x_1, x_2, x_3)$ by a real number k is the point $k\vec{x}=(kx_1, kx_2, kx_3)$, clearly _____ for all k and $\vec{x}$.
Answer (Please select your correct option)
(a) $k\vec{x}=\vec{0}$
(b) $0\vec{x}=\vec{0}$
(c) $k\vec{0}=\vec{0}$
(d) both (b) and (c)
76:00
Time Left




MTH632 Complex Analysis and Differential Geometry
In the trigonometric functions of complex numbers Cos x =
Answer (Please select your correct option)
$\frac{e^{ix}+e^{-ix}}{2i}$
$\frac{e^{ix}+e^{-ix}}{2}$
$\frac{e^{-ix}-e^{ix}}{2i}$
$\frac{e^{-ix}-e^{ix}}{2}$
Start Time: 5:03 PM
81:00
Time Left

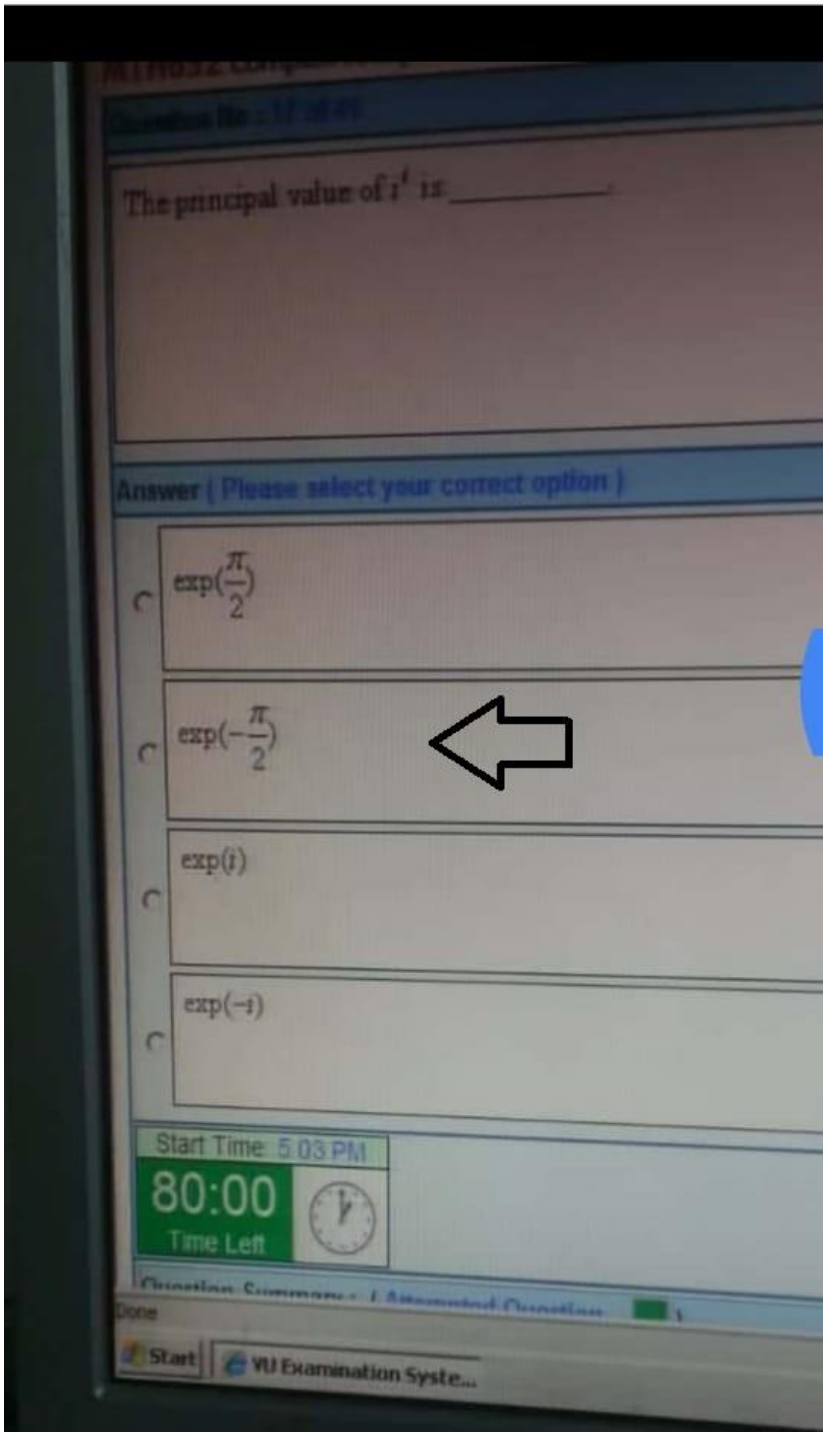


Sca

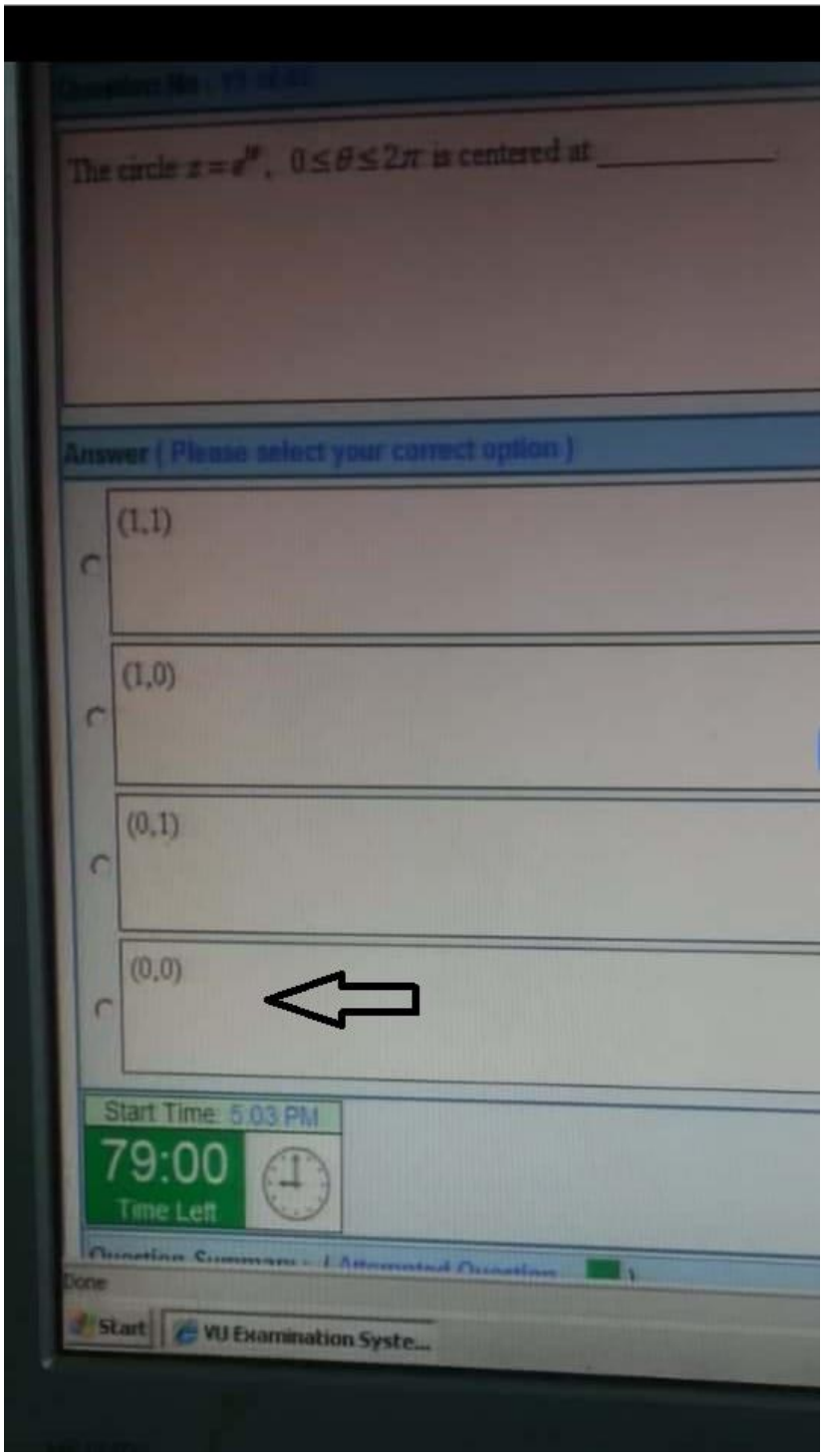

Let Z be a complex valued function then arcsin (Z) =
Answer (Please select your correct option)
log(iz +(1−z²)^1/2)
−i log(iz+(1−z²)^1/2)
log(iz −(1−z²)^1/2)
−i log(iz−(1−z²))
Start Time: 5:03 PM
79:00
Time Left
Start
VU Examination Syste...

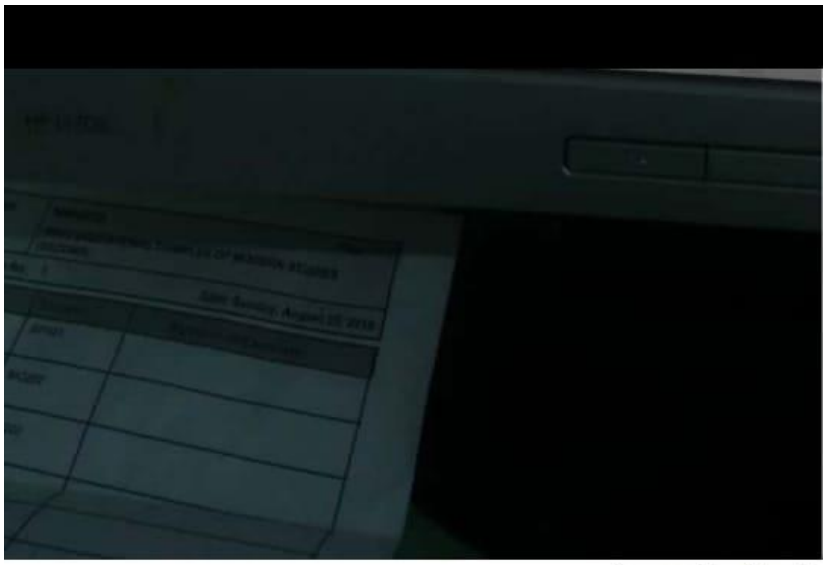


Sca

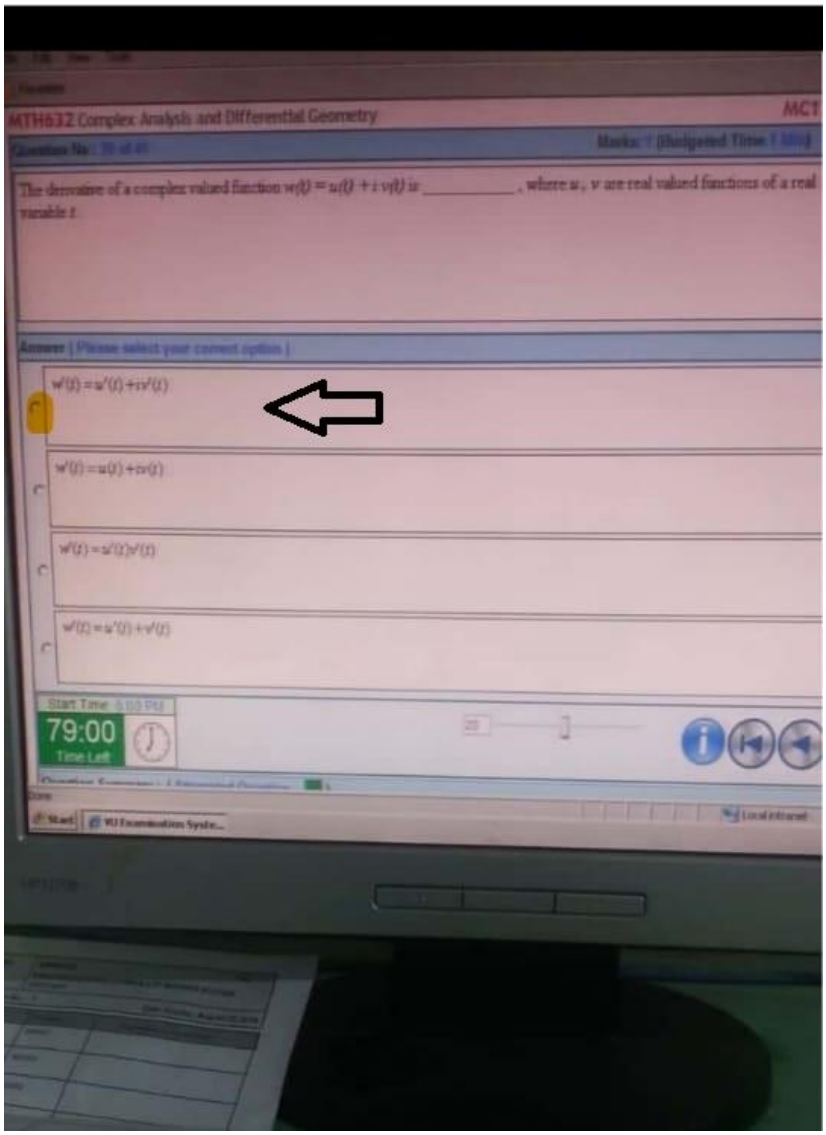

The principal value of i^i is _____
Answer (Please select your correct option)
$\exp(\frac{\pi}{2})$
$\exp(-\frac{\pi}{2})$
$\exp(i)$
$\exp(-i)$
Start Time 5:03 PM
80:00
Time Left
Done
Start
VU Examination Syste...

Sca

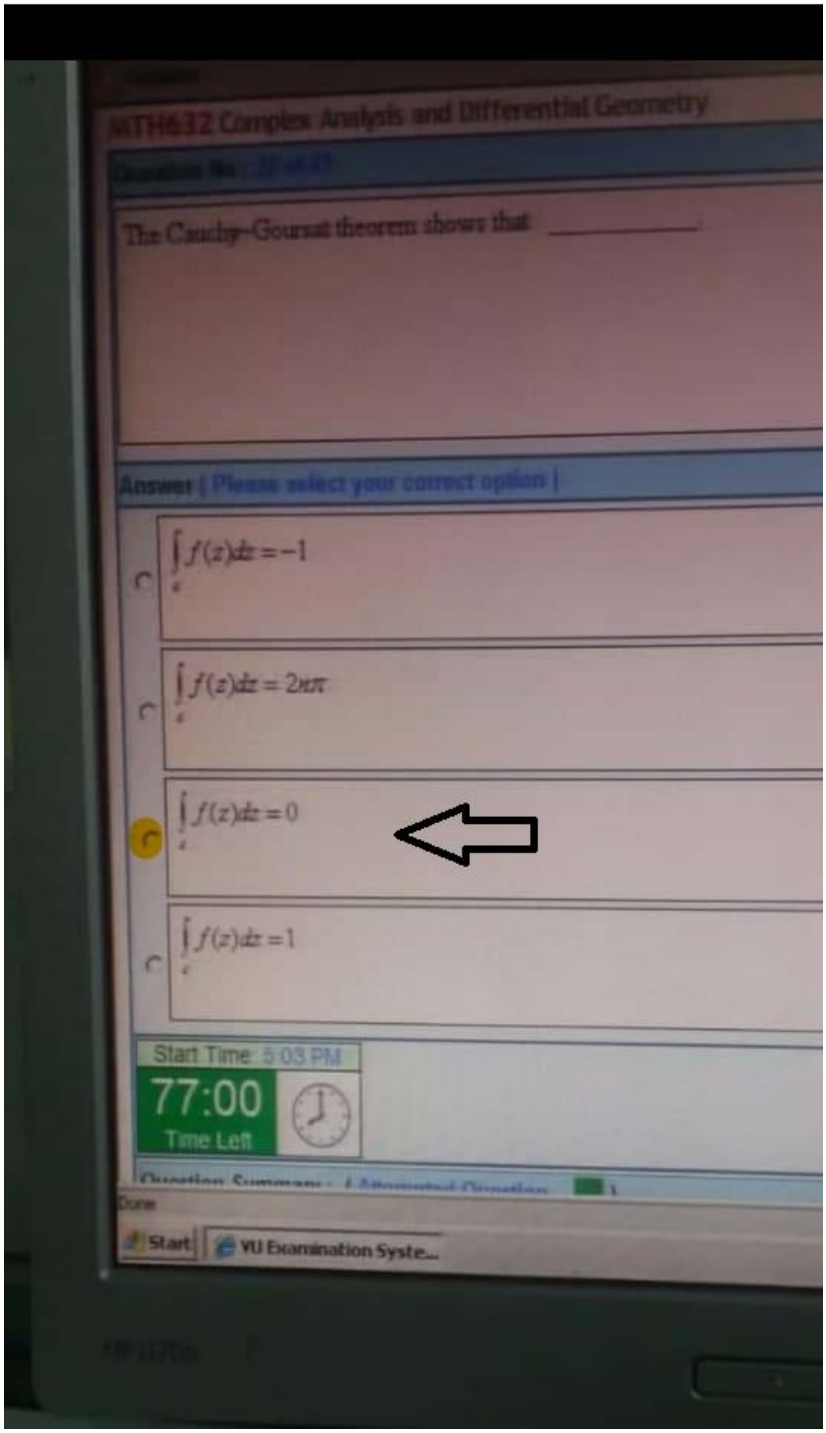

The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 5:03 PM
79:00
Time Left
Done
Start
VU Examination Syste...

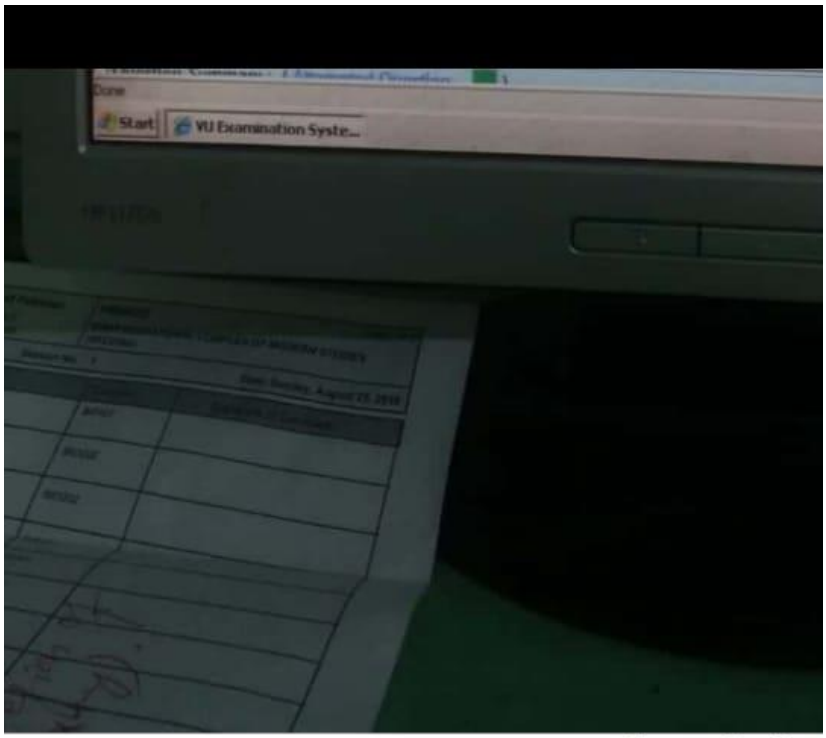


Scanned by CamSc

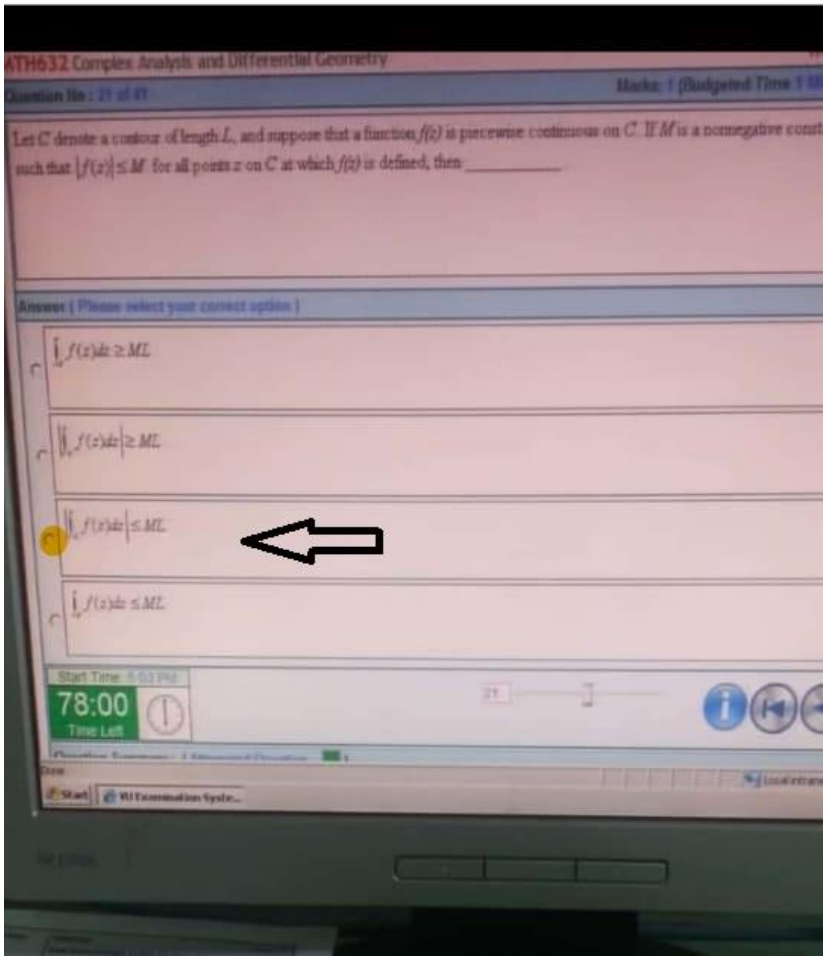


Scar


MTH632 Complex Analysis and Differential Geometry
The Cauchy-Gourst theorem shows that _____
Answer [Please select your correct option]
$\int_c f(z)dz = -1$
$\int_c f(z)dz = 2\pi i$
$\int_c f(z)dz = 0$
$\int_c f(z)dz = 1$
Start Time: 5:03 PM
77:00
Time Left
Start
VU Examination Syste...

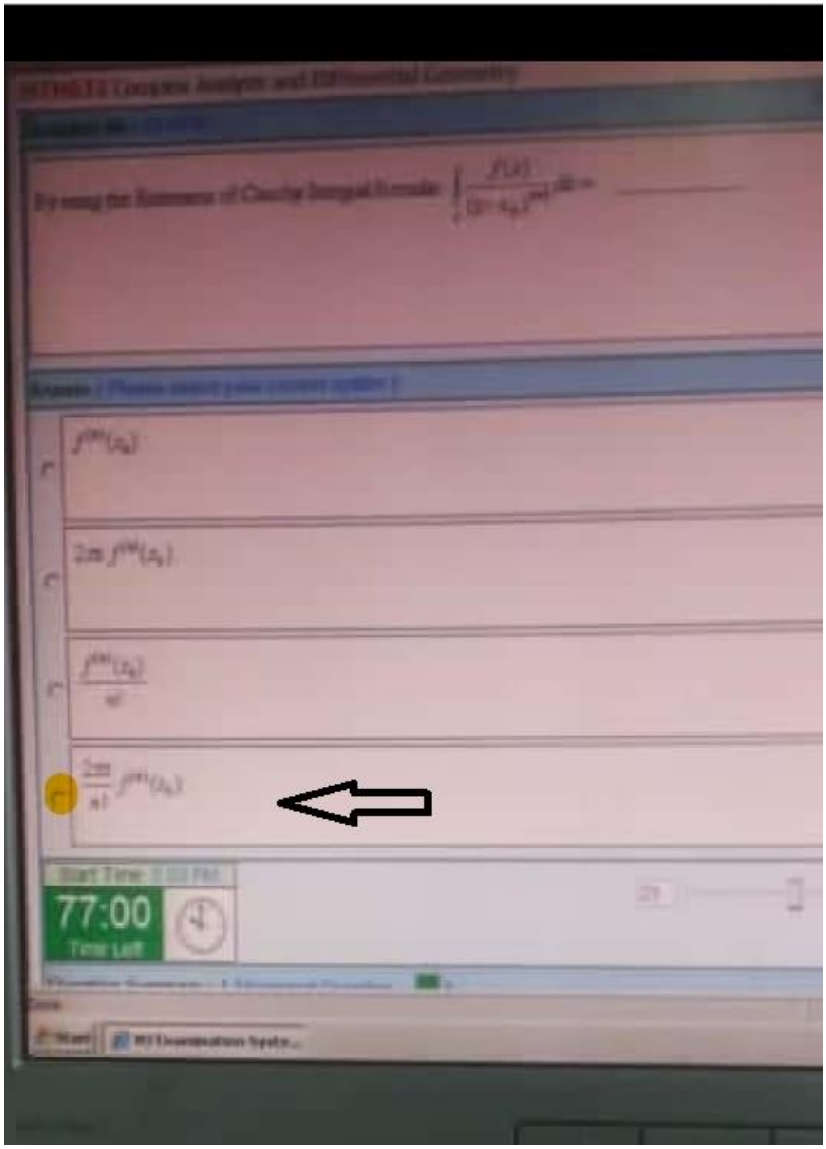


Scanned by Cam



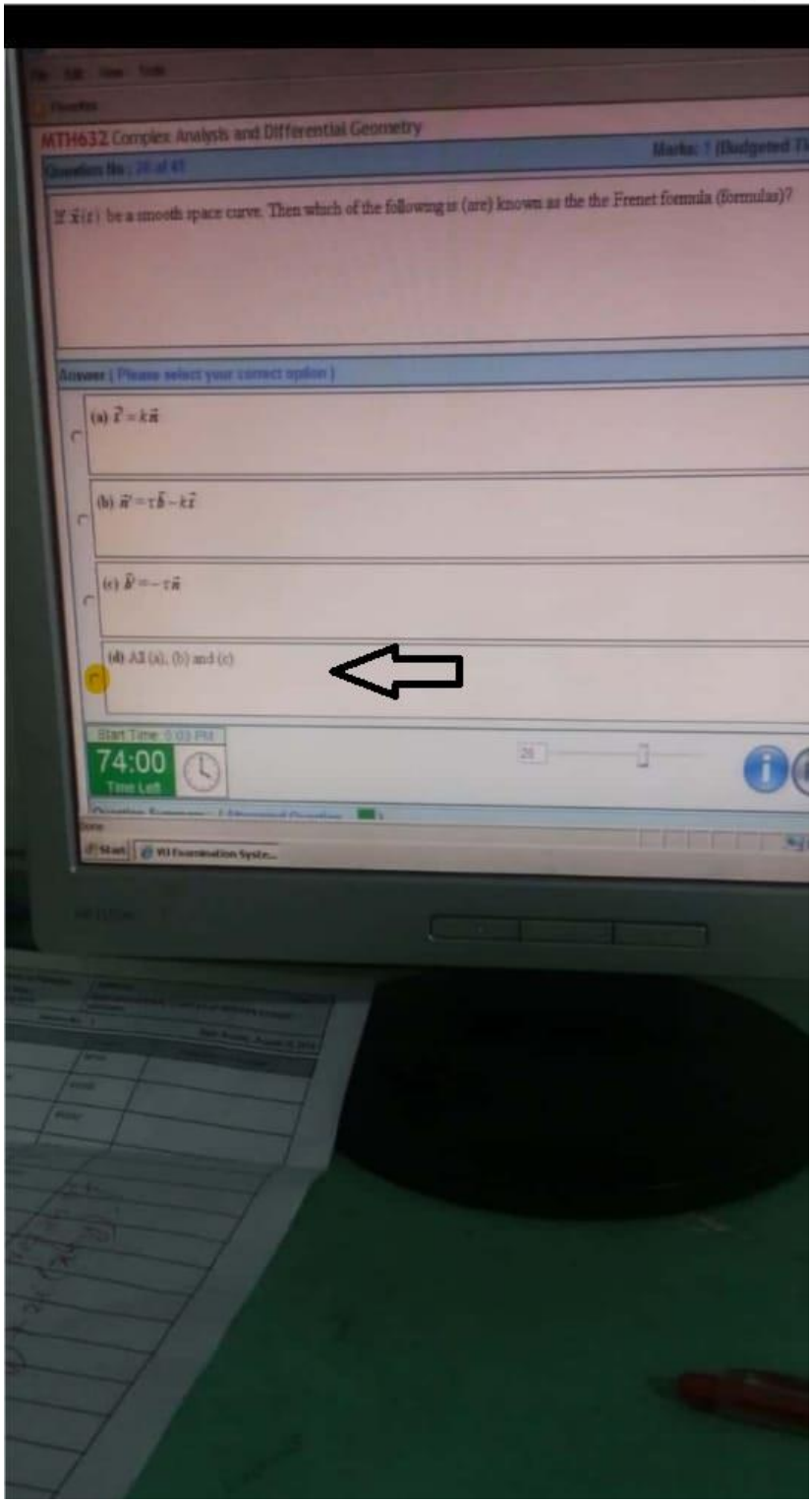


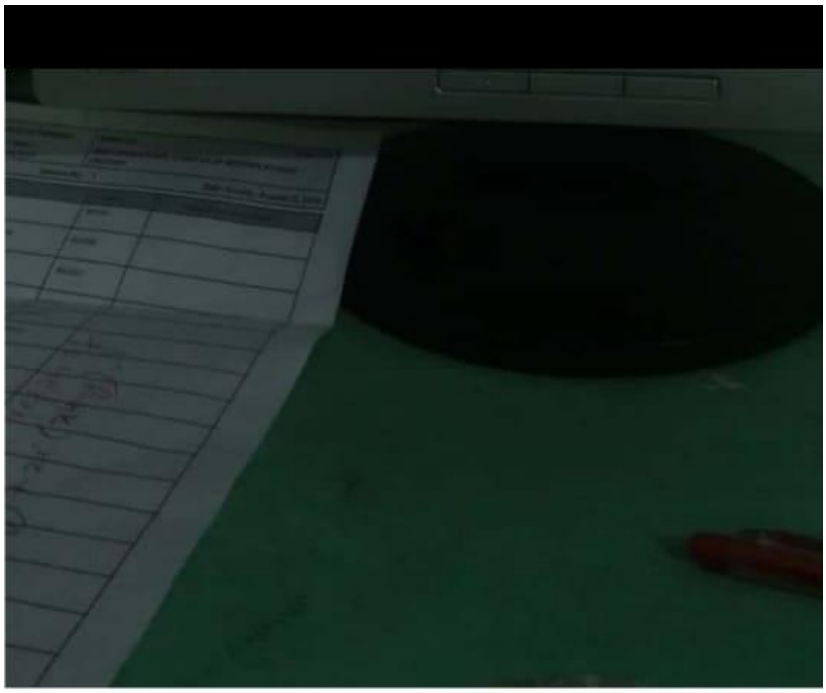
Scanned



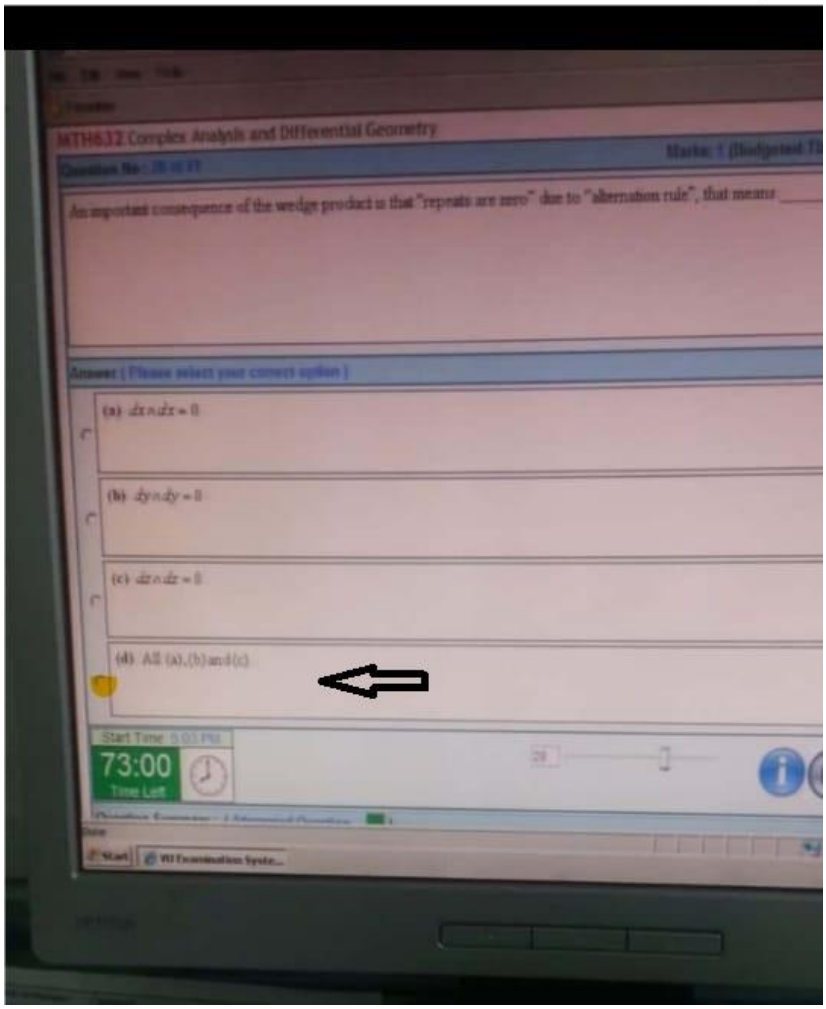


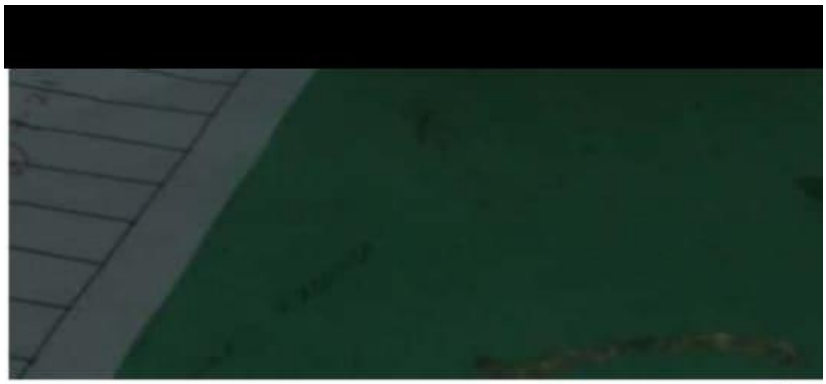
Scanned by Cam



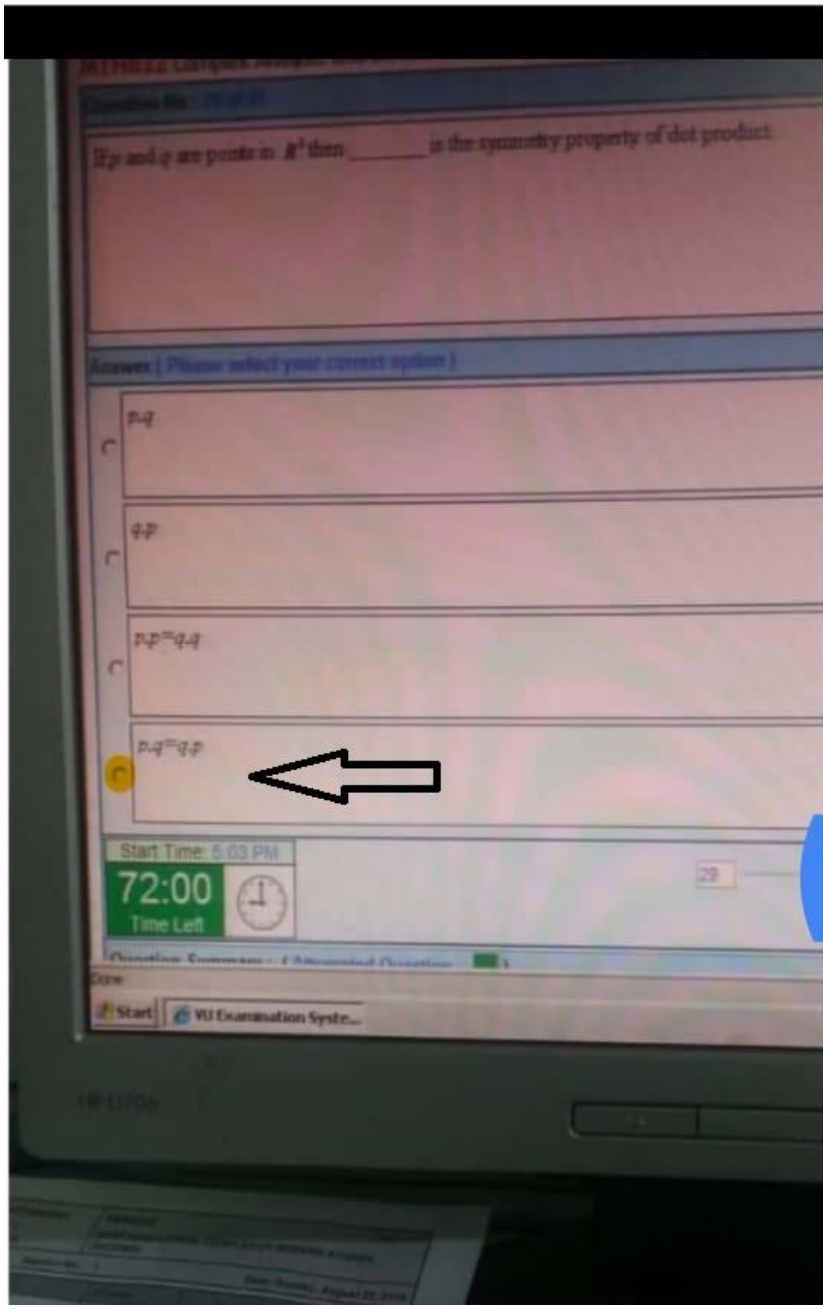


Scanned by Cam



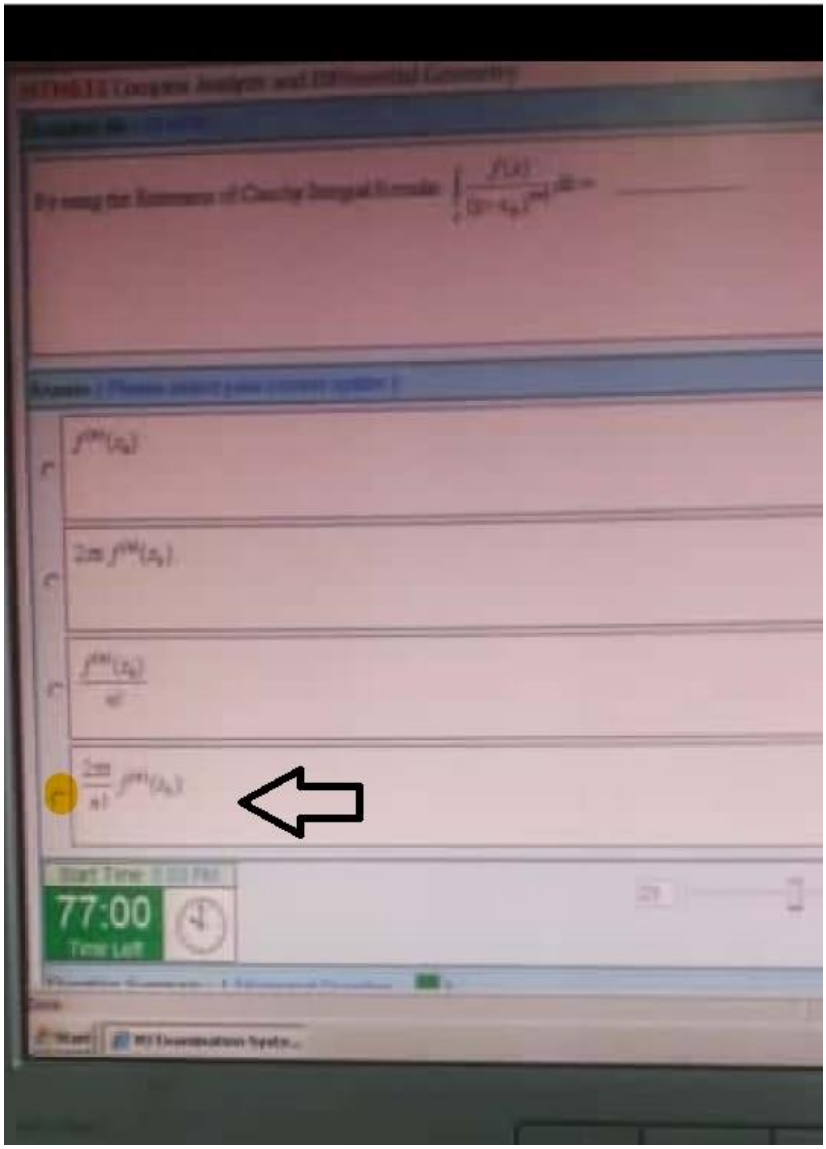


Scan


If p and q are points in R^3 then _____ is the symmetry property of dot product
Answer [Please select your correct option]
p.q
q.p
p.p=q.q
p.q=q.p
Start Time: 5:03 PM
72:00
Time Left
29
Start
VU Examination Syste...

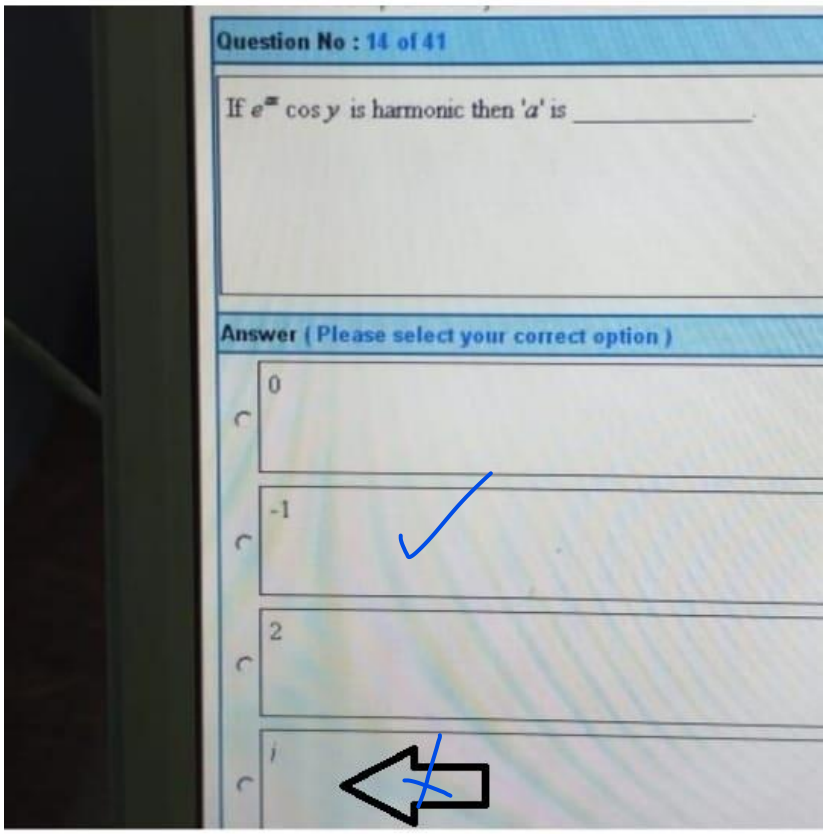


Scanned

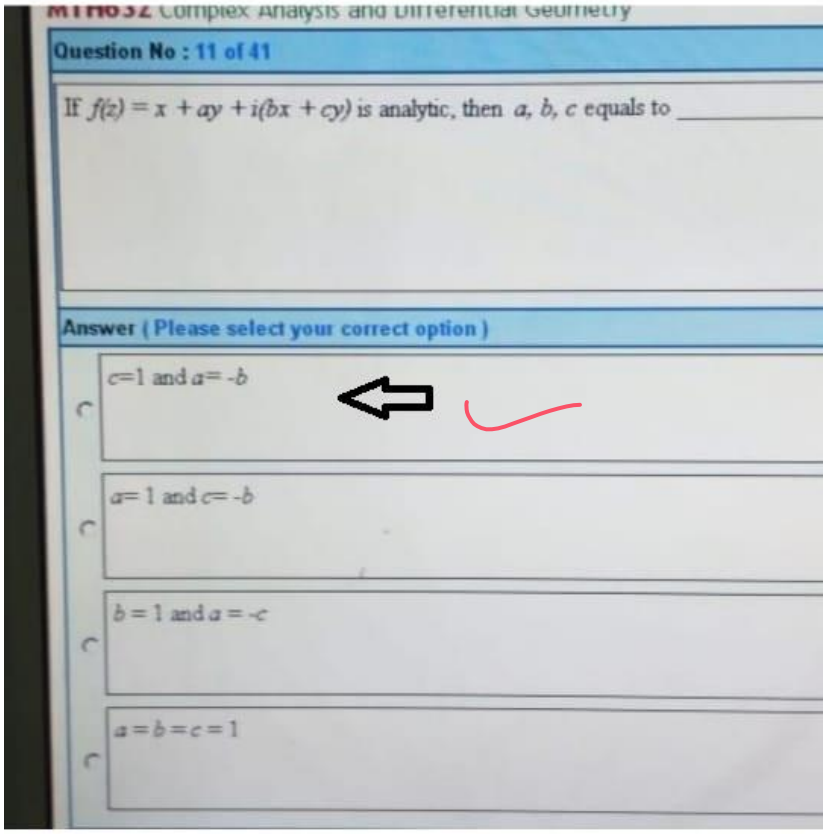


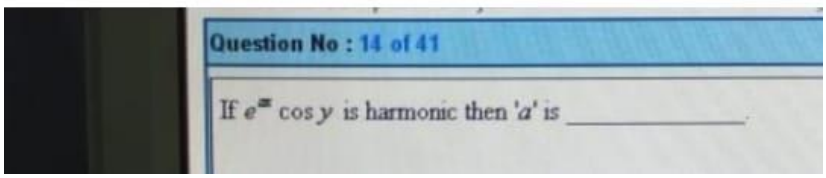
[redacted]

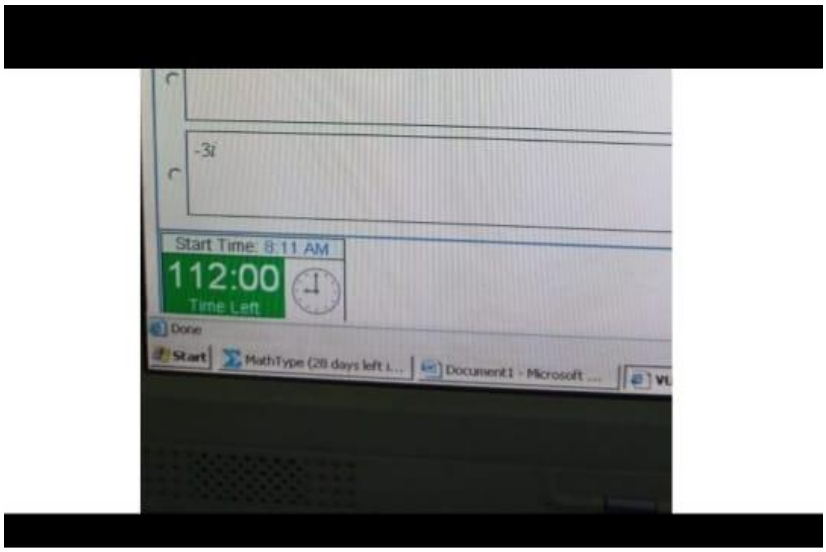
[redacted]

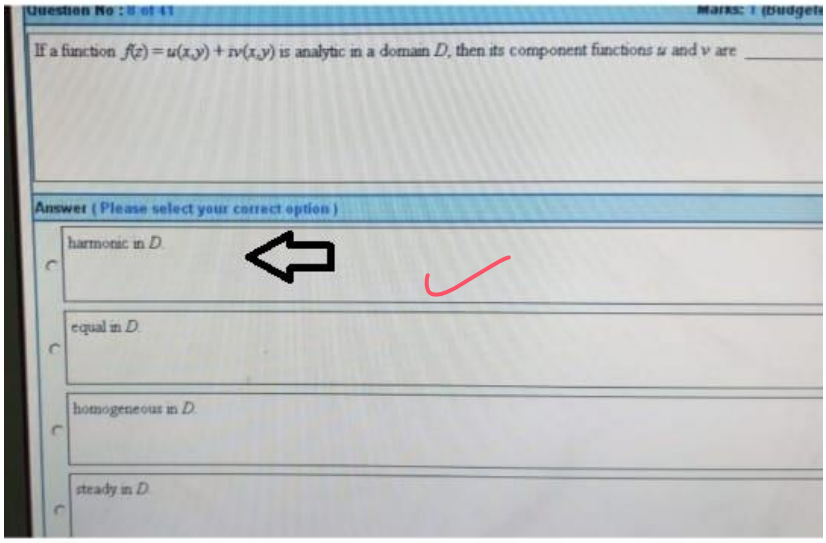


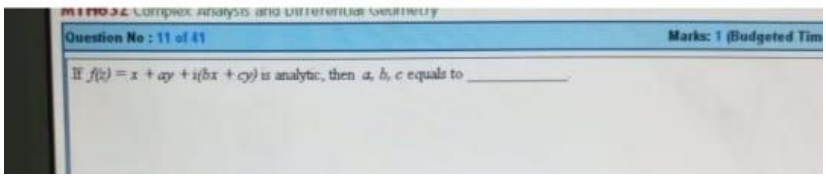
[redacted]

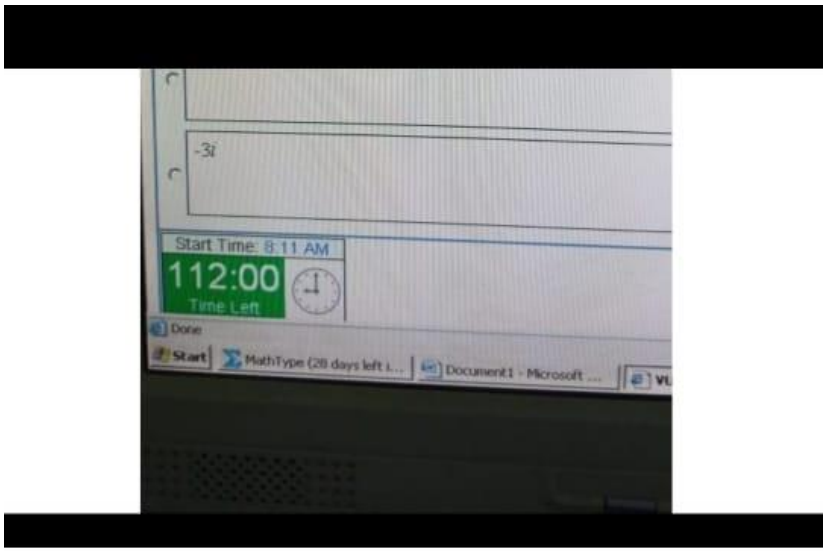

MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 41
If f(z) = x + ay + i(bx + cy) is analytic, then a, b, c equals to _____
Answer (Please select your correct option)
c=1 and a= -b
a= 1 and c= -b
b = 1 and a = -c
a = b = c = 1

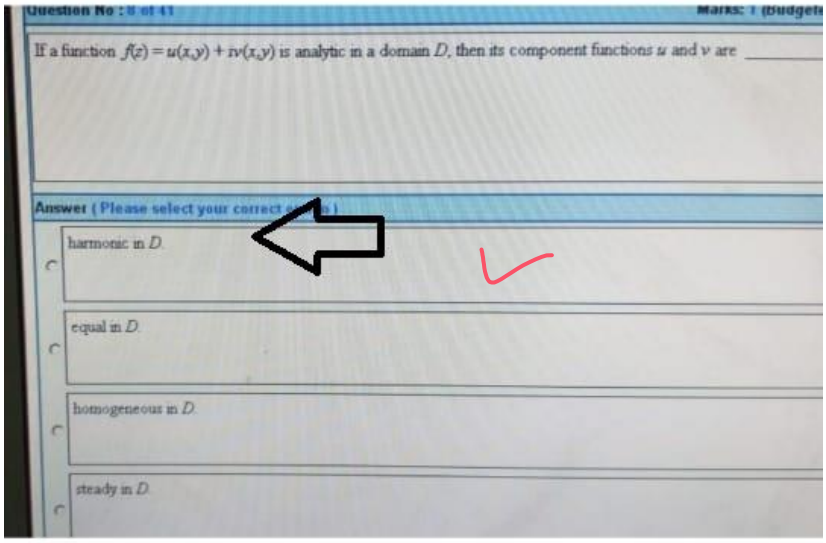

Question No : 14 of 41
If e^{ax} cos y is harmonic then 'a' is _____

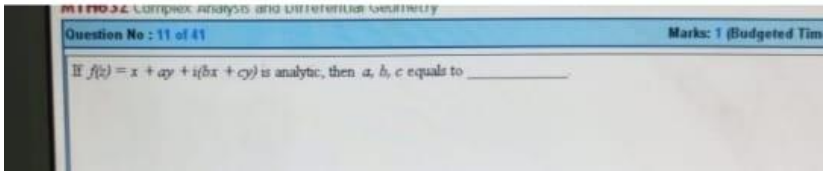

-3i
Start Time: 8:11 AM
112:00
Time Left
Done
Start
MathType (28 days left i...
Document1 - Microsoft ...


If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.


Question No : 11 of 41
Marks: 1 (Budgeted Tim
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.


-3i
Start Time: 8:11 AM
112:00
Time Left
Done
Start
MathType (28 days left i...
Document1 - Microsoft ...


If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____
Answer (Please select your correct
harmonic in D.
equal in D.
homogeneous in D.
steady in D.


Question No : 11 of 41
Marks: 1 (Budgeted Tim
If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.

$w'(t) = u'(t)v'(t)$

$w'(t) = u'(t) + v'(t)$

art Time: 8:11 AM

07:00

21

Marks: 1 (Budg

If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ ____

Answer (Please select your correct option)

- $\vec{b}$
- $\vec{t}$
- $\vec{n}$
- $-\vec{n}$

Start Time: 8:11 AM

106:00

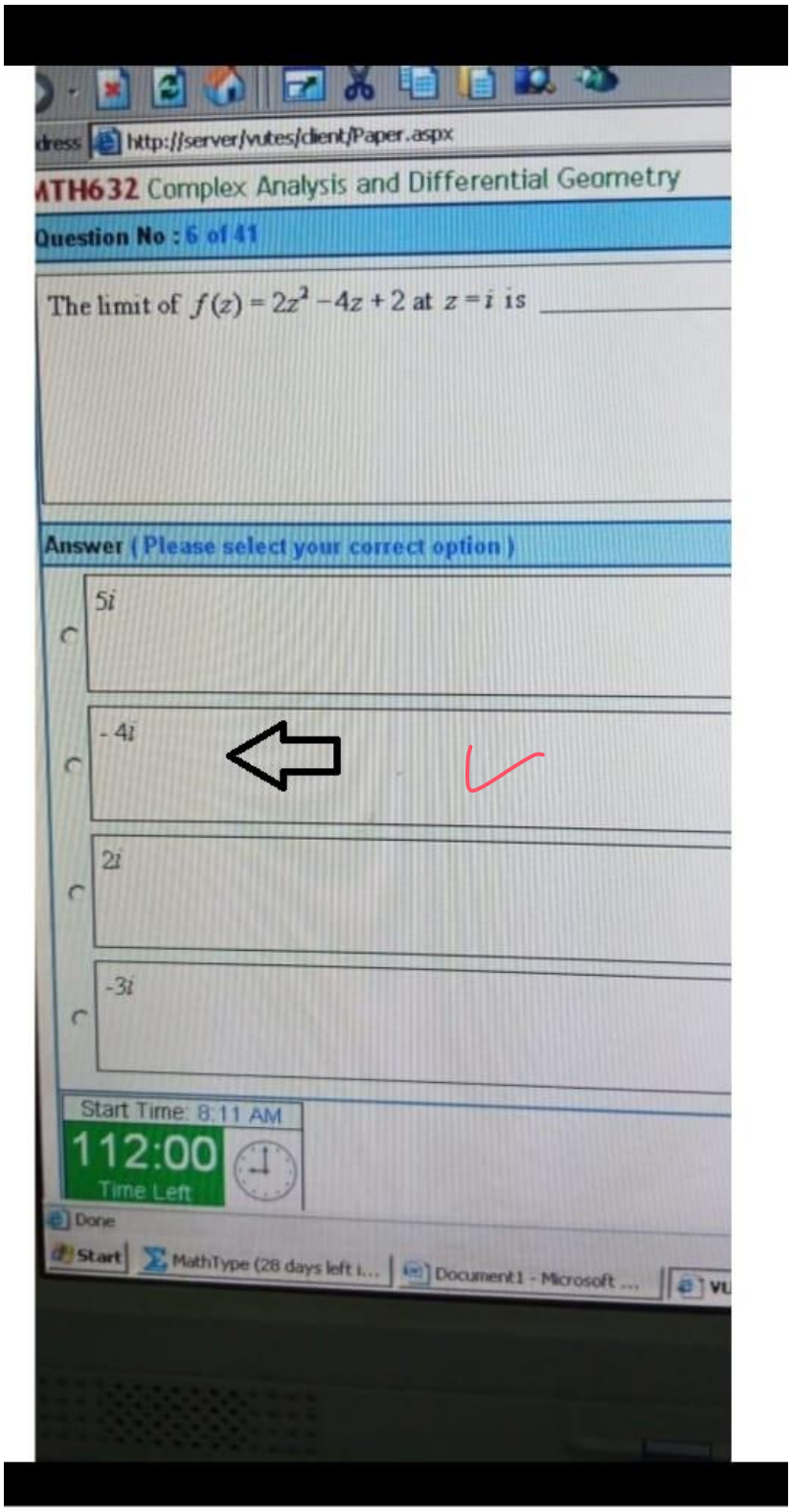
Time Left

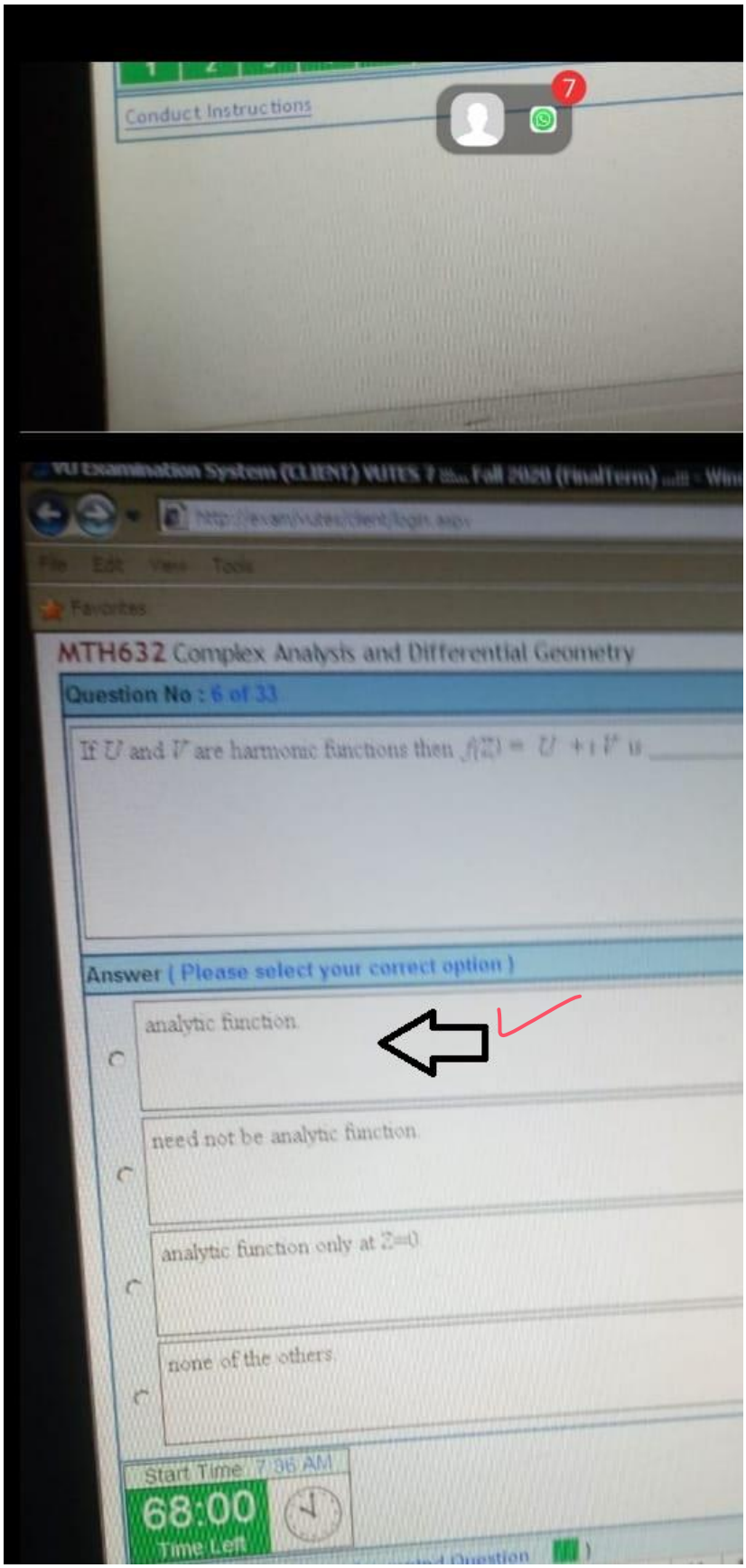
27

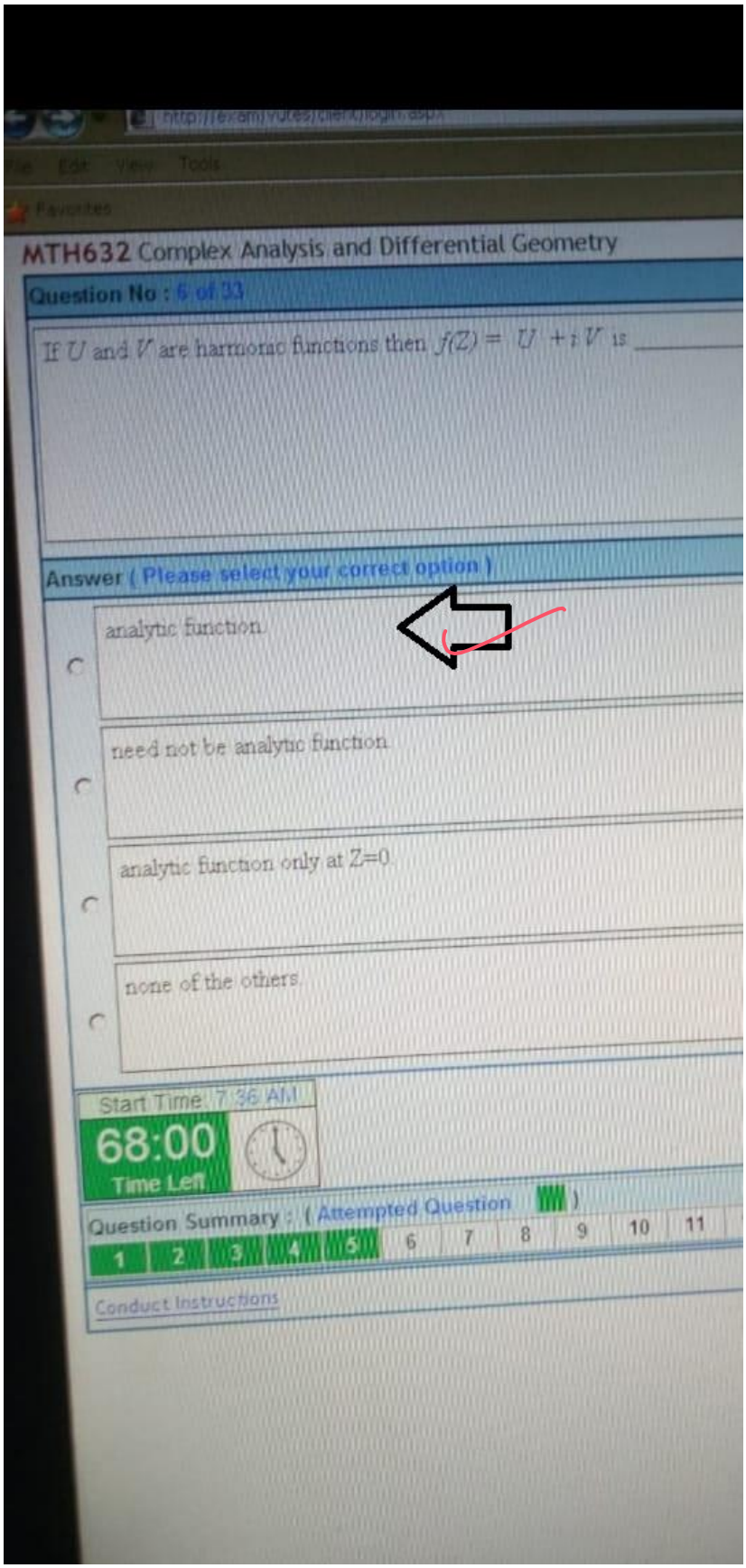
lo : 22 of 41

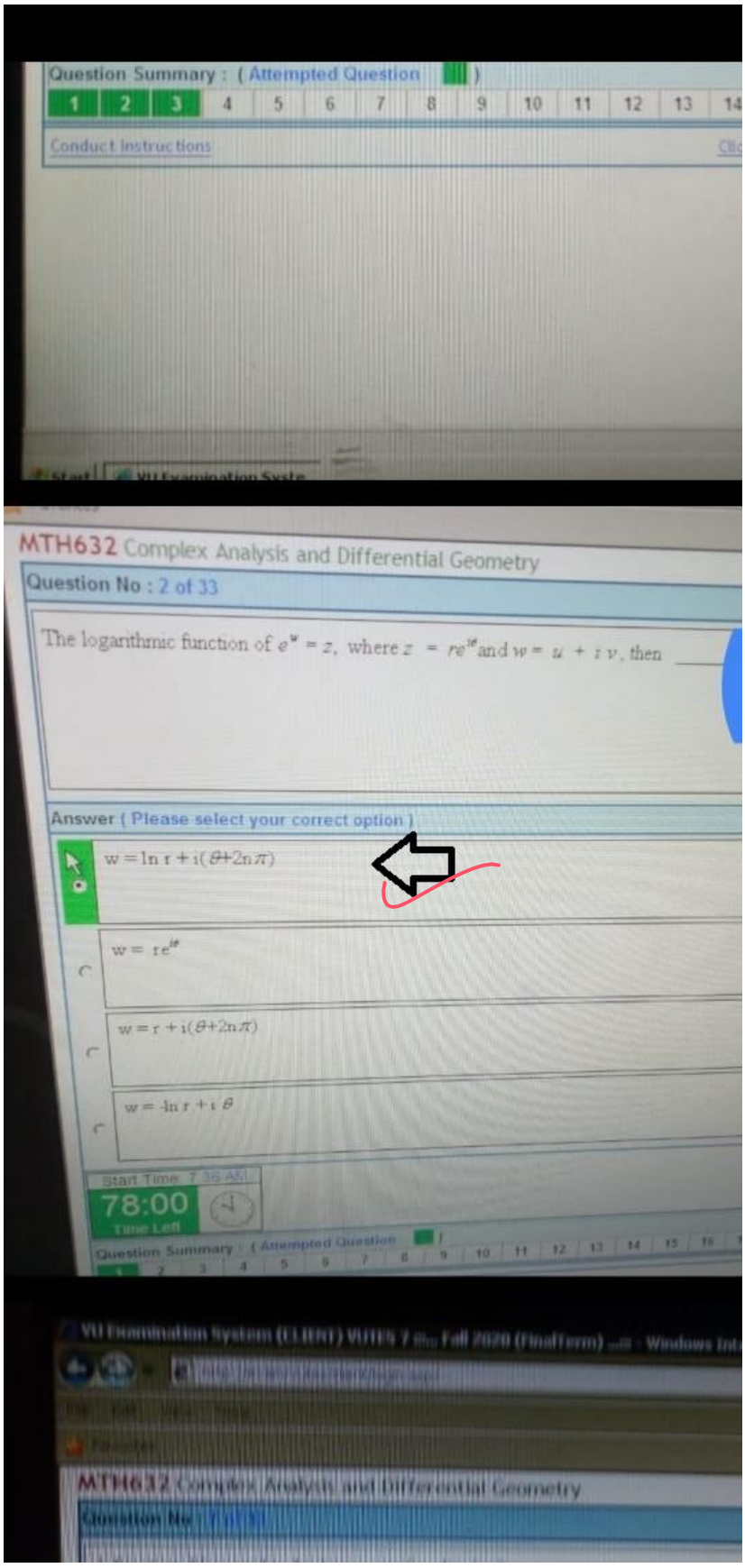
Marks: 1 (

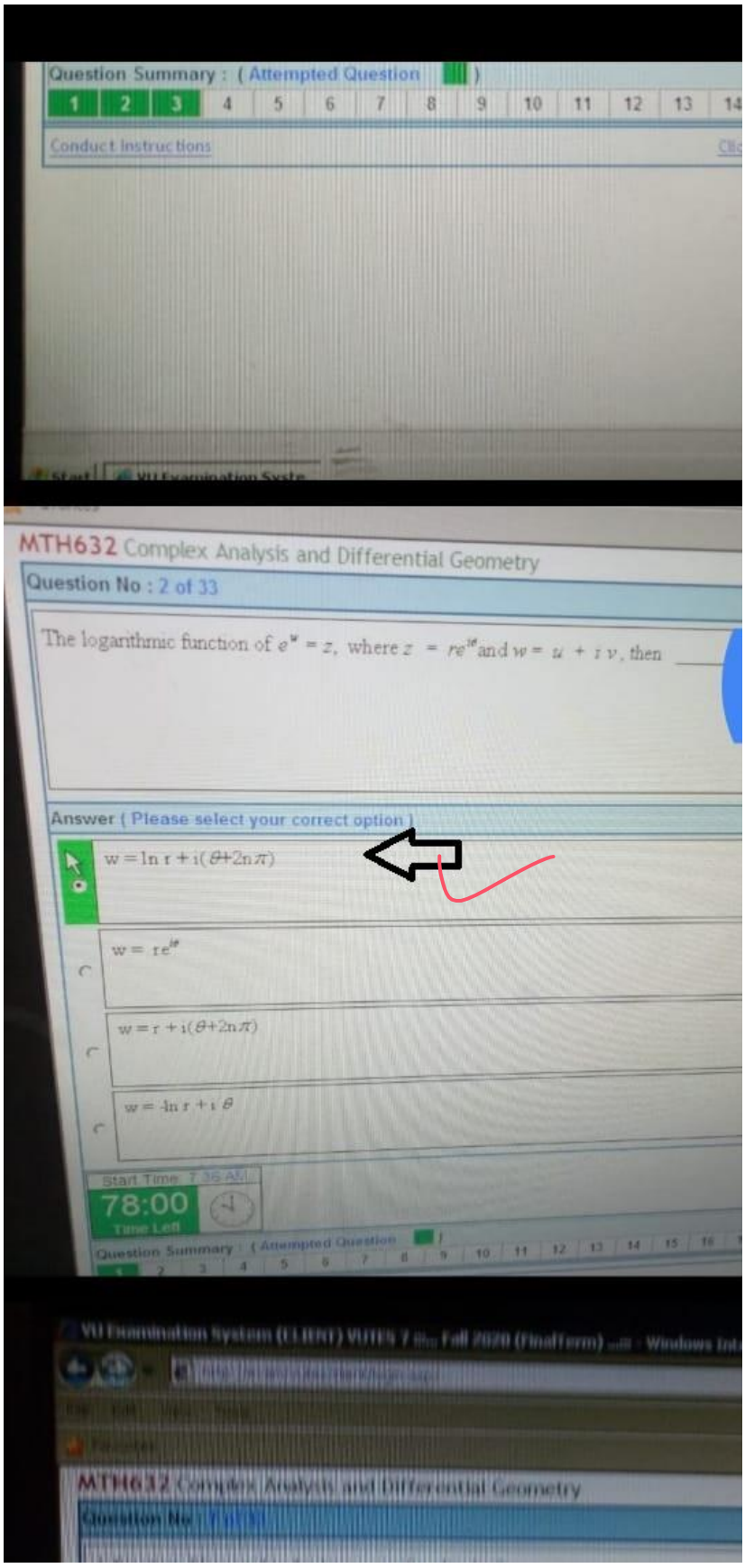
a piecewise continuous complex-valued function defined on an interval $a \le t \le b$, then _____.

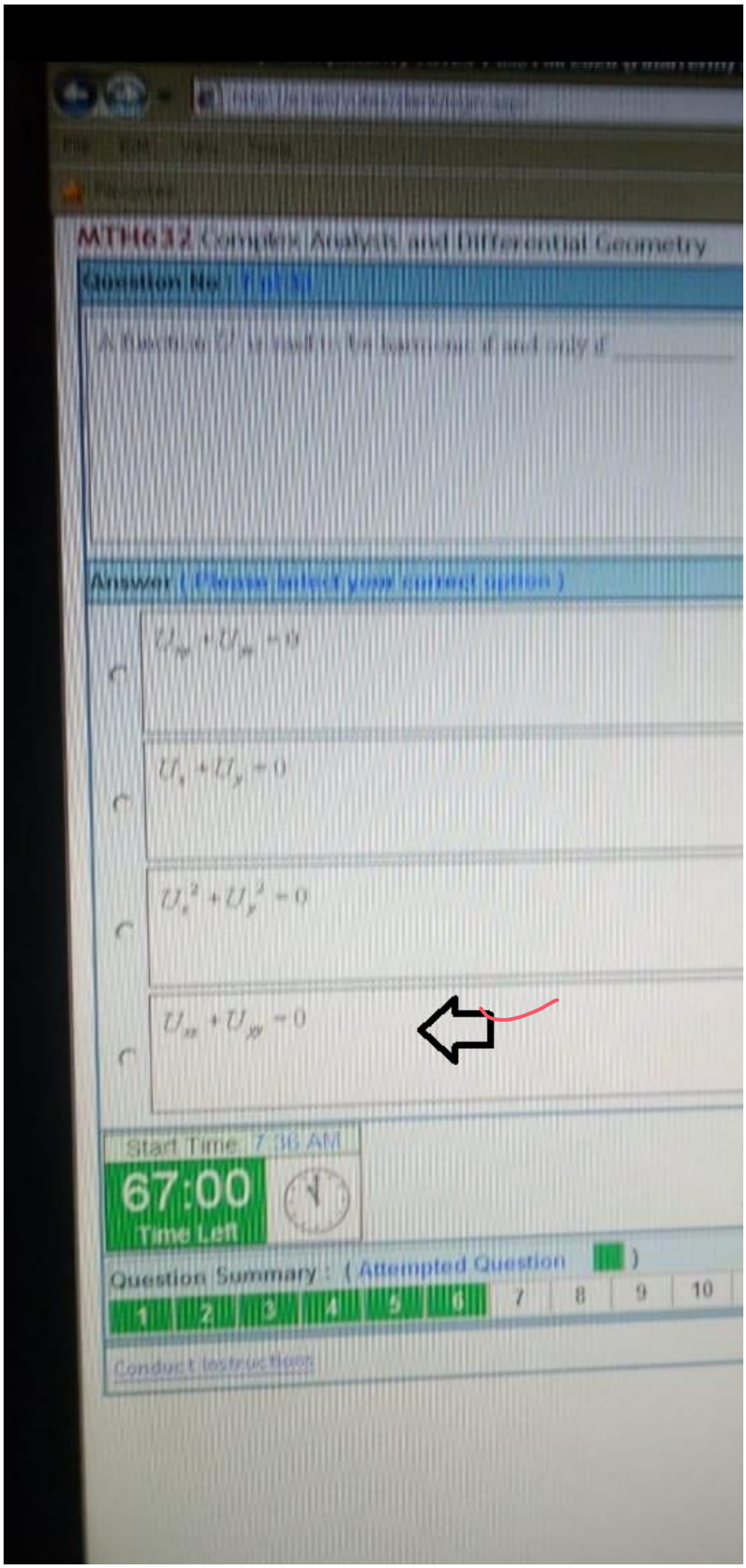

http://server/vutes/client/Paper.aspx
ATH632 Complex Analysis and Differential Geometry
Question No : 6 of 41
The limit of $f(z) = 2z^2 - 4z + 2$ at $z = i$ is _____
Answer (Please select your correct option)
5i
- 4i
2i
-3i
Start Time: 8:11 AM
112:00
Time Left
Done
Start
MathType (28 days left i...
Document1 - Microsoft ...

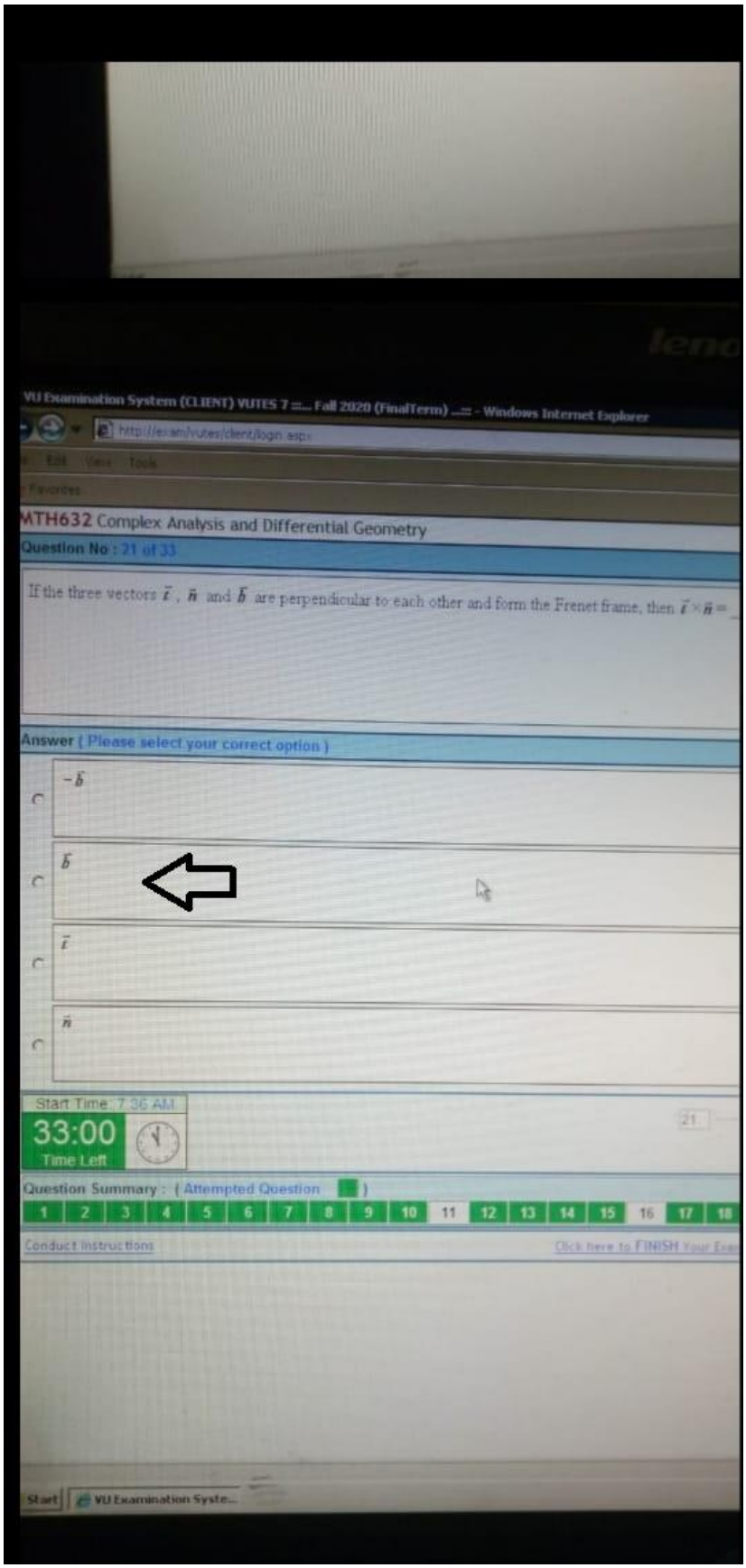

Conduct Instructions
7
MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
If U and V are harmonic functions then
Answer (Please select your correct option)
analytic function
need not be analytic function
analytic function only at
none of the others
Start Time: 7:36 AM
68:00
Time Left

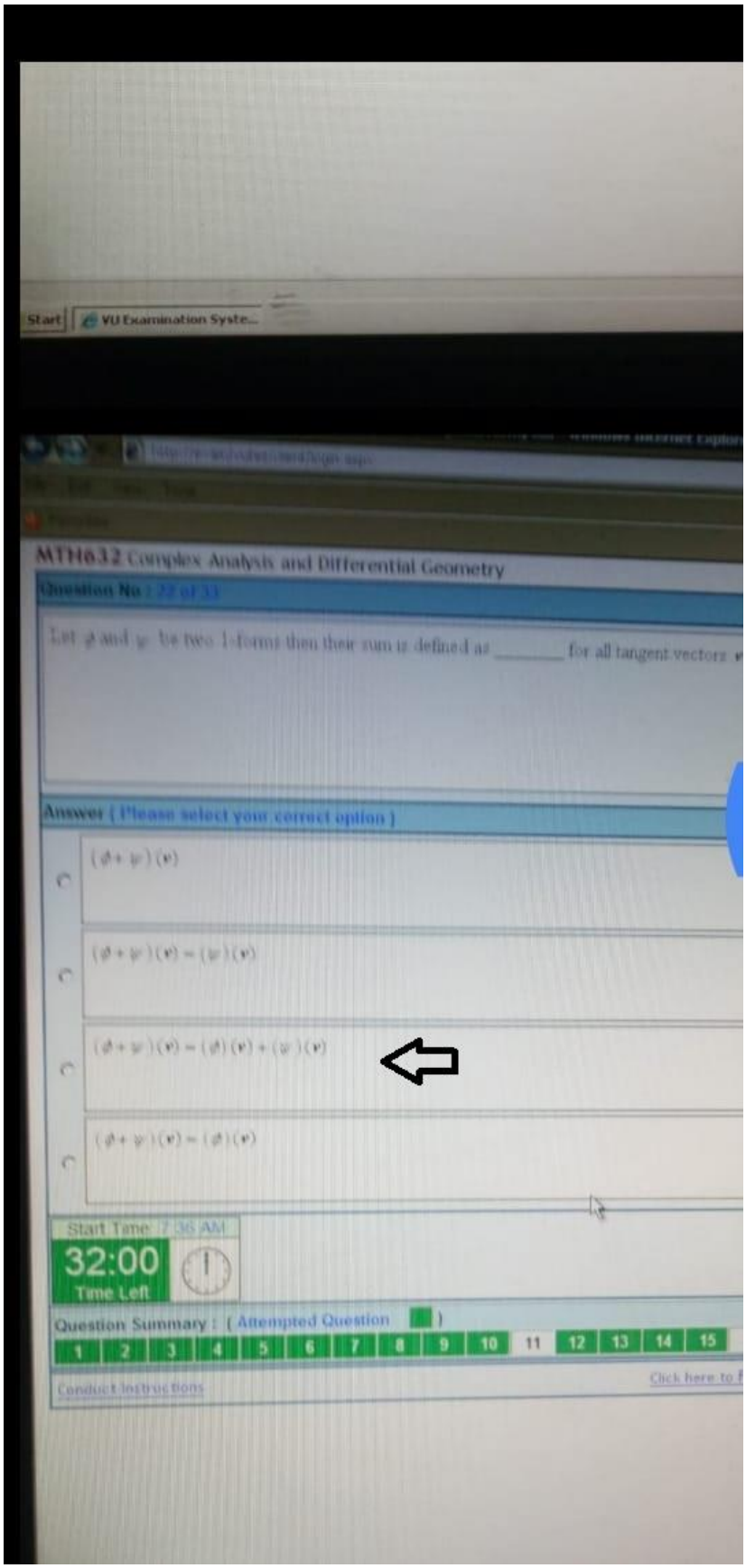

MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
If U and V are harmonic functions then f(Z) = U + i V is
Answer (Please select your correct option)
analytic function
need not be analytic function
analytic function only at Z=0
none of the others
Start Time: 7:36 AM
68:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

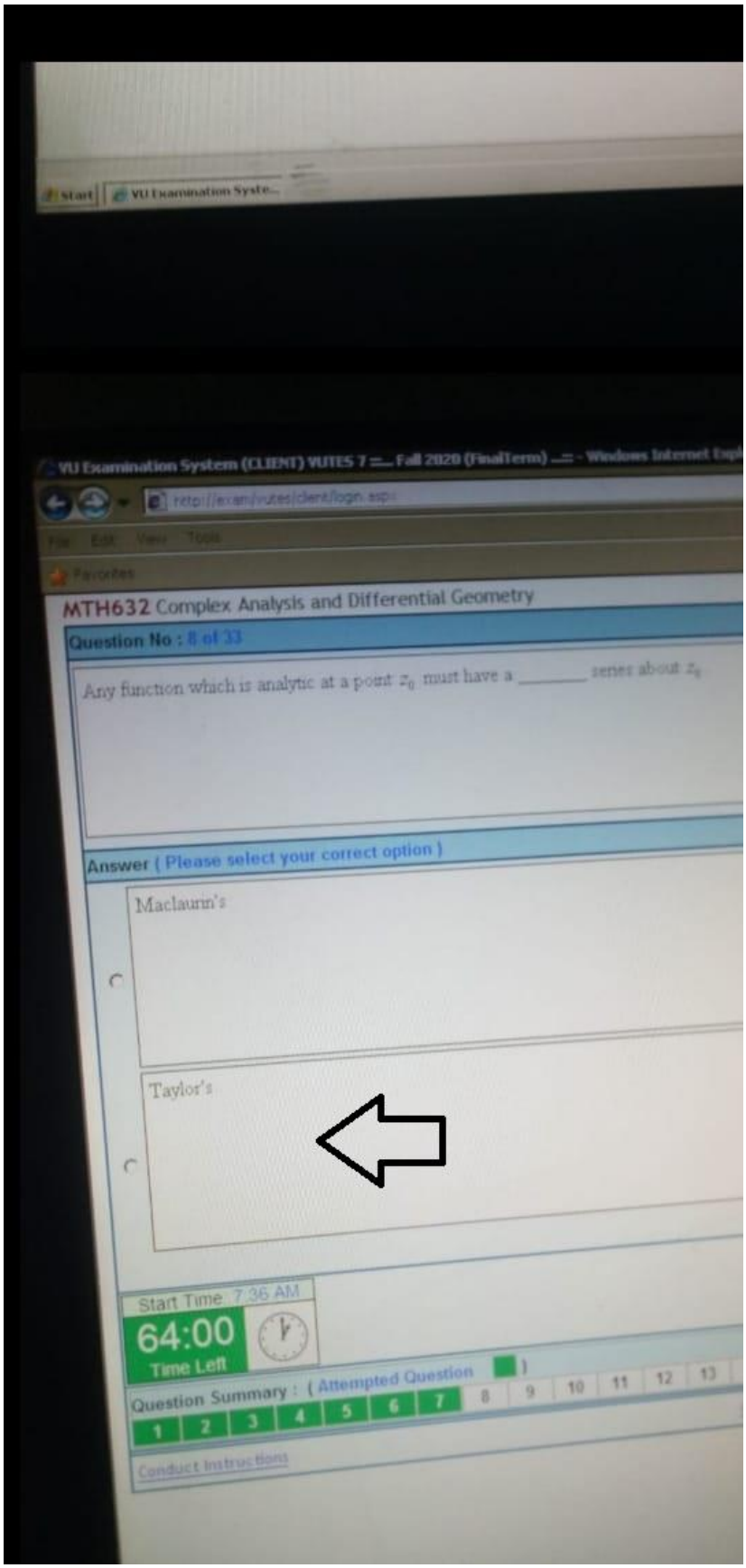

Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = \ln r + i\theta$
Start Time: 7:36 AM
78:00
Time Left
Question Summary (Attempted Question)
MTH632 Complex Analysis and Differential Geometry

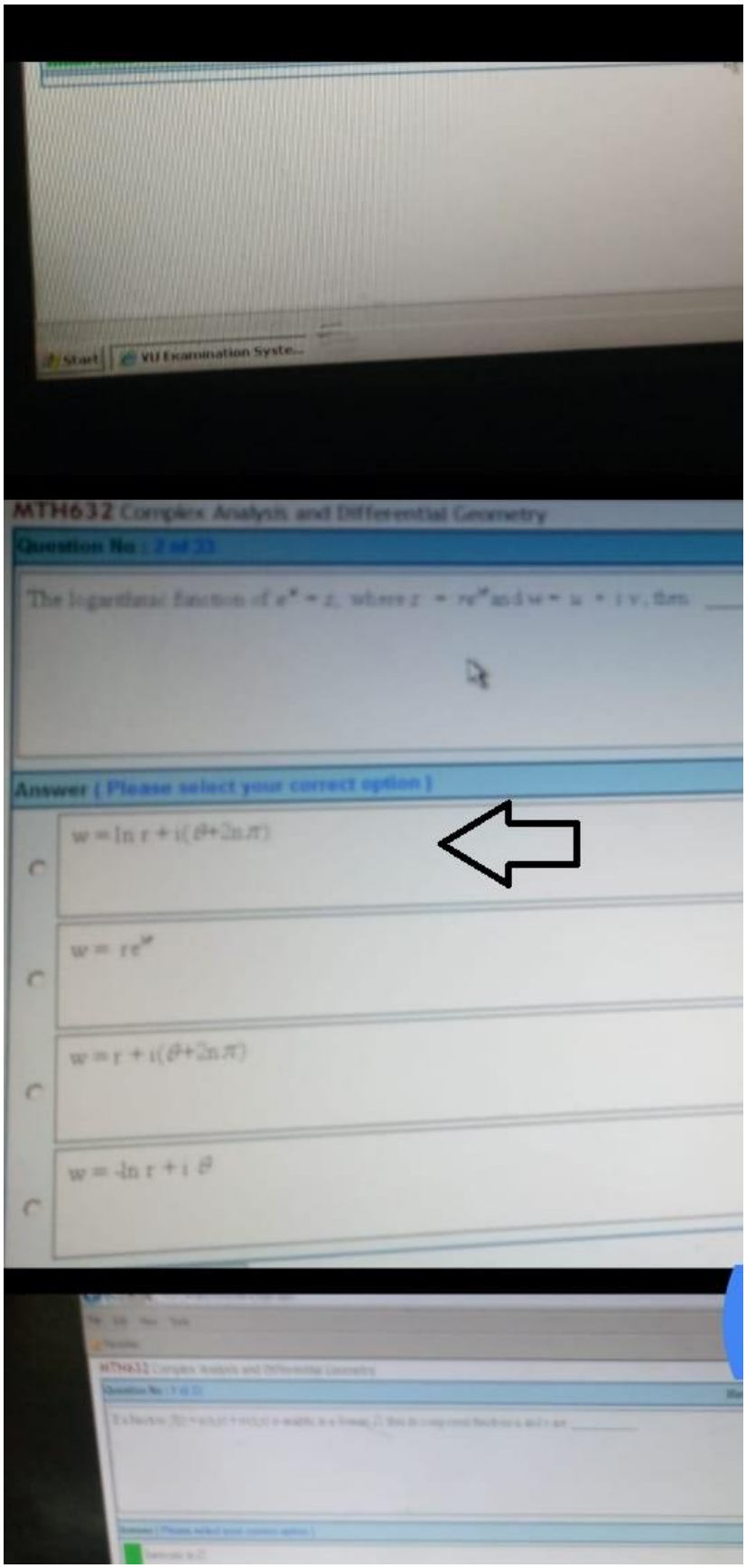

Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = \ln r + i\theta$
Start Time 7:36 AM
78:00
Time Left
Question Summary (Attempted Question)
MTH632 Complex Analysis and Differential Geometry

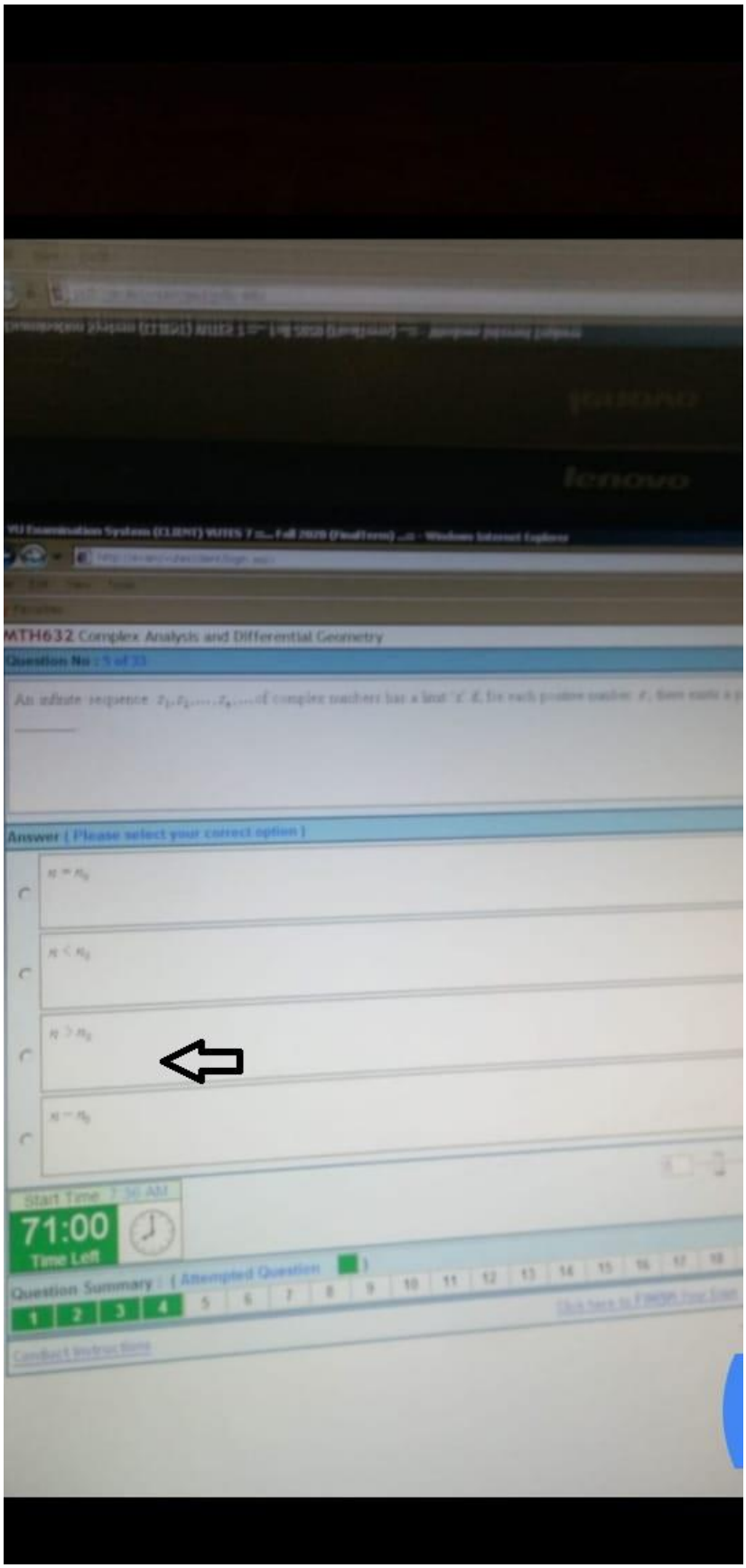

MTH632 Complex Analysis and Differential Geometry
Question No :
Answer (Please select your correct option)
$U_{xx} + U_{yy} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time: 7:36 AM
67:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10
Conduct Instructions

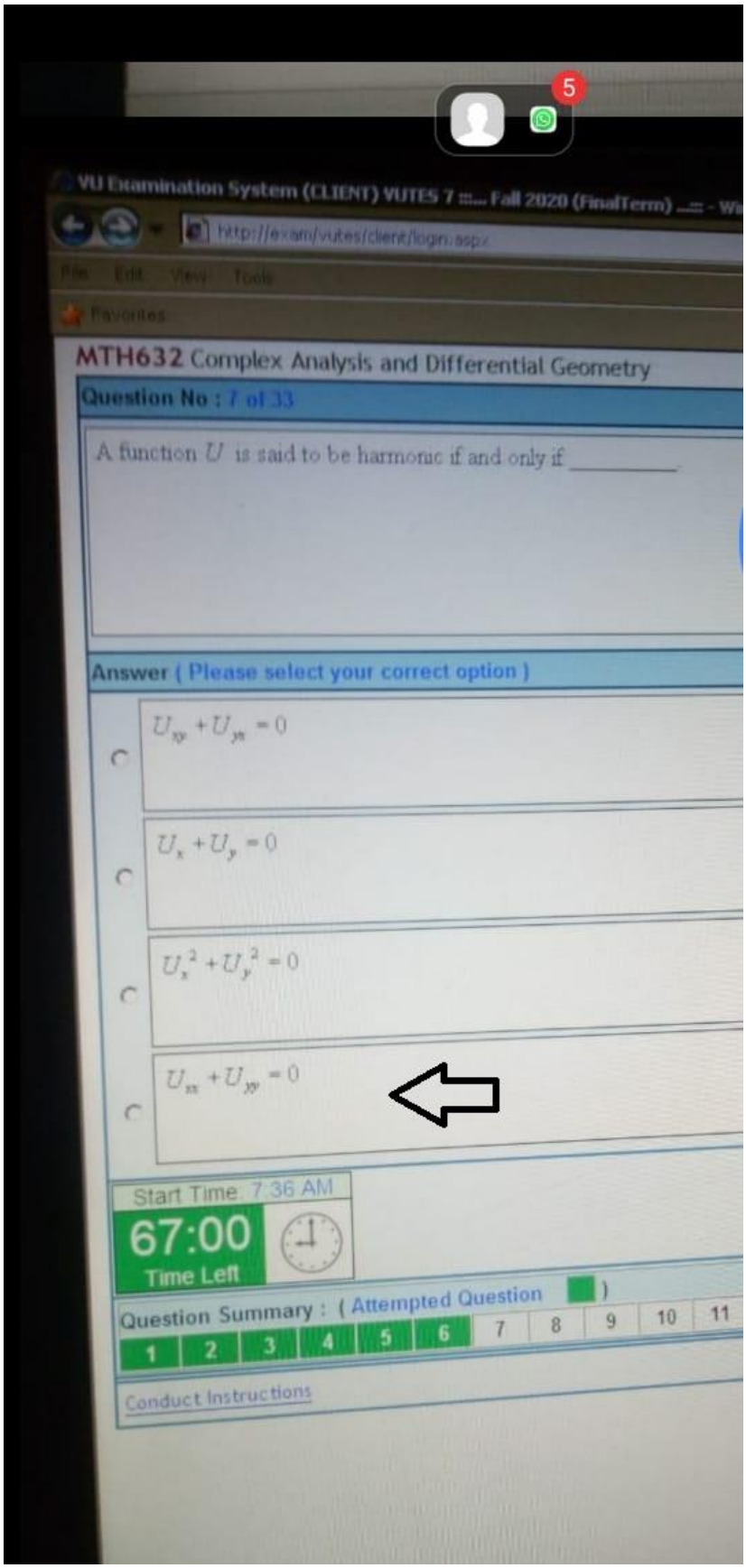

VU Examination System (CLIENT) VUTES 7 ... Fall 2020 (FinalTerm) ... - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 21 of 33
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$
Answer (Please select your correct option)
$-\vec{b}$
$\vec{b}$
$\vec{t}$
$\vec{n}$
Start Time: 7:36 AM
33:00
Time Left
21
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
Click here to FINISH Your Exam
Start
VU Examination Syste...

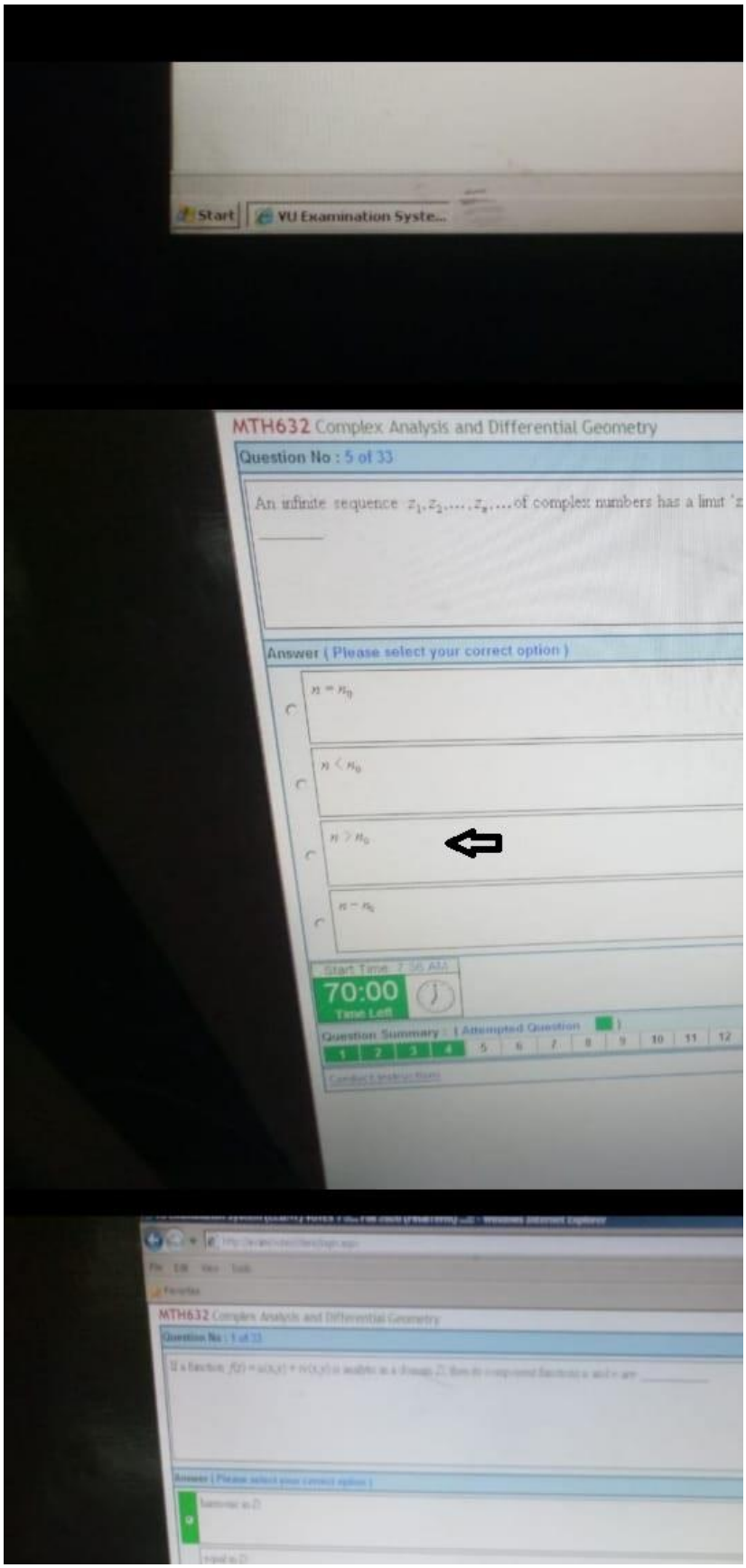

Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 22 of 33
Answer (Please select your correct option)
Start Time: 7:36 AM
32:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions

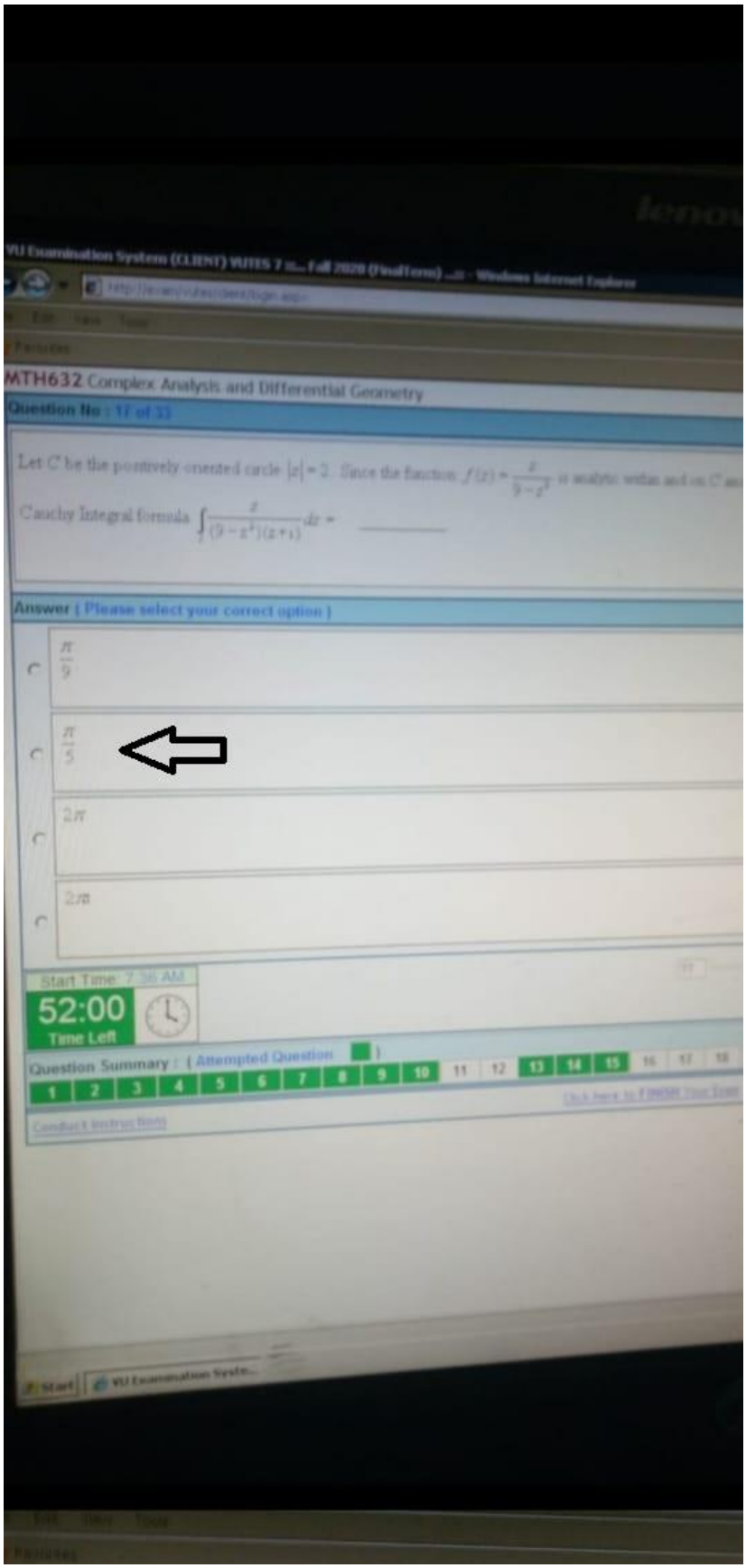

VU Examination System (CLIENT) VUTES 7 — Fall 2020 (FinalTerm)
MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
Any function which is analytic at a point z_0 must have a _____ series about z_0
Answer (Please select your correct option)
Maclaurin's
Taylor's
Start Time 7:36 AM
64:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

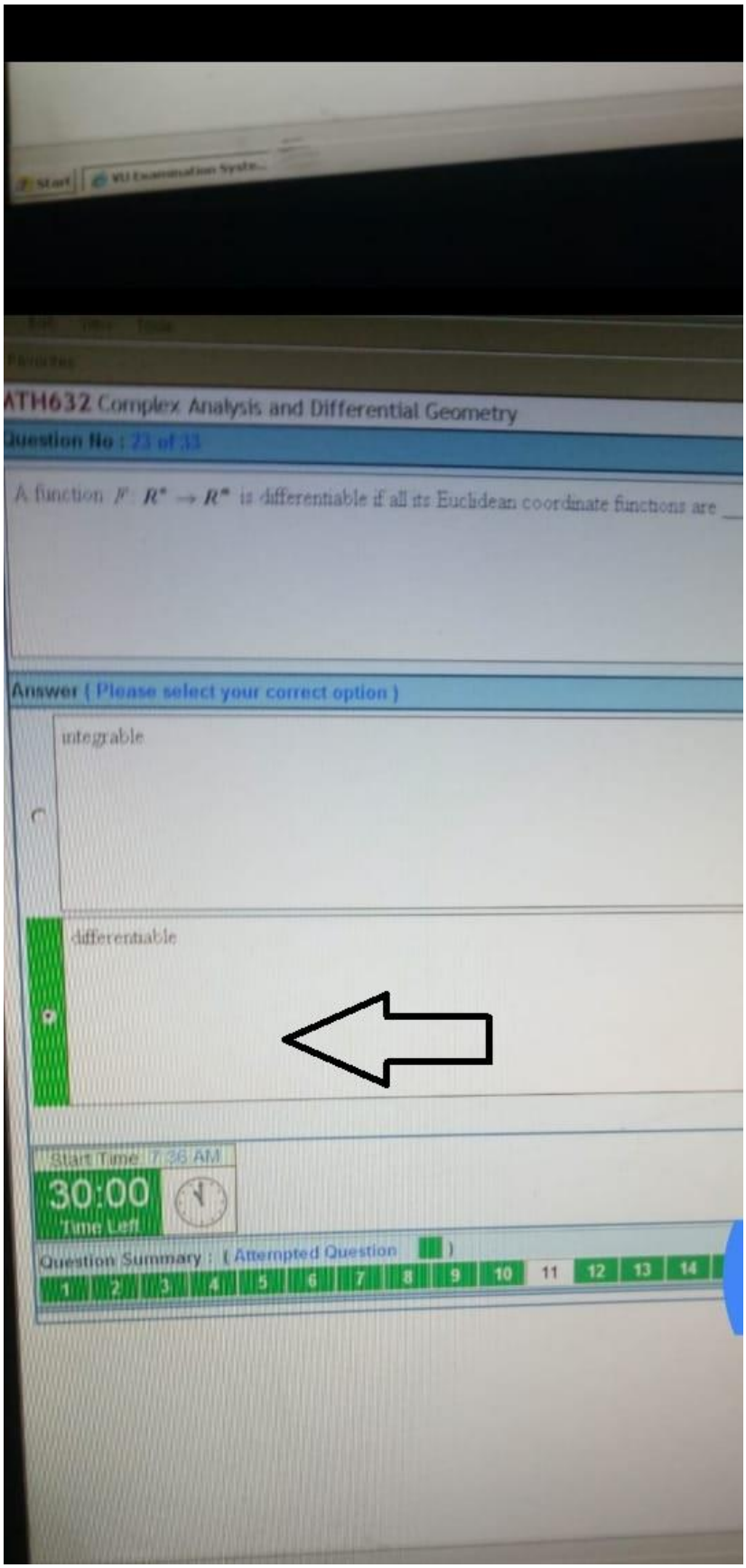

Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 23
The logarithmic function of $e^z = z$, where $z = re^{\theta}$ and $w = u + iv$, then
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\,\theta$

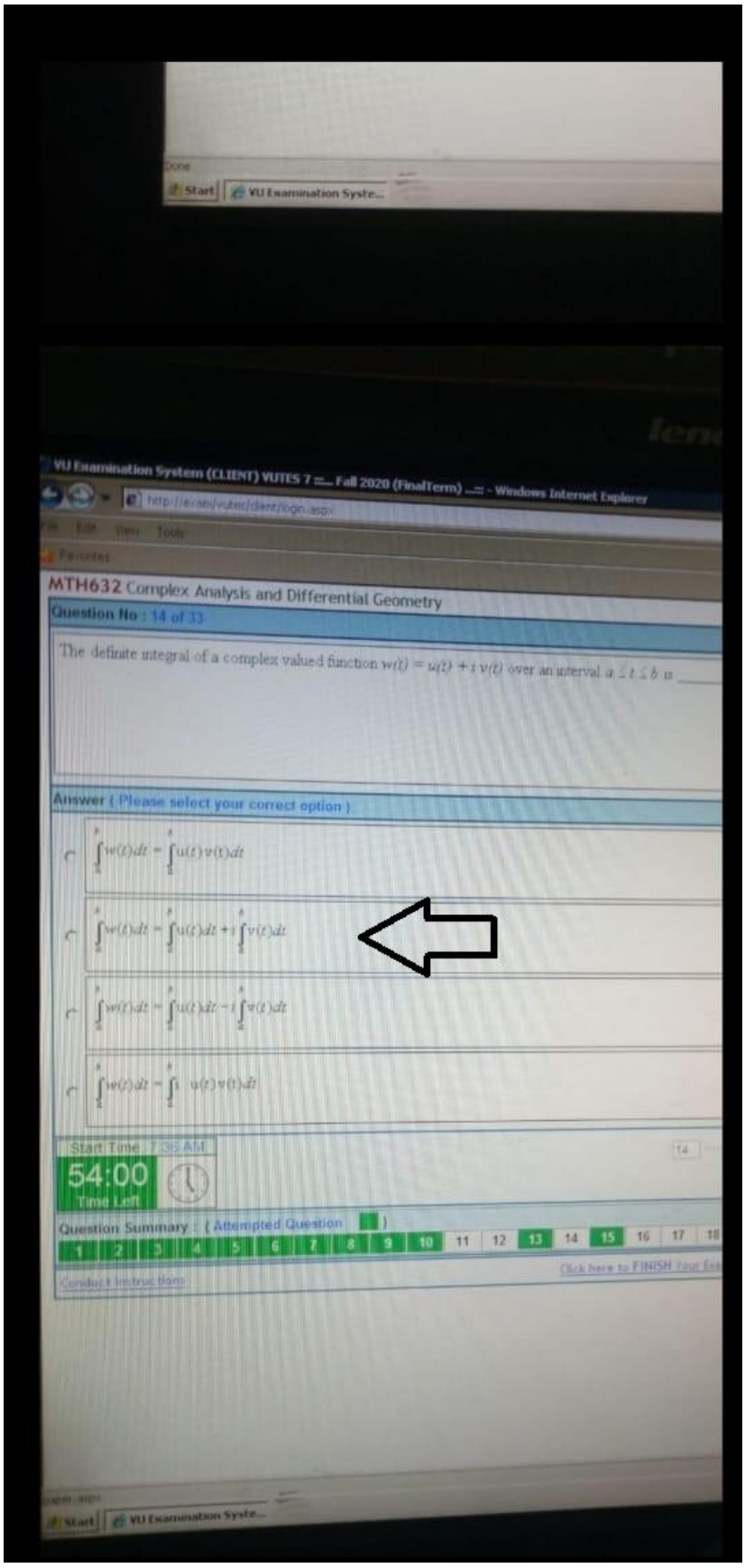

MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
Answer [Please select your correct option]
Start Time
71:00
Time Left
Question Summary : [Attempted Question]
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

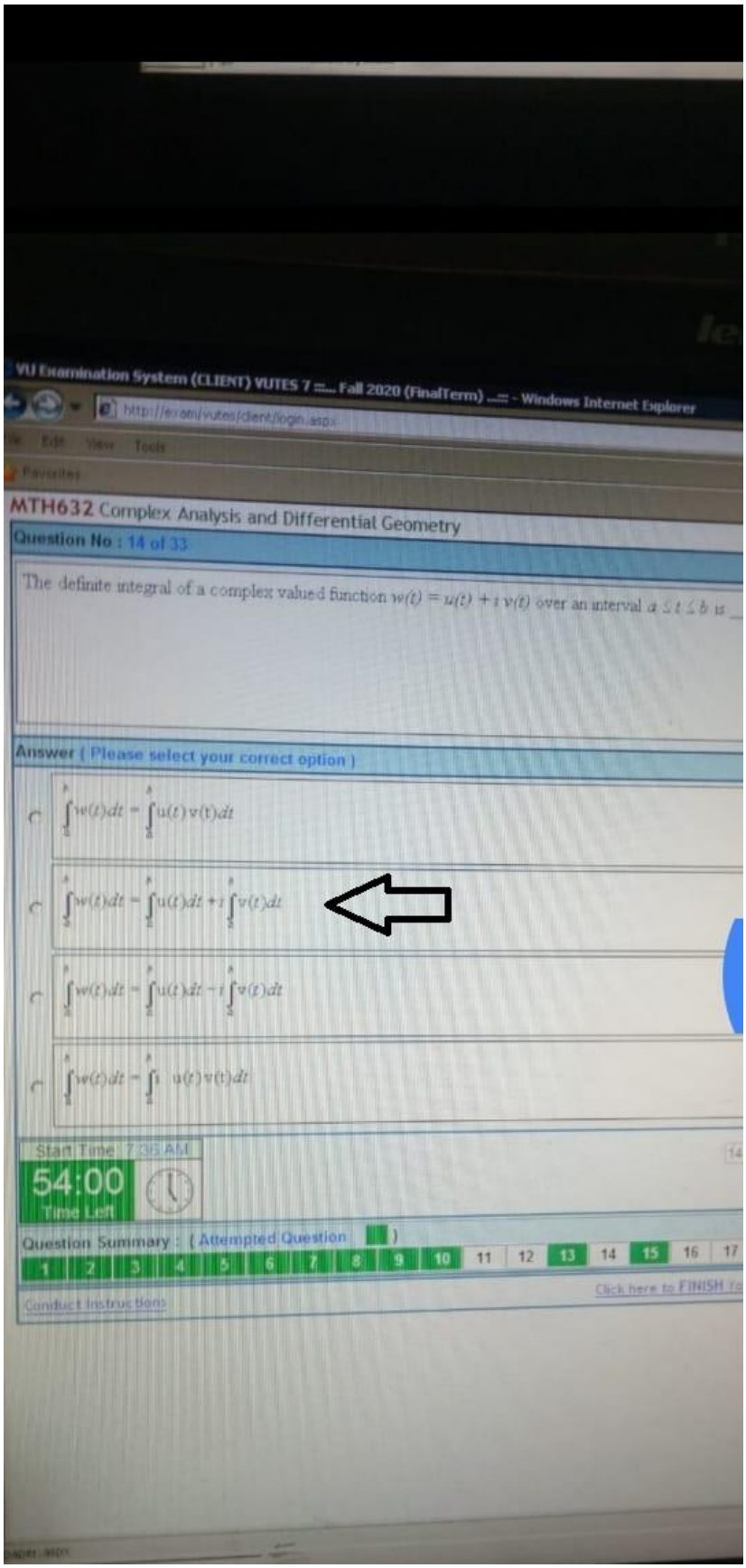

VU Examination System (CLIENT) VUTES 7 ::... Fall 2020 (FinalTerm) ...:: - Wi
http://exam/vutes/client/login.aspx
File Edit View Tools
Favorites
MTH632 Complex Analysis and Differential Geometry
Question No : 7 of 33
A function U is said to be harmonic if and only if _____.
Answer (Please select your correct option)
$U_{xy} + U_{yx} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Start Time 7:36 AM
67:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

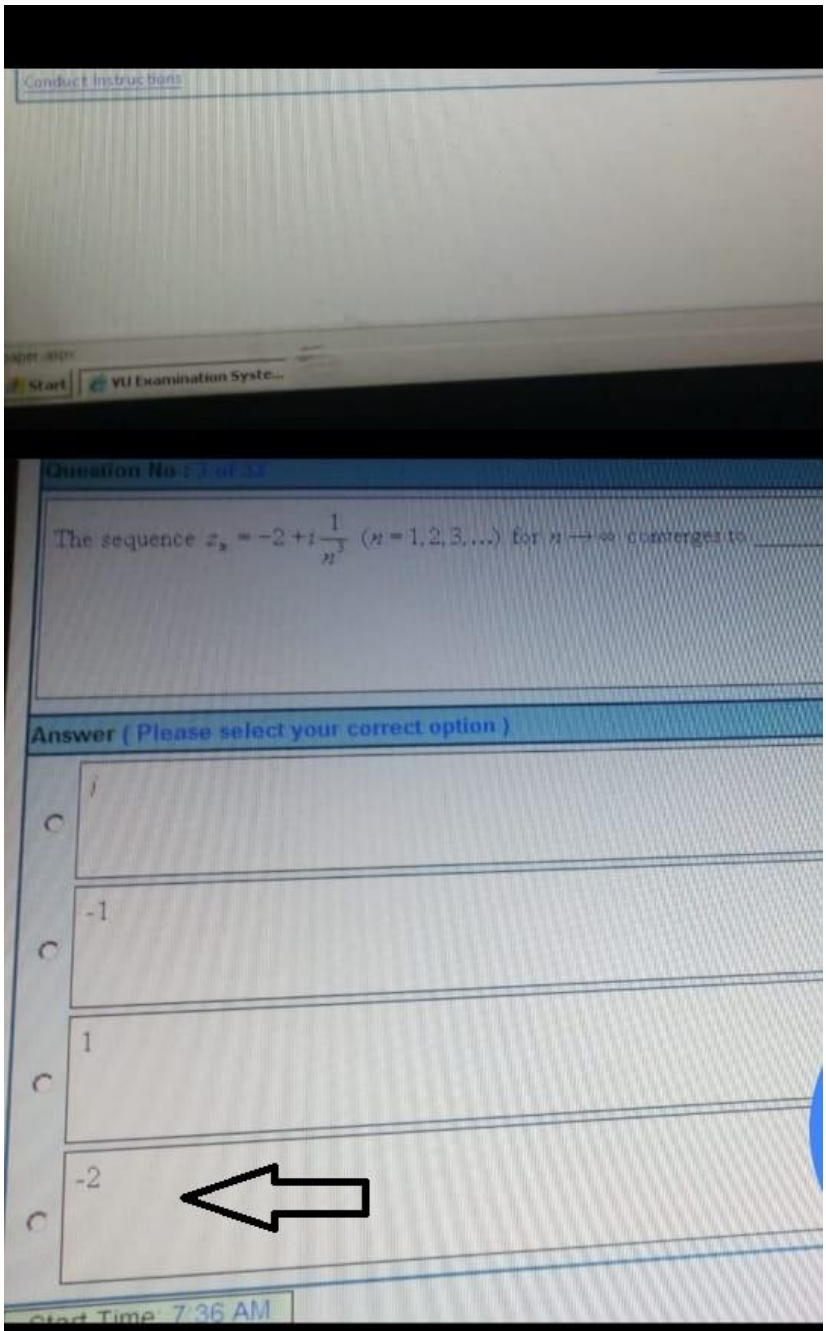

Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
An infinite sequence $z_1, z_2, \ldots, z_n, \ldots$ of complex numbers has a limit
Answer (Please select your correct option)
$n = n_0$
$n < n_0$
$n > n_0$
$n = n_0$
70:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
MTH632 Complex Analysis and Differential Geometry

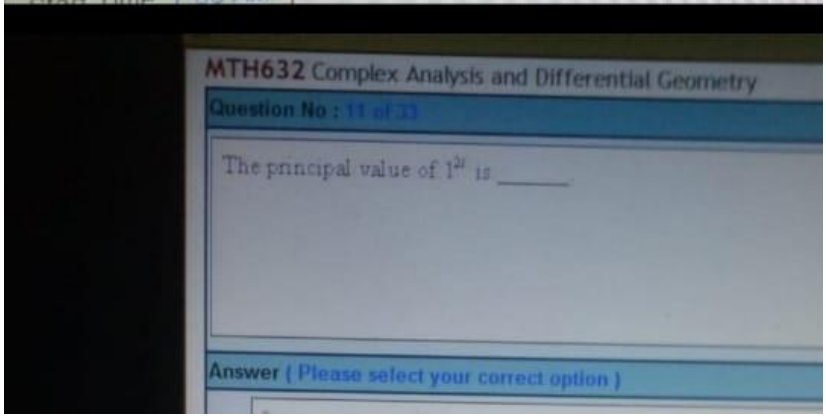

VU Examination System (CLIENT) VUTES 7 ... Fall 2020 (FinalTerm) ... - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
Let C be the positively oriented circle |z| = 2. Since the function f(z) = z/(9 − z²) is analytic within and on C and
Cauchy Integral formula ∫_C z/((9 − z²)(z + 1)) dz = _____
Answer (Please select your correct option)
π/9
π/5
2π
2πi
Start Time: 7:36 AM
52:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Click here to FINISH Your Exam
Conduct Instructions
Start
VU Examination Syste...

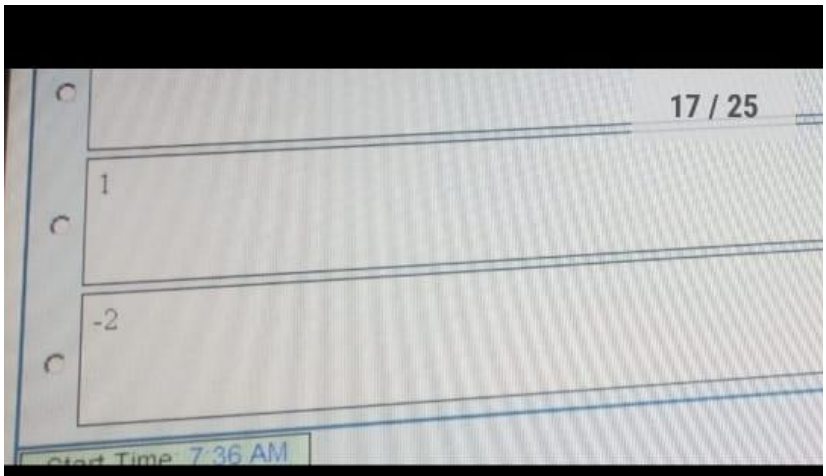

ATH632 Complex Analysis and Differential Geometry
Question No : 23 of 33
A function $F: R^n \to R^m$ is differentiable if all its Euclidean coordinate functions are
Answer (Please select your correct option)
integrable
differentiable
Start Time: 7:36 AM
30:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14

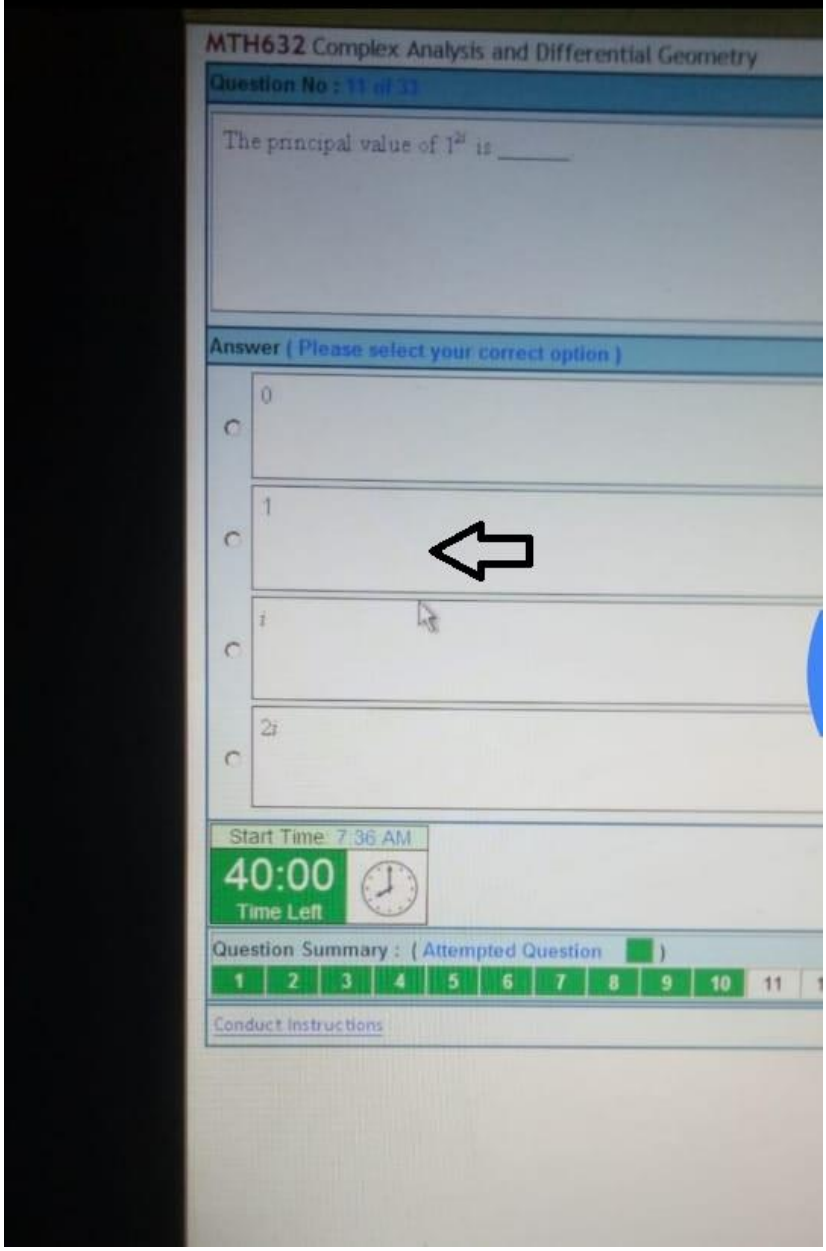

Done
Start
VU Examination Syste...
VU Examination System (CLIENT) VUTES 7 ... Fall 2020 (FinalTerm) ... - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is
Answer (Please select your correct option)
$\int_a^b w(t)\,dt = \int_a^b u(t)\,v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,dt - i\int_a^b v(t)\,dt$
$\int_a^b w(t)\,dt = \int_a^b u(t)\,v(t)\,dt$
Start Time : 9:35 AM
54:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
Click here to FINISH Your Exa
Start
VU Examination Syste...

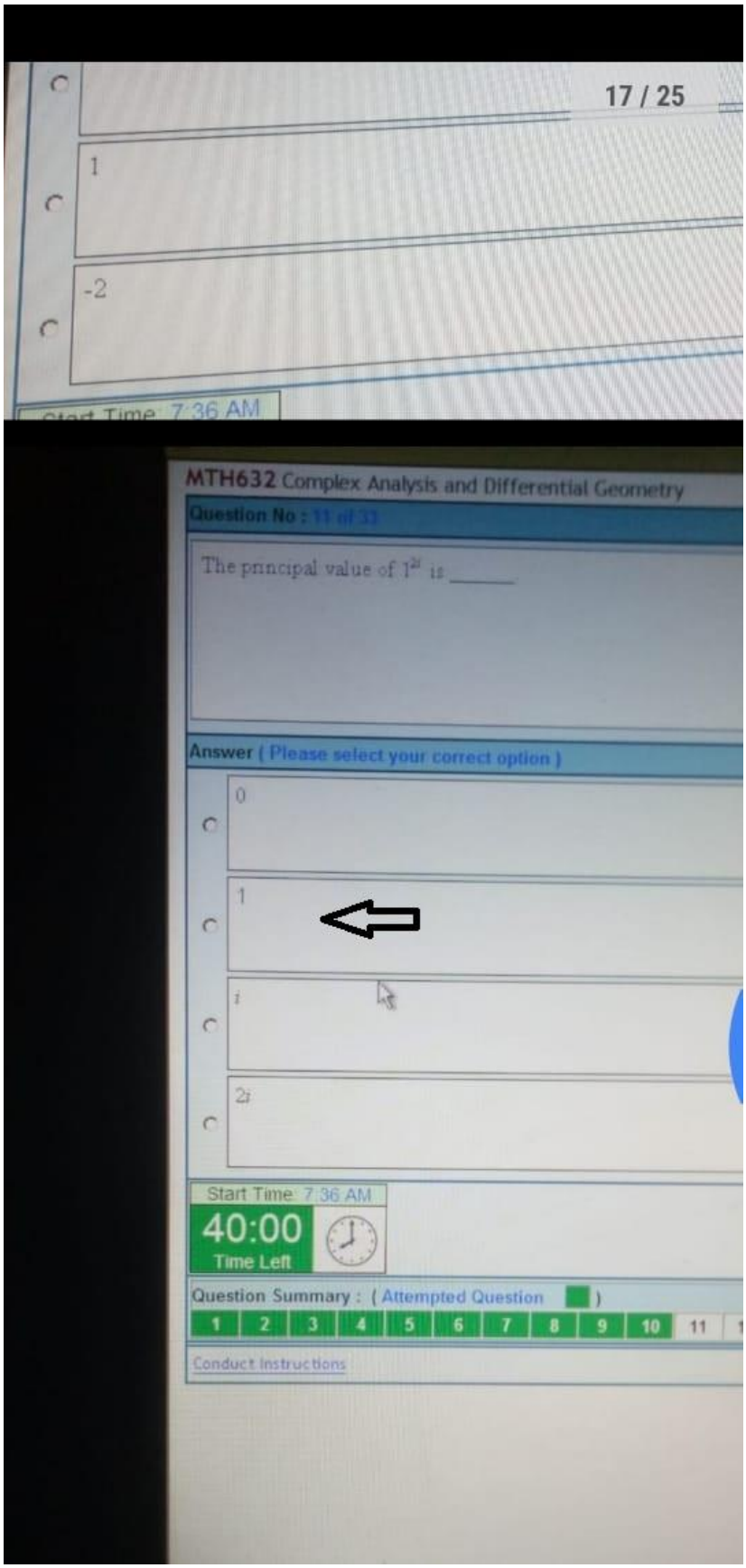

VU Examination System (CLIENT) VUTES 7 ::... Fall 2020 (FinalTerm) ...:: - Windows Internet Explorer
MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
Start Time 7:35 AM
54:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Conduct Instructions
Click here to FINISH

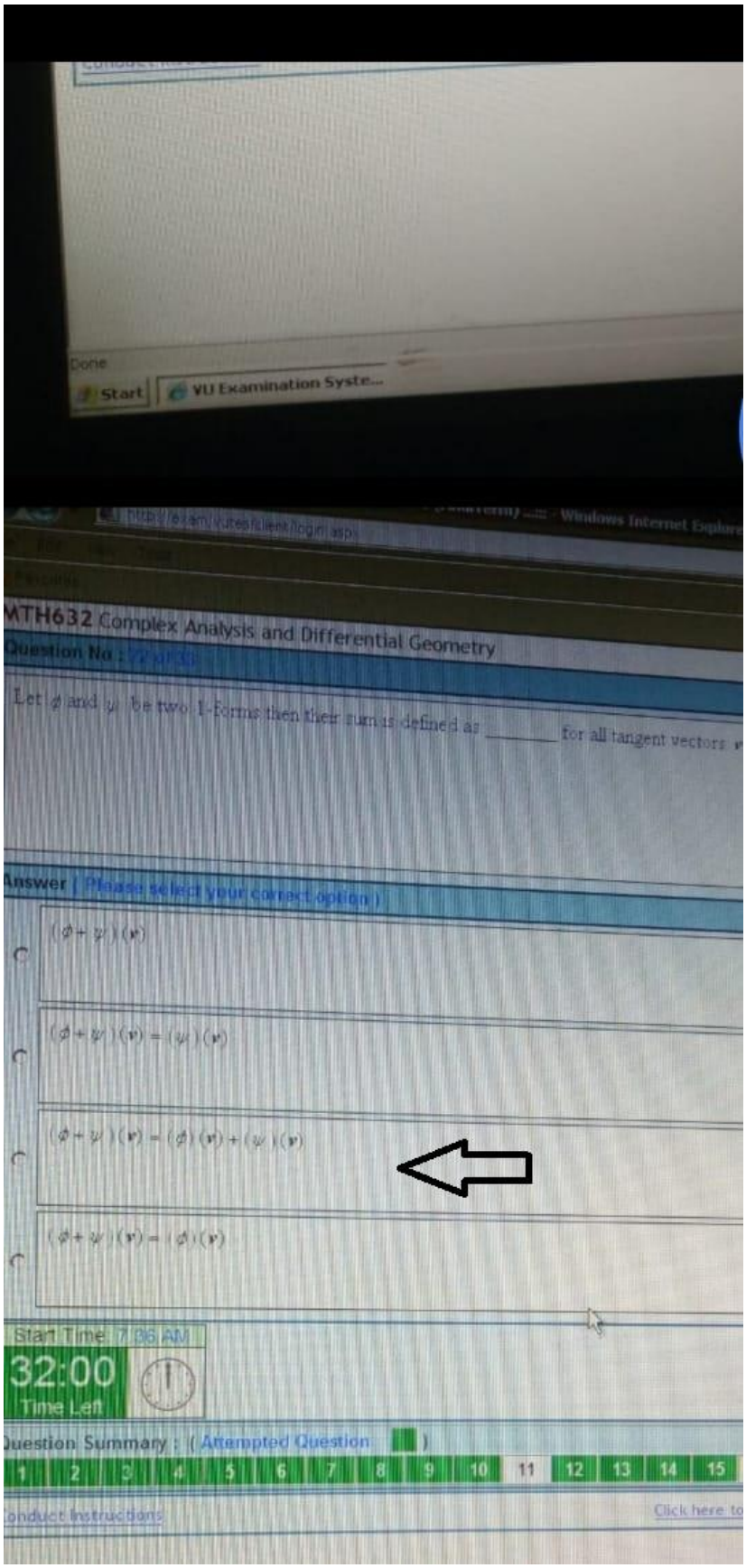

Conduct Instructions
Start
VU Examination Syste...
Question No : 3 of 33
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 7:36 AM

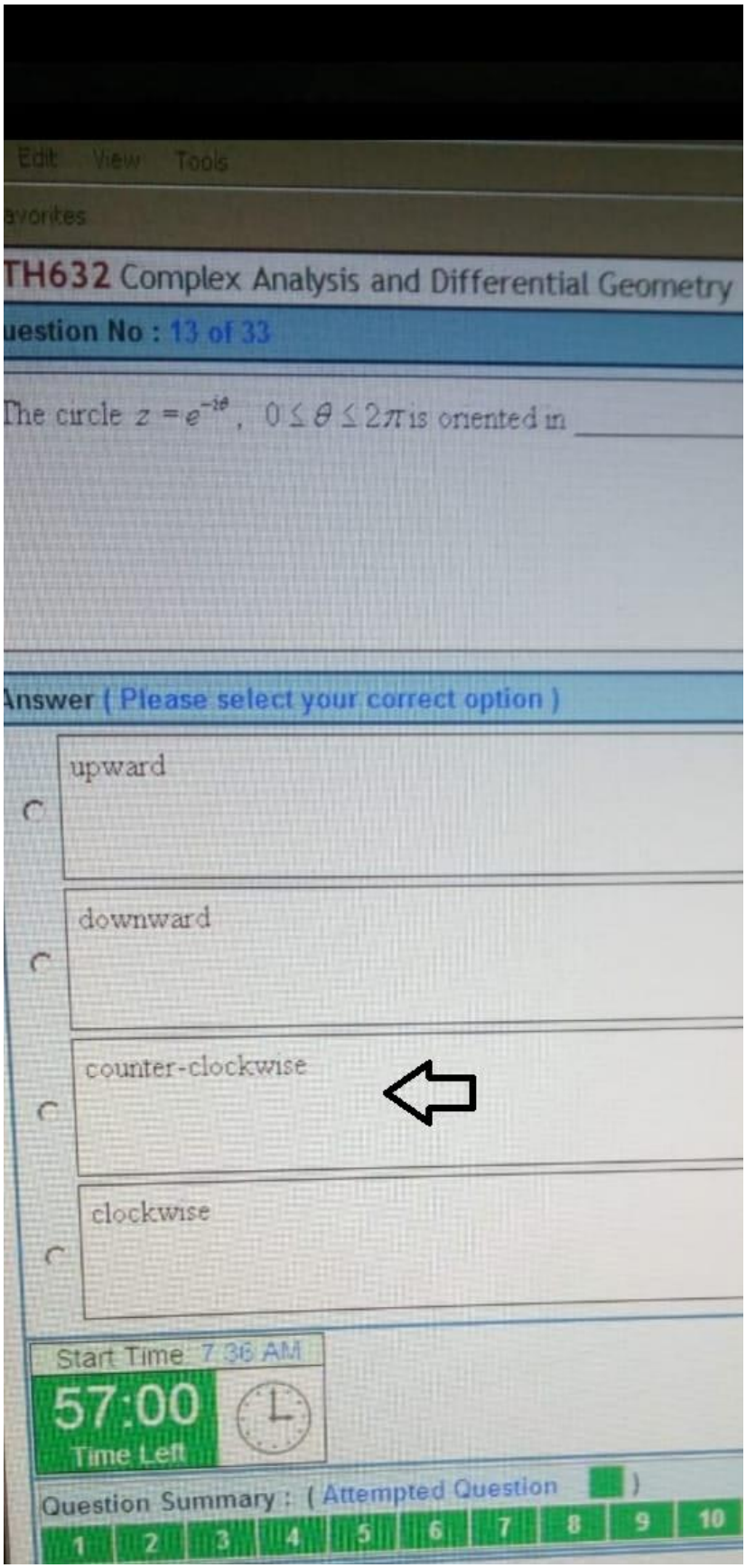

MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)

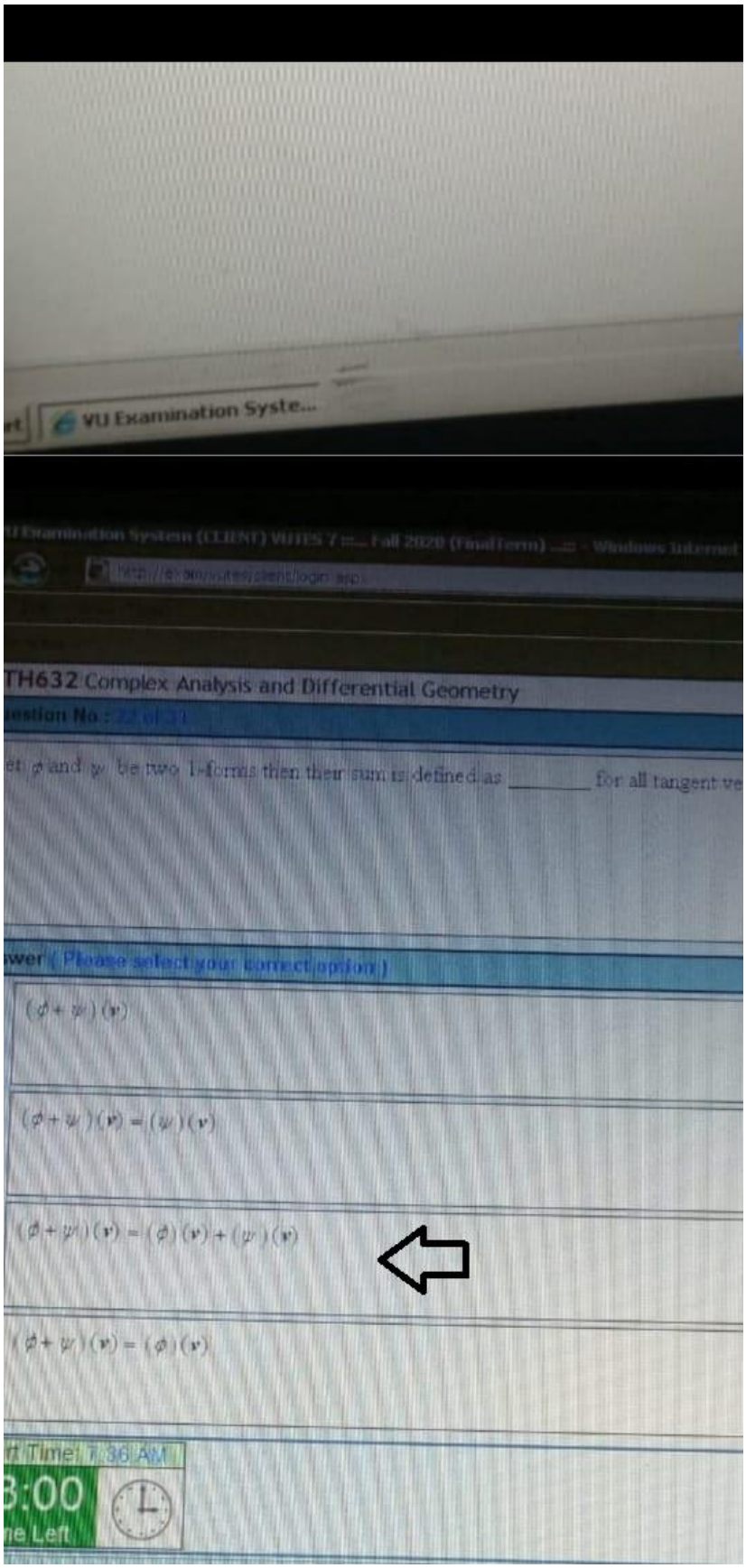

17 / 25
1
-2
Start Time: 7:36 AM

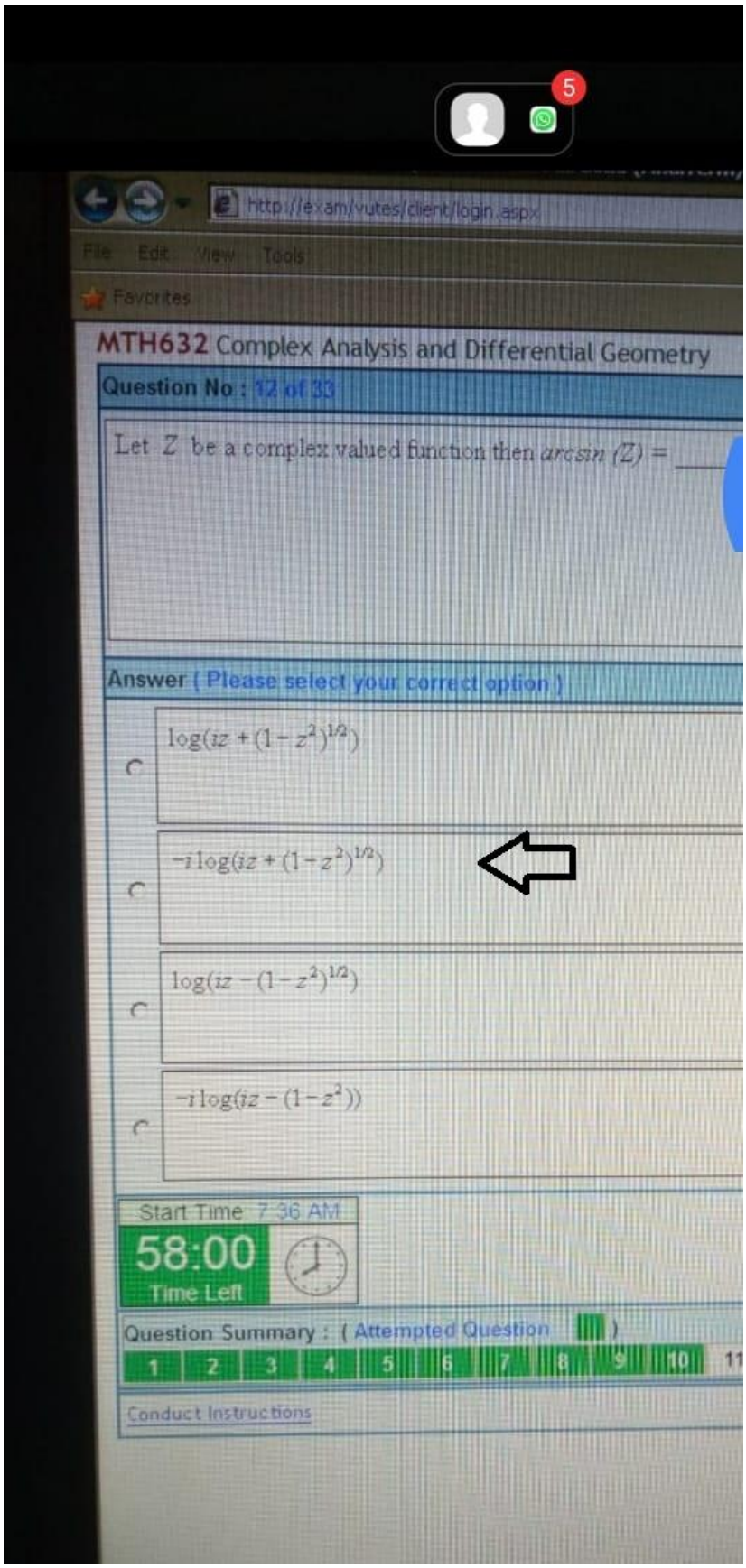

MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)
0
1
i
2i
Start Time: 7:36 AM
40:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

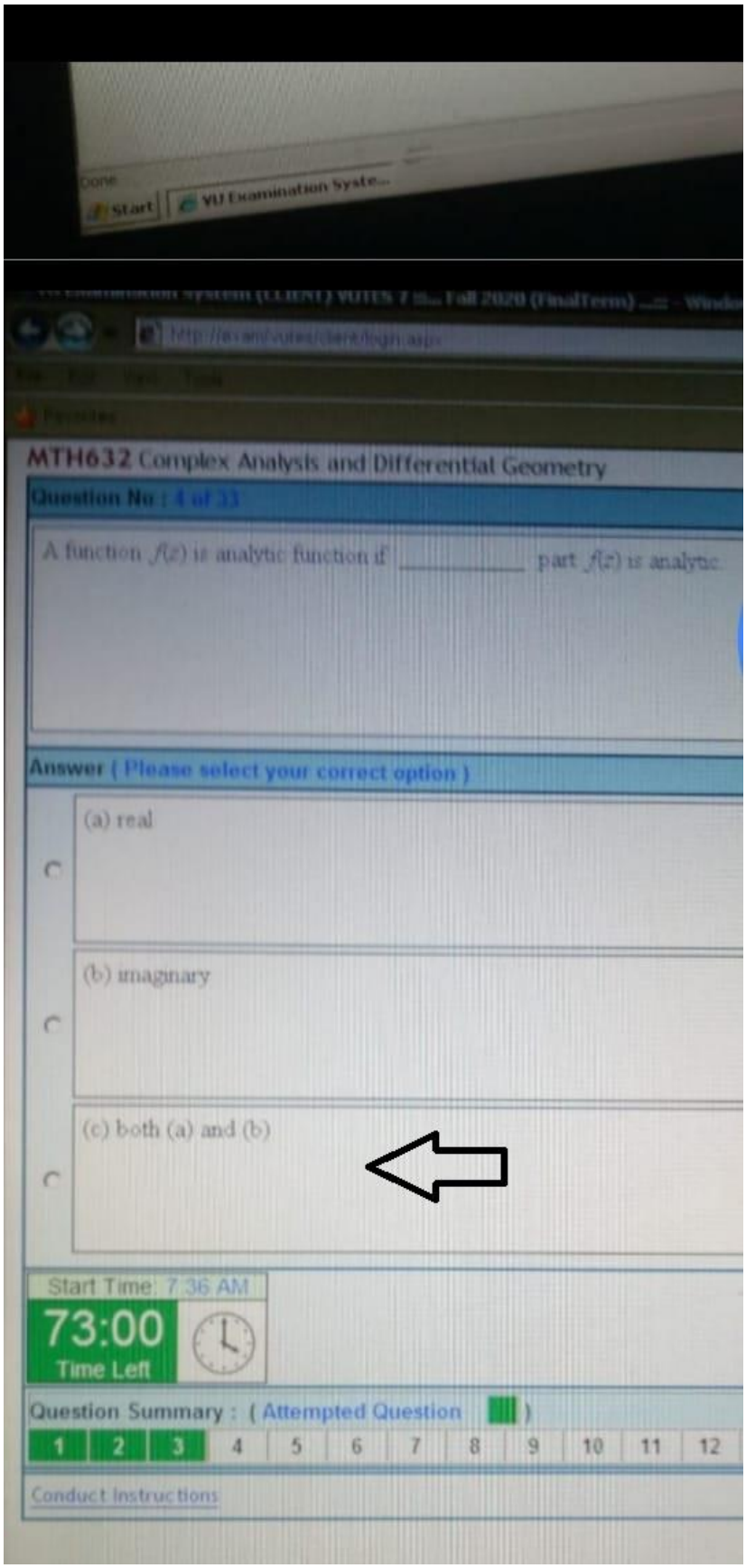

17 / 25
1
-2
Start Time: 7:36 AM
MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____
Answer (Please select your correct option)
0
1
i
2i
Start Time: 7:36 AM
40:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

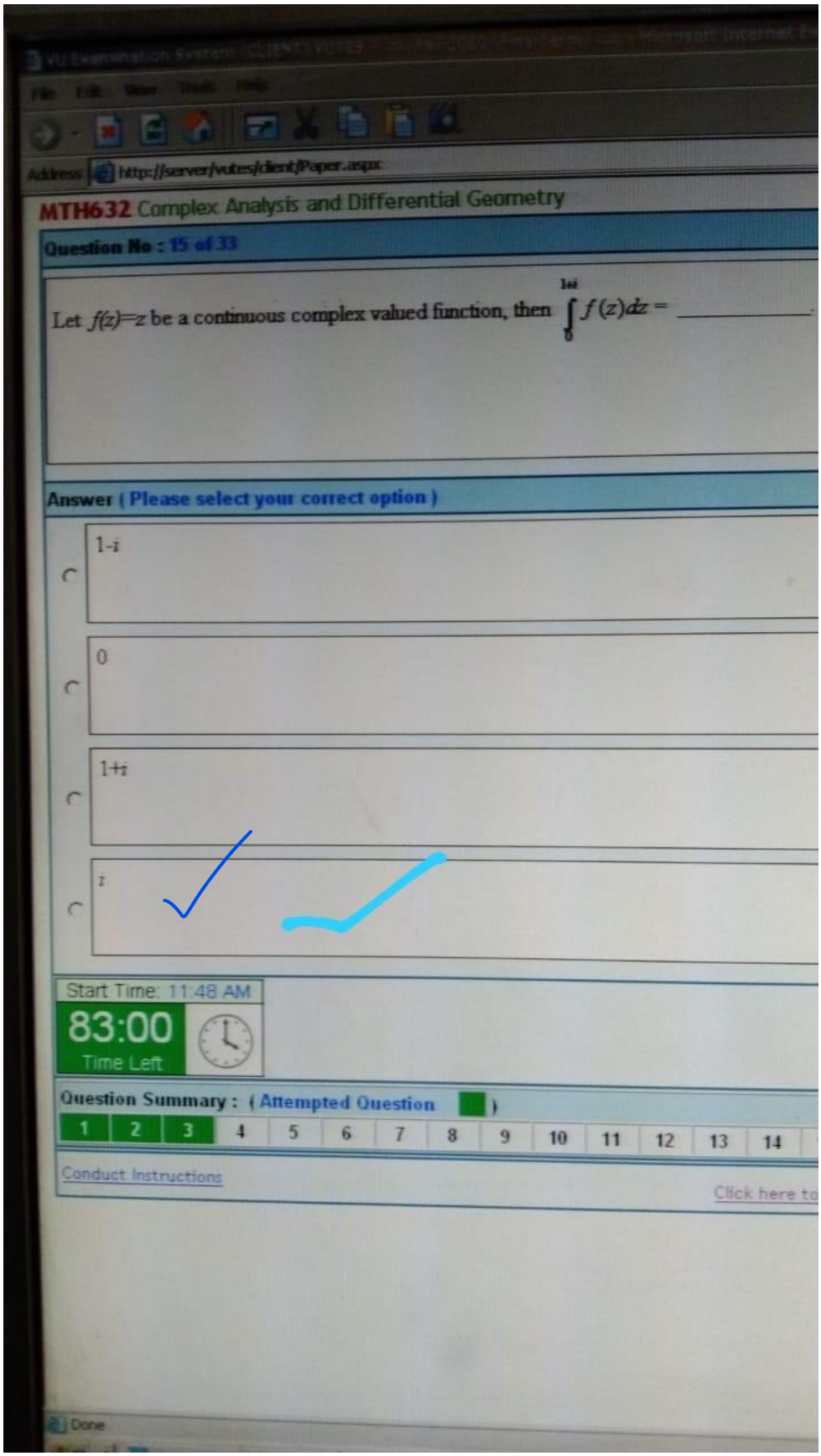

Done
Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No :
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 7:06 AM
32:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to

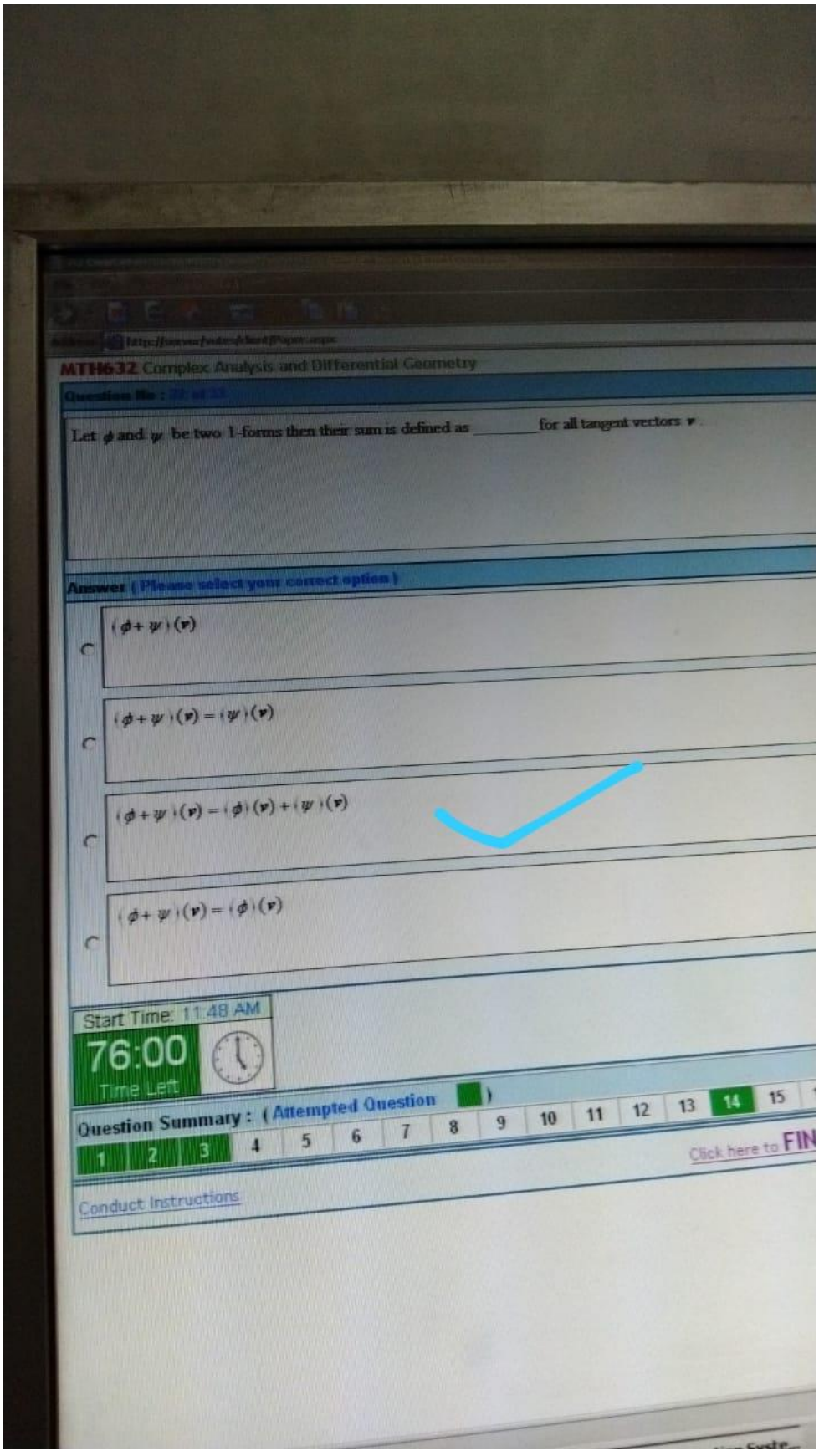

Edit View Tools
avorites
TH632 Complex Analysis and Differential Geometry
uestion No : 13 of 33
The circle $z = e^{-i\theta}$, $0 \le \theta \le 2\pi$ is oriented in _____
Answer (Please select your correct option)
upward
downward
counter-clockwise
clockwise
Start Time: 7:36 AM
57:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10

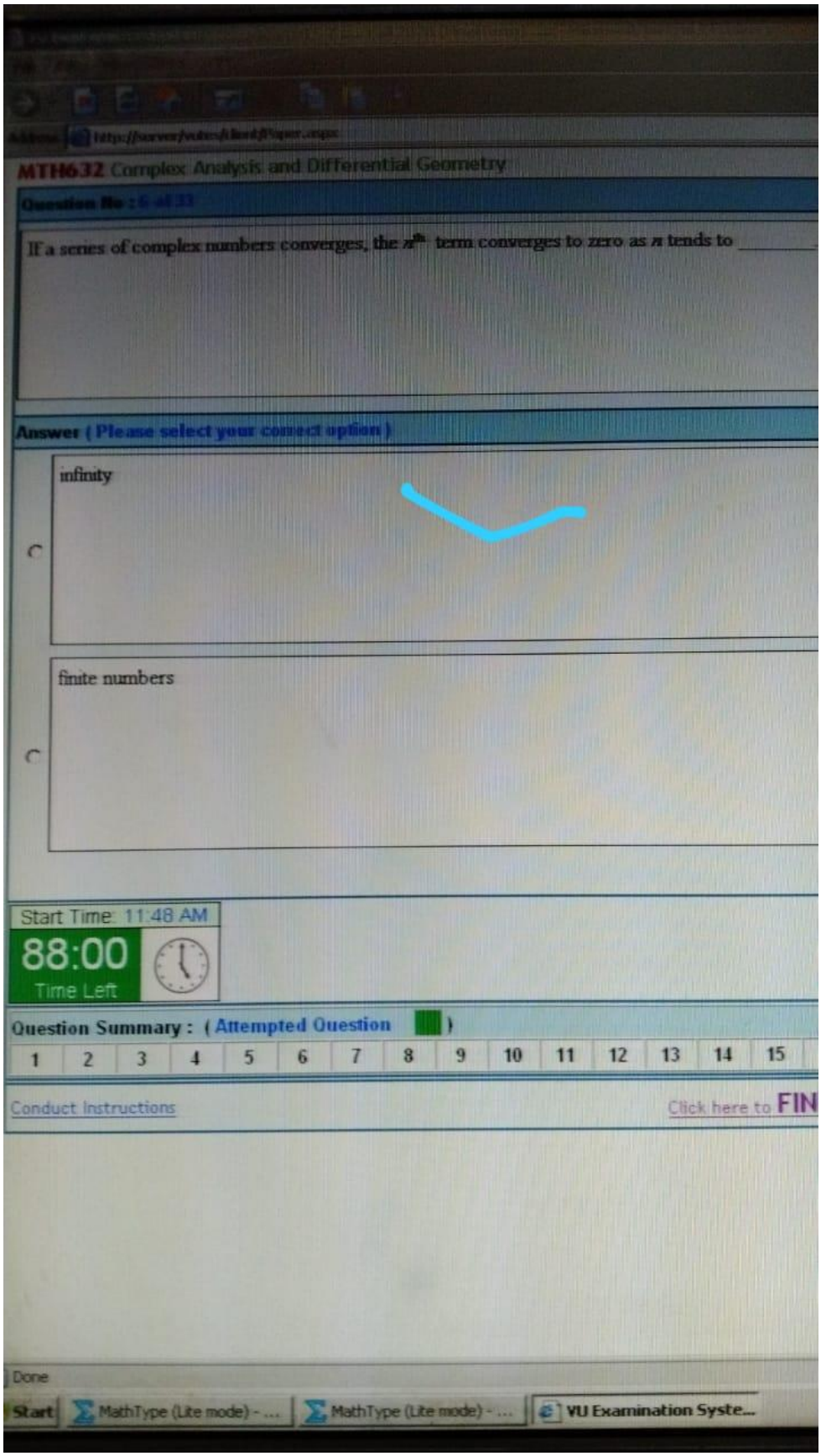

VU Examination Syste...
TH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent ve
Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Time: 7:36 AM
3:00
ne Left

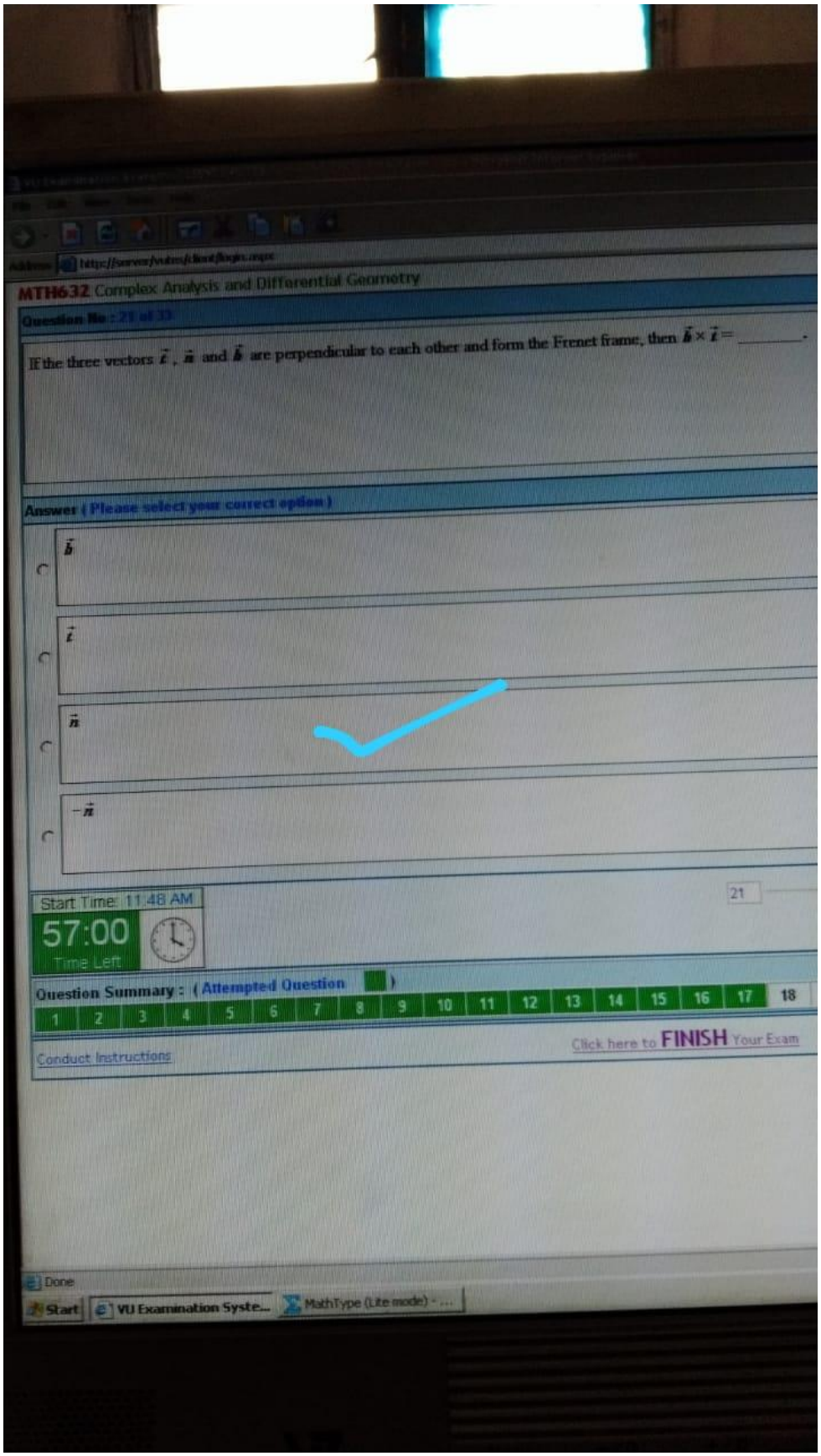

5
http://exam/vutes/client/login.aspx
File Edit View Tools
Favorites
MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = ____
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time 7:36 AM
58:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

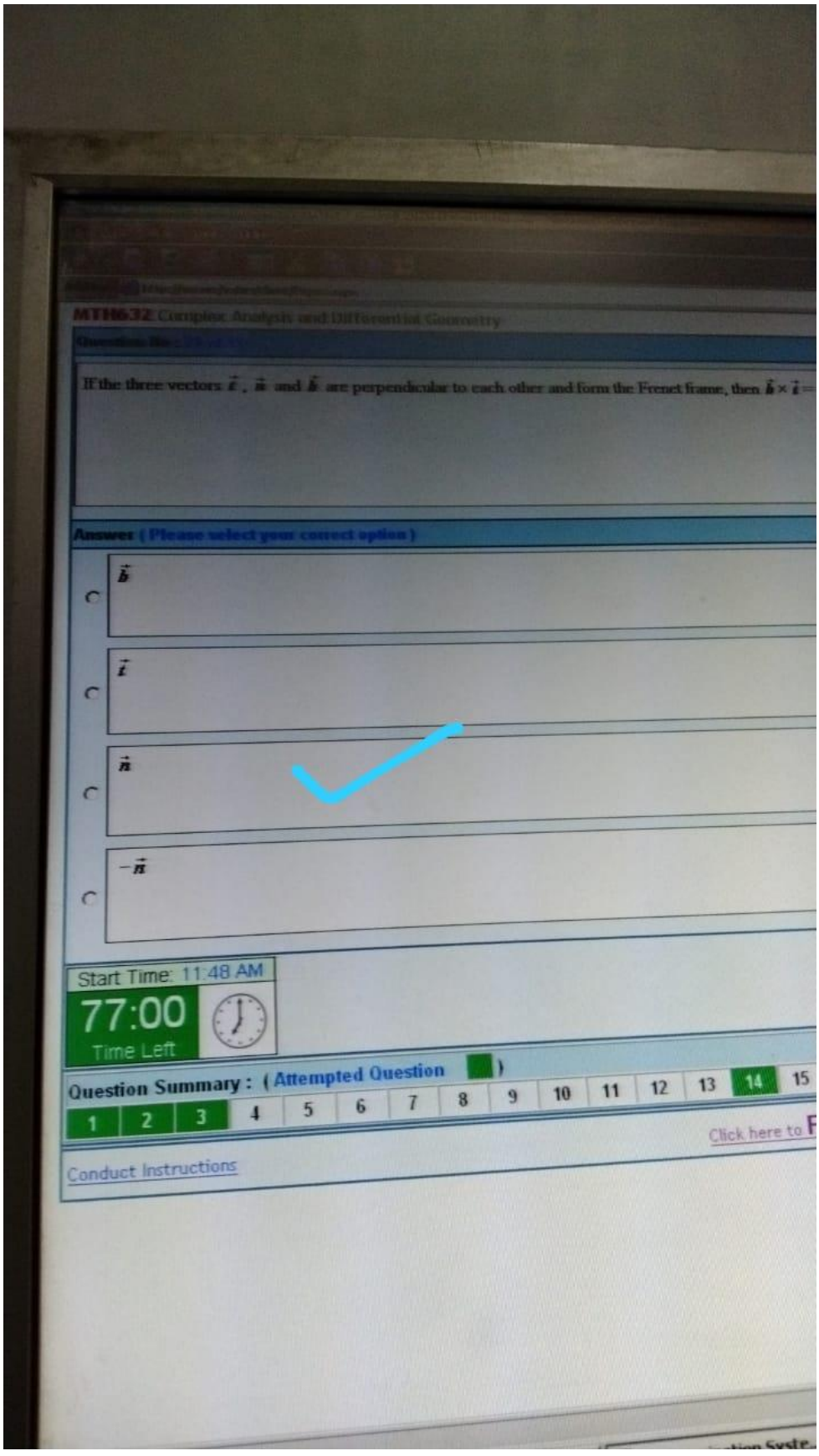

Done
Start
VU Examination Syste...
MTH632 Complex Analysis and Differential Geometry
Question No : 4 of 33
A function f(z) is analytic function if _____ part f(z) is analytic
Answer (Please select your correct option)
(a) real
(b) imaginary
(c) both (a) and (b)
Start Time: 7:36 AM
73:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

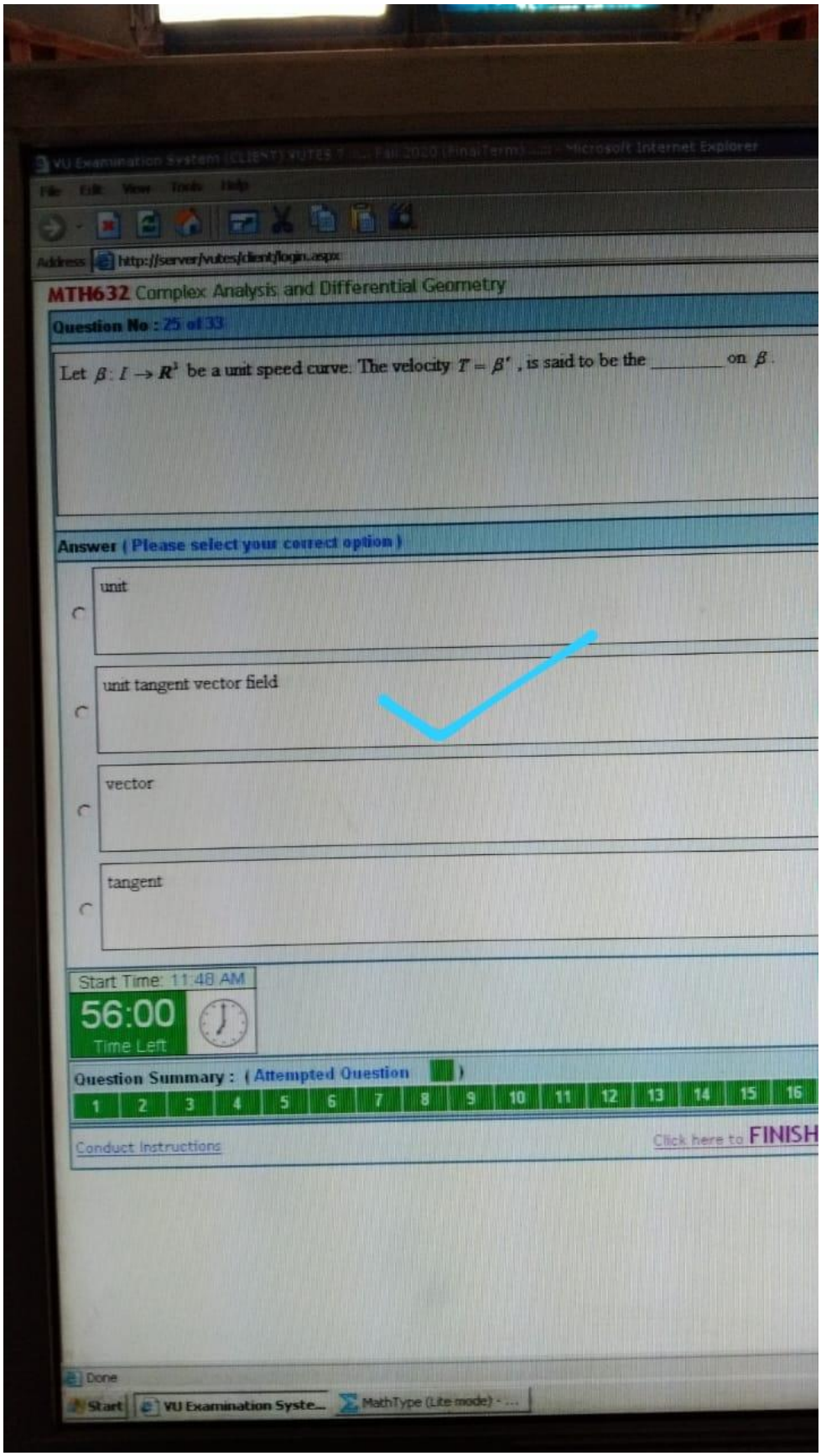

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done


MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to FIN
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
88:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
57:00
Time Left
21
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
Click here to FINISH Your Exam
Done
Start
VU Examination Syste...
MathType (Lite mode) - ...


MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Click here to
Conduct Instructions


Microsoft Internet Explorer
Address http://server/vutes/client/login.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
56:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Conduct Instructions
Click here to FINISH
Done
Start
VU Examination Syste...
MathType (Lite mode) - ...

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

Start Time: 11:48 AM

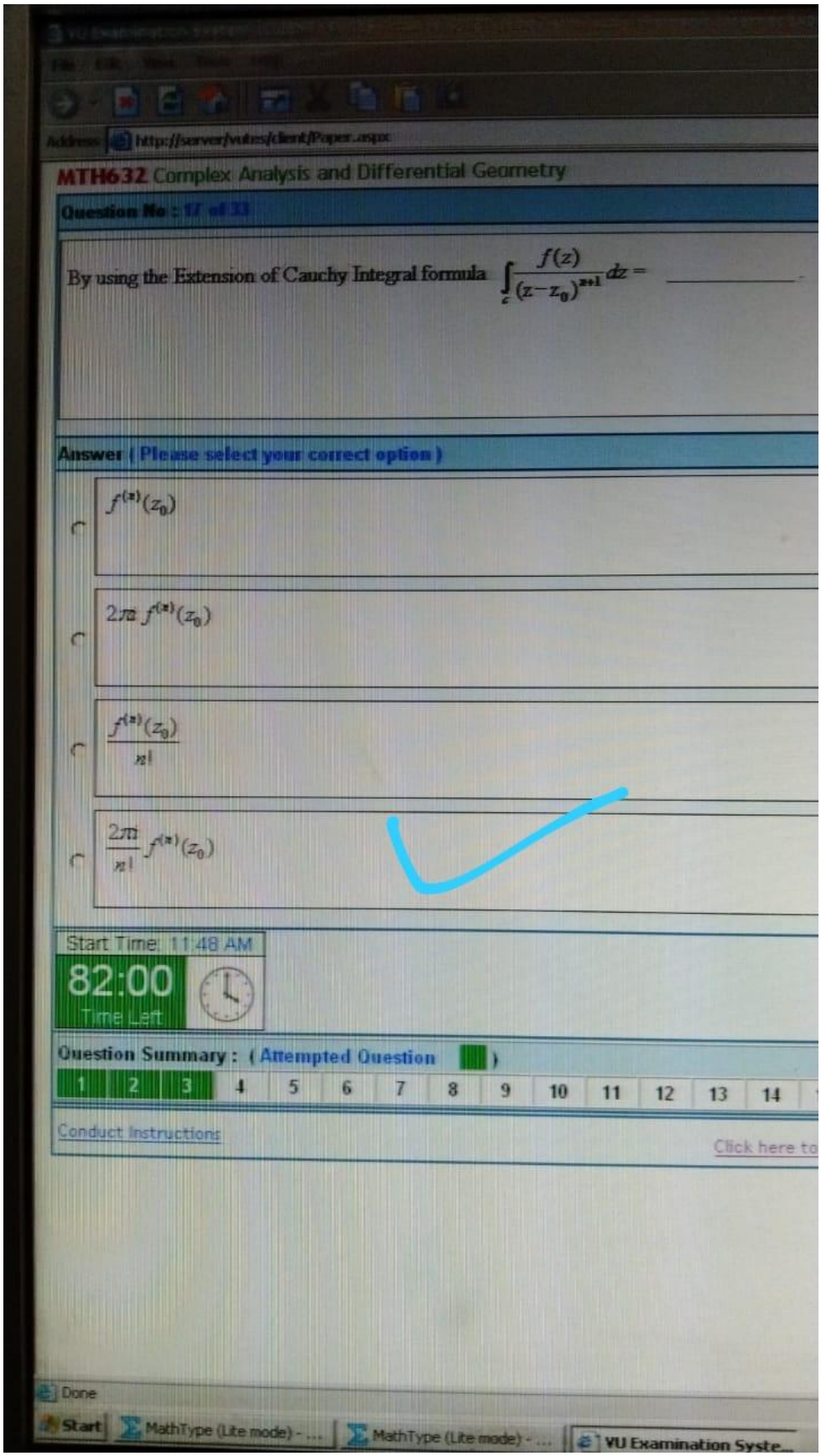
88:00

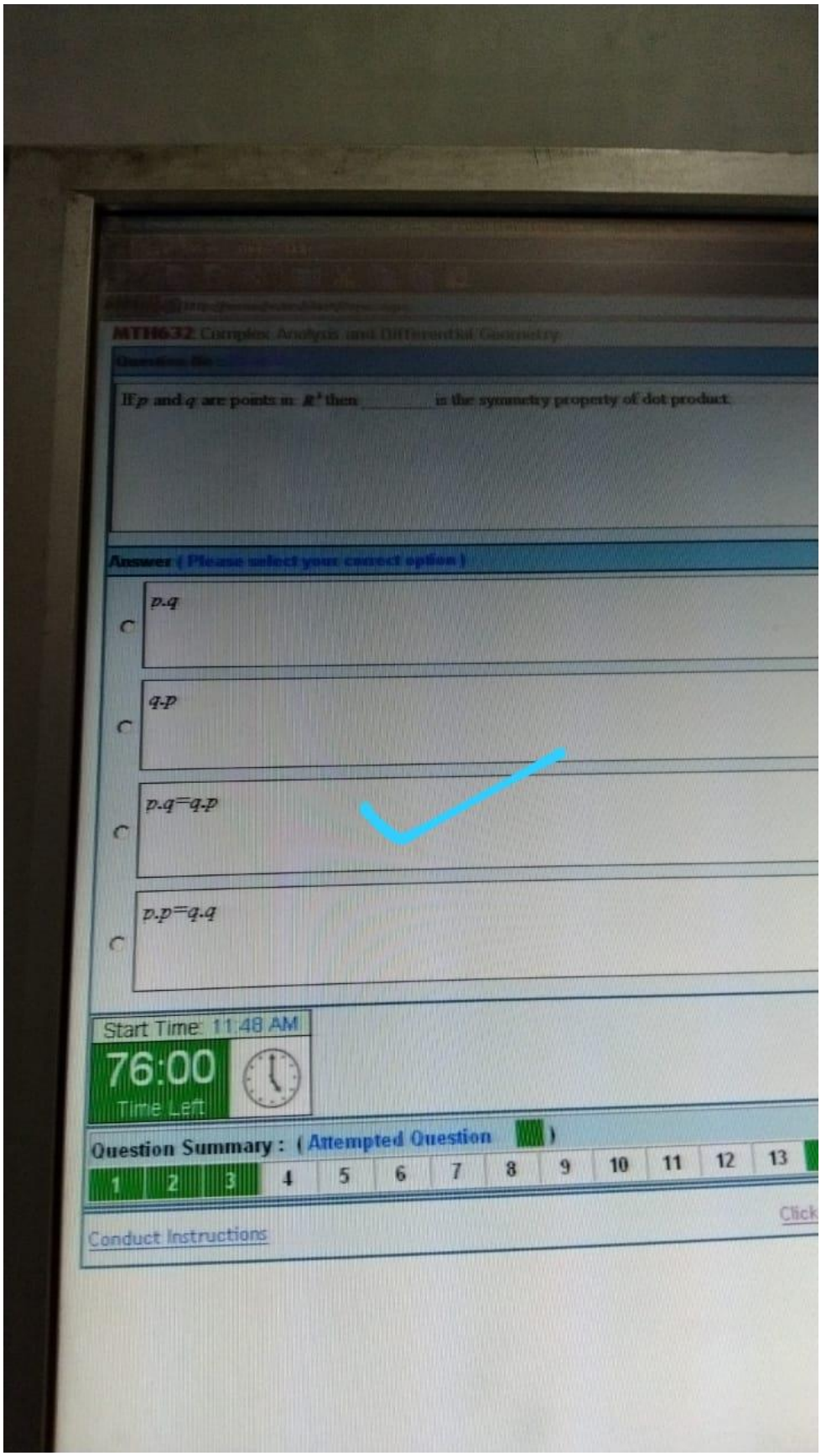
Time Left

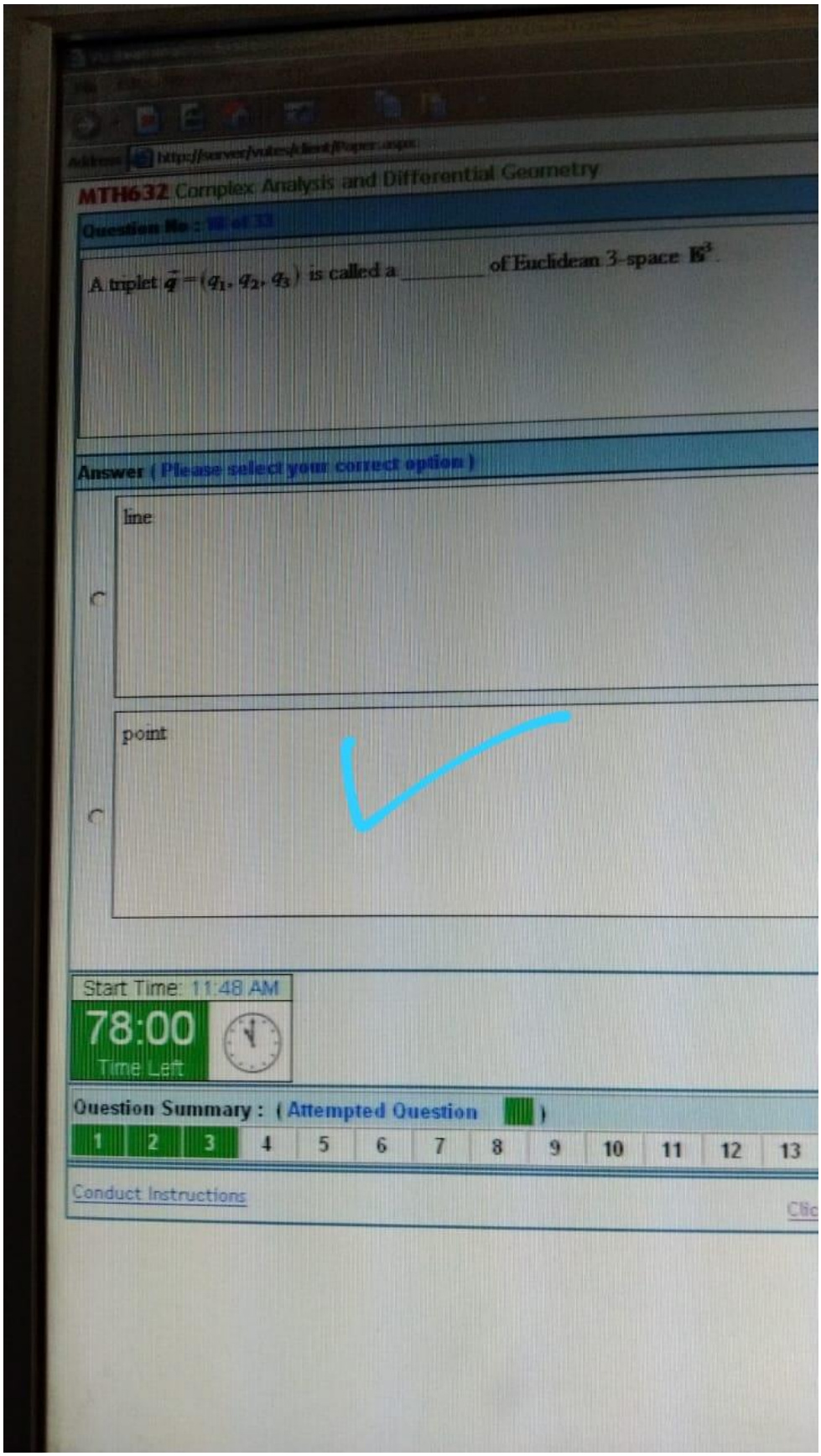
Question Summary : (Attempted Question)

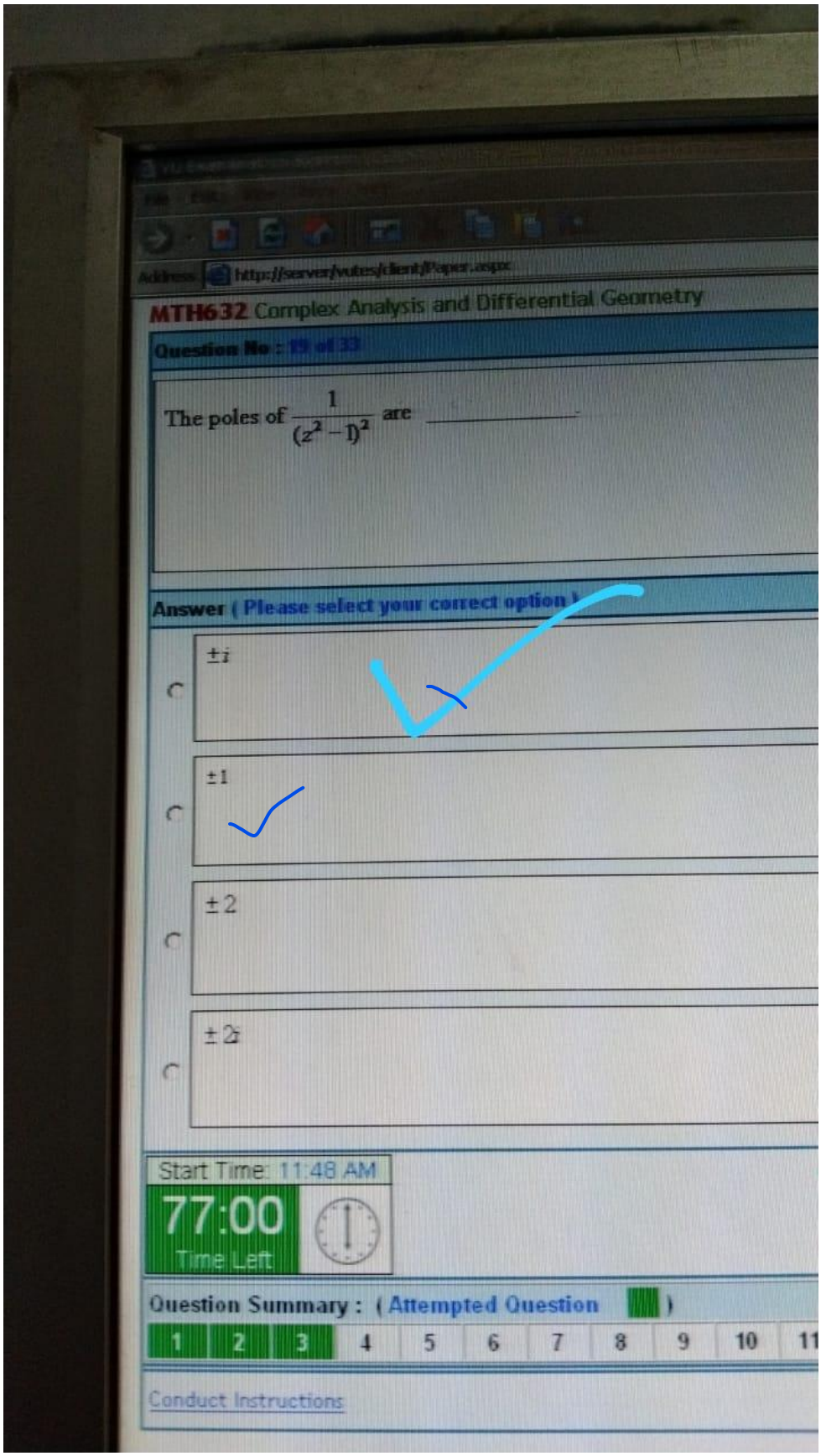
1	2	3	4	5	6	7	8	9	10

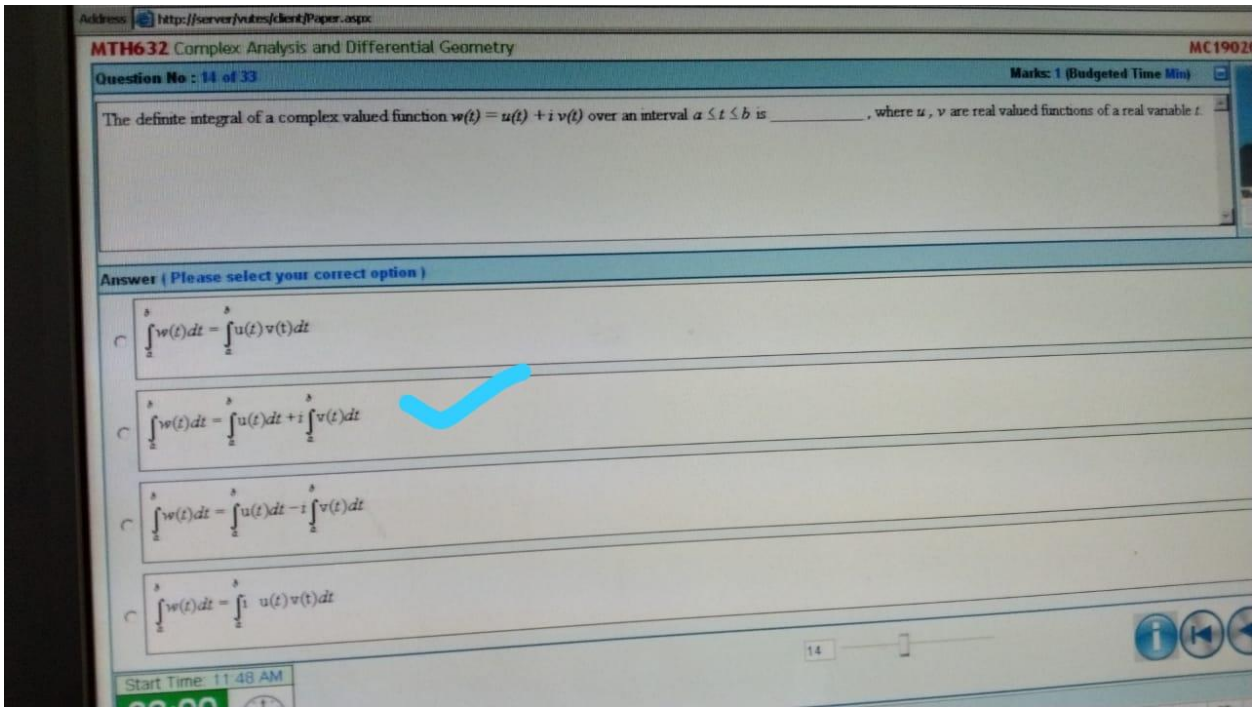
Conduct Instructions

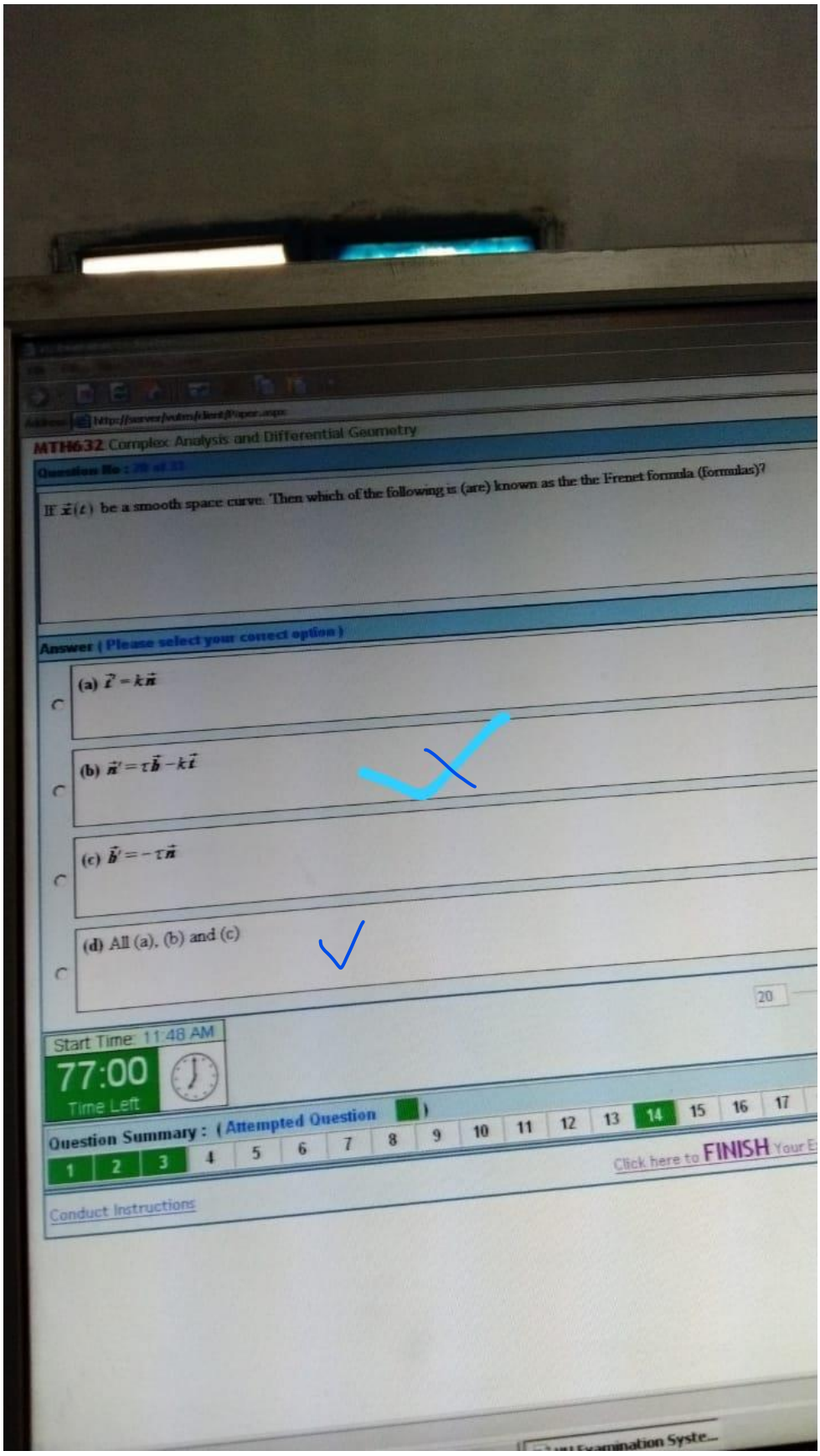

MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
Click here to
Done
Start
MathType (Lite mode) - ...
MathType (Lite mode) - ...
VU Examination Syste...

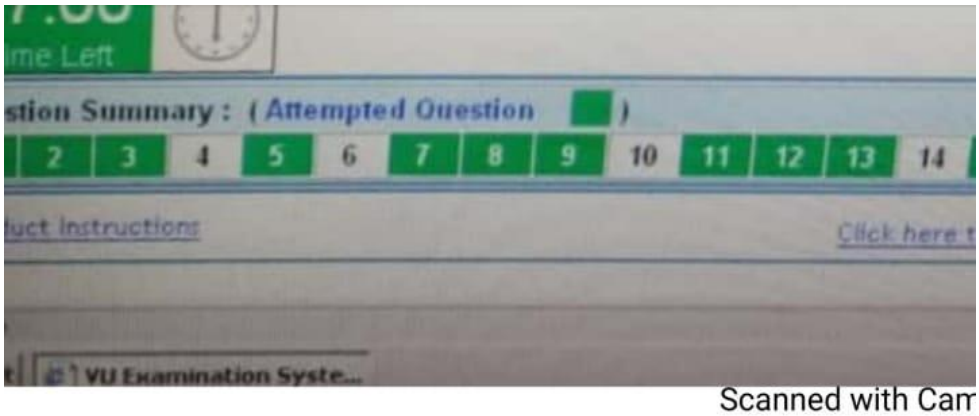

MTH632 Complex Analysis and Differential Geometry
If p and q are points in R³ then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time: 11:48 AM
78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

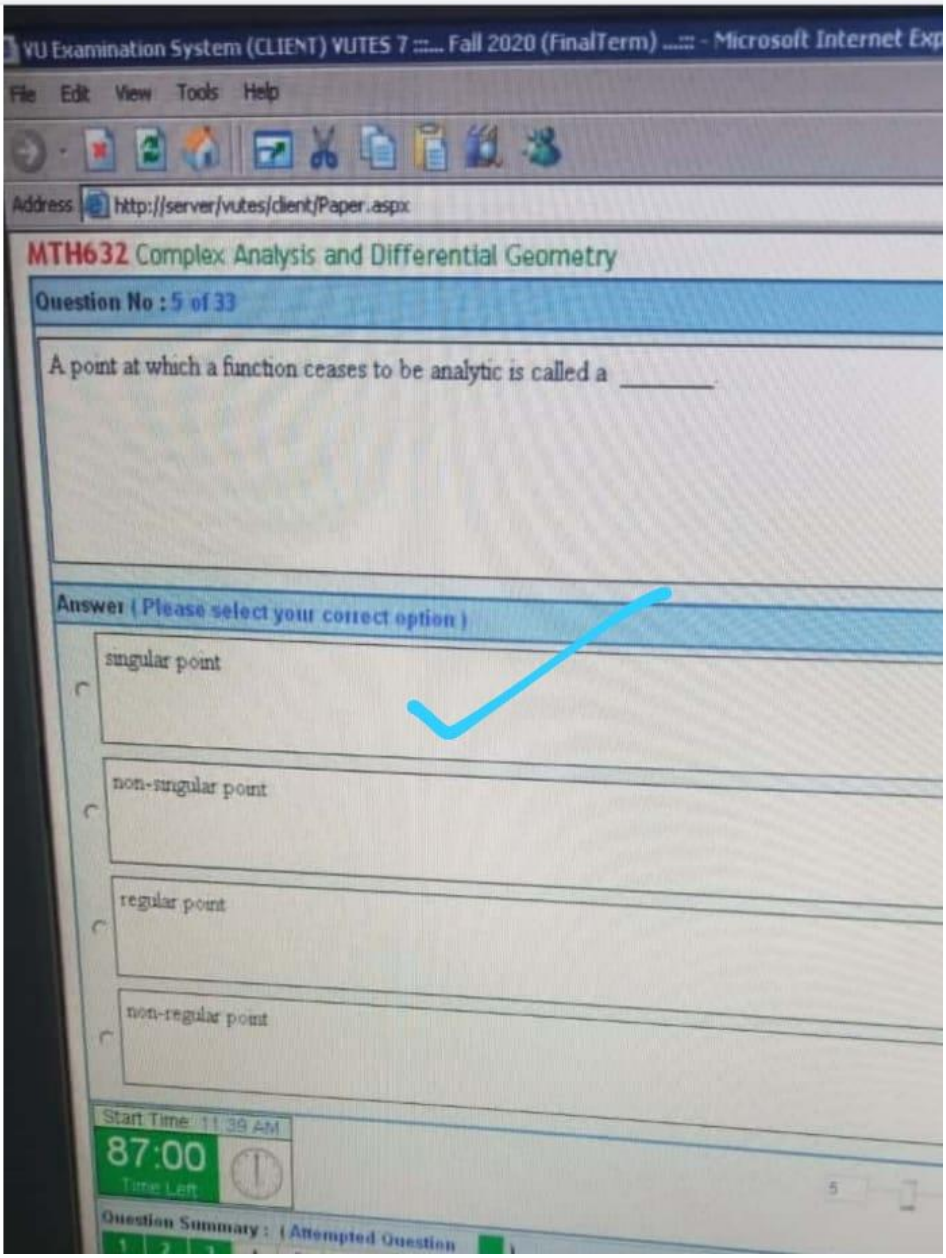

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

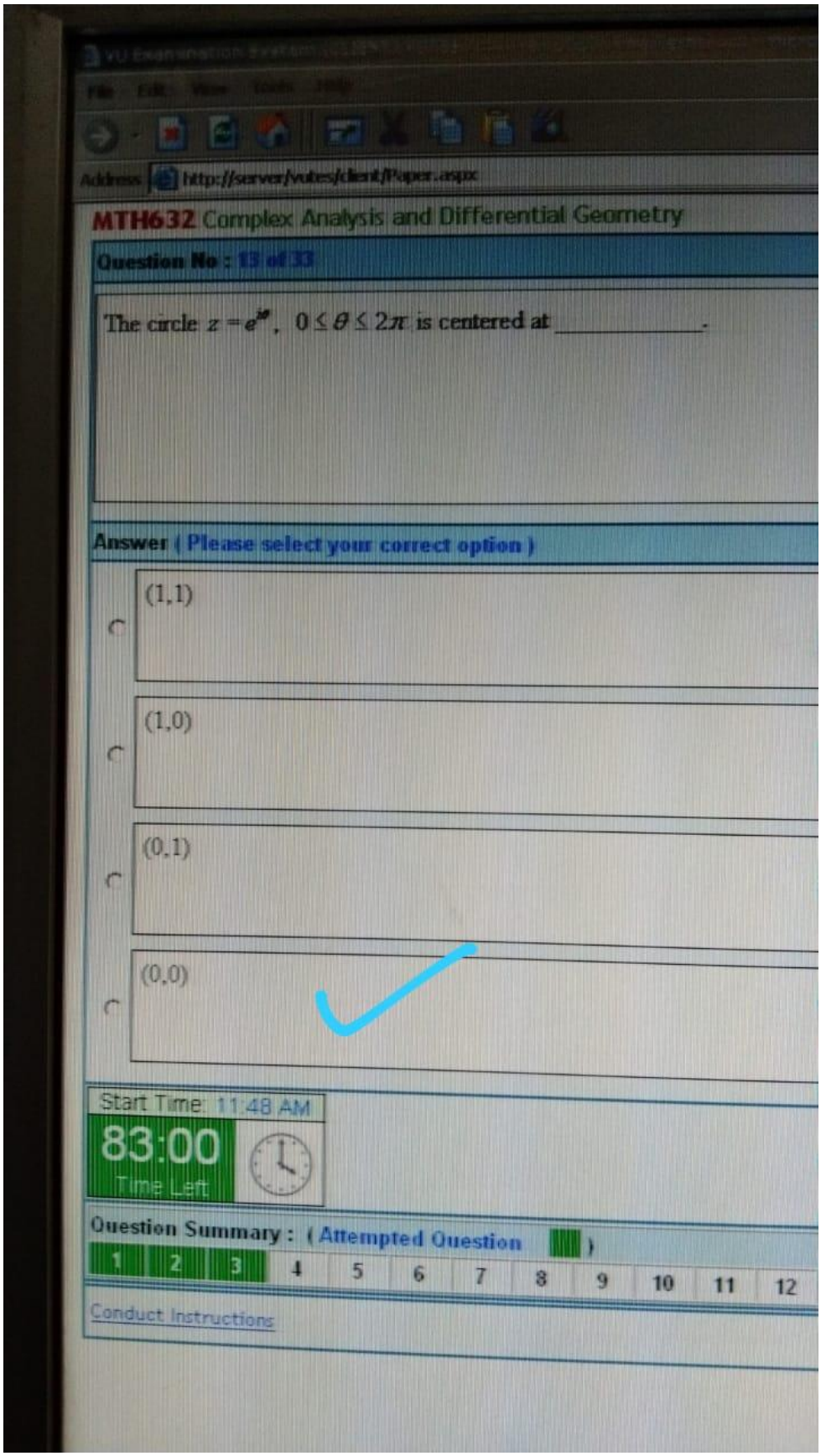

Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$
14
Start Time: 11:48 AM


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH
Conduct Instructions

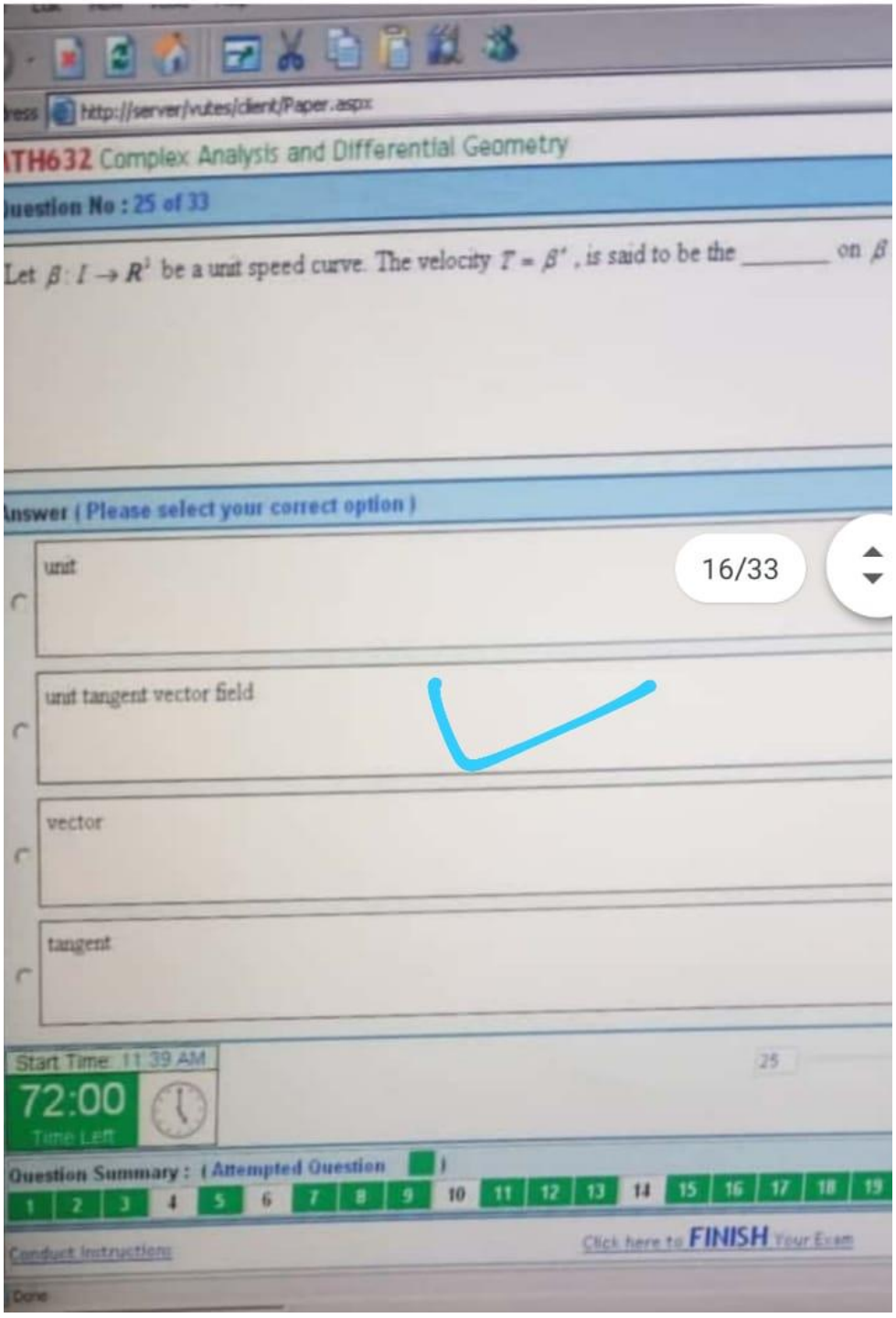


Scanned with Cam

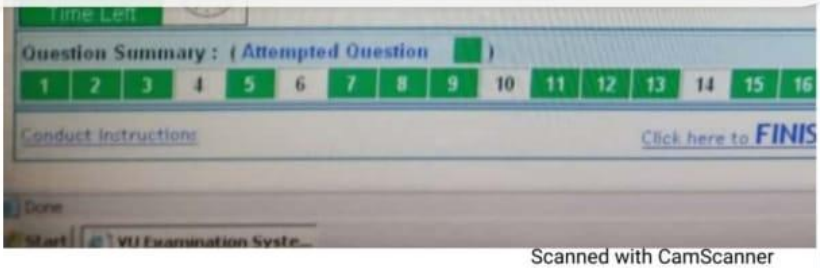



Address http://server/votes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

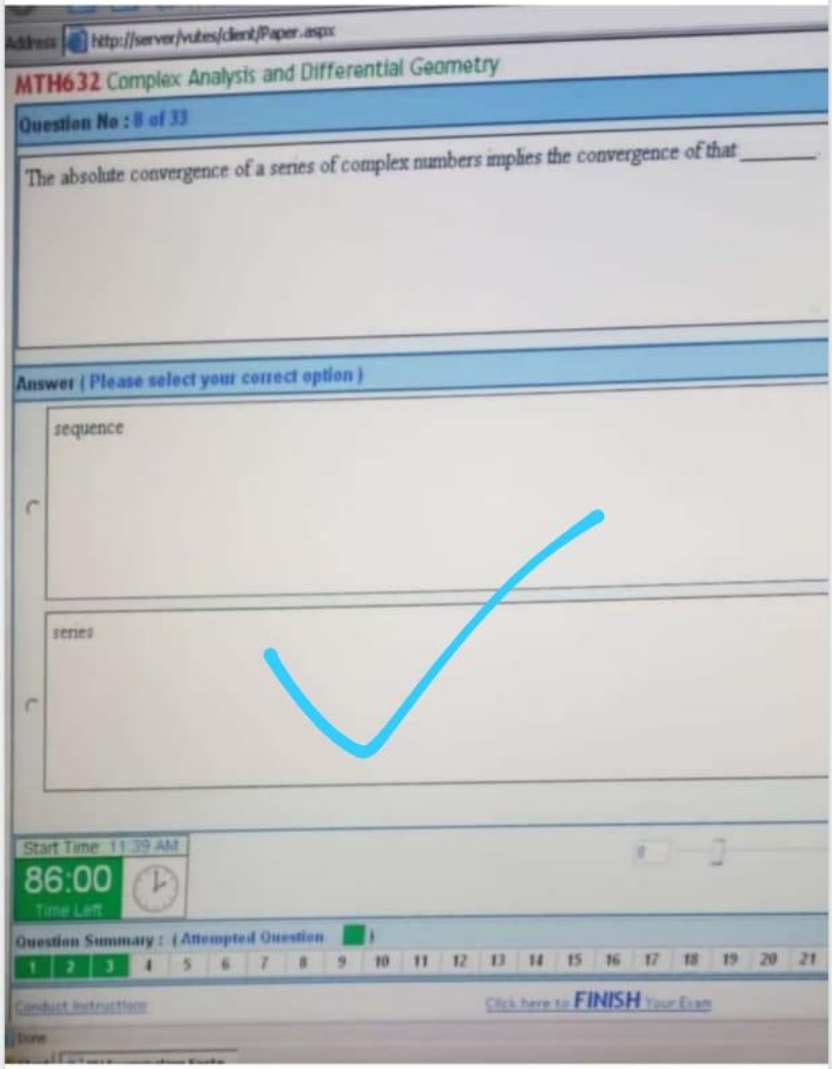

Conduct Instructions
Click here to
Done
Start
VU Examination Syste...


http://server/vutes/client/Paper.aspx
TH632 Complex Analysis and Differential Geometry
uestion No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β
nswer (Please select your correct option)
unit
16/33
unit tangent vector field
vector
tangent
Start Time: 11:39 AM
72:00
Time Left
25
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
Click here to FINISH Your Exam
Done

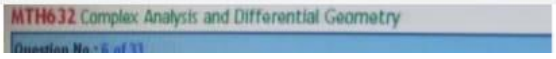
Ringtone



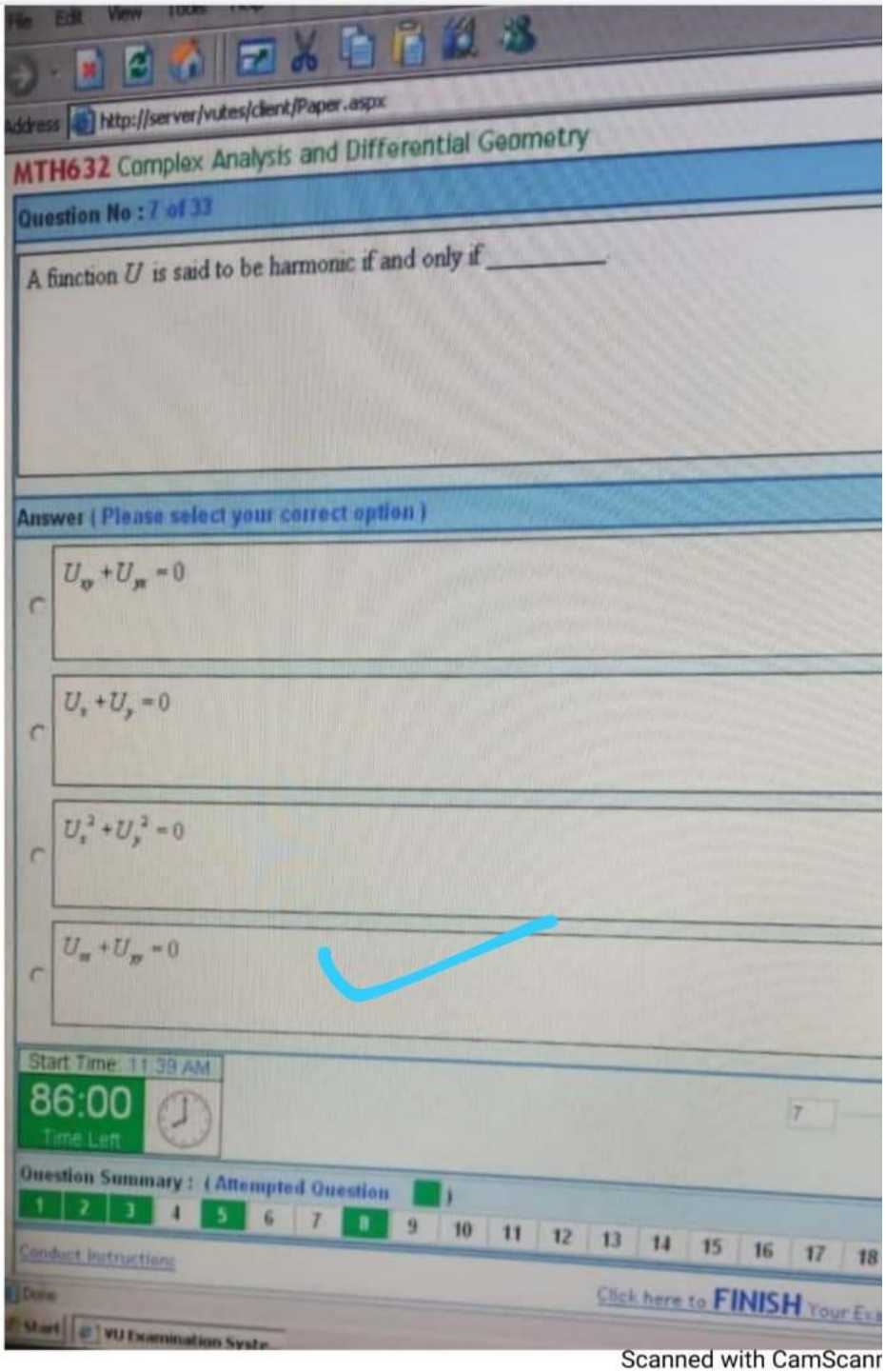
Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner



Scanned w

MTH632 Complex Analysis and Differential Geometr

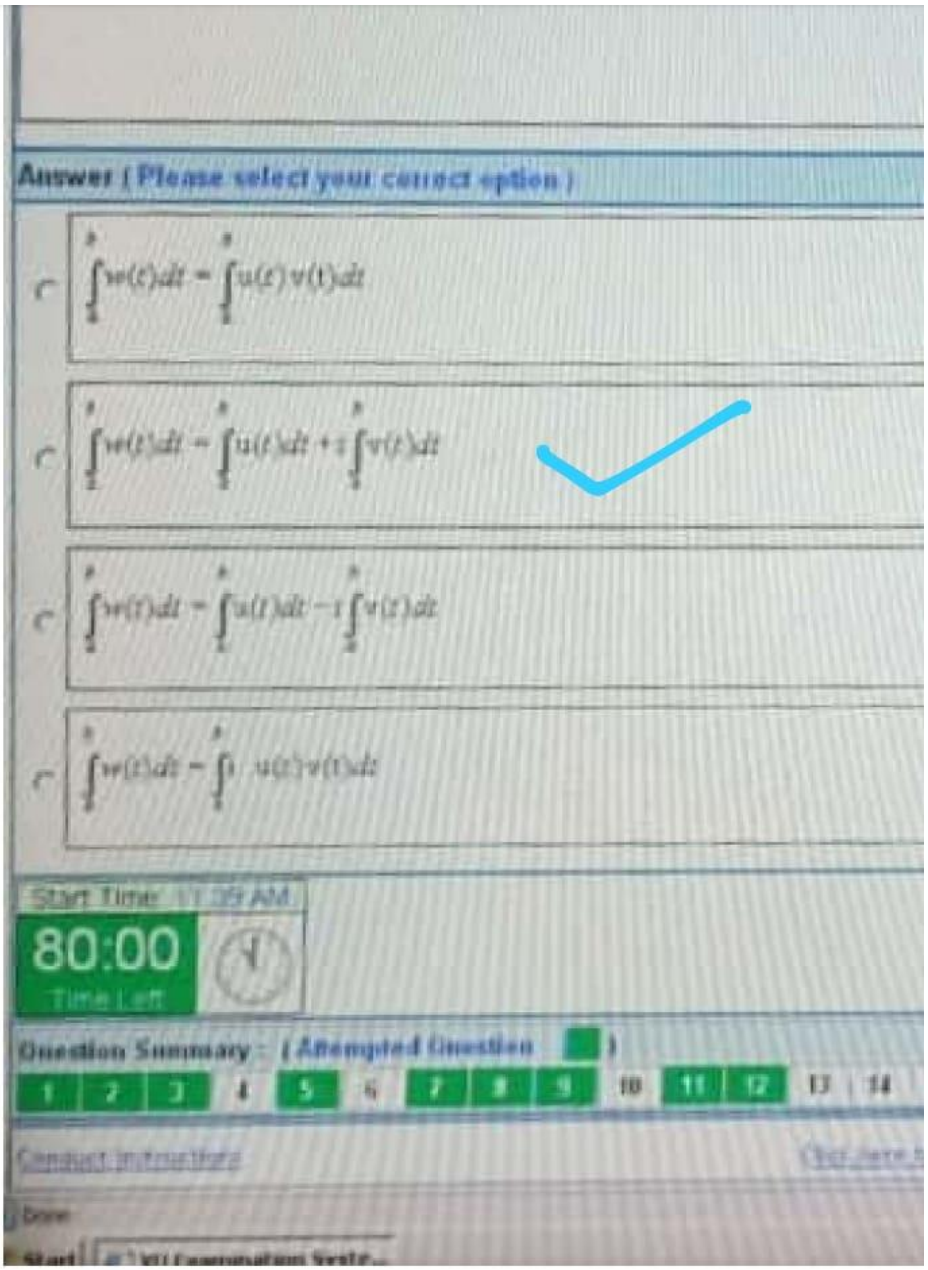
Question No : 11 of 33

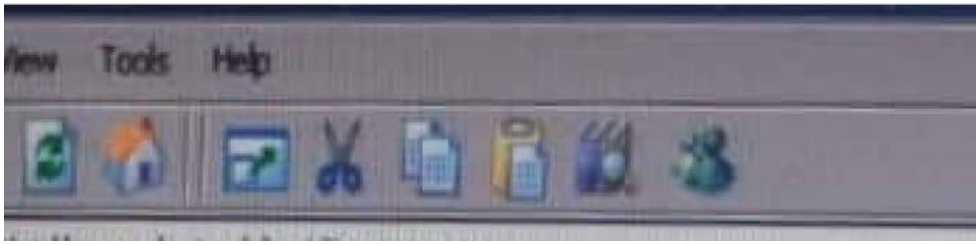
The principal value of 1^{2i} is _____.

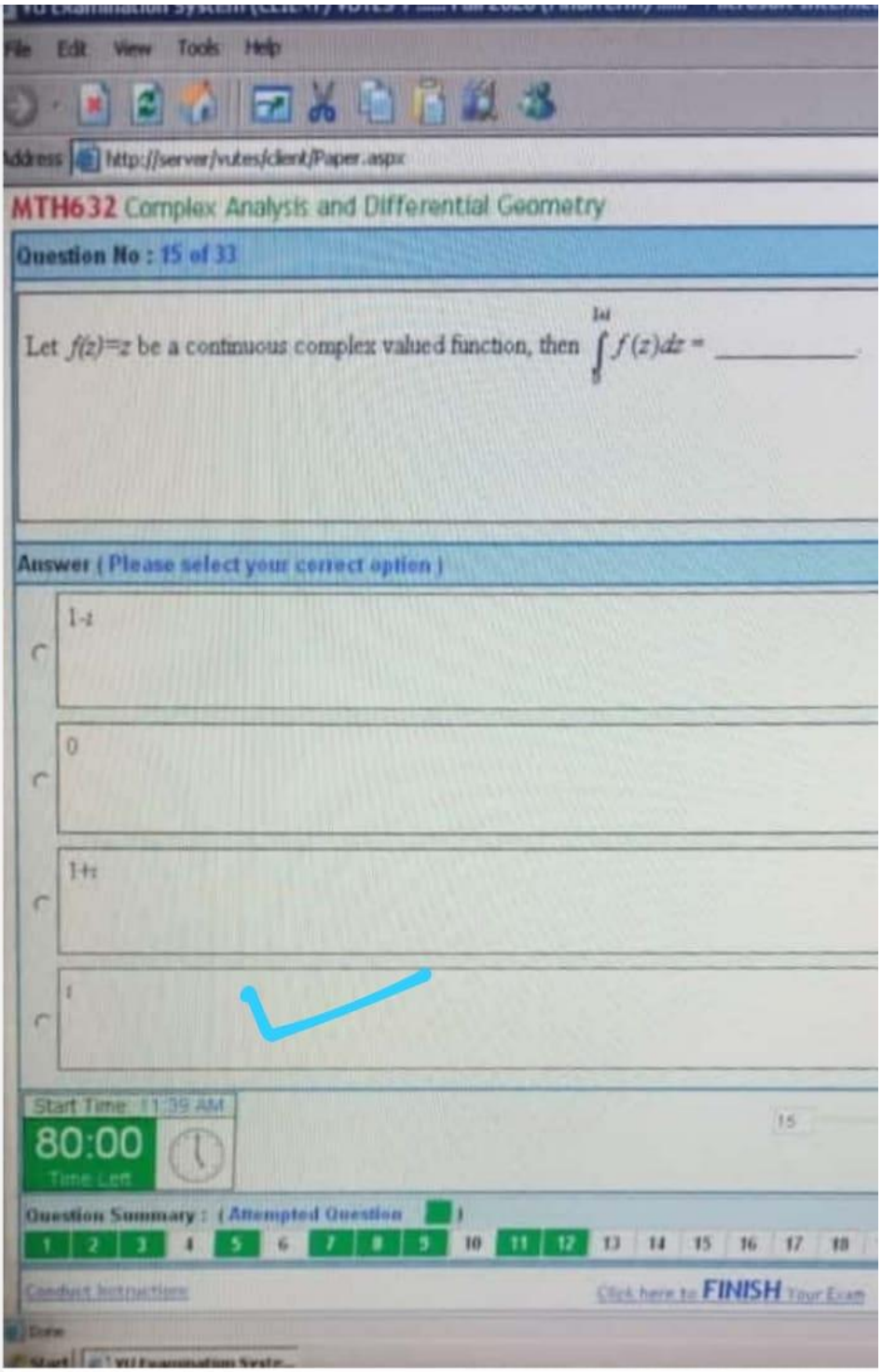
Answer (Please select your correct option)

- 0
- 1
- i
- 2i

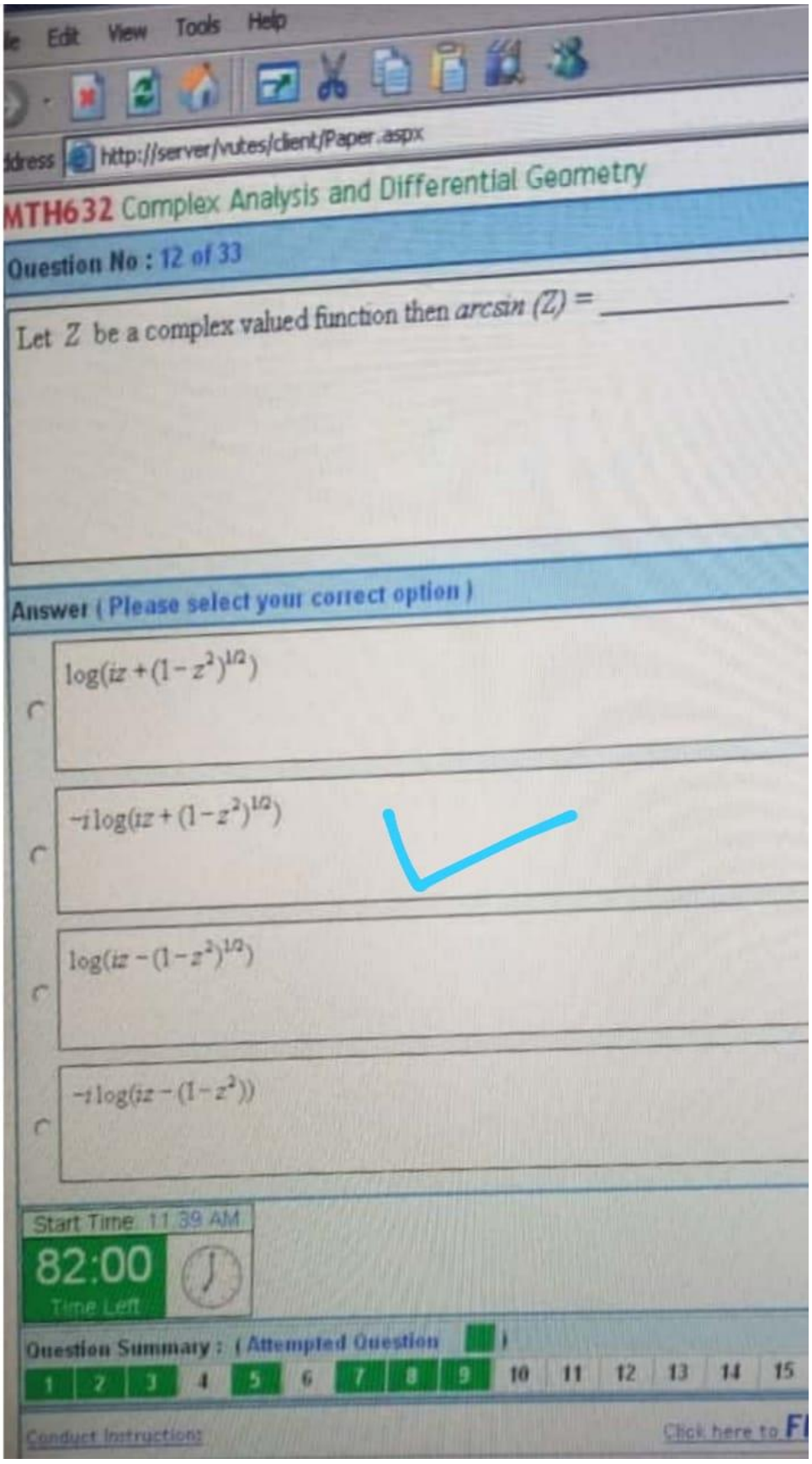
Start Time 11:39 AM


Answer (Please select your correct option)
Start Time:
80:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14


View Tools Help


File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:39 AM
80:00
Time Left
15
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
Conduct Instructions
Click here to FINISH Your Exam

Scanned w


Edit View Tools Help
http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time 11:39 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to